RESIDUAL TYPE I SEREGIN PROFILES FOR THE NAVIER–STOKES EQUATIONS: DECOMPOSITION AND EXCLUSION

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Abstract. This paper proves the literal raw-residual closure theorem required by the companion setup paper for local pointwise Type I Navier–Stokes blow-up. Starting from the setup Seregin extraction, every remaining bounded centered ancient profile with retained local velocity L^3-concentration is assigned to an ordered residual decomposition. Axisymmetric bounded-circulation profiles are excluded by a centered angular-circulation absorption argument. Rotational and stationary-hull profiles are reduced to a terminal concentration theorem. The generic terminal branch is closed by a local concentration-set decomposition, diffuse-defect and critical-tail compactifications, a minimal mesoscopic-scale rigidity theorem, exclusion of separated retained profiles, and an endpoint sequence-L^3 Liouville argument for bounded mild ancient solutions. No global critical-norm estimate or global Liouville theorem for bounded centered profiles is used.

Contents

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1. Introduction and main theorem

The incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ are written as

$$\partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \qquad \nabla \cdot u = 0. \tag{1.1}$$

The residual analysis in this paper begins after the standard Type I blow-up construction of Seregin has been performed [10]. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla)V + \nabla P = 0, \qquad \nabla \cdot V = 0, \tag{1.2}$$

on $\mathbb{R}^3_y \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative (U, Q) by

$$y = \frac{x}{\sqrt{-t}}, \qquad \tau = -\log(-t), \qquad V(y,\tau) = \sqrt{-t}\, U(x,t), \qquad P(y,\tau) = (-t)Q(x,t), \tag{1.3}$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$U(x,t) = (-t)^{-1/2} V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t < 0. \tag{1.4}$$

The purpose of this paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the companion setup paper. In this two-paper presentation, the current paper depends only on the imported setup results stated below and on cited Navier–Stokes regularity and Liouville literature [6, 10]. The rotational-flux estimates, terminal concentration-set decompositions, exclusions of local compactness failure and diffuse exterior concentration, and finite separated-profile exclusions are proved in this paper from local estimates rather than assumed as global estimates.

This paper does not reconstruct the Seregin extraction or the endpoint mildness theory. Those are imported from the setup paper. The sole new task is to prove that the residual alternative left open there is empty, namely

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

1.1. **Proof architecture.** The retained normalized residual class $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ of Definition 1.9 is the complement, inside the normalized Seregin collection $\mathcal{S}$, of the alternatives already removed in the setup paper. After the previously treated small, stationary L^3, uniformly L^3-tight, and closed structured/decay alternatives have been removed (Definition 1.8), the proof first has four analytic modules: the axisymmetric bounded-circulation module $\mathcal{C}_{\mathrm{ax}}$, the rotational relative-equilibrium module $\mathcal{C}_{\mathrm{rot}}$, the degenerate stationary-hull module $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$, and the terminal residual module. The terminal module is refined into the affine/parasitic quotient stratum $\mathcal{C}_{\mathrm{aff}}$, the log-diffuse and Young/defect critical-tail alternatives $\mathcal{C}_{\mathrm{log\,diff}}, \mathcal{C}_{\mathrm{Young}}$, the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives $\mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$, and the final ordered remainder $\mathcal{C}_{\mathrm{gen}}$. Thus the four-module description and the ten-state list in Definition 1.10 are respectively the coarse analytic and refined terminal views of the same residual argument.

The closure of these classes is organized as follows.

(1) $\mathcal{C}_{\mathrm{ax}}$ is excluded directly by a centered angular-circulation equation and an axis-absorption Liouville theorem (Theorem 4.4).
(2) $\mathcal{C}_{\mathrm{rot}}$ is reduced by a local Coriolis-flux identity to either lower classes already excluded by the setup paper or the terminal stratification assembly (Proposition 5.8); its closure is obtained only after the terminal assembly has excluded retained velocity concentration profiles.
(3) $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$ is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local velocity concentration, and is then excluded by the terminal stratification assembly theorem (Proposition 7.6).
(4) $\mathcal{C}_{\mathrm{gen}}$ and the critical-tail subclasses are closed by covariant tail normalization, concentration-set extraction, diffuse-defect compactness, critical-tail compactification, recurrence for chains of concentration profiles, exclusion of finite separated concentration profiles, indecomposable-profile sequence-L^3 completeness, parasitic-free mildness inheritance, and the Albritton–Barker endpoint Liouville theorem.

The paper distinguishes statements inherited from the setup paper, local compactness estimates and reduction results proved here, and the minimal mesoscopic-scale reduction theorem (Theorem 9.86) which excludes the critical-tail boundary defects. The affine stratum and the coherent critical-tail alternatives are closed here; the log-diffuse and Young alternatives are both reduced to the same minimal mesoscopic-scale theorem, which is then proved by pressure compactness and a bounded ancient variance Liouville argument. No global Coriolis–Leray Liouville theorem for bounded profiles or global critical-norm estimate is used in the residual closure. A short reader's guide is given in Section 1.3 below.

Definition 1.1 (Type I convention used in the chain)**.** In this paper and in the final two-paper chain, a terminal singular point $z_* = (x_*, T)$ is called *local pointwise Type I* if there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_*))} \le M.$$

This is the local pointwise Type I hypothesis used in the setup paper and in Section 1.9. For suitable weak solutions this envelope lies in the endpoint weak Serrin class $L^{2,\infty}_t L^\infty_x$. The Albritton–Barker weak-Serrin local Type I estimate, recorded below as Theorem 1.26, upgrades it on smaller cylinders to the scale-invariant local energy bounds needed for terminal velocity no-escape. Thus the terminal no-escape condition used by the setup paper is automatic for the local pointwise Type I singularities considered here. The paper does not use the phrase "Type I" to mean an unrelated global L^∞, local CKN, weak-L^3, or Serrin scale condition unless that condition is explicitly stated.

Remark 1.2 (Translation table for standard Type I hypotheses). The global pointwise condition

$$\sup_{t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty$$

implies Definition 1.1 at every terminal point. Any stronger local criterion that yields the displayed L^∞ envelope on a neighborhood of x_* enters the same argument, provided the solution is suitable in the corresponding cylinder. The endpoint weak Serrin estimate then gives the local scale-invariant energy bounds and terminal no-escape. Purely CKN-type, weak-L^3, or other critical criteria are not silently identified with this hypothesis; they require their own implication into the local pointwise envelope, or directly into the local Type I estimates, before the present exclusion applies.

1.2. **Imported setup results and the residual target.**

Theorem 1.3 (Imported setup results)**.** *Fix a finite-energy suitable weak solution (u,p), a local pointwise Type I singular point z_*, and normalization data (R_0,η_*) from an admissible setup extraction at z_*. Every statement below is instantiated at this fixed singular point and extraction. Whenever a normalized profile is mentioned, it is the current profile selected from $\mathcal{S}_{\mathrm{raw\text{-}gen}}(u,p,z_*;R_0,\eta_*)$ and carries this fixed setup ledger. The companion setup paper proves the following facts, in the exact theorem numbers listed below.*

(1) *The fixed singular point z_* supplies positive terminal local velocity concentration.*
  Theorem 5.7 of the setup paper
  Corollary 5.8 of the setup paper
(2) *The local pointwise Type I control at z_* gives the fixed admissible Seregin extraction sequence with pressure gauges, no-escape, and retained unit-scale velocity concentration.*
  Theorem 8.4 of the setup paper
  Corollary 8.5 of the setup paper
  Lemma 8.7 of the setup paper
(3) *The normalized profile selected from this fixed extraction is a bounded ancient suitable weak solution and carries the pressure atlas of the setup paper. If this profile lies in the setup mild stratum, its projected mild identity is stable under the locally smooth limiting operations used along its realized branch and gives the finite-shift physical pullback used at the endpoint.*
  Theorem 9.1 of the setup paper
  Proposition 11.12 of the setup paper
  Proposition 12.2 of the setup paper
  Lemma 11.2 of the setup paper
  Lemma 11.3 of the setup paper
  Lemma 11.4 of the setup paper
  Definition 13.14 of the setup paper
  Lemma 13.17 of the setup paper
  Lemma 15.3 of the setup paper
  Lemma 20.4 of the setup paper
  Lemma 25.4 of the setup paper
(4) *If the current profile is selected into the small, stationary L^3, uniformly L^3-tight, or known structured/decay state, the corresponding setup theorem closes that state for this profile.*
  Theorem 15.7 of the setup paper
  Theorem 15.8 of the setup paper

Theorem 15.9 of the setup paper
Theorem 26.1 of the setup paper
Theorem 25.8 of the setup paper

(5) *For a realized terminal profile whose local route has produced the mild sequence-L^3 interface, the endpoint result supplies the final contradiction with that profile's retained local velocity concentration.*
Theorem 25.5 of the setup paper

The purpose of the present paper is to prove the literal raw-residual closure

$$\mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big) = \varnothing.$$

For an arbitrary incoming profile V carrying the fixed setup ledger, the proof establishes the implication

$$V \in \mathcal{R}_{\text{setup}}(\mathcal{S}) \quad \Longrightarrow \quad \perp$$

by routing the antecedent through the affine or retained row and closing the selected row. Thus residual membership is only the antecedent of the contrapositive argument: if it is false there is nothing to prove for V, and if it is true the exhaustive routing yields a contradiction. No residual profile is assumed to exist, and the setup paper's residual set is not replaced by a different object.

Definition 1.4 (Exact setup-to-residual handoff)**.** Fix a finite-energy suitable weak solution, a terminal point z_*, and normalization data (R_0, η_*) obtained from an admissible setup extraction. In this paper

$$\mathcal{S} := \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)$$

denotes the same raw generated descendant hull as in Theorem 16.1 of the setup paper. It is not a newly generated state space. Let

$$I_{\text{setup}} := \{\text{small}, \text{stat-}L^3, L^3\text{-tight}, \text{sd}\}$$

record the four setup-paper lower classes, and let

$$J_{\text{res}} := \{\text{ax}, \text{rot}, \text{stat-hull}, \text{aff}, \text{log diff}, \text{Young}, \text{homcrit}, \text{logper}, \text{ap}, \text{gen}\}$$

record the ordered refinement used here. The indices are bookkeeping data; they impose no additional analytic condition on $\mathcal{S}$.

The setup residual set

$$\mathcal{R}_{\text{setup}}(\mathcal{S}) := \mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big)$$

is the complement of the four classes indexed by I_{setup}. The notation $\mathcal{R}_{\text{pre}}(\mathcal{S}; I_{\text{setup}}, J_{\text{res}})$ denotes this same set at the entry to Paper II. Its canonical affine/parasitic normalization has two ordered outcomes: the candidate is closed by routing to the affine/parasitic row $\mathcal{C}_{\text{aff}}$, or it produces a retained canonical representative in $\mathcal{R}^{\#}(\mathcal{S}; I_{\text{setup}}, J_{\text{res}})$. Thus the superscript $\#$ records the representative convention used after the exact setup handoff; it does not replace the underlying raw generated hull or its normalization data. In particular, the object to be proved empty remains $\mathcal{R}_{\text{setup}}(\mathcal{S})$, exactly as in Theorem 16.1 of the setup paper.

Proposition 1.5 (Completeness of the setup residual handoff)**.** *Suppose, contrapositively, that $V \in \mathcal{R}_{\text{setup}}(\mathcal{S})$. The canonical affine normalization assigns this same candidate and its canonical representative to exactly one of the two ordered rows*

$$\mathcal{C}_{\text{aff}}, \qquad \mathcal{R}^{\#}(\mathcal{S}; I_{\text{setup}}, J_{\text{res}}).$$

In the first row, the canonical quotient assigns the candidate to the already closed affine/parasitic lower stratum. This closes the incoming branch; it is not a claim that an unnormalized homogeneous mode loses its local mass. In the second row the centered profile, the fixed retained compact velocity obstruction, the pressure atlas, the normalization data (R_0, η_*), and the setup ancestry *are unchanged. Conversely, every retained representative in $\mathcal{R}^{\#}(\mathcal{S}; I_{\mathrm{setup}}, J_{\mathrm{res}})$ has an ancestor in $\mathcal{R}_{\mathrm{setup}}(\mathcal{S})$. Consequently, if the affine alternative is closed and the retained row is empty, then no such V exists and hence*

$$\mathcal{R}_{\mathrm{setup}}(\mathcal{S}) = \varnothing.$$

Proof. The first assertion is the canonical affine-normalization dichotomy in Theorems 8.21 and 9.99, applied after the four setup lower classes have been removed. The affine alternative is routed to the already closed affine/parasitic lower row by Definition 8.16, Remark 8.17, and Theorem 9.99; otherwise the canonical representative retains positive non-affine activity and belongs to the retained residual class. The affine operation is an exact centered symmetry or the associated homogeneous quotient operation, so it does not create a new raw state space or discard the ancestry of the candidate under the antecedent. For the retained alternative, the root compact velocity lower bound, pressure atlas, and fixed normalization data are those of the setup candidate. Their transport under later realized operations is Theorems 1.13 and 10.12. The converse follows from the definition of the retained class as the canonical representative of a pre-quotient setup residual candidate. Since the two alternatives exhaust the affine-normalization dichotomy, closing both leaves no possible element of the pre-quotient set. This is precisely the displayed raw-residual conclusion. □

Table 1. Setup-paper export interface for the fixed singular point, extraction, and ancestry ledger used by the residual argument. The first column gives the exact setup label; imported producers appear first and the final downstream consumers appear in the last row. The third column identifies the Paper II statement that records, consumes, or supplies the corresponding local interface.

Setup label	**Exported datum or consumer**	**Paper II consumer**	**Transport rule**
Theorem 5.7 of the setup paper; Corollary 5.8 of the setup paper	The fixed singular point z_* supplies the selected sequence of scales with positive local scale-invariant concentration and a fixed positive velocity lower bound.	The physical entry is recorded in Definition 1.1 and Section 1.9; Theorem 1.7 records the retained compact velocity mass after extraction.	Converted by admissible extraction into the fixed compact velocity obstruction.
Theorem 8.4 of the setup paper; Corollary 8.5 of the setup paper; Lemma 8.7 of the setup paper	The local pointwise Type I envelope at z_* gives terminal velocity no-escape. The selected positive concentration sequence is admissible and yields the current smooth bounded centered Seregin profile with nonzero compact local L^3 mass.	Theorem 1.7 is the public residual-entry statement; Definition 1.9 then selects the residual part of the generated Seregin collection.	The extracted profile and its generating sequence form the root residual state.
Theorem 9.1 of the setup paper; Proposition 11.12 of the setup paper; Proposition 12.2 of the setup paper; Lemma 11.2 of the setup paper; Lemma 11.3 of the setup paper; Lemma 11.4 of the setup paper	The normalized profile selected from the fixed extraction is a bounded ancient suitable solution, is nonzero by retained velocity mass, and carries compatible local pressure representatives modulo the permitted gauges.	Theorems 1.7 and 1.13 record the root state and its realized limiting operations; Lemma 8.15 records the pressure interface used by the local descendants.	Velocity retention is transported by descendant heredity; pressure is transported chartwise by the named atlas lemmas.

Setup label	**Exported datum or consumer**	**Paper II consumer**	**Transport rule**
Definition 13.14 of the setup paper; Lemma 13.17 of the setup paper; Lemma 15.3 of the setup paper; Lemma 20.4 of the setup paper; Lemma 25.4 of the setup paper	If the current profile lies in $\mathcal{S}_{\text{mild}}$, its projected mild identity is stable under the locally smooth limiting operations along its realized branch, and its physical pullback is mild after each finite terminal shift used at the endpoint.	Theorem 10.8 is the local descendant interface used in Lemma 10.10; it applies after the current profile has entered the mild branch and the named inheritance theorem has transported that interface.	Transported only along the mild branch and then by the named local inheritance theorem after affine routing.
Theorem 15.7 of the setup paper; Theorem 15.8 of the setup paper; Theorem 15.9 of the setup paper; Theorem 26.1 of the setup paper; Theorem 25.8 of the setup paper	If the current fixed-ledger profile is selected into the small, stationary L^3, uniformly L^3-tight, or closed structured/decay state, the corresponding setup theorem closes that state for this profile.	Definition 1.8 records these local closures, and Definition 1.9 enters the residual route only when none is selected for the current profile.	Later reductions of the same profile or a realized descendant may return to a named closed lower state.
Theorem 25.5 of the setup paper	For a realized terminal profile whose local route has produced finite-shift mildness and sequence-L^3, the setup paper supplies the endpoint input.	The endpoint input is recorded in Theorem 10.9 and is invoked only for that terminal profile after Lemma 10.6 and Theorem 10.8 have produced its hypotheses.	Contradiction with that profile's retained local velocity concentration.
Theorem 16.1 of the setup paper; Theorem 33.1 of the setup paper	These are downstream setup-paper consumers, not imported premises of the residual proof.	Theorem 12.1 and Corollary 12.2 prove that exact local residual statement, and Corollary 12.3 invokes the resulting assembly.	Final return to the setup paper.

Remark 1.6 (Use of imported labels). Below, the phrase "by the imported setup results" means that the terminal profile is bounded, locally suitable, pressure-normalized by the setup pressure atlas, and carries the setup ancestry and retained normalization data. When a mild representative is used, membership in the setup mild stratum or the local inheritance theorem is cited separately. The exact producers and consumers are listed in Theorem 1.3 and Table 1. The affine/parasitic quotient, descendant realization, and terminal sequence-L^3 property are established in this paper by their named local results. Paper II closes the affine routing row and proves $\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$. Applied conditionally to an arbitrary incoming profile carrying the fixed ledger, the exact-handoff proposition turns these exhaustive local closures into the literal conclusion $\mathcal{R}_{\text{setup}}(\mathcal{S}) = \varnothing$.

(1) Local concentration and terminal no-escape entry: positive local concentration depends on CKN theory and the setup paper; terminal no-escape follows from the endpoint weak Serrin local Type I bounds in Theorems 1.26 and 1.28.
(2) Seregin extraction is imported from the setup paper. Affine/parasitic normalization and its descendant interfaces are proved in this paper.
(3) Small and stationary L^3 strata: handled in the setup paper.
(4) Uniformly tight stratum: handled in the setup paper.
(5) Closed structured/decay strata: handled in the setup paper.
(6) Axisymmetric and rotational residual strata: proved in this paper.

(7) Terminal concentration-set decomposition, recurrence for concentration chains, and diffuse-defect compactness: proved in this paper.
(8) Critical-tail compactification and realized critical-tail decomposition: proved in this paper.
(9) Residual indecomposable-profile closure: proved in this paper, with the final sequence-L^3 endpoint supplied by Albritton–Barker.

The complete entry-to-closure graph is given in the numbered proof-flow guide in Section 1.3. That graph separates state selection from theorem production and displays the terminal alternatives individually.

Theorem 1.7 (Admissible Type I Seregin extraction theorem)**.** *An admissible finite-energy Type I singular point produces a normalized Seregin ancient limit* (V, P) *satisfying* (1.2)*, with*

$$\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le M < \infty,$$

local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3\times\mathbb{R}$ *for which*

$$\iint_K |V|^3 \, dy \, d\tau \ge \eta_0 > 0. \tag{1.5}$$

The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$*.*

Proof. This is the base admissible Seregin extraction theorem included in Theorem 1.3: velocity nonvanishing, local suitability, and local pressure compactness are imported setup conclusions. The present paper uses this inherited reduction and does not reprove the Type I blow-up compactness theorem. □

Definition 1.8 (Earlier alternatives inherited from the setup paper)**.** Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary L^3, uniformly L^3-tight, and closed structured/decay classes are the non-residual alternatives already established in the setup paper. The small-amplitude and stationary L^3 alternatives are closed there; the uniformly tight class is excluded there; and the closed structured/decay class is precisely the part proved empty there. Any structured profile not belonging to that closed class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

Definition 1.9 (Coarse residual class)**.** Let $\mathcal{S}$ be the setup hull fixed in Definition 1.4, and put $I = I_{\mathrm{setup}}$, $J = J_{\mathrm{res}}$. The pre-quotient coarse residual class

$$\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$$

is exactly $\mathcal{R}_{\mathrm{setup}}(\mathcal{S})$, the complement in $\mathcal{S}$ of the already removed alternatives in Definition 1.8. The retained normalized residual class

$$\mathcal{R}^{\#}(\mathcal{S}; I, J)$$

is obtained from $\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$ by passing to the canonical affine/parasitic representative and by removing any representative assigned to the affine/parasitic lower stratum. Unless the word unreduced is used explicitly, $\mathcal{R}(\mathcal{S}; I, J)$ denotes this retained normalized residual class. Thus an "empty affine/parasitic stratum" below always means that no retained normalized residual representative remains in that quotient stratum.

Definition 1.10 (Refined residual decomposition)**.** The refined residual decomposition is an ordered definition, not a separate compactness theorem. Starting with the pre-quotient coarse residual class $\mathcal{R}_{\mathrm{pre}}(\mathcal{S}; I, J)$, one removes the following alternatives in order; the affine/parasitic alternative is then recorded as a quotient stratum with no retained normalized representative in $\mathcal{R}^{\#}(\mathcal{S}; I, J)$. After the first nine alternatives have been removed, whatever remains is by definition the generic terminal residual $\mathcal{C}_{\mathrm{gen}}$:

$$\mathcal{C}_{\mathrm{ax}}, \qquad \mathcal{C}_{\mathrm{rot}}, \qquad \mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}, \qquad \mathcal{C}_{\mathrm{aff}},$$
$$\mathcal{C}_{\mathrm{log\,diff}}, \qquad \mathcal{C}_{\mathrm{Young}}, \qquad \mathcal{C}_{\mathrm{homcrit}}, \qquad \mathcal{C}_{\mathrm{logper}}, \qquad \mathcal{C}_{\mathrm{ap}}, \qquad \mathcal{C}_{\mathrm{gen}}.$$

Here $\mathcal{C}_{\mathrm{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\mathrm{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$ is the degenerate stationary-hull class with retained local velocity concentration, $\mathcal{C}_{\mathrm{aff}}$ is the affine/parasitic lower stratum selected by affine-normalized oscillatory entry, $\mathcal{C}_{\mathrm{log\,diff}}$ and $\mathcal{C}_{\mathrm{Young}}$ are respectively the log-diffuse and Young/defect critical-tail alternatives, $\mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$ are the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives, and $\mathcal{C}_{\mathrm{gen}}$ is the generic terminal residual class. The inclusion below is part of this ordered definition, written for the retained normalized residual class; the appearance of $\mathcal{C}_{\mathrm{aff}}$ records the quotient alternative before retained normalization, and $\mathcal{C}_{\mathrm{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$ is proved in Theorem 9.99. The substantive content is the closure of each non-quotient alternative, especially $\mathcal{C}_{\mathrm{gen}}$. Equivalently,

$$\mathcal{R}(\mathcal{S}; I, J) \subset \mathcal{C}_{\mathrm{ax}} \cup \mathcal{C}_{\mathrm{rot}} \cup \mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{log\,diff}} \cup \mathcal{C}_{\mathrm{Young}} \cup \mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} \cup \mathcal{C}_{\mathrm{gen}}. \tag{1.6}$$

The paper uses five complementary views of this definition. The entry view connects a physical local pointwise Type I singularity to the residual Seregin state. The coarse analytic view separates the three structural modules from the terminal module. The refined terminal view gives the ten states in Definition 1.10. The dependency view in Tables 4 and 16 records theorem production order. The closure view in Section 12 records the theorem that discharges each state. The routing interface between these views is summarized in Table 2.

TABLE 2. Routing interface for the refined residual states. The obstruction column records the datum transported from the current state; a normalized support profile is identified separately from a retained velocity descendant.

State	Selecting criterion	Local producer and obstruction	Next route or closure
$\mathcal{C}_{\mathrm{ax}}$	Axisymmetry with bounded centered circulation after the setup lower classes have been removed.	Theorem 4.4; the imported compact velocity obstruction and residual ancestry remain attached to the profile.	Direct axis-absorption closure, with the no-swirl output in an already closed structured class.
$\mathcal{C}_{\mathrm{rot}}$	Stationarity modulo a rotation, hence a local Coriolis transport structure.	Proposition 5.8; retained local velocity concentration in the co-rotating frame.	An imported lower closure or a realized terminal concentration profile; Corollary 10.68 closes the class after terminal assembly.
$\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$	A degenerate stationary hull contains a sequence-realized stationary element with retained velocity concentration.	Propositions 6.12 and 7.6; stationary-hull ancestry and retained velocity concentration.	An imported lower closure or a realized stationary terminal profile; Corollary 10.69 closes the class after terminal assembly.
$\mathcal{C}_{\mathrm{aff}}$	Affine normalization removes all non-affine oscillation from the selected representative.	Theorems 8.21 and 9.99; the affine output is a quotient state, while the retained branch keeps positive non-affine activity.	Routing to the affine/parasitic lower stratum; no retained normalized representative remains in this row.
$\mathcal{C}_{\mathrm{log\,diff}}$	The realized critical-tail compactification has a nonzero logarithmic escape component.	Theorem 9.56; a normalized log-window support profile carrying positive defect support.	Hidden-scale reduction together with the already closed coherent-tail rows; Corollary 9.57.
$\mathcal{C}_{\mathrm{Young}}$	A Young/defect critical-tail coordinate survives, including the non-Dirac variance alternative.	Theorems 9.86 and 9.88; a normalized defect support profile.	Minimal mesoscopic reduction and defect-coordinate closure; Corollary 9.100.

State	Selecting criterion	Local producer and obstruction	Next route or closure
$\mathcal{C}_{\mathrm{homcrit}}$	A realized coherent tail is (-1)-homogeneous.	Theorem 9.104; a realized coherent support profile.	Bounded-origin incompatibility; Corollary 9.111.
$\mathcal{C}_{\mathrm{logper}}$	A realized coherent logarithmic tail has a nonzero log period.	Theorem 9.104; a realized coherent support profile.	Bounded-origin decay and log-periodicity; Corollary 9.111.
$\mathcal{C}_{\mathrm{ap}}$	A realized coherent logarithmic hull is nonzero, minimal, and aperiodic.	Theorem 9.104; a realized coherent support profile.	Bounded-origin realization and log-hull minimality; Corollary 9.111.
$\mathcal{C}_{\mathrm{gen}}$	Every preceding lower residual and critical-tail state has been removed.	Theorem 11.2; the original retained velocity obstruction is carried through realized terminal descendants.	Paired terminal extraction, separated-profile and chain exclusion, terminal indecomposability, and endpoint closure in Theorem 11.3.

Definition 1.11 (Profiles realized from the original blow-up sequence)**.** Fix a normalized Seregin collection $\mathcal{S}$ and a retained residual candidate $(V,P)\in\mathcal{R}(\mathcal{S};I,J)$. An auxiliary profile or limiting measure used in the residual proof is *realized from the original blow-up sequence* if it is obtained from (V,P), or from the terminal sequence which generated (V,P), by a finite or diagonal sequence of the following operations:

(i) exact centered symmetries T_a, Θ_s, and the covariant observer maps $\mathcal{R}_z$;
(ii) unreduced parabolic recenterings before passing to centered variables, with the local pressure gauges supplied by Seregin compactness;
(iii) retained tail limits or concentrating successors with $\mathcal{V}_1\ge\eta_*$ after the fixed admissible pressure gauges;
(iv) normalized diffuse or critical-tail defect limits obtained by restricting the associated defect measure to a positive-measure window and renormalizing it;
(v) minimal mesoscopic blow-downs selected from such retained or normalized defect windows.

When the shorter phrase "realized from the original sequence" is used below, it refers to this definition and to the original singularity-generated candidate (V,P), not merely to the immediately preceding normalized profile.

Definition 1.12 (Retained descendants and normalized support profiles)**.** A *retained descendant* is a descendant with a genuine inherited positive unit velocity concentration bound

$$\liminf_n \mathcal{V}_1(\cdot;0)\ge\eta_*>0.$$

A *normalized support profile*, also called a *support profile descendant*, is a normalized profile in the support of a defect measure or normalized window measure associated with the original blow-up sequence. It may arise from unnormalized mass tending to zero and therefore need not carry the original velocity threshold. Log-window and Young/defect normalizations are called retained only when the displayed unit velocity concentration bound is explicitly proved; otherwise they are normalized support profiles.

Table 3. Stability of the equation, gauges, and concentration bounds under limiting operations.

Operation	Equation	Boundedness	Pressure gauge	Mildness	Threshold
Centered time shift	yes	yes after finite shift	shifted gauge	setup mildness	yes
Spatial recentering	covariant	yes	translated gauge	yes	yes

Operation	**Equation**	**Boundedness**	**Pressure gauge**	**Mildness**	**Threshold**
Parabolic rescaling	yes	normalized scale	scaled gauge	yes	yes
Tail recentering	quotient covariant	locally yes	tail gauge	setup closure	support profile or retained
Log-window normalization	normalized defect equation	normalized	quotient gauge	not automatic	support profile
Affine pressure quotient	modulo $\mathcal{C}_{\mathrm{aff}}$	yes	affine removed	yes outside $\mathcal{C}_{\mathrm{aff}}$	specified
Critical-tail blow-down	limiting equation	locally yes	scaled/quotient gauge	after bounded-origin realization	support profile

Theorem 1.13 (Stability under realized limiting operations)**.** *Every descendant, tail limit, concentrating successor, log-window normalization, critical-tail profile, and minimal-scale blow-down used in the proof of Theorem 1.16 is realized from the original blow-up sequence in the sense of Definition 1.11. More precisely, each such limit satisfies one of the following alternatives.*

(a) *It is a retained Seregin descendant of the original candidate, with the fixed descendant lower bound $\mathcal{V}_1 \geq \eta_*$.*

(b) *It is a normalized defect profile whose unnormalized defect measure has positive total mass on the selected window. If every normalized window descendant is excluded, then the original defect is excluded.*

(c) *It lies in a previously closed lower stratum, and the retained descendant heredity theorem applies.*

(d) *It generates a retained descendant or a normalized defect descendant satisfying* (a), (b), *or* (c).

Consequently a contradiction obtained for a realized limiting profile is a contradiction for the original retained residual candidate, not merely for an auxiliary compactification limit. The theorem is used only through the concrete support-transfer statements proved for the individual compactifications below.

Proof. It suffices to verify the assertion for the operations listed in Definition 1.11. Exact centered symmetries preserve the centered equation, local suitability, pressure gauges, and retained local velocity concentration; this is recorded in Lemma 8.2 and Theorem 8.6. Unreduced terminal recenterings before centering are the recenterings in the Seregin extraction theorem of the setup paper and are compact after the gauges in Proposition 1.21. Retained tail limits and concentrating successors are realized from the original sequence by diagonal lifting: Lemma 10.11 and Theorem 10.12 show that the limit remains in the normalized compactness class and keeps the fixed retained velocity threshold through its velocity part, up to the once-for-all threshold loss in Definition 3.4.

Normalized defect profiles are obtained by restricting a positive defect measure to a window and dividing by its mass. The mass of the chosen window may depend on the index and may tend to zero; this does not detach the normalized profile from the original sequence. It records a point in the support of the defect compactification. The renormalized-window heredity lemma Lemma 9.55 below states explicitly that if all such support profiles are assigned to closed lower alternatives, then the defect measure itself has no surviving mass outside those alternatives.

Minimal-scale blow-downs are selected from retained or normalized defect windows; their centers and scales therefore come with the same defect measure or retained velocity lower bound. The pressure/gauge normalization for those blow-downs is treated in Lemmas 9.70 and 9.77

and Proposition 9.73. Thus every contradiction later in the proof is applied either to a retained descendant of the original profile or to a normalized support profile whose closure removes the corresponding defect. This proves the stated principle. □

Lemma 1.14 (Support transfer in compact metrizable defect spaces)**.** *Let X be a compact metrizable defect space and let*

$$\lambda_n = \frac{\mu_n \lfloor E_n}{\mu_n(E_n)} \qquad (\mu_n(E_n) > 0)$$

be normalized realized defect measures converging weakly to $\lambda \in \mathcal{P}(X)$. Let $A \subset X$ be closed. If every normalized support profile in $\operatorname{supp}\lambda$ *belongs to A, then the original normalized defect has no surviving mass in $X \setminus A$. Equivalently, if a positive amount of the normalized defect survives outside A, then there exists a support profile in $X \setminus A$.*

Proof. If positive normalized defect survives outside A, then $\lambda(X \setminus A) > 0$. Since X is compact metrizable and A is closed, $X \setminus A$ is open. By inner regularity choose a compact set $K \subset X \setminus A$ with $\lambda(K) > 0$. Then $K \cap \operatorname{supp}\lambda \neq \varnothing$, and every point of this intersection is a normalized support profile outside A, contradicting the support hypothesis. Hence λ is supported on A, which is exactly the assertion that no normalized defect mass survives outside the closed excluded set. □

Lemma 1.15 (Transfer of contradictions to the original blow-up sequence)**.** *Let $(V,P) \in \mathcal{R}(\mathcal{S}; I, J)$ be a retained residual candidate, and let $\mathcal{A}$ be a closed union of lower alternatives which is stable under the descendant operations used in this paper. Suppose that every limit realized from (V,P) by the relevant construction has one of the following two properties:*

(i) *it is a retained Seregin descendant lying in $\mathcal{A}$;*
(ii) *it is a normalized support profile of a defect measure associated with the original sequence and lies in $\mathcal{A}$.*

If $\mathcal{A}$ has already been excluded in the ordered residual decomposition, then the original candidate (V,P) cannot exist.

Proof. In case (i), retained descendant heredity Theorem 10.12 identifies the descendant as an admissible successor of the original residual candidate with the velocity threshold fixed in Definition 3.4. Since $\mathcal{A}$ is closed under the allowed limiting operations and is excluded, the retained descendant cannot occur.

In case (ii), the normalized support profile is obtained by restricting a positive defect measure to a window and renormalizing. If all such normalized support profiles lie in the closed excluded set $\mathcal{A}$, then regularity of the compactified defect measure implies that the defect measure has no mass outside $\mathcal{A}$; this is the support-transfer mechanism of Lemma 1.14, instantiated below for the individual compactifications. Hence the defect has no surviving support in the residual class after $\mathcal{A}$ is removed. In both cases the construction cannot produce a retained residual profile from (V,P), contradicting the defining retained velocity concentration of the residual candidate. □

Theorem 1.16 (Refined residual closure for the complete ordered decomposition)**.** *Relative to the inherited alternatives in Theorem 1.7 and Definition 1.8, one has*

$$\mathcal{C}_{\mathrm{ax}} = \mathcal{C}_{\mathrm{rot}} = \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} = \mathcal{C}_{\mathrm{gen}} = \varnothing,$$
$$\mathcal{C}_{\mathrm{log\,diff}} = \mathcal{C}_{\mathrm{Young}} = \mathcal{C}_{\mathrm{homcrit}} = \mathcal{C}_{\mathrm{logper}} = \mathcal{C}_{\mathrm{ap}} = \varnothing.$$

Moreover,

$$\mathcal{C}_{\mathrm{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing,$$

that is, no retained normalized residual representative remains in the affine/parasitic quotient stratum. Consequently

$$\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

Proof. Deferred. The proof is given in Section 12, after the terminal decomposition theorem and the exclusion of all reduced cases. The dependency order is recorded in Table 4. □

TABLE 4. Theorem-production order for the residual closure proof. This is the acyclic dependency view; state-selection criteria are recorded separately in Table 2.

Result	**Uses**	**Does not use**
Imported setup results	Setup paper only: local concentration, local Type I implication, normalized Seregin extraction, pressure atlas, lower classes, mildness, and local final assembly.	Paper II residual closure.
R1 axisymmetric closure	Centered circulation equation, axis regularity, killed-strip comparison, and setup lower classes.	Terminal stratification assembly theorem.
R2 reduction	Rotational flux identities and finite unit-cylinder velocity concentration cover; closure is obtained only after terminal stratification.	R2 closure or terminal stratification conclusion.
R2 closure	R2 reduction, terminal stratification assembly, finite separated-family exclusion, and no atomic active profile.	Standalone rotational Liouville theorem.
R3 reduction	Hull realization, centered scale reselection, and retained velocity heredity; closure is obtained only after terminal stratification.	R3 closure or terminal stratification conclusion.
R3 closure	R3 reduction, terminal stratification assembly, sequence-L^3, and endpoint atomic contradiction.	Premature stationary-hull exclusion.
Concentration-set extraction	Local compactness, raw atlas defect measures, compact observer windows, and pressure atlas.	Endpoint Albritton–Barker theorem.
Diffuse-defect compactification	Concentration-set residual measure, observer compactification, and amenability of $\mathbb{R}^3 \rtimes \mathbb{R}$.	Generic closure.
Affine routing	Affine normalization dichotomy and retained oscillation functional.	Critical-tail closure.
Coherent critical-tail closure	Bounded-origin realization and log-hull minimality.	Log-diffuse exclusion, Young exclusion, hidden-scale exclusion.
Minimal mesoscopic reduction	Local quotient blow-down, corrected scaled pressure compactness, and variance Liouville theorem.	Log-diffuse exclusion.
No hidden-scale variance	Minimal mesoscopic reduction and coherent closure.	Log-diffuse exclusion.

Result	Uses	Does not use
Young branch exclusion	No hidden-scale variance for the non-Dirac subcase and defect-coordinate closure for pressure/viscous subcases by definition of $\mathcal{C}_{\mathrm{Young}}$.	Log-diffuse exclusion.
Log-diffuse branch exclusion	Log-window support profiles, no hidden-scale variance, and coherent closure.	Young branch exclusion except through the hidden-scale theorem.
Complete critical-tail exclusion	Regenerated concentration routing, affine routing, log-diffuse exclusion, Young exclusion, and coherent closure.	Generic final assembly.
Finite separated-profile exclusion	Diagonal lifting, finite directed graph recurrence, no atomic profile, and no infinite retained velocity concentration chain.	Terminal assembly theorem and global L^3 bound.
Atomic sequence-L^3	Indecomposability and absence of diffuse, tail, separated, pressure-loss, and critical-tail alternatives.	Albritton–Barker endpoint conclusion.
Endpoint atomic contradiction	Bounded mild finite terminal shifts, sequence-L^3, and retained nonzero velocity concentration.	Residual decomposition.
Final assembly	All previous rows and the ordered decomposition.	Itself.

Corollary 1.17 (Raw residual output for the setup paper)**.** *For every normalized Seregin collection $\mathcal{S}$ generated by an admissible finite-energy Type I singularity in the setup paper, the setup remaining class is empty.*

Proof deferred. This is proved in Section 12, after Theorem 1.16 has been established and inserted into the setup paper's exact raw-residual interface. □

Corollary 1.18 (Local pointwise Type I exclusion after residual closure)**.** *No finite-energy local pointwise Type I singularity can generate a normalized Seregin collection $\mathcal{S}$.*

Proof deferred. This is the final two-paper consequence recorded in Corollary 12.3; it is stated here only to identify the consequence of residual closure. □

1.3. **Architecture and reading guide.** A reader interested only in the residual closure can navigate the paper as follows. This is a reading order by analytic module. The state transitions are listed in Table 2, while the acyclic theorem-production order is listed in Tables 4 and 16.

(1) *Setup interface and Type-I implication* (Sections 1.2 and 1.9). Records the setup results used here as imported hypotheses and verifies that local pointwise Type-I singularities enter the admissible Seregin compactness setting.
(2) *Residual decomposition* (Definition 1.10). The coarse analytic view separates $\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$ from the terminal module. The refined terminal view is the ordered list
$$\mathcal{C}_{\mathrm{ax}},\ \mathcal{C}_{\mathrm{rot}},\ \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}},\ \mathcal{C}_{\mathrm{aff}},\ \mathcal{C}_{\mathrm{log\,diff}},\quad \mathcal{C}_{\mathrm{Young}},\ \mathcal{C}_{\mathrm{homcrit}},\ \mathcal{C}_{\mathrm{logper}},\ \mathcal{C}_{\mathrm{ap}},\ \mathcal{C}_{\mathrm{gen}}.$$
The inherited list of allowed limiting operations is recorded in Theorem 1.13.
(3) *Class $\mathcal{C}_{\mathrm{ax}}$.* Centered angular-circulation equation (Lemma 4.1), one-dimensional killed radial absorption, axis-absorption Liouville theorem, closure (Theorem 4.4).

(4) *Class* $\mathcal{C}_{\mathrm{rot}}$. Coriolis Bernoulli identity, local flux dichotomy, reduction proposition (Proposition 5.8). Closure deferred until the terminal stratification assembly.
(5) *Class* $\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$. Stationary phase space, hull realization, centered scale reselection, reduction (Propositions 6.12 and 7.6). Closure deferred until the terminal stratification assembly.
(6) *Terminal module and final generic remainder*. Affine routing, tail geometry and covariant observer calculus, coherent-tail and hidden-scale closure, terminal concentration profiles, diffuse defects, separated-profile theory, recurrence, and the Albritton–Barker endpoint produce the terminal stratification and generic exhaustion (Theorems 10.66 and 11.3).
(7) *Final assembly* (Section 12). Combines all class closures and the deferred lower-class arguments.

A reader who only wants to see the central technical core can read in this order: the asymptotic hull and recurrent-core rigidity, the terminal concentration-set extraction, the diffuse-defect compactification, the critical-tail compactification, and the minimal mesoscopic-scale reduction theorem (Theorem 9.86). Everything else is either tracking for the residual decomposition or reduction of the four analytic classes into this terminal part of the argument.

The diagrams below form one directed proof graph rooted at the fixed candidate singular point. Rectangles are live states or interfaces, diamonds are exhaustive selections, and solid ellipses are branch contradictions. A dashed ellipse is only a visual reference to the numbered solid terminal ellipse drawn in another part. Facts carried along an arrow remain attached to the selected datum recorded in the monotone ledger; an arrow never assigns a fact to every element of the raw hull.

Residual membership appears only as the antecedent tested at node **[9]**. If that antecedent is false, the raw-residual branch is absent. If it is true, the exact handoff at node **[10]** routes the same profile. The graph therefore neither postulates a residual element nor changes the raw hull. Every solid terminal means that the selected branch contradicts the retained obstruction belonging to the fixed ledger. The residual and setup assembly theorems consume these branch contradictions only after the displayed producers have been established.

1.4. **Diagram map.**

Table 5. Map of the nine logical parts of the nine-panel Type I proof-flow graph.

Part	**Nodes**	**Branch resolved**	**Principal internal sources**
I	**[1]**–**[12]**	Receives the profile the setup paper hands over and pins down exactly what is still open. The four classes the setup paper already closed are discharged on sight; what survives is the raw residual class, and this part fixes the statement of it that the rest of the paper must empty.	Theorems 1.3 and 1.32 and Proposition 1.5.

Part	Nodes	Branch resolved	Principal internal sources
II	**[13]**–**[37]**	Splits the residual class into ten refined states and works through them in order. Axisymmetric profiles are killed outright by the axis closure; the rotational and stationary-hull states are reduced to terminal forms that later parts dispose of. Only the affine and generic states leave the panel alive.	Definition 1.10, Lemmas 4.1 to 4.3, 5.3, 5.6 and 6.6, Theorems 4.4 and 11.3, and Propositions 5.8 and 7.6.
III	**[38]**–**[57]**	Asks what a surviving profile can look like far from the origin. A covariant calculus makes the question independent of the observer, affine normalization removes the parasitic pressure freedom, and a pressure atlas keeps the gauge consistent chart to chart. What comes out is a short list of tail geometries and recentering alternatives.	Theorems 8.6 and 8.21, Proposition 8.26, and Corollary 9.11.
IV	**[58]**–**[63]**, **[76]**, **[141]**–**[146]**, **[149]**–**[153]**	The dispatcher. Everything the earlier parts leave alive arrives here as one of eight terminal alternatives, and each is routed to the panel that closes it: pressure-only data goes chartwise, critical tails take the direct bridge, and retained chains are followed until a critical shell or a moving cell reduces them to the sparse current-saturation branch.	Theorems 9.24, 10.17, 10.60 to 10.62, 10.65 and 10.66, Lemmas 9.5, 9.114, 10.23, 10.29 and 10.63, and Proposition 10.21.
V	**[64]**–**[75]**	Handles the case where the defect spreads out instead of concentrating. Compactifying such a profile makes it recurrent, and recurrence forces one of three outcomes, each of which is sent on to a closure.	Theorems 9.29, 9.34 and 9.38.

Part	Nodes	Branch resolved	Principal internal sources
VI	**[77]**–**[98]**	Does the same for a profile whose mass sits in a critical tail. Compactifying it and reading off a virial identity classifies the defect coordinate, and the classification decides which closure the branch belongs to.	Theorems 9.40 and 9.91 and Lemmas 9.45 and 9.50.
VII	**[99]**–**[124]**	Attacks a tail that hides its failure at some intermediate scale. Choosing the smallest scale at which control is lost makes the pressure compact there; the variance of the profile is then rigid, and that rigidity closes the log-periodic and Young alternatives.	Lemmas 9.65, 9.77 and 9.84 and Theorem 9.86.
VIII	**[125]**–**[140]**	Deals with tails that stay coherent rather than dispersing. Such a profile can be realized with a bounded origin, and once it is there are exactly three coherent critical-tail shapes left, all three of which are excluded.	Theorems 9.97 and 9.104 and Corollary 9.111.
IX	**[147]**–**[148]**, **[154]**–**[159]**	The last route out. A profile that is not separated into several pieces must be a single atom; testing its retained tail produces an endpoint estimate, and that estimate contradicts the atom's own concentration. With this the residual class is empty and the setup paper's argument resumes.	Lemmas 10.4, 10.6 and 10.10, Proposition 10.5, and Theorems 10.8 and 10.9.

1.5. **Proof-dependency diagram.**

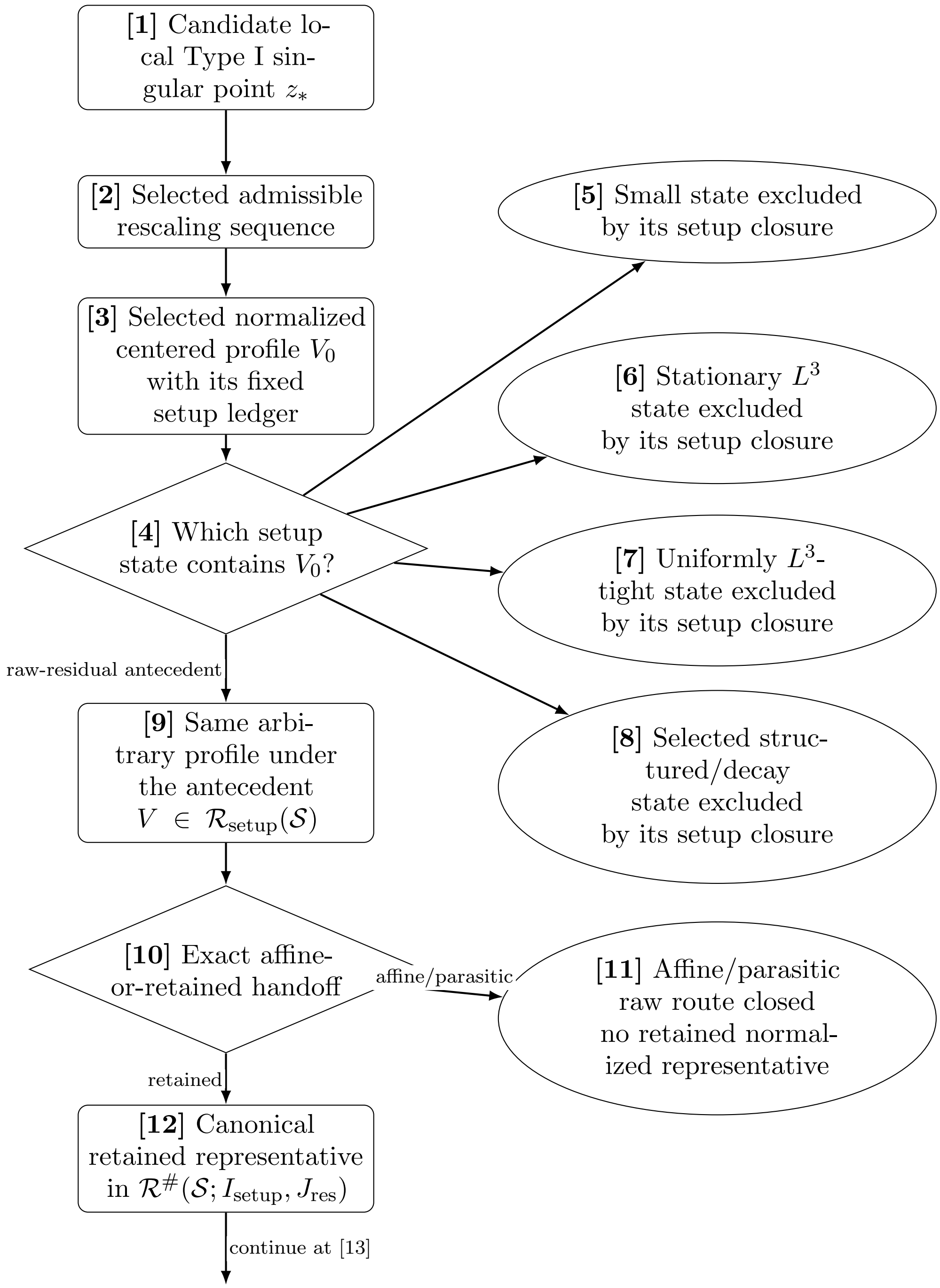

FIGURE 1. Proof flow, Part I: fixed-point entry and exact handoff. Nodes [**1**]–[**3**] concern the fixed point, selected sequence, and selected profile respectively. Node [**9**] is an implication antecedent, not an existence assertion.

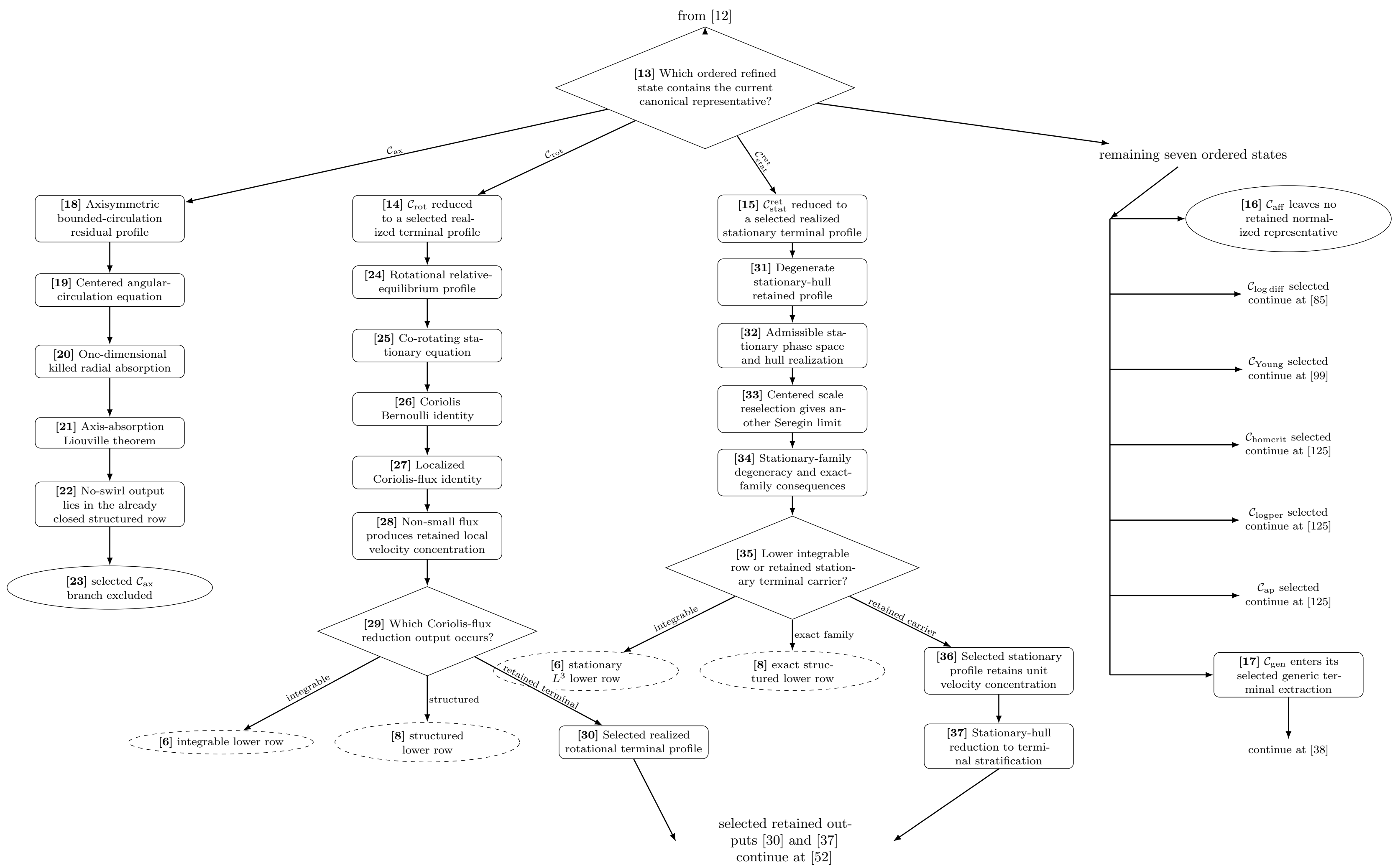

FIGURE 2. Proof flow, Part II: the complete ordered refined selection and its three lower-class modules. The $\mathcal{C}_{\mathrm{ax}}$ route closes at **[23]**; the retained R2 and R3 outputs meet only at the displayed continuation to **[52]**. The remaining-state rail and the common continuation annotation are unnumbered layout aids, not states or hypotheses. Neither R2 nor R3 receives sequence-L^3 as an entry hypothesis.

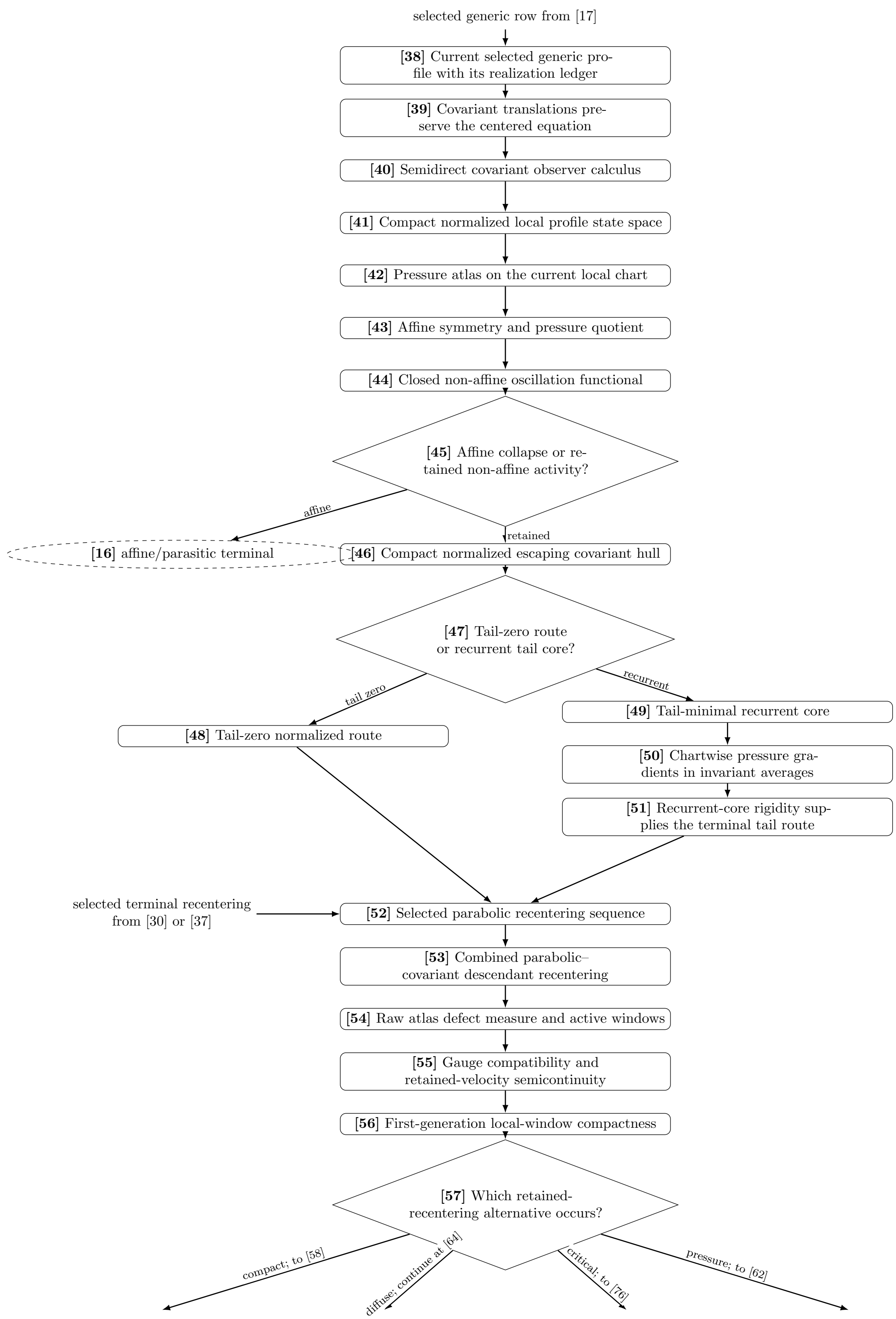

FIGURE 3. Proof flow, Part III: covariant tail geometry, affine normalization, and first-generation retained recentering. The internal routes [48] and [51] enter [52] directly; the unnumbered arrow supplies only the Part II inputs [30] and [37]. Pressure remains chartwise at [42] and [55].

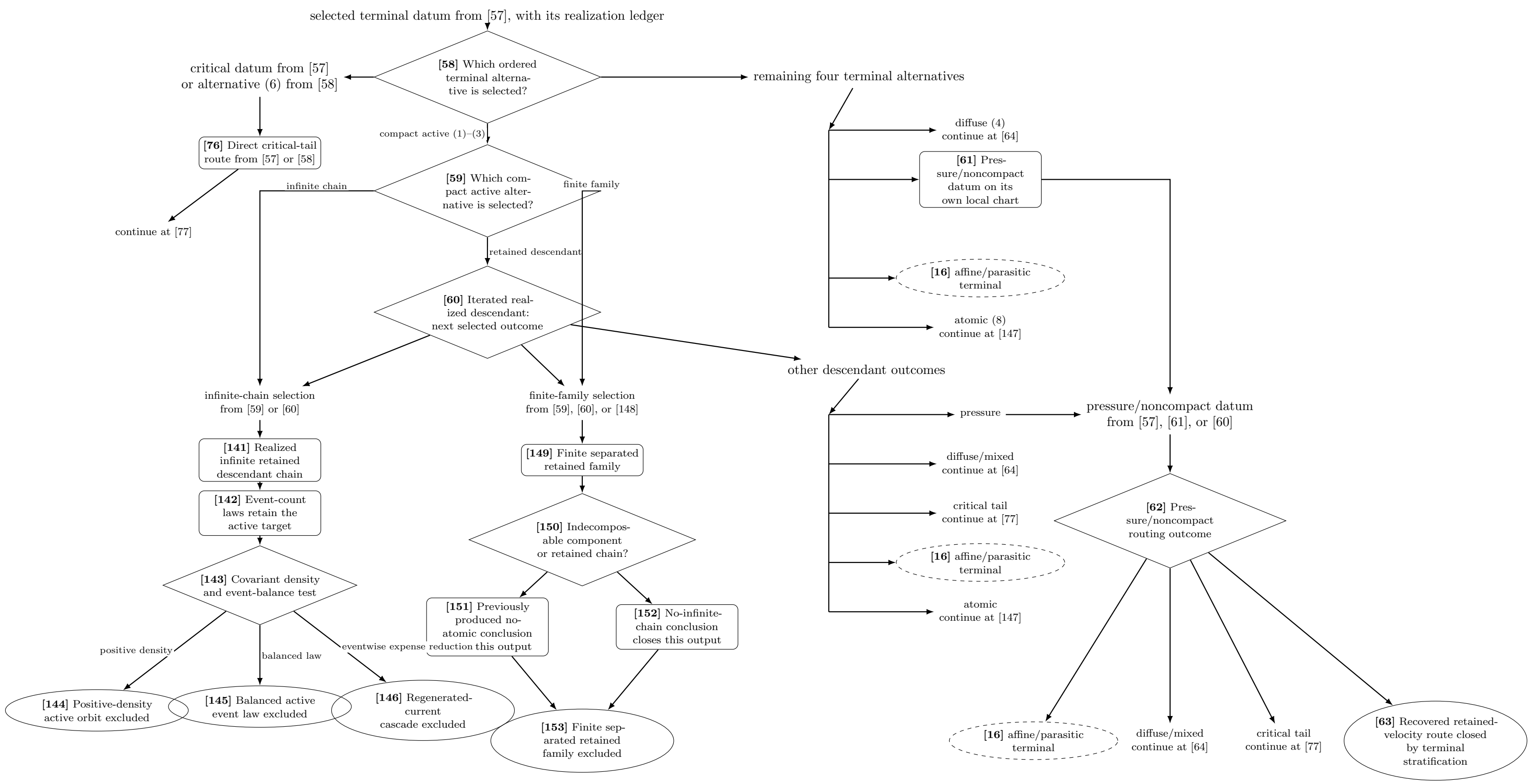

FIGURE 4. Proof flow, Part IV: terminal selection and compact-active analysis. Nodes **[141]**–**[146]** display the infinite-chain exclusion through the critical-shell, moving-cell expense, marked recurrence, and realized critical-tail closure. Nodes **[149]**–**[153]** display the finite-family analysis; nodes **[62]**–**[63]** display the complete pressure/noncompact route on the chart that produces it. Node **[76]** is the direct critical-tail bridge for the datum selected at **[57]** or **[58]**. The finite-family input also includes **[148]**; the two rails and the two input annotations are only case distributors.

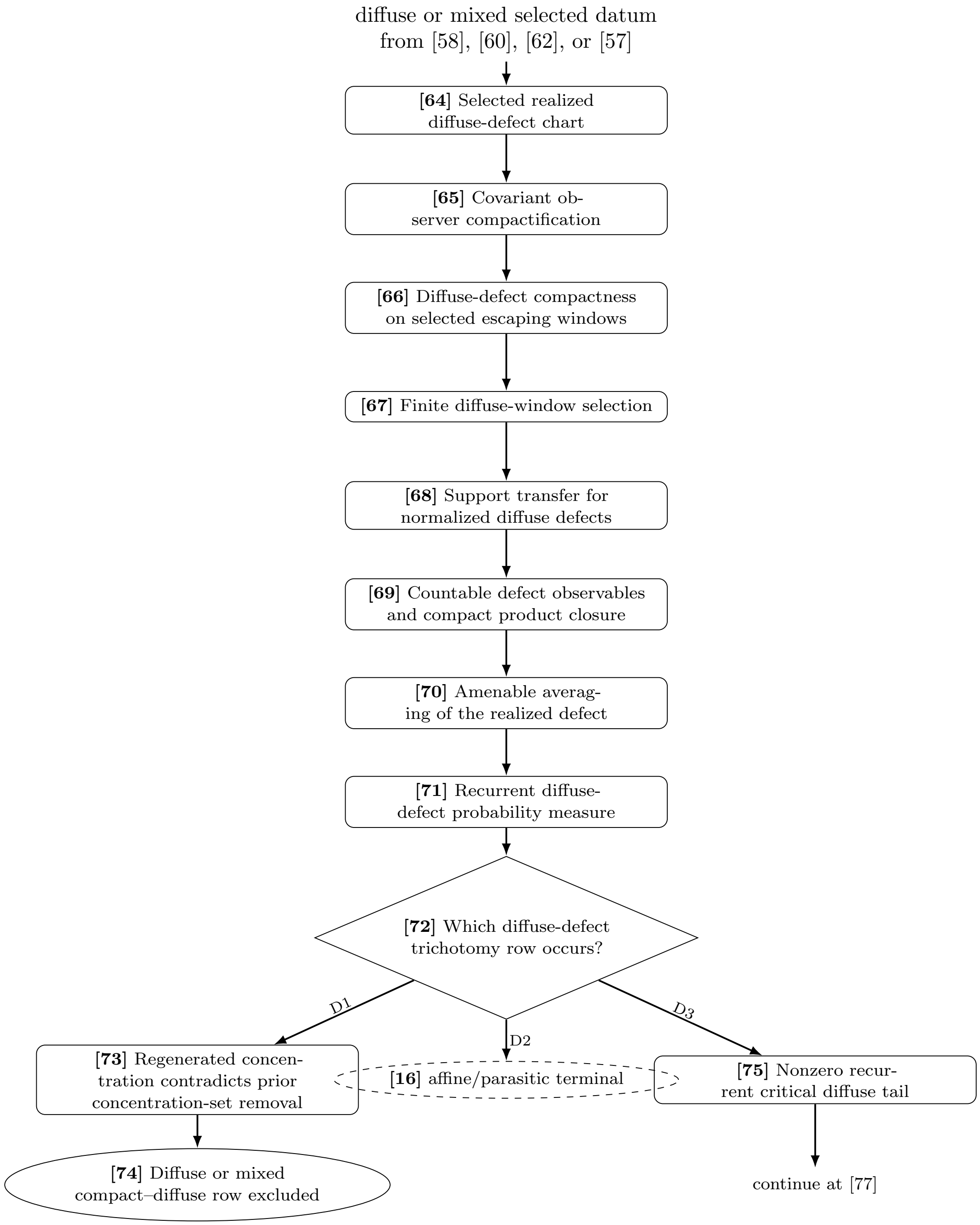

FIGURE 5. Proof flow, Part V: diffuse-defect compactification and the exact D1–D3 trichotomy. Regenerated concentration, affine collapse, and a critical diffuse tail are displayed separately.

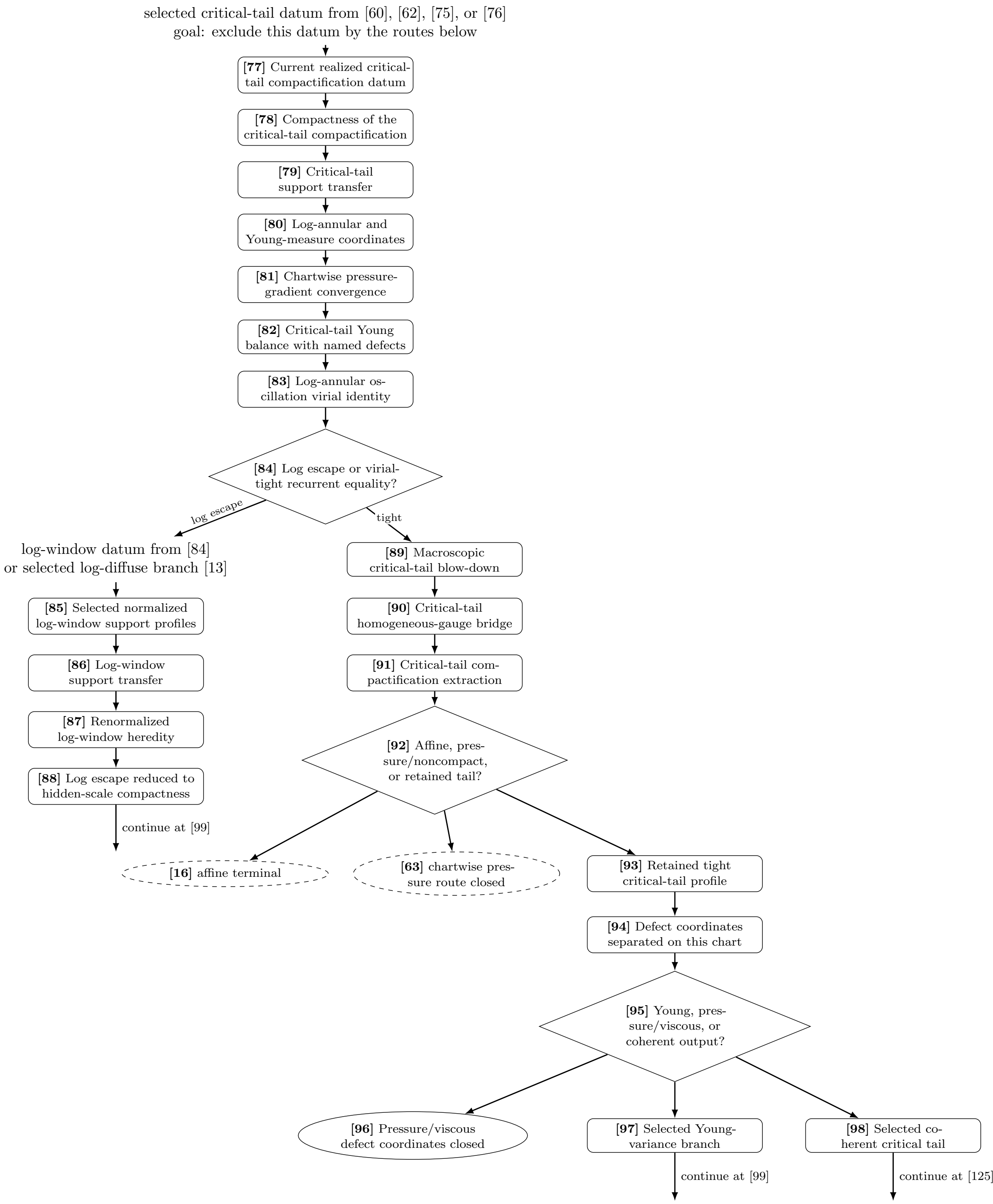

FIGURE 6. Proof flow, Part VI: critical-tail compactification, virial routing, and defect-coordinate classification. Pressure alternatives are closed only on the chart that produces them. The unnumbered log-window annotation only combines the two displayed inputs to node **[85]**. The input states the goal of the routing, not an exclusion hypothesis; the complete exclusion is produced only after the displayed and named downstream closures.

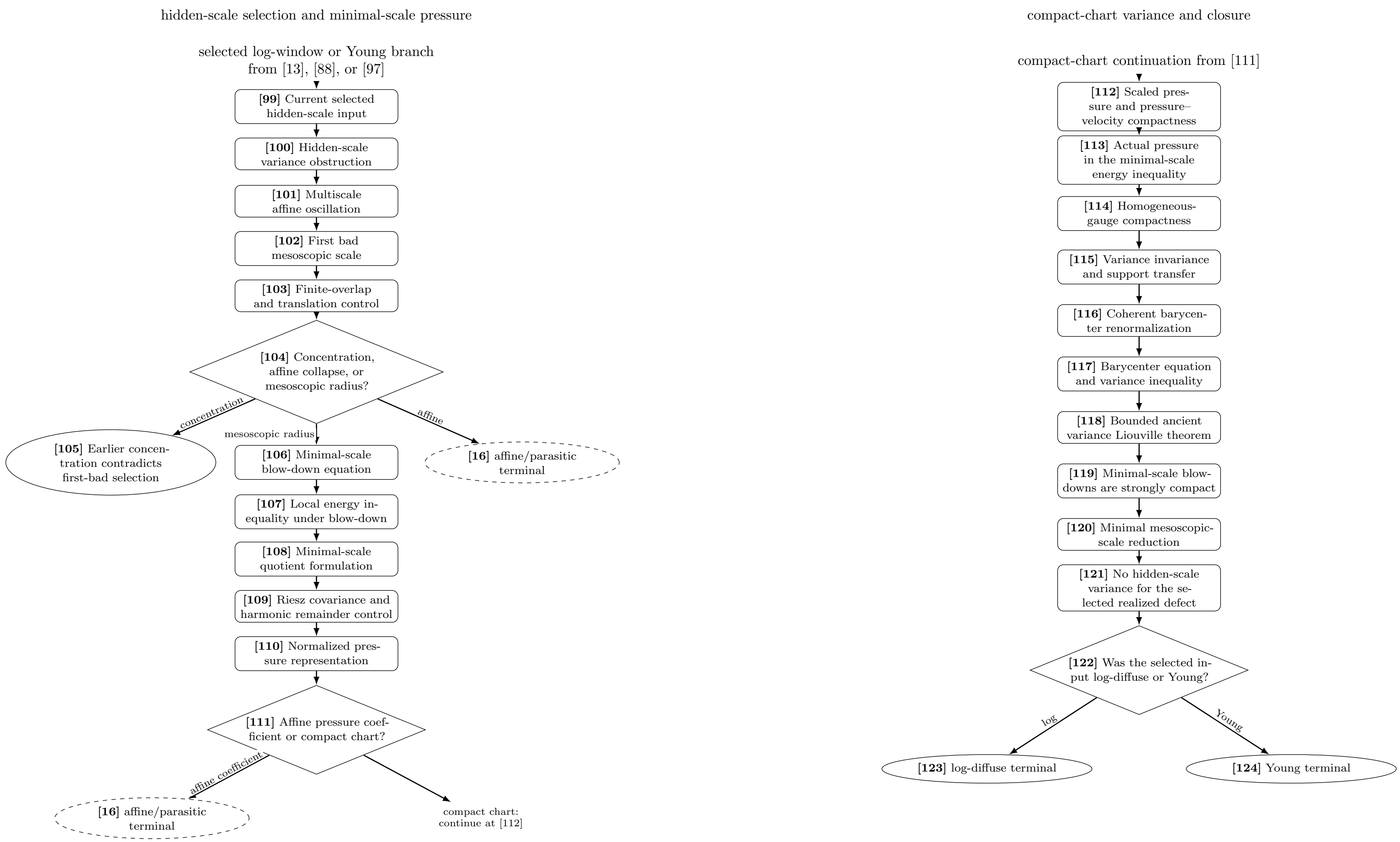

FIGURE 7. Proof flow, Part VII: the complete minimal-mesoscopic-scale pressure and variance argument, displayed as two top-down diagrams. The left diagram runs continuously from **[99]** through **[111]**, and the right diagram runs continuously from **[112]** through the terminal split at **[122]**. Their sole handoff is the compact-chart output **[111]**–**[112]**. The pressure representation is chartwise, and the log/Young closures appear only after no hidden-scale variance is produced.

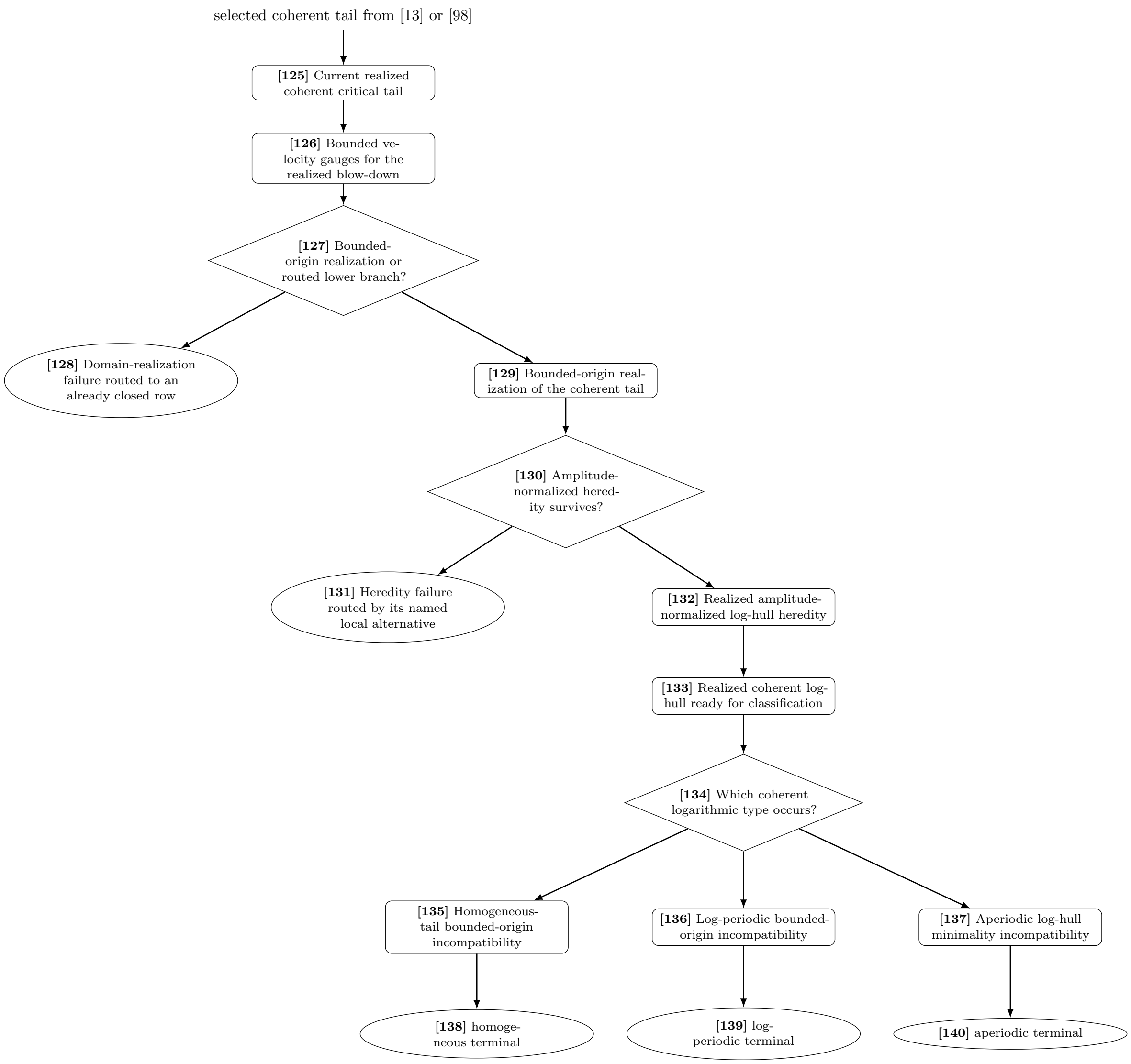

FIGURE 8. Proof flow, Part VIII: bounded-origin realization and the three coherent critical-tail closures. The homogeneous, log-periodic, and aperiodic outputs remain distinct through their terminal contradictions.

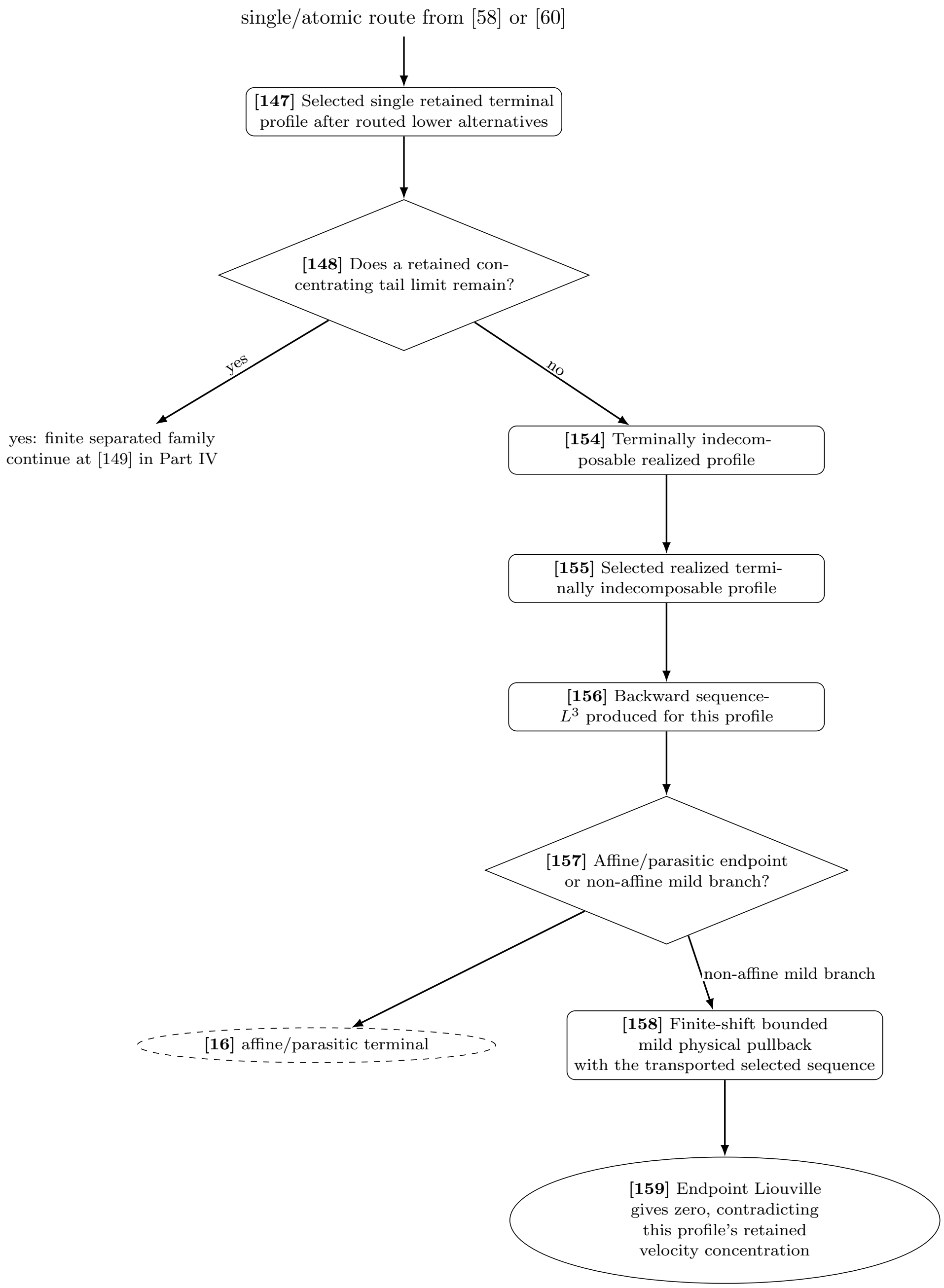

FIGURE 9. Proof flow, Part IX: the complete single/atomic and endpoint route. The yes branch at **[148]** enters the fully displayed finite-family branch at **[149]** in Part IV. On the no branch, sequence-L^3 is produced for the selected realized profile at **[156]**, and finite-shift mildness is produced only on the non-affine branch at **[158]**. The endpoint theorem and the final residual/setup assemblies are downstream consumers of the displayed closures.

1.6. **Node-by-node audit table.**

TABLE 6. Source audit for the numbered Type I proof-flow nodes.

Node	State or test	Exact proof source	Successor or closure
[1]	Fixed candidate singular point and local pointwise Type I envelope.	Definition 1.1 and Theorem 1.32.	Selected admissible sequence [2].
[2]	Fixed rescaling sequence with concentration and terminal no-escape.	Theorems 1.26 and 1.28.	Seregin extraction [3].
[3]	Selected normalized centered profile with the fixed setup ledger.	Theorems 1.3 and 1.7 and Table 1.	Setup-state decision [4].
[4]	Exhaustive setup lower-state or raw-residual selection.	Definitions 1.4 and 1.8.	[5]–[9].
[5]	Selected small setup state.	Theorem 1.3 and Definition 1.8.	Corresponding setup closure.
[6]	Selected stationary L^3 setup state.	Theorem 1.3 and Definition 1.8.	Corresponding setup closure.
[7]	Selected uniformly L^3-tight setup state.	Theorem 1.3 and Definition 1.8.	Corresponding setup closure.
[8]	Selected closed structured/decay setup state.	Theorem 1.3 and Definition 1.8.	Corresponding setup closure.
[9]	Raw-residual membership tested for the same arbitrary profile.	Definition 1.4 and Theorem 12.1.	Exact handoff [10].
[10]	Affine-or-retained handoff for the current candidate.	Proposition 1.5.	[11] or [12].
[11]	Raw affine/parasitic quotient route.	Proposition 1.5 and Theorem 9.99.	No retained representative.
[12]	Retained canonical representative in the refined class.	Proposition 1.5 and Definition 1.9.	Ordered selection [13].
[13]	Ten-state ordered refined selection.	Definition 1.10 and Eq. (1.6).	[14]–[18], [85], [99], or [125].
[14]	Current profile selected into $\mathcal{C}_{\mathrm{rot}}$.	Proposition 5.8.	Localized rotational reduction [24].
[15]	Current profile selected into $\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$.	Propositions 6.12 and 7.6.	Stationary-hull reselection [31].
[16]	Current representative selected into $\mathcal{C}_{\mathrm{aff}}$.	Theorems 8.21 and 9.99.	No retained representative.
[17]	Current profile selected into $\mathcal{C}_{\mathrm{gen}}$.	Theorem 11.2.	Covariant tail and recentering route [38].
[18]	Selected axisymmetric bounded-circulation profile.	Definition 1.10 and Theorem 4.4.	Circulation equation [19].
[19]	Centered angular-circulation equation.	Lemma 4.1.	Killed absorption [20].
[20]	One-dimensional killed radial absorption.	Lemma 4.2.	Axis Liouville [21].
[21]	Axis-absorption Liouville output.	Lemma 4.3.	No-swirl routing [22].
[22]	No-swirl output in the closed structured row.	Theorem 4.4.	Terminal [23].
[23]	Current profile selected into $\mathcal{C}_{\mathrm{ax}}$.	Theorem 4.4.	Axisymmetric branch contradiction.
[24]	Selected rotational relative-equilibrium profile.	Definition 5.1 and Proposition 5.8.	Co-rotating equation [25].

Node	State or test	Exact proof source	Successor or closure
[25]	Co-rotating stationary equation.	Lemma 5.2.	Bernoulli identity [26].
[26]	Coriolis Bernoulli identity.	Lemma 5.3.	Localized flux [27].
[27]	Localized Coriolis-flux identity.	Lemma 5.4.	Active concentration [28].
[28]	Non-small flux produces retained local velocity concentration.	Lemma 5.5.	Reduction test [29].
[29]	Exhaustive local Coriolis-flux reduction.	Lemma 5.6 and Proposition 5.8.	Lower terminals [6]/[8] or [30].
[30]	Selected realized rotational terminal profile.	Proposition 5.8.	Common recentering interface [52].
[31]	Selected degenerate stationary-hull profile.	Definition 6.4.	Phase-space realization [32].
[32]	Stationary phase space and hull realization.	Definitions 6.1 and 6.2.	Scale reselection [33].
[33]	Centered scale reselection produces a Seregin limit.	Lemma 6.6.	Family degeneracy [34].
[34]	Stationary-family degeneracy and exact-family consequences.	Lemmas 6.7 and 6.8 and Corollary 6.9.	Test [35].
[35]	Lower integrable/exact row or retained stationary carrier.	Propositions 6.12 and 6.13.	Lower terminals [6]/[8] or [36].
[36]	Selected stationary Seregin limit retains unit velocity concentration.	Propositions 6.10 and 7.1.	Reduction [37].
[37]	Stationary-hull reduction to terminal stratification.	Proposition 7.6.	Common recentering interface [52].
[38]	Selected generic profile with its fixed realization ledger.	Definition 11.1 and Theorem 11.2.	Covariance [39].
[39]	Covariant translation invariance.	Definition 8.1 and Lemma 8.2.	Observer calculus [40].
[40]	Semidirect covariant observer calculus.	Definition 8.4, Lemma 8.5, and Theorem 8.6.	State space [41].
[41]	Compact normalized local profile state space.	Definition 8.7 and Lemmas 8.9 and 8.11.	Pressure atlas [42].
[42]	Pressure representative on the current local chart.	Definition 8.14 and Lemma 8.15.	Affine quotient [43].
[43]	Affine symmetry and pressure quotient.	Definition 8.12 and Lemmas 8.13 and 8.18.	Activity [44].
[44]	Closed non-affine oscillation functional.	Theorem 8.19 and Corollary 8.20.	Affine test [45].
[45]	Affine collapse versus retained non-affine activity.	Theorem 8.21.	[16] or [46].
[46]	Compact normalized escaping covariant hull.	Definition 8.22, Theorem 8.23, and Lemma 8.24.	Tail test [47].
[47]	Tail-zero route versus recurrent tail core.	Definition 8.25 and Proposition 8.26.	[48] or [49].
[48]	Tail-zero normalized route.	Proposition 8.26.	Recentring interface [52].
[49]	Tail-minimal recurrent core.	Definition 8.27.	Pressure averages [50].
[50]	Pressure gradients in invariant local averages.	Lemma 8.28.	Core rigidity [51].
[51]	Recurrent profile tail-core rigidity.	Theorem 8.29.	Recentring interface [52].
[52]	Selected parabolic recentering sequence.	Definition 9.1.	Combined recentering [53].
[53]	Combined parabolic–covariant descendant recentering.	Definition 9.2.	Raw measures [54].

Node	State or test	Exact proof source	Successor or closure
[54]	Raw atlas defect measure and active windows.	Definition 9.3.	Gauge/velocity semicontinuity [55].
[55]	Gauge compatibility and retained-velocity semicontinuity.	Lemma 9.4.	Local-window compactness [56].
[56]	First-generation local-window compactness.	Definition 9.7 and Lemma 9.8.	Alternative test [57].
[57]	Compact, diffuse, critical, or pressure recentering alternative.	Lemmas 9.9 and 9.10 and Corollary 9.11.	[58], [64], [77], or [62].
[58]	Eight terminal alternatives for the selected realized extraction.	Theorem 10.66; for the generic owner, Theorem 11.2.	[59], [61], [16], [64], [77], or [147].
[59]	Compact active alternatives: descendant, finite family, or chain.	Theorem 10.66 and Lemma 9.14.	[60], [149], or [141].
[60]	Iterated retained descendant with realization and heredity.	Theorems 10.12, 10.17 and 10.66.	[149], [141], [64], [77], [16], [62], or [147].
[61]	Pressure/noncompact datum on its own local chart.	Corollary 9.11 and Lemmas 9.5 and 9.114.	Routing decision [62].
[62]	Pressure/noncompact routing outcome.	Lemmas 9.5 and 9.114.	[16], [64], [77], or recovered-velocity terminal [63].
[63]	Retained velocity recovered from the pressure/product route.	Theorem 10.66.	Selected terminal input excluded.
[64]	Selected realized diffuse-defect chart.	Definition 9.28 and Theorem 9.29.	Observer compactification [65].
[65]	Covariant observer compactification.	Definition 9.28.	Diffuse compactness [66].
[66]	Diffuse-defect compactness on escaping windows.	Theorem 9.29.	Window selection [67].
[67]	Finite diffuse-window selection.	Lemma 9.30.	Support transfer [68].
[68]	Support transfer for the normalized diffuse defect.	Lemma 9.32.	Observable closure [69].
[69]	Countable defect observables and compact product closure.	Definition 9.33 and Theorem 9.34.	Averaging [70].
[70]	Amenable averaging for the selected realized defect.	Proposition 9.36.	Recurrence [71].
[71]	Recurrent diffuse-defect probability measure.	Theorem 9.37.	Trichotomy [72].
[72]	Exact regenerated/affine/critical diffuse trichotomy.	Theorem 9.38.	[73], [16], or [75].
[73]	Regenerated concentration after concentration-set removal.	Theorems 9.38 and 9.113.	Diffuse terminal [74].
[74]	Diffuse or mixed compact–diffuse selected row.	Theorems 9.27, 9.38 and 9.113.	Selected row excluded.
[75]	Nonzero recurrent critical diffuse tail.	Theorem 9.38.	Critical compactification [77].
[76]	Direct critical-tail route from terminal recentering.	Lemma 9.114 and Theorem 9.24.	Critical compactification [77].
[77]	Current realized critical-tail datum; the routing goal is to exclude this selected datum.	Definition 9.39 and Theorem 9.91.	Compactification [78].
[78]	Compactness of the critical-tail compactification.	Theorem 9.40.	Support transfer [79].

Node	State or test	Exact proof source	Successor or closure
[79]	Critical-tail support transfer.	Lemma 9.41.	Critical coordinates [80].
[80]	Log-annular and Young-measure coordinates.	Definitions 9.42 and 9.43.	Pressure convergence [81].
[81]	Distributional pressure-gradient convergence on the selected critical chart.	Definition 9.44.	Young balance [82].
[82]	Critical-tail Young balance with named pressure/viscous defects.	Lemma 9.45.	Virial identity [83].
[83]	Log-annular oscillation virial identity.	Lemma 9.46.	Equality test [84].
[84]	Log escape versus virial-tight recurrent equality.	Definition 9.48 and Lemma 9.50.	[85] or [89].
[85]	Selected normalized log-window support profiles.	Definition 9.52 and Lemma 9.53.	Support transfer [86].
[86]	Log-window support transfer.	Lemma 9.54.	Heredity [87].
[87]	Renormalized log-window heredity.	Lemma 9.55.	Hidden-scale reduction [88].
[88]	Logarithmic escape reduced to hidden-scale compactness.	Theorem 9.56.	Hidden-scale input [99].
[89]	Macroscopic critical-tail blow-down.	Definition 9.89.	Gauge bridge [90].
[90]	Critical-tail homogeneous-gauge bridge.	Lemma 9.90.	Extraction [91].
[91]	Critical-tail compactification extraction.	Theorem 9.91.	Routing test [92].
[92]	Affine, pressure/noncompact, or retained critical-tail route.	Theorem 9.91.	[16], [63], or [93].
[93]	Retained tight critical-tail profile.	Theorem 9.91.	Defect separation [94].
[94]	Defect coordinates separated on the current chart.	Definition 9.92 and Lemma 9.95.	Test [95].
[95]	Young, pressure/viscous, or coherent critical output.	Lemma 9.95 and Corollary 9.96.	[96]–[98].
[96]	Selected pressure/viscous defect coordinates.	Lemmas 9.93 to 9.95.	Branch contradiction.
[97]	Selected Young-variance branch.	Definition 9.92.	Hidden-scale input [99].
[98]	Selected coherent critical tail.	Corollary 9.96.	Coherent realization [125].
[99]	Current selected log-window or Young hidden-scale input.	Definition 9.58 and Lemma 9.59.	Variance [100].
[100]	Hidden-scale variance obstruction for this selected datum.	Definition 9.58 and Lemma 9.59.	Oscillation [101].
[101]	Multiscale affine oscillation.	Definition 9.60.	First bad scale [102].
[102]	Minimal first-bad mesoscopic oscillation scale.	Definition 9.61 and Lemma 9.65.	Overlap control [103].
[103]	Finite-overlap and spacetime translation control.	Lemmas 9.62 and 9.63.	Translation test [104].
[104]	Concentration, affine collapse, or mesoscopic-radius alternative.	Lemma 9.64.	[105], [16], or [106].
[105]	Earlier concentration contradicts first-bad selection.	Lemmas 9.64 and 9.65.	Branch contradiction.
[106]	Minimal-scale blow-down equation.	Definition 9.66 and Lemma 9.68.	Local energy [107].
[107]	Local energy inequality under spatial/logarithmic blow-down.	Lemma 9.69.	Quotient [108].

Node	State or test	Exact proof source	Successor or closure
[108]	Minimal-scale quotient formulation.	Lemma 9.70.	Pressure structure [109].
[109]	Riesz covariance and harmonic-remainder control.	Lemmas 9.71 and 9.72.	Representation [110].
[110]	Normalized pressure representation with blow-down coefficient.	Proposition 9.73.	Coefficient test [111].
[111]	Affine pressure coefficient versus compact pressure chart.	Lemma 9.75.	[16] or [112].
[112]	Scaled pressure and pressure–velocity compactness.	Lemmas 9.74, 9.76 and 9.77.	Actual pressure [113].
[113]	Actual pressure in the minimal-scale energy inequality.	Lemma 9.78.	Gauge compactness [114].
[114]	Compactness modulo homogeneous gauges.	Lemma 9.79.	Variance transfer [115].
[115]	Variance invariance and Young-defect support transfer.	Lemmas 9.80 and 9.81.	Barycenter [116].
[116]	Coherent barycenter renormalization.	Lemma 9.82.	Variance equation [117].
[117]	Barycenter equation and variance inequality.	Lemma 9.83.	Variance Liouville [118].
[118]	Bounded ancient variance Liouville theorem.	Lemma 9.84.	Strong compactness [119].
[119]	Minimal-scale blow-downs are strongly compact.	Lemma 9.85.	Mesoscopic reduction [120].
[120]	Minimal mesoscopic oscillation-scale reduction.	Theorem 9.86.	No hidden variance [121].
[121]	No hidden-scale variance for this realized defect.	Theorem 9.88.	Owner test [122].
[122]	Log-diffuse versus Young owner of the selected hidden-scale input.	Corollaries 9.57 and 9.100.	[123] or [124].
[123]	Current realized log-diffuse defect and its selected windows.	Lemmas 9.54 and 9.55 and Corollary 9.57.	Selected defect excluded.
[124]	Current realized Young or pressure/viscous defect datum.	Theorem 9.88, Lemma 9.95, and Corollary 9.100.	Selected datum excluded.
[125]	Current realized coherent critical tail.	Corollary 9.96 and Theorem 9.98.	Bounded gauges [126].
[126]	Bounded velocity gauges for the realized blow-down.	Lemma 9.102.	Domain test [127].
[127]	Bounded-origin realization versus routed lower branch.	Lemma 9.103.	[128] or [129].
[128]	Domain-realization failure routed to a named lower row.	Lemma 9.103.	Branch contradiction.
[129]	Bounded-origin realization of the coherent tail.	Theorem 9.104.	Heredity test [130].
[130]	Amplitude-normalized heredity survives or fails by a routed alternative.	Lemma 9.106.	[131] or [132].
[131]	Amplitude-normalized heredity failure routed locally.	Lemma 9.106.	Branch contradiction.
[132]	Realized amplitude-normalized log-hull heredity.	Lemma 9.107.	Realized hull [133].
[133]	Realized coherent log-hull ready for classification.	Theorem 9.97.	Type test [134].

Node	State or test	Exact proof source	Successor or closure
[134]	Homogeneous, log-periodic, or aperiodic coherent type.	Theorem 9.97 and Corollary 9.111.	[135]–[137].
[135]	Homogeneous-tail bounded-origin incompatibility.	Lemma 9.108.	Terminal [138].
[136]	Log-periodic bounded-origin incompatibility.	Lemma 9.109.	Terminal [139].
[137]	Aperiodic log-hull minimality incompatibility.	Lemma 9.110.	Terminal [140].
[138]	Current realized homogeneous coherent tail.	Theorem 9.104 and Corollary 9.111.	Selected tail excluded.
[139]	Current realized log-periodic coherent tail.	Theorem 9.98 and Corollary 9.111.	Selected tail excluded.
[140]	Current realized aperiodic coherent tail.	Theorem 9.98 and Corollary 9.111.	Selected tail excluded.
[141]	Selected realized infinite retained descendant chain.	Theorem 10.12, Definition 10.15, and Lemma 10.18.	Event laws [142].
[142]	Event-count measures whose weak limits retain $\Phi \geq \eta$.	Theorem 10.17.	Covariant test [143].
[143]	Positive covariant activity density, vanishing event covariance defect, or sparse active-chain output.	Definitions 10.19 and 10.22, Theorem 10.25, and Lemma 10.59.	[144]–[146].
[144]	Current active profile has positive covariant activity density.	Proposition 10.21.	Selected density branch excluded.
[145]	Event-count laws have a covariantly balanced subsequence, or a coherent critical-shell family has positive mean production and sublinear interscale current.	Lemmas 10.23 and 10.29 and Corollary 10.30.	Selected balanced or positive-production/sublinear-current branch excluded.
[146]	Zero-density active chain reduced to persistent current-saturated moving cells after transverse antichain routing.	Definition 10.24, Lemmas 10.33 and 10.36, Proposition 10.43, Theorems 10.41 and 10.61, and Corollary 10.50.	Selected sparse chain excluded.
[147]	Selected single retained terminal profile after routed lower alternatives.	Proposition 10.5.	Tail test [148].
[148]	Retained concentrating tail limit remains or not.	Lemma 10.4 and Proposition 10.5.	[149] or [154].
[149]	Selected finite separated retained family.	Definition 9.115 and Lemma 10.13.	Alternative test [150].
[150]	Indecomposable component versus retained chain.	Lemma 10.63.	[151] or [152].
[151]	No-atomic conclusion applied to the selected indecomposable component.	Lemma 10.10.	Finite-family terminal [153].
[152]	No-infinite-chain conclusion applied to the selected chain output.	Theorem 10.62.	Finite-family terminal [153].
[153]	Selected finite separated retained family.	Theorem 10.65.	Selected family excluded unless its successor graph produces the output at [146].
[154]	Terminally indecomposable realized profile.	Proposition 10.5.	Endpoint production [155].

Node	State or test	Exact proof source	Successor or closure
[155]	Remaining selected single retained profile.	Proposition 10.5.	Terminal indecomposable state [156].
[156]	Backward sequence-L^3 for the selected terminal profile.	Lemma 10.6.	Endpoint branch decision [157].
[157]	Affine/parasitic or non-affine mild endpoint branch.	Theorem 10.8.	[16] or [158].
[158]	Finite-shift bounded mild pullback and transported selected sequence.	Proposition 10.7 and Lemma 10.10.	Endpoint contradiction [159].
[159]	Endpoint zero conclusion contradicts the retained obstruction of the same realized profile.	Theorem 10.9 and Lemma 10.10.	Atomic branch contradiction.

1.7. **Monotone retained-fact ledger.**

TABLE 7. Ownership and transport of facts carried by the proof-flow graph.

Current owner	Named producer	Fact carried for that owner	No inference made
Fixed singular point z_*.	Theorem 1.3.	Positive local velocity concentration at the fixed point.	No property of another point or solution.
Selected rescaling sequence.	Theorems 1.26 and 1.28.	Admissibility, local energy control, and terminal no-escape for that sequence.	No assertion about an unselected sequence.
Selected normalized profile V_0.	Theorem 1.7 and Table 1.	Bounded ancient suitability, retained compact velocity concentration, fixed normalization data, and the setup pressure atlas.	Mildness only if the selected profile enters the mild branch.
Current raw-residual candidate under the antecedent at [9].	Proposition 1.5.	The same setup ledger through the affine-or-retained handoff.	No residual element is postulated and no replacement state space is formed.
Realized retained descendant.	Theorems 1.13 and 10.12.	Ancestry and retained velocity at the stated descendant threshold.	No retained mass from descendant status alone.
Selected normalized defect window or support profile.	Lemmas 1.14 and 9.55.	Normalized positive defect mass and a support point when the cited support theorem produces it.	No conversion of defect mass into retained velocity concentration.
Current pressure-dependent profile or window.	Lemma 8.15 and Table 10.	The representative and time gauge valid on that local chart.	No transport of one representative through an uncited recentering or blow-down.
Selected first-bad mesoscopic blow-down.	Lemmas 9.70 and 9.77 and Proposition 9.73.	The quotient equation, pressure coefficient, and energy inequality on the selected normalized blow-down chart.	No pressure coefficient or compactness assertion for another scale.
Selected coherent critical tail.	Theorem 9.104 and Lemma 9.107.	Bounded-origin realization and the realized amplitude-normalized log hull for that tail.	No bounded-origin assertion for an unrealized support point.
Selected finite family or retained descendant chain.	Theorem 10.12, Definition 10.15, and Lemma 10.18.	The named descendant thresholds, observer witnesses, and gauges at each realized successor.	No profile-only closedness and no retained mass for an uncited descendant.
Selected terminally indecomposable profile.	Lemma 10.6.	A backward sequence with uniformly bounded L^3 norm for that profile.	No sequence-L^3 input at R2 or R3 entry.

Current owner	Named producer	Fact carried for that owner	No inference made
Same terminal profile on the non-affine branch.	Theorem 10.8 and Proposition 10.7.	Finite-shift bounded mild physical pullbacks.	No mild identity assigned to the affine branch or to unrelated profiles.
Selected pullback and selected backward sequence.	Lemma 10.10 and Theorem 10.9.	The endpoint zero conclusion for that pullback, returned to the same centered profile.	No endpoint theorem used before nodes [156] and [158] produce its inputs.

1.8. **Branch-closure audit table.**

TABLE 8. Closure provenance for every terminal route in the proof-flow graph.

Terminal route	Selected owner	Closure source	Downstream consumer only
Nodes [5]–[8]	Current profile in one of the four setup lower states.	Theorem 1.3 and Definition 1.8, using the corresponding setup closure for the selected state.	Setup final assembly.
Nodes [11] and [16]	Current affine/parasitic representative.	Theorems 8.21 and 9.99.	Theorem 1.16 and Proposition 1.5.
Node [23]	Current $\mathcal{C}_{\mathrm{ax}}$ profile.	Theorem 4.4.	Theorem 1.16.
Node [123]	Current log-diffuse defect and selected windows.	Corollary 9.57.	Corollary 9.112 and Theorem 11.3.
Node [124]	Current Young or pressure/viscous defect datum.	Corollary 9.100.	Corollary 9.112 and Theorem 11.3.
Nodes [138]–[140]	Current realized coherent critical tail.	Corollary 9.111.	Corollary 9.112 and Theorem 11.3.
Node [74]	Current selected diffuse or mixed terminal datum.	Theorems 9.27, 9.38 and 9.113.	Theorems 10.66 and 11.3.
Input at node [77]	Current selected critical-tail datum; exclusion is the conclusion of the complete routing, not an input hypothesis.	Theorem 9.98 and Corollary 9.112.	Theorems 10.66 and 11.3.
Node [153]	Current finite separated retained family.	Theorem 10.65.	Theorems 10.66 and 11.3.
Nodes [144]–[145]	Current positive-density, balanced-event-law, or positive-production/sublinear-critical-current output of an infinite retained descendant chain.	Proposition 10.21, Lemmas 10.23 and 10.29, and Corollary 10.30.	Selected output excluded.
Node [146]	Persistent current-saturation output of an infinite retained descendant chain after transverse antichain routing.	Lemmas 10.33, 10.36, 10.40 and 10.55, Theorems 10.41, 10.56 and 10.61, and Corollary 10.57.	Concentration-free current enters the critical-tail closure; regenerated current is converted by marked recurrence to a realized critical tail and is excluded.
Node [63]	Current terminal extraction after pressure routing recovers retained velocity concentration.	Theorem 10.66; the R2 and R3 owners are then discharged by Corollaries 10.68 and 10.69, while the generic owner is discharged by Theorem 11.3.	Theorem 1.16.
Node [159]	Current realized terminally indecomposable profile.	Lemmas 10.6 and 10.10 and Theorems 10.8 and 10.9.	Theorems 10.66 and 11.3 and Corollaries 10.68 and 10.69.

Terminal route	**Selected owner**	**Closure source**	**Downstream consumer only**
Node [96]	Current selected pressure or viscous critical-tail defect coordinate.	Lemmas 9.93 to 9.95.	Corollaries 9.100 and 9.112.
Node [105]	Current first-bad-scale branch in which earlier concentration reappears.	Lemmas 9.64 and 9.65.	Theorem 9.86.
Nodes [128] and [131]	Current coherent-tail realization whose domain or amplitude heredity fails.	Lemmas 9.103 and 9.106.	Theorem 9.98 and Corollary 9.112.
All residual terminals	The same arbitrary incoming profile and fixed setup ledger from nodes [9]–[12].	Theorems 1.16 and 12.1.	Corollaries 12.2 and 12.3, in that order.

1.9. **Local Type I singularities enter the terminal concentration setting.** The purpose of this subsection is to make explicit the local reduction that is implicit in the two-paper proof architecture here. There are two logically different issues. The local pointwise Type I envelope gives Seregin compactness on every compact negative-time cylinder

$$B_R \times [-S, -\sigma], \qquad \sigma > 0,$$

but this compactness is strictly away from the terminal slice $t = 0$. It does not alone imply that the positive L^3-mass in $B_1 \times (-1, 0)$ survives away from the terminal layer $(-\sigma, 0)$. The latter is the terminal velocity no-escape condition used in the setup paper.

The implication below first records the admissible form used by the unified setup paper. It then uses the endpoint weak Serrin local Type I estimate of Albritton–Barker to derive the required scale-invariant L^2 bound on the terminal interval. Interpolation with the Type I envelope gives terminal velocity no-escape.

For this subsection, if $z_* = (x_*, T)$ and $r_n \downarrow 0$, write

$$u_n(x,t) := r_n u(x_* + r_n x, T + r_n^2 t), \qquad p_n(x,t) := r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Also write $Q_R := B_R(0) \times (-R^2, 0)$ and

$$Q_1(0,0) = B_1(0) \times (-1, 0).$$

Definition 1.19 (Admissible local Type I concentration sequence)**.** Let $z_* = (x_*, T)$ satisfy the local Type I bound (1.10), and let $r_n \downarrow 0$. The corresponding rescaled velocities u_n form an admissible local Type I concentration sequence if there are $\eta_v > 0$ such that

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \geq \eta_v \qquad \text{for every } n, \tag{1.7}$$

and the terminal velocity no-escape condition holds:

$$\lim_{\sigma \downarrow 0} \sup_n \iint_{B_1 \times (-\sigma, 0)} |u_n|^3 \, dx \, dt = 0. \tag{1.8}$$

Equivalently, after lowering the mass threshold, there are $\sigma_0 \in (0,1)$ and $\eta_1 > 0$ such that

$$\iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 \, dx \, dt \geq \eta_1 \qquad \text{for all } n. \tag{1.9}$$

Lemma 1.20 (No-escape leaves positive mass away from the terminal layer)**.** *If u_n is an admissible local Type I concentration sequence in the sense of Definition 1.19, then (1.9) holds for some $\sigma_0 \in (0,1)$ and some $\eta_1 > 0$.*

Proof. Choose $\sigma_0 \in (0,1)$ so small that

$$\sup_n \iint_{B_1\times(-\sigma_0,0)} |u_n|^3\,dx\,dt \le \frac{1}{2}\eta_v,$$

which is possible by (1.8). Subtracting this terminal-layer contribution from (1.7) gives

$$\iint_{B_1\times(-1,-\sigma_0)} |u_n|^3\,dx\,dt \ge \frac{1}{2}\eta_v.$$

Take $\eta_1 = \eta_v/2$. □

Proposition 1.21 (Local Type I rescalings have Seregin compactness)**.** *Let (u,p) be a suitable Leray–Hopf solution on $\mathbb{R}^3\times(0,T)$, let $z_* = (x_*,T)$ be a singular point, and assume that there exist $\rho>0$ and $M<\infty$ such that*

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_*))} \le M. \tag{1.10}$$

Let $r_n \downarrow 0$, and let (u_n,p_n) be the rescaled sequence above. Then the following hold.

(i) *For every compact cylinder*

$$K = B_R\times[-S,-\sigma] \Subset \mathbb{R}^3\times(-\infty,0), \qquad R,S<\infty, \quad \sigma>0,$$

there exists $n_0(K)$ such that

$$\|u_n\|_{L^\infty(K)} \le M\sigma^{-1/2}, \qquad n\ge n_0(K). \tag{1.11}$$

(ii) *On every compact cylinder $K \Subset \mathbb{R}^3\times(-\infty,0)$ there are time-dependent pressure gauges $a_{n,K}(t)$ and a constant $C_K<\infty$ such that, for $n\ge n_0(K)$,*

$$\|u_n\|_{L^\infty_t L^2_x(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \le C_K. \tag{1.12}$$

The constant C_K may depend on the fixed Leray energy level of the original solution, but no concentration or no-escape assumption enters it.

(iii) *After subtracting these local pressure gauges, the family (u_n,p_n) is locally precompact in the standard Seregin topology on $\mathbb{R}^3\times(-\infty,0)$. Along every subsequence there is a further subsequence and an ancient suitable weak solution (U,Q) such that, on every compact $K\Subset\mathbb{R}^3\times(-\infty,0)$,*

$$u_n \to U \quad \textit{strongly in } L^q(K), \qquad 1\le q<\infty,$$

and

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \textit{weakly in } L^{3/2}(K).$$

The limit is bounded and satisfies

$$|U(x,t)| \le M(-t)^{-1/2}, \qquad t<0.$$

(iv) *More generally, let $\mathfrak{c} = (z_n)$, $z_n = (y_n,\tau_n)$, be a parabolic recentering sequence for (u_n,p_n) with $\tau_n\le 0$. Assume that $\mathfrak{c}$ remains in the local Type I window in the following precise sense: for every $R,S<\infty$, all sufficiently large n satisfy*

$$r_n(|y_n|+R) < \rho, \qquad T + r_n^2(\tau_n - S) > T-\rho^2.$$

If $\mathfrak{c}$ carries retained unit velocity concentration, then after passing to a subsequence it has no loss of local compactness in the sense of Definition 9.3, and its recentered fields converge locally to a bounded suitable terminal profile. Recenterings which do not satisfy this local-window condition are assigned by the compact-window-to-expanding-region protocol to the exterior, diffuse, pressure/noncompact, or critical-tail terminal alternatives rather than to the compact local profile class.

Proof. Fix
$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0).$$
For n sufficiently large,
$$r_n R < \rho/2, \qquad r_n^2 S < \rho^2/2.$$
Hence, for a.e. $(x,t) \in K$,
$$x_* + r_n x \in B_\rho(x_*), \qquad T + r_n^2 t \in (T - \rho^2, T).$$
The local Type I bound (1.10) gives
$$|u_n(x,t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \le r_n M \left(T - (T + r_n^2 t)\right)^{-1/2} = M(-t)^{-1/2} \le M\sigma^{-1/2}.$$
This proves (1.11). In particular,
$$\operatorname*{ess\,sup}_{-S<t<-\sigma} \int_{B_R} |u_n(x,t)|^2 \, dx \le |B_R| M^2 \sigma^{-1},$$
and
$$\iint_K |u_n|^3 \, dx\, dt \le |K| M^3 \sigma^{-3/2}.$$
We next localize the nonlocal pressure contribution. The solution in this proposition is a whole-space suitable Leray–Hopf solution. Its finite-energy bounds imply, for almost every physical time, that $u(t) \in L^2(\mathbb{R}^3) \cap L^6(\mathbb{R}^3)$, and hence $u_i(t)u_j(t) \in L^{3/2}(\mathbb{R}^3)$. The whole-space pressure-recovery theorem therefore permits the canonical representative [2]. After rescaling, for almost every time,
$$p_n(\cdot, t) = \mathcal{R}_i \mathcal{R}_j \big(u_{n,i} u_{n,j}\big)(\cdot, t)$$
modulo a function of time. This representative is used only for rescalings of the original whole-space finite-energy solution in the present proposition.

Fix a cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on B_{4R}, and write in B_{4R}
$$p_n = q_n + h_n, \qquad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$
The local term satisfies
$$\|q_n\|_{L^{3/2}(B_{4R} \times [-S,-\sigma])} \le C_{R,S,\sigma,M}.$$
For the harmonic part choose the gauge $a_{n,K}(t) = h_n(0,t)$. Thus the quantity estimated below is the spatial oscillation $h_n(x,t) - h_n(0,t)$, rather than an absolute representative of the far-field pressure. If
$$K_{ij}(x) = c_3 \, \text{p.v.} \, \frac{3x_i x_j - \delta_{ij}|x|^2}{|x|^5}$$
is the Calderón–Zygmund pressure kernel, then for $|x| \le 2R$ and $|z| \ge 4R$,
$$|K_{ij}(x - z) - K_{ij}(-z)| \le C_R |z|^{-4}.$$
The subtraction of the spatial gauge gains one power of far-field decay; this is the standard local pressure-expansion mechanism [3]. Decomposing the exterior region into dyadic annuli gives, for $x \in B_{2R}$,
$$|h_n(x,t) - a_{n,K}(t)| \le C_R \sum_{m \ge 2} 2^{-4m} \int_{2^m R < |z| < 2^{m+1} R} |u_n(z,t)|^2 \, dz.$$
For annuli whose physical image remains in $B_\rho(x_*)$, namely while $2^{m+1} R r_n \le \rho$, the local Type I bound and $-t \ge \sigma$ on K give
$$\int_{2^m R < |z| < 2^{m+1} R} |u_n(z,t)|^2 \, dz \le C M^2 \sigma^{-1} (2^m R)^3.$$

Multiplication by 2^{-4m} leaves a summable 2^{-m} factor, so the corresponding part of the series is bounded by a geometric sum depending only on R, M, σ. For the remaining outer annuli one uses only the finite Leray energy of the original solution:

$$\int_{\mathbb{R}^3} |u_n(z,t)|^2 \, dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 \, dy.$$

If m_n is the first index for which $2^{m_n} R r_n \gtrsim \rho$, then

$$\sum_{m \geq m_n} 2^{-4m} \int_{\mathbb{R}^3} |u_n(z,t)|^2 \, dz \leq C_{\rho,R} \, r_n^3 \|u\|^2_{L^\infty(0,T;L^2)} \leq C_{\rho,R,u}.$$

In fact, this remote contribution is $O(r_n^3)$ and therefore vanishes along the blow-up sequence. Thus $h_n - a_{n,K}$ is uniformly bounded on $B_{2R} \times [-S, -\sigma]$. Since this cylinder has finite measure,

$$\|h_n - a_{n,K}\|_{L^{3/2}(B_{2R} \times [-S,-\sigma])} \leq C_K.$$

Together with the Calderón–Zygmund bound for q_n, this gives

$$\|p_n - a_{n,K}(t)\|_{L^{3/2}(B_{2R} \times [-S,-\sigma])} \leq C_K.$$

This is a local pressure-gauge estimate. The local Type I envelope controls the annuli whose physical images remain in $B_\rho(x_*)$, while the original Leray energy controls the remaining annuli after the kernel cancellation. No uniform whole-space estimate for p_n, and no global critical norm of u_n, enters the bound.

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder $K' \Subset \mathbb{R}^3 \times (-\infty, 0)$. The already obtained local L^2, L^3, and pressure-gauge bounds control every term on the right-hand side, and therefore

$$\iint_K |\nabla u_n|^2 \, dx \, dt \leq C_K.$$

This proves (1.12).

The Navier–Stokes equation now bounds $\partial_t u_n$ locally in a negative Sobolev space, for instance in $L_t^{3/2} H_x^{-2}(K)$. Aubin–Lions compactness and the uniform L^∞ bound give strong local convergence of the velocity in every finite L^q, after passing to a subsequence. The pressure bound gives weak $L^{3/2}$ convergence after gauges. Passing to the limit in the distributional equations and in the local energy inequality gives an ancient suitable weak solution (U, Q). The pointwise Type I bound passes to the limit, so U is bounded on every compact negative-time cylinder. This proves (iii).

Finally, let $\mathfrak{c} = (y_n, \tau_n)$ be as in (iv), and set

$$u_n^{\mathfrak{c}}(y,s) = u_n(y + y_n, s + \tau_n), \qquad p_n^{\mathfrak{c}}(y,s) = p_n(y + y_n, s + \tau_n).$$

The local-window condition, together with $\tau_n \leq 0$, is what is needed to repeat the preceding estimates on each fixed backward compact cylinder in the (y,s)-variables: the physical cylinders remain in $B_\rho(x_*) \times (T - \rho^2, T)$, the local Type I envelope gives the same L^∞ bound on compact windows, and the pressure decomposition has the same local and finite-energy tail estimates. Thus the two failure modes in Definition 9.3, blow-up of the local dissipation measure or failure of local pressure-gauge control, cannot occur for such a retained local recentering sequence after subsequence extraction. The recentered fields therefore converge locally to a bounded suitable terminal profile. If the local-window condition fails, then the recentering leaves every bounded terminal window attached to the singular point, or its critical mass persists only through expanding regions; those are precisely the exterior and diffuse terminal alternatives rather than loss of local compactness inside the local Type I implication.

The whole-space Riesz representative is used only in this compactness argument on cylinders with $t \le -\sigma < 0$. It is not assigned to arbitrary local or ancient profiles produced later in the decomposition; those profiles carry only the local pressure gauges supplied by the pressure atlas. At the terminal layer $t = 0$, pressure is removed by the local Stokes projection in Lemma 1.22, and terminal no-escape is obtained from that local estimate together with the endpoint input recorded below. $\square$

1.10. **Pressure-projected terminal energy estimate.** The preceding compactness result is stated away from the terminal slice. The next estimate supplies the terminal-layer bound required for velocity no-escape. The local pressure projection removes the nonlocal pressure flux from the local energy inequality; the remaining estimate is an interior Caccioppoli propagation bound on balls contained in the local Type I window.

Let $v = u^{(r)}$, $\pi = p^{(r)}$, and $\Lambda = \rho/r$. After translating z_* to $(0,0)$, the local Type I bound gives

$$|v(x,s)| \le M(-s)^{-1/2} \qquad (x,s) \in B_\Lambda \times (-4,0),$$

for all sufficiently small r. For $a \in \mathbb{R}^3$, $L \ge 1$, and $-4 < s < t < 0$, write

$$A_L(a,s) := \int_{B_L(a)} |v(x,s)|^2\,dx, \qquad D_L(a;s,t) := \int_s^t \int_{B_L(a)} |\nabla v|^2\,dx\,d\sigma.$$

Lemma 1.22 (Local pressure-projected Caccioppoli inequality)**.** *There is a universal constant C with the following property. Let (v,π) be a suitable weak solution in $B_{8L}(a) \times (s_-,0)$, $L \ge 1$, and assume*

$$|v(x,s)| \le M(-s)^{-1/2} \qquad \text{on } B_{8L}(a) \times (s_-,0).$$

If $s_0 \in (s_-,0)$ is a Lebesgue time for v and ∇v, then for a.e. $t \in (s_0,0)$,

$$\begin{aligned}\int_{B_L(a)} |w(x,t)|^2\,dx + D_L(a;s_0,t) &\le C A_{4L}(a,s_0)\\ &\quad + C\int_{s_0}^t \left(L^{-2} + ML^{-1}(-s)^{-1/2}\right) A_{4L}(a,s)\,ds.\end{aligned} \tag{1.13}$$

The estimate is entirely local on $B_{8L}(a)$; no pressure tail from outside $B_{8L}(a)$ appears. Here $w = v + \nabla p_h$ is the local projected velocity defined in the proof below.

Proof. The proof uses the standard local pressure-projection argument in the scale L normalization. Let $B = B_{8L}(a)$. Denote by E_B^* the Stokes pressure projection on B: for $F \in W^{-1,q}(B)$, $1 < q < \infty$, $\nabla E_B^*(F)$ is the gradient part of the stationary Stokes solution in B with zero boundary velocity. It satisfies the scale-invariant estimates

$$\|\nabla E_B^*(\nabla\cdot F)\|_{L^q(B)} \le C_q \|F\|_{L^q(B)}, \qquad \|\nabla E_B^*(\Delta w)\|_{L^2(B)} \le C\|\nabla w\|_{L^2(B)}.$$

Set

$$\nabla p_h = -\nabla E_B^*(v), \qquad w := v + \nabla p_h.$$

The harmonic pressure p_h is harmonic in B, and the projection estimates give

$$\|w\|_{L^2(B)} + \|\nabla p_h\|_{L^2(B)} \le C\|v\|_{L^2(B)}, \qquad \|\nabla w\|_{L^2(B_{6L}(a))} \le C\|\nabla v\|_{L^2(B)}.$$

The projected local energy inequality for w, obtained by applying the Stokes projection to the distributional Navier–Stokes equations and testing with $\zeta^2 w$, reads

$$\begin{aligned}\int |w(t)|^2\zeta^2 + \int_{s_0}^t \int |\nabla v|^2\zeta^2 \\ \leq C\int |w(s_0)|^2\zeta^2 + C\int_{s_0}^t\int |w|^2\left(|\partial_s\zeta| + |\Delta\zeta| + |\nabla\zeta|^2\right) \\ + C\int_{s_0}^t\int |v|\,|w|^2\,|\nabla\zeta| + C\int_{s_0}^t\int |v|\,|\nabla p_h|^2\,|\nabla\zeta|.\end{aligned} \tag{1.14}$$

Here $\zeta \in C_c^\infty(B_{4L}(a))$, $\zeta \equiv 1$ on $B_L(a)$, and

$$|\nabla\zeta| \leq CL^{-1}, \qquad |\Delta\zeta| + |\partial_s\zeta| \leq CL^{-2}.$$

The pressure-projection estimates allow w and ∇p_h to be bounded by v in $L^2(B_{4L}(a))$. Therefore the cutoff terms in (1.14) are bounded by

$$CL^{-2}\int_{s_0}^t A_{4L}(a,s)\,ds.$$

The two convective terms are bounded, using the Type I envelope, by

$$CL^{-1}\int_{s_0}^t \|v(s)\|_{L^\infty(B_{4L}(a))}A_{4L}(a,s)\,ds \leq CML^{-1}\int_{s_0}^t (-s)^{-1/2}A_{4L}(a,s)\,ds.$$

The initial projected energy is controlled by

$$\int |w(s_0)|^2\zeta^2\,dx \leq CA_{4L}(a,s_0).$$

Combining the preceding bounds gives (1.13). □

Definition 1.23 (Interior terminal energy estimate)**.** The local pointwise Type I window at z_* is said to satisfy the interior terminal energy estimate if, for every fixed $R < \infty$, there are

$$r_R > 0, \qquad B_R < \infty,$$

such that every rescaling $v = u^{(r)}$, $0 < r < r_R$, with $\Lambda = \rho/r$, satisfies

$$\operatorname*{ess\,sup}_{-1<s<0}\int_{B_{8R}} |v(x,s)|^2\,dx + \int_{-1}^0\int_{B_{8R}} |\nabla v|^2\,dx\,ds \leq B_R. \tag{1.15}$$

Equivalently, the dependence on A_{4L} in Lemma 1.22 can be stopped inside a fixed multiple of the target ball, uniformly as the boundary $\partial B_{\rho/r}$ of the local Type I window recedes to infinity.

Proposition 1.24 (Pressure-projected reduction to an interior energy bound)**.** *Assume the local pointwise Type I bound* (1.10) *at z_*. If the interior energy estimate* (1.15) *holds with $R = 1$, then the velocity no-escape condition* (1.8) *holds for every positive local Type I concentration sequence at z_*. Consequently such a sequence is admissible in the sense of Definition 1.19.*

Proof. Let $u_n = u^{(r_n)}$ be any positive local Type I concentration sequence. By (1.15) with $R = 1$, after discarding finitely many indices,

$$\operatorname*{ess\,sup}_{-1<s<0}\int_{B_1} |u_n(x,s)|^2\,dx \leq B_1.$$

The local Type I envelope gives

$$\|u_n(s)\|_{L^\infty(B_1)} \leq M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\int_{-\sigma}^{0}\int_{B_1}|u_n|^3\,dx\,ds \le \int_{-\sigma}^{0}\|u_n(s)\|_{L^\infty(B_1)}\int_{B_1}|u_n(x,s)|^2\,dx\,ds \le 2MB_1\sigma^{1/2}.$$

The finitely many discarded indices have absolutely continuous $|u_n|^3$-integrals on $B_1 \times (-1,0)$, so taking the supremum over the whole sequence and then letting $\sigma \downarrow 0$ gives (1.8). The positive Q_1 mass hypothesis is the other part of Definition 1.19. □

Remark 1.25 (Pressure-projected reduction). Lemma 1.22 shows, independently of the Albritton–Barker endpoint input used below, that terminal no-escape reduces to an interior terminal energy estimate: after local pressure projection, the local energy inequality controls the projected velocity and the dissipation up to the larger-scale quantity A_{4L}. The estimate (1.15) prevents terminal-layer concentration entering from the boundary of the local Type I window. In the present paper this bound is supplied instead by the endpoint weak Serrin local Type I bounds in Theorem 1.26.

Theorem 1.26 (Setup local Type-I implication). *Let (v,q) be a suitable weak solution in a parabolic cylinder*

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0).$$

Assume the local pointwise Type I envelope

$$\operatorname*{ess\,sup}_{t_0-\rho^2<t<t_0} (t_0 - t)^{1/2}\|v(t)\|_{L^\infty(B_\rho(x_0))} \le M < \infty.$$

Then the setup paper proves that the terminal local energy, pressure-gauge, and no-escape bounds stated in Proposition 1.21 hold on every smaller cylinder $Q_{\theta\rho}(z_0)$ with $0 < \theta < 1$, with constants depending only on ρ, θ, M and on the suitability constants of (v,q) on $Q_\rho(z_0)$. In particular every rescaling sequence centered at a local pointwise Type I terminal singularity satisfies Definition 1.19.

Proof. This theorem is imported verbatim from the companion setup paper; see Theorem 1.3 (2). Paper II does not reprove the weak-Serrin-to-local-Type I estimate or the terminal no-escape estimate. Its only role here is to place first-generation terminal sequences inside the admissible local-window class and to supply the local energy, pressure, and no-escape bounds used by Definition 9.7. □

Remark 1.27 (Albritton–Barker as the underlying input). The setup proof of Theorem 1.26 relies on the Albritton–Barker endpoint weak-Serrin estimate, which gives finite scale-invariant A, C, D, E-quantities on every $Q_{\theta\rho}(z_0)$ under the local pointwise Type I envelope. The constants depend on local suitable-solution background quantities on $Q_\rho(z_0)$, with no circularity because the input envelope and the suitable-solution data are part of the hypothesis data of the cylinder $Q_\rho(z_0)$, while the conclusion is restricted to $Q_{\theta\rho}(z_0)$. The detailed weak-Serrin discussion belongs to the setup paper; it is recalled here only to identify the source of the constants used in Proposition 1.21.

Theorem 1.28 (Local pointwise Type I implies terminal no-escape). *Let (u,p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$, let $z_* = (x_*, T)$, and assume that*

$$u \in L^\infty(0,T; L^2(\mathbb{R}^3)) \cap L^2(0,T; \dot H^1(\mathbb{R}^3)).$$

Assume that z_* satisfies the local pointwise Type I bound: there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\,\|u(t)\|_{L^\infty(B_\rho(x_*))} \le M. \tag{1.16}$$

For $0<r<\rho$, set

$$u^{(r)}(x,s)=r\,u(x_*+rx,T+r^2s),\qquad p^{(r)}(x,s)=r^2p(x_*+rx,T+r^2s).$$

Then for every $0<R<\infty$ there are constants

$$r_R\in(0,\rho/R),\qquad B_R<\infty,$$

and time-dependent pressure gauges $a_{r,R}(s)$, depending on R,M,ρ,T and the local suitable-solution background quantities of (u,p) in $Q_\rho(z_*)$, such that the following terminal local energy estimate holds for every $0<r<r_R$:

$$\begin{aligned}\operatorname*{ess\,sup}_{-1<s<0}\int_{B_R}|u^{(r)}(x,s)|^2\,dx+\int_{-1}^0\int_{B_R}|\nabla u^{(r)}|^2\,dx\,ds\\ +\int_{-1}^0\int_{B_R}|p^{(r)}(x,s)-a_{r,R}(s)|^{3/2}\,dx\,ds\le B_R.\end{aligned}\tag{1.17}$$

Consequently, for every $0<\sigma<1$ and every $0<r<r_R$,

$$\int_{-\sigma}^0\int_{B_R}|u^{(r)}(x,s)|^3\,dx\,ds\le 2MB_R\sigma^{1/2},\tag{1.18}$$

and in particular

$$\lim_{\sigma\downarrow 0}\sup_{0<r<r_R}\int_{-\sigma}^0\int_{B_R}|u^{(r)}(x,s)|^3\,dx\,ds=0.\tag{1.19}$$

Proof. Fix once and for all $0<\theta<1$, for instance $\theta=1/2$, and put $L_R:=\max\{R,1\}$. By decreasing r_R, we may assume

$$B_{L_Rr}(x_*)\times(T-L_R^2r^2,T)\subset Q_{\theta\rho}(z_*)\qquad(0<r<r_R).$$

The pointwise Type I bound implies the endpoint weak Serrin estimate on $Q_\rho(z_*)$. Indeed, after writing

$$f(t)=\|u(t)\|_{L^\infty(B_\rho(x_*))},$$

one has

$$|\{t\in(T-\rho^2,T):f(t)>\alpha\}|\le M^2\alpha^{-2},\qquad\alpha>0,$$

and therefore $u\in L_t^{2,\infty}L_x^\infty(Q_\rho(z_*))$. Applying Theorem 1.26 gives a finite constant K_θ such that $A,C,D,E\le K_\theta$ on all cylinders contained in $Q_{\theta\rho}(z_*)$.

Taking the cylinder $Q_{L_Rr}(z_*)$ gives

$$(L_Rr)^{-1}\operatorname*{ess\,sup}_{T-L_R^2r^2<t<T}\int_{B_{L_Rr}(x_*)}|u(x,t)|^2\,dx\le K_\theta,$$

and similarly the D- and E-bounds give

$$(L_Rr)^{-1}\int_{T-L_R^2r^2}^T\int_{B_{L_Rr}(x_*)}|\nabla u|^2\,dx\,dt\le K_\theta,$$

$$(L_Rr)^{-2}\int_{T-L_R^2r^2}^T\int_{B_{L_Rr}(x_*)}|p-a_{r,R}^{\mathrm{phys}}(t)|^{3/2}\,dx\,dt\le K_\theta$$

for suitable time-dependent pressure gauges $a_{r,R}^{\mathrm{phys}}(t)$. Scaling these three estimates and restricting to

$$B_R\times(-1,0)\subset B_{L_R}\times(-L_R^2,0)$$

gives (1.17) with

$$a_{r,R}(s):=r^2a_{r,R}^{\mathrm{phys}}(T+r^2s),$$

after enlarging the constant B_R by a harmless factor depending on R and K_θ.

After translating (x_*, T) to $(0, 0)$, the rescaled fields also satisfy

$$|u^{(r)}(x,s)| \le M(-s)^{-1/2} \qquad (x,s) \in B_R \times (-1,0)$$

for all $r < r_R$, after decreasing r_R if necessary. The terminal local energy estimate established above gives

$$\operatorname*{ess\,sup}_{-1<s<0} \int_{B_R} |u^{(r)}(x,s)|^2 \, dx \le B_R.$$

For $0 < \sigma < 1$, the local Type I bound above and the kinetic bound give

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}|^3 \, dx \, ds &\le \int_{-\sigma}^0 \|u^{(r)}(s)\|_{L^\infty(B_R)} \int_{B_R} |u^{(r)}(x,s)|^2 \, dx \, ds \\ &\le M B_R \int_{-\sigma}^0 (-s)^{-1/2} \, ds = 2 M B_R \sigma^{1/2}. \end{aligned}$$

This proves (1.18) and (1.19). ☐

Corollary 1.29 (No diffuse terminal-layer mass)**.** *Under the hypotheses of Theorem 1.28, for every fixed $R < \infty$ there do not exist $r_n \downarrow 0$, $\sigma_n \downarrow 0$, and $\varepsilon_0 > 0$ such that*

$$r_n^{-2} \int_{T-\sigma_n r_n^2}^{T} \int_{B_{Rr_n}(x_*)} |u(x,t)|^3 \, dx \, dt \ge \varepsilon_0 \qquad \textit{for all } n.$$

Proof. For $u_n = u^{(r_n)}$, the left hand side is

$$\int_{-\sigma_n}^0 \int_{B_R} |u_n(x,s)|^3 \, dx \, ds.$$

By (1.18), this is bounded by

$$2 M B_R \sigma_n^{1/2} \to 0,$$

which contradicts any fixed positive lower bound. ☐

Corollary 1.30 (Local pointwise Type I concentration sequences are admissible)**.** *Let (u, p) and $z_* = (x_*, T)$ satisfy the hypotheses of Theorem 1.28. Let $r_n \downarrow 0$ be any velocity concentration sequence satisfying*

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \ge \eta_v > 0, \qquad u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

Then u_n is an admissible local Type I concentration sequence in the sense of Definition 1.19.

Proof. Apply Theorem 1.28 with $R = 1$. After discarding finitely many n, all $r_n < r_1$, and

$$\lim_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 \, dx \, dt = 0.$$

The finitely many discarded indices do not affect the limit as $\sigma \downarrow 0$, because each fixed suitable weak solution has $|u_n|^3 \in L^1(B_1 \times (-1, 0))$. The positive mass hypothesis is the other condition in Definition 1.19. ☐

Lemma 1.31 (Local Type I concentration enters the terminal dichotomy)**.** *Under the hypotheses of Proposition 1.21, let $r_n \downarrow 0$ be a velocity concentration sequence at z_* satisfying*

$$\iint_{B_1 \times (-1,0)} |u_n|^3 \, dx \, dt \ge \eta_v > 0.$$

Then r_n is admissible by Corollary 1.30. Let σ_0, η_1 be given by Lemma 1.20. For the associated terminal extraction sequence (V_N, P_N) generated by a Seregin limit of this sequence, choose the small-data threshold $\varepsilon_{\mathrm{sd}} \le \eta_1/2$. Then the origin extraction produces a bounded ancient Seregin profile with retained local velocity concentration away from the terminal layer. In particular the retained velocity concentration generated from the origin recentering has no loss of local compactness. Any retained local recentering which stays in the local Type I window has the same compactness property by Proposition 1.21; any concentration leaving that window is assigned to the exterior or diffuse terminal remainder. The paired terminal decomposition of Theorem 9.24, together with the mixed-profile exclusions and concentration-chain exclusions Theorems 9.113 and 10.62, reduces the retained possibilities to a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles.

Proof. By Lemma 1.20,

$$\iint_{B_1\times(-1,-\sigma_0)} |u_n|^3\,dx\,dt \ge \eta_1 \qquad \text{for all } n.$$

The cylinder $B_1 \times [-1, -\sigma_0]$ is compactly contained in $\mathbb{R}^3 \times (-\infty, 0)$. Therefore Proposition 1.21 gives, after passing to a subsequence, strong local L^3 convergence on this cylinder:

$$u_n \to U \qquad \text{in } L^3(B_1 \times (-1, -\sigma_0)).$$

Consequently

$$\iint_{B_1\times(-1,-\sigma_0)} |U|^3\,dx\,dt \ge \eta_1.$$

Cover $B_1 \times (-1, -\sigma_0)$ by finitely many unit terminal cylinders strictly away from $t = 0$. At least one such cylinder carries a fixed positive velocity lower bound, and since the raw atlas defect measure dominates the velocity part, choosing $\varepsilon_{\mathrm{sd}} \le \eta_1/2$ after adjusting by this finite cover makes the corresponding origin-frame recentering raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$. This argument is precisely where no-escape is used: without (1.8), the lower bound could remain entirely in $B_1 \times (-\sigma, 0)$ for every fixed $\sigma > 0$.

The distinguished recentering satisfies the local-window condition in Proposition 1.21, so it is not in the local-compactness-failure class. Other retained local recenterings inside the same Type I window have the same compactness. Recenterings leaving the window are exterior or diffuse terminal remainders. The paired decomposition Theorem 9.24 records these remainders alongside the concentration set rather than treating them as mutually exclusive alternatives. The sole-source radiative cases are excluded by Theorem 9.27, mixed compact–diffuse or escaping noncompact coexistence is handled by Theorem 9.113 and Lemma 9.114, and infinite chains of retained concentrating descendants are excluded by Theorem 10.62. Hence Corollary 9.25 leaves only a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles. □

Theorem 1.32 (Local Type I singularities enter the terminal concentration setting)**.** *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that z_* satisfies the local Type I bound (1.10). Then there exists a bounded centered Type I Seregin profile with retained local velocity concentration generated from (u, p, z_*). Consequently the local terminal concentration setting of Theorem 10.66 applies after the residual closure theorem is proved.*

Proof. By the local velocity concentration theorem in the setup paper, choose $r_n \downarrow 0$ and $\eta_v > 0$ with

$$\iint_{B_1\times(-1,0)} |u_n|^3\,dx\,dt \ge \eta_v, \qquad u_n(x,t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

Theorem 1.28 and Corollary 1.30 show that this concentration sequence satisfies the terminal velocity no-escape condition required by the setup paper. Apply Lemma 1.31 to this sequence. The local Type I rescalings therefore produce at least one bounded terminal profile carrying retained local velocity concentration. After passing to the centered variables, the bound $|U(x,t)| \le M(-t)^{-1/2}$ becomes the global centered Type I bound

$$|V(y,\tau)| \le M,$$

so each retained profile generated by this implication is a bounded centered Type I Seregin profile.

If every such retained profile belongs to one of the previously closed lower classes, then the contradiction is supplied by the setup paper: small-amplitude and stationary L^3 profiles are excluded there, the uniformly tight class is excluded there, and the closed structured/decay class is excluded there. Otherwise choose a retained profile outside those lower classes. It is then a refined residual candidate, so Theorem 10.66 applies. In that residual setting, a finite separated family of retained concentrating terminal profiles is excluded by Theorem 10.65, infinite chains of retained concentrating descendants are excluded by Theorem 10.62, and mixed compact–diffuse tails are excluded by Theorem 9.113. The remaining single retained profile is the terminal case closed in Section 12. Thus the present theorem supplies the entry into the residual decomposition; the contradiction is the final two-paper corollary proved after the residual closure. □

Corollary 1.33 (Local pointwise Type I exclusion after setup plus residual closure)**.** *Let (u,p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0,T)$. Combining the residual closure Theorem 1.16 of this paper with the setup paper, the following holds: if $z_* = (x_*, T)$ is a singular point satisfying the local Type I bound (1.10), then the completed case closures give a contradiction. Equivalently, in the local Type I/local Type II dichotomy of the setup paper, every finite-energy singular point belongs to the local Type II alternative. The conclusion is the joint output of the two-paper chain; it is not asserted as a hidden hypothesis inside Paper II alone.*

Proof. The entry assertion is Theorem 1.32. The contradiction is deferred to the final assembly, where the residual closure is proved and then inserted into the setup paper's local final theorem. The equivalence is the local Type I/local Type II dichotomy of the setup paper. □

2. Ordered residual profile decomposition

The first diagram is the physical entry view: it ends when the setup-paper alternatives leave a coarse residual Seregin state. The second diagram is the refined classification view of that residual state. Its diamonds are ordered selection criteria, not theorem dependencies. The local Type II alternative is the complementary physical regime after the local pointwise Type I case is excluded. Each Type I state is closed by an inherited lower theorem, a local closure, or a reduction followed by terminal assembly, as recorded in Tables 2 and 9.

The dashed edge continues the ordered classification from the first unresolved lower criterion. It carries no profile transformation and asserts no logical dependence on a later closure theorem. Once the preceding states have been assigned and removed, $\mathcal{C}_{\mathrm{gen}}$ is the remaining residual class by definition. For the table, write

$$\mathfrak{T}_\rho(u; z_*) := \operatorname*{ess\,sup}_{T-\rho^2<t<T} \sqrt{T-t}\, \|u(t)\|_{L^\infty(B_\rho(x_*))}.$$

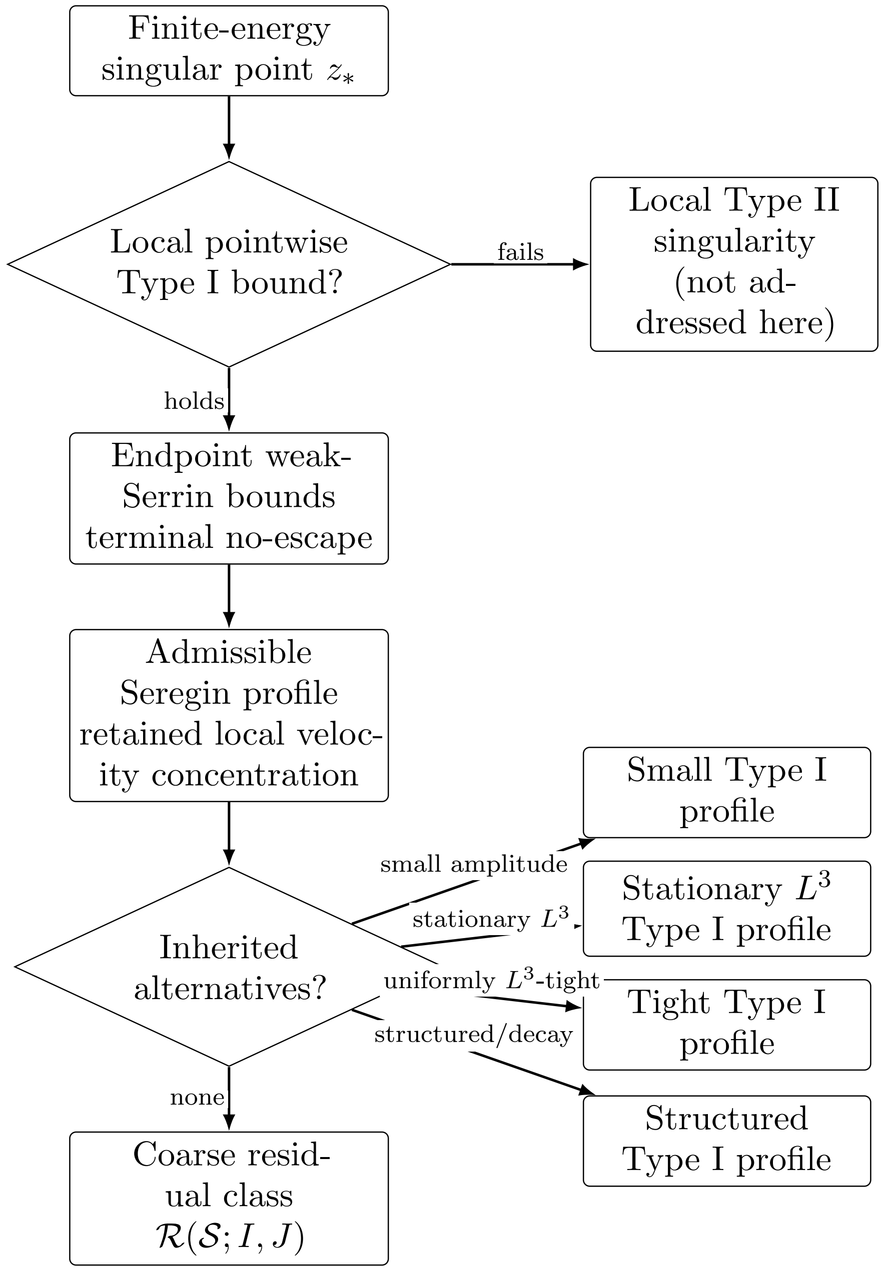

FIGURE 10. Physical entry view. Local pointwise Type I enters the admissible Seregin compactness setting through endpoint weak Serrin estimates and terminal no-escape. Failure of the bound selects the local Type II alternative, which is not addressed here. This diagram ends at the coarse residual state and does not encode the later theorem dependency order.

Terminal alternative	Criterion, exclusion mechanism, and reference

TABLE 9. Detailed criterion-and-closure view of the residual decomposition. Each row gives the criterion selecting the alternative, the mechanism that excludes or routes it, and the statement used at that point. This table describes state selection and closure provenance; theorem-production order is recorded in Tables 4 and 16. For the inherited alternatives, the displayed condition is a representative form of the inherited definition; the precise definitions and closures are those of the companion setup paper, as summarized in Theorem 1.3.

Terminal alternative	Criterion, exclusion mechanism, and reference
Local Type II singularity	**Criterion.** For every $\rho > 0$, $\mathfrak{T}_\rho(u; z_*) = \infty$. **Role.** This is the complementary alternative left by the local Type I/Type II dichotomy after local pointwise Type I is excluded. **Link.** Corollary 1.33.
Small Type I profile	**Criterion.** Inherited small-amplitude class; in the setup normalization, $\|V\|_{L^\infty(\mathbb{R}^3\times\mathbb{R})} \le \varepsilon_{\mathrm{sm}}$. **Exclusion.** Small-data/regularity closure recorded in the setup paper. **Link.** Definition 1.8.
Stationary L^3 Type I profile	**Criterion.** Inherited stationary class; representative condition $\partial_\tau V = 0$ and $V \in L^3(\mathbb{R}^3)$. **Exclusion.** Stationary L^3 Liouville/regularity closure recorded in the setup paper. **Link.** Definition 1.8.
Tight Type I profile	**Criterion.** Inherited tight class; representative condition $\lim_{R\to\infty} \sup_\tau \int_{\lvert y\rvert>R} \lvert V(y,\tau)\rvert^3\, dy = 0$. **Exclusion.** Endpoint L^3 rigidity for uniformly tight ancient dynamics. **Link.** Definition 1.8.
Structured Type I profile	**Criterion.** Inherited closed structured/decay class, $V \in \mathcal{C}^{\mathrm{setup}}_{\mathrm{struct/decay}}$. **Exclusion.** Closed structured/decay alternatives recorded in the setup paper. **Link.** Definition 1.8.
$\mathcal{C}_{\mathrm{ax}}$ residual profile	**Criterion.** Axisymmetric residual profile with bounded centered circulation $\Gamma(\rho, Z, \tau) = \rho V_\theta$ and $\|\Gamma\|_{L^\infty} < \infty$. **Exclusion.** Centered circulation equation plus axis-absorption Liouville theorem; the profile becomes no-swirl and falls into an earlier structured class. **Link.** Theorem 4.4.
$\mathcal{C}_{\mathrm{rot}}$ residual profile	**Criterion.** Rotational relative equilibrium $V(y,\tau) = Q(\tau)W(Q(\tau)^T y)$, $Q(\tau) = e^{\tau\Omega}$. **Reduction/closure.** Localized Coriolis-flux reduction sends the profile into lower classes already excluded by the setup paper or the terminal velocity-concentration decomposition; closure is only after terminal stratification assembly. **Link.** Proposition 5.8 and Corollary 10.68.
$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$ residual profile	**Criterion.** Degenerate stationary-hull retained profile: along time translations $T_{s_k}V \to W$, where W is stationary and $\mathcal{V}_1(W; 0) \ge \eta_*$. **Exclusion.** Scale reselection reduces to a stationary retained concentration profile, then terminal concentration excludes it. **Link.** Propositions 6.12 and 7.6 and Corollary 10.69.
$\mathcal{C}_{\mathrm{aff}}$ affine/parasitic routing row	**Criterion.** The affine-normalized oscillation vanishes: $\Phi(S_c U) = 0$ for an admissible affine symmetry S_c. **Routing.** Oscillatory entry normalization assigns such profiles to the affine/parasitic quotient row before retention. **Link.** Theorem 9.99.
$\mathcal{C}_{\log\mathrm{diff}}$ critical tail	**Criterion.** Logarithmic escape component in the realized critical-tail compactification: $\Lambda \ne 0$. **Exclusion.** Log-window descendant selection plus minimal mesoscopic-scale reduction and hidden-scale compactness. **Link.** Corollary 9.57.
$\mathcal{C}_{\mathrm{Young}}$ tail	**Criterion.** Tight annular critical-tail profile with non-Dirac Young measure: $\Lambda = 0$ and $\mathcal{Y}_{x,s} \ne \delta_{\bar W(x,s)}$ on a set of positive measure. **Exclusion.** No hidden-scale variance gives strong L^3_{loc} compactness, hence a Dirac Young measure. **Link.** Corollary 9.100.

Terminal alternative	**Criterion, exclusion mechanism, and reference**
$\mathcal{C}_{\mathrm{homcrit}}$ coherent tail	**Criterion.** Coherent (-1)-homogeneous tail: $W(r\omega, s) = r^{-1}A(\omega, s)$. **Exclusion.** Bounded-origin realization gives $W \in L^\infty_{\mathrm{loc}}$ near $r = 0$, forcing $A = 0$. **Link.** Corollary 9.111.
$\mathcal{C}_{\mathrm{logper}}$ coherent tail	**Criterion.** Nonzero log-periodic coherent tail: $Z(\rho + L, \omega, s) = Z(\rho, \omega, s)$, $L \neq 0$, where $Z = e^\rho W(e^\rho\omega, s)$. **Exclusion.** Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \to 0$ as $\rho \to -\infty$; periodicity forces $Z \equiv 0$. **Link.** Corollary 9.111.
$\mathcal{C}_{\mathrm{ap}}$ coherent tail	**Criterion.** Nonzero aperiodic minimal logarithmic hull: $\mathcal{H}(Z)$ is minimal, nonperiodic, and nonzero. **Exclusion.** Bounded-origin realization puts 0 in the compact log-translation hull; minimality forces the hull to be $\{0\}$. **Link.** Corollary 9.111.
$\mathcal{C}_{\mathrm{gen}}$ generic terminal residual	**Criterion.** Ordered complement $$V \in \mathcal{R}(\mathcal{S}; I, J) \setminus \bigcup_{\mathcal{B} \in \mathfrak{B}_{\mathrm{lower}}} \mathcal{B},$$ where $\mathfrak{B}_{\mathrm{lower}}$ is the list of all preceding Type I alternatives. **Exclusion.** Paired concentration-set decomposition, finite separated-family exclusion, terminal indecomposability, sequence-L^3, parasitic-free mildness, and the Albritton–Barker endpoint Liouville theorem. **Link.** Theorems 11.2 and 11.3.

3. Shared local notation

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

Definition 3.1 (Scale-invariant CKN concentration and defect measures)**.** For a centered profile (V, P), define the velocity concentration

$$\mathcal{V}_r(V; z_0) := r^{-2} \iint_{Q_r(z_0)} |V|^3 \, dy \, d\tau, \tag{3.1}$$

and the quotient pressure concentration

$$D_r(P; z_0) := r^{-2} \inf_{c \in L^{3/2}(\tau_0 - r^2, \tau_0)} \iint_{Q_r(z_0)} |P(y, \tau) - c(\tau)|^{3/2} \, dy \, d\tau. \tag{3.2}$$

The scale-invariant local CKN quantity is

$$\mathcal{E}_r(V, P; z_0) := \mathcal{V}_r(V; z_0) + D_r(P; z_0). \tag{3.3}$$

Thus $\mathcal{E}_r$ is parabolically scale-invariant under the Navier–Stokes scaling.

If an explicit pressure atlas $\mathfrak{a}$ has been chosen, with local representative $a_{z_0,r}(\tau)$, we also write

$$\mathcal{E}^{\mathfrak{a}}_r(V, P; z_0) := r^{-2} \iint_{Q_r(z_0)} \left(|V|^3 + |P - a_{z_0,r}(\tau)|^{3/2} \right) dy \, d\tau. \tag{3.4}$$

Then

$$\mathcal{E}_r(V, P; z_0) \leq \mathcal{E}^{\mathfrak{a}}_r(V, P; z_0).$$

Thresholds and regularity statements use the quotient quantity $\mathcal{E}_r$; compactness statements use a fixed setup pressure atlas when an explicit representative is needed.

For a terminal sequence (V_n, P_n) with chosen pressure gauges $a_n(\tau)$, the atlas-dependent Radon defect measure is

$$d\mu_n := \left(|V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy \, d\tau. \tag{3.5}$$

The measure μ_n is not denoted by $\mathcal{E}_r$. On unit cylinders the factor 1^{-2} is invisible, so unit-scale atlas mass and unit-scale CKN concentration are compatible, with the atlas quantity dominating

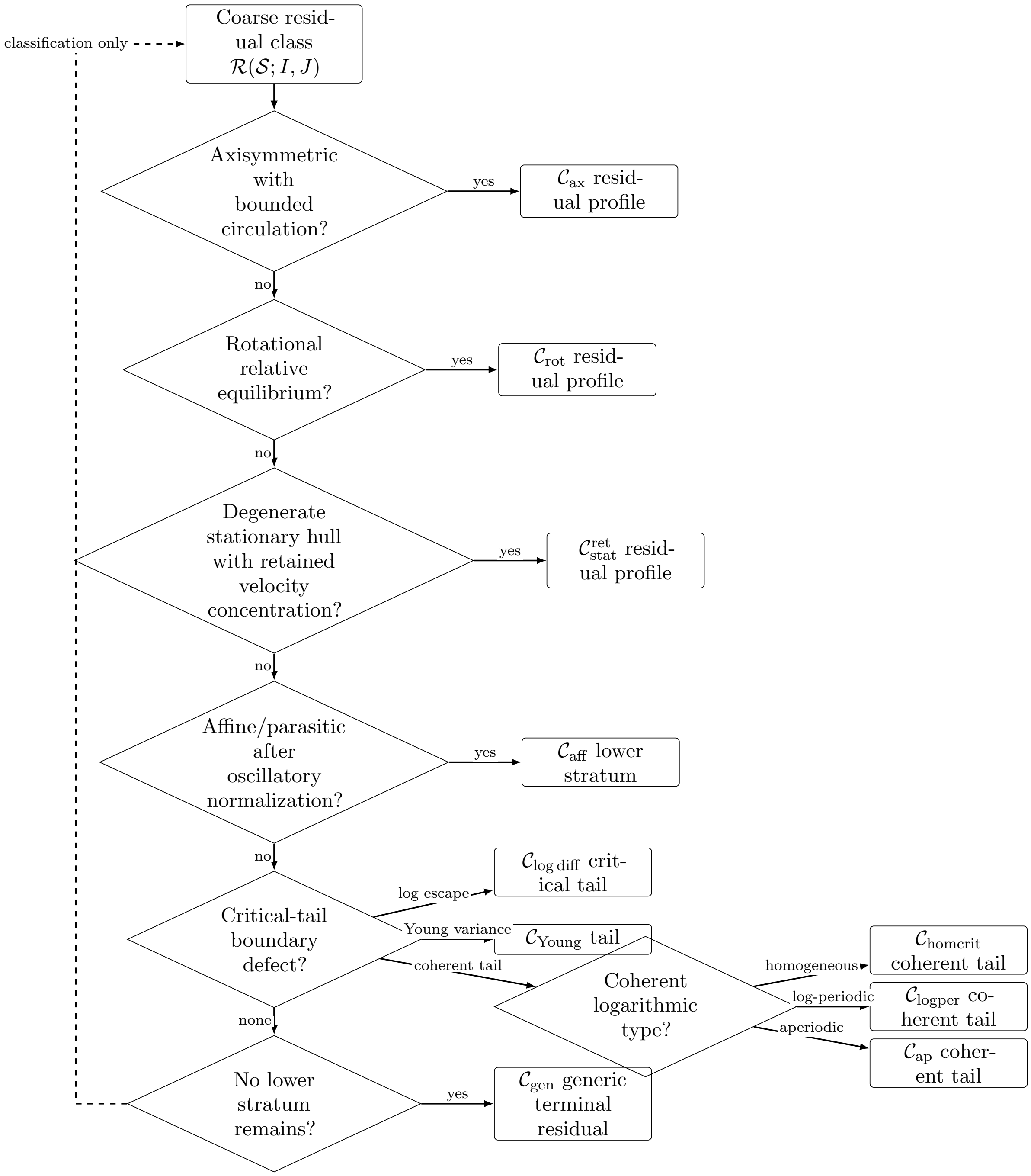

FIGURE 11. Refined residual classification view. Solid arrows select a residual state. The dashed return edge records continuation of the ordered classification when a lower state has not yet been assigned; it is not a theorem dependency or a descendant-construction edge. After all lower states are removed, the remaining class is $\mathcal{C}_{\text{gen}}$.

the quotient quantity. The velocity quantity $\mathcal{V}_r$ is the part used to certify nonzero retained limits under weak pressure convergence; the pressure-inclusive quantity $\mathcal{E}_r$ is used for CKN regularity and pressure-normalized smallness.

Remark 3.2 (Two-scale dictionary)*.* We use two different normalizations. The quantity $\mathcal{E}_r$ is parabolic and CKN-critical, hence carries the factor r^{-2}. By contrast, the tail quantities $\mathcal{A}_n(z,r)$ are spatial/logarithmic averaged oscillations on fixed centered time windows; their natural normalization is r^{-3}. These two quantities are never identified.

Definition 3.3 (Velocity retention at a level and unit CKN activity)**.** Fix $0 < \eta_*, \varepsilon_* < \infty$. For $\eta > 0$, a terminal profile (U, Π) has retained local velocity concentration at level η if, after the terminal normalization under consideration,

$$\mathcal{V}_1(U; 0) \geq \eta.$$

It has positive retained local velocity concentration if this holds for some $\eta > 0$. When no level is displayed, the setup-level retained descendant threshold is η_*. It has unit CKN activity if

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Only retained velocity concentration is used to pass nontriviality to limits from approximating sequences. Pressure-inclusive CKN activity is used for local regularity, pressure compactness, and threshold smallness, and any pressure-only lower bound is treated through the pressure-gauge alternatives below.

Definition 3.4 (Threshold hierarchy)**.** The constants used in the terminal concentration argument are chosen in the following order. The compact velocity lower bounds η_0 and η_v are supplied by the imported setup results and the local concentration input of the companion setup paper, respectively. The local small-data threshold $\varepsilon_{\mathrm{sd}}$ is then fixed with

$$0 < \varepsilon_{\mathrm{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\mathrm{CKN}}\},$$

where $\varepsilon_{\mathrm{CKN}}$ denotes the universal CKN regularity threshold whenever that threshold is invoked. The retained descendant velocity threshold η_* is chosen after the pressure gauges and compactness topology are fixed, with

$$0 < \eta_* \leq \frac{1}{2}\varepsilon_{\mathrm{sd}}.$$

The pressure-inclusive unit CKN activity threshold ε_* is then chosen with

$$0 < \varepsilon_* \leq \frac{1}{2}\varepsilon_{\mathrm{sd}}.$$

Whenever a compactness or lower-semicontinuity passage loses a harmless fixed factor, the relevant threshold, η_* for velocity retention or ε_* for pressure-inclusive CKN activity, is replaced once and for all by a smaller positive value. No later argument chooses a new concentration threshold depending on a previously extracted iterated profile extraction.

Definition 3.5 (Refined residual candidate)**.** A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the admissible Seregin extraction theorem in the companion setup paper, which carries the positive compact local velocity concentration in (1.5), and which is not in any alternative already closed before the refined residual decomposition is entered. The definition is local: the only persistent lower bound attached to the candidate by the setup paper is the fixed compact velocity L^3 lower bound.

Remark 3.6 (No hidden global L^3 assertion)*.* The terminal arguments below do not claim that a generic residual profile is globally L^3-tight or globally bounded in $L^\infty_\tau L^3_y$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence-L^3 completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

3.1. **Notation, threshold, and gauge directories.** The following compact directories collect the persistent symbols, thresholds, and gauges used in the residual closure. They are not new definitions; each entry refers to the location where the symbol is first defined.

Velocity, pressure, and profile-space symbols.

- (V, P): centered Seregin ancient profile in (1.2); first used in Section 1.
- (U, Q): physical ancient representative when (V, P) is treated under physical pullback (1.4); also used in $\mathcal{R}_z$-recentered form throughout Sections 8 and 9. The intended meaning is fixed locally in each section by the explicit definition of the immediately preceding rescaling.
- (W, Q_W): blown-down, co-rotating, or tail profile, depending on the section; defined locally in Sections 5 and 9.3.
- $\mathcal{E}_r(V, P; z_0)$: quotient, scale-invariant parabolic CKN concentration, see Definition 3.1.
- $\mathcal{V}_r(V; z_0)$: velocity part of the scale-invariant local concentration; retained descendants and nonzero limiting profiles are certified with this quantity.
- $\mathcal{E}_1$: pressure-normalized unit CKN concentration, fixed at $r = 1$; used for local regularity, CKN smallness, and pressure estimates once the velocity lower bound has been identified.
- μ_n: atlas-dependent Radon defect measure in (3.5); used for concentration-set and defect compactness, not as a scale-invariant quantity.
- $\mathcal{A}_n(z, r)$: spatial/logarithmic averaged affine-quotient oscillation with r^{-3} normalization, distinct from $\mathcal{E}_r$.
- $\mathfrak{X}_M$: bounded terminal profile class with pressure gauges, Definition 8.7.
- $\mathfrak{X}_M^{\#}$: compact normalized descendant profile class, Definition 8.10.
- $\mathfrak{T}_M$: critical-tail compactification of Young/log profiles together with pressure-defect, viscous-defect, affine-quotient, virial, and support coordinates; abbreviated as $(\bar{W}, \mathcal{Y}, \Lambda)$ only when the extra coordinates are zero or irrelevant, Definition 9.39.
- $\mathcal{C}_{\text{aff}}$: affine/parasitic reduction/quotient stratum, Definition 8.16; not a usual nonempty/empty PDE class but an exclusion label, see Remark 8.17 and Theorem 9.99.
- $\text{osc}_{3,\text{sp}}$: spatial oscillation modulo time-dependent homogeneous gauges, Definition 9.60.
- $\text{osc}_{3,\text{ct}}$ and Φ: constant-mode/non-affine retained oscillation, Definition 8.8 and Theorem 8.19.

Threshold ordering.

- η_0: compact velocity L^3 lower bound supplied by the base Seregin extraction Theorem 1.7.
- η_v: compact velocity lower bound from the local concentration input of the setup paper, used as initial mass for terminal extraction; see Definition 3.4.
- ε_{sd}: local small-data threshold, $0 < \varepsilon_{\text{sd}} \le \min\{\eta_0, \eta_v, \varepsilon_{\text{CKN}}\}$, Definition 3.4.
- η_*: retained descendant velocity threshold, $0 < \eta_* \le \varepsilon_{\text{sd}}/2$, used in Definition 3.3 and in concentration-set extraction.
- ε_*: unit CKN activity threshold, $0 < \varepsilon_* \le \varepsilon_{\text{sd}}/2$, used only for pressure-inclusive CKN activity and smallness tracking.
- ε_0: minimal mesoscopic-scale oscillation threshold; depends only on the hidden-scale defect lower bound and on the finite-overlap constants of Lemma 9.62; first appears in Definition 9.61.
- η_K: compact oscillation lower bound on the recurrent concentration core, first introduced in Theorem 10.17.

The thresholds are fixed in the order $\eta_0, \eta_v \Rightarrow \varepsilon_{\text{sd}} \Rightarrow (\eta_*, \varepsilon_*) \Rightarrow \varepsilon_0, \eta_K$, and no later argument chooses a new concentration threshold depending on a previously extracted iterated profile extraction. Phrases of the form "after lowering the threshold" refer either to the once-and-for-all reduction of η_* or ε_* recorded in Definition 3.4, or to a slight reduction of the mesoscopic threshold ε_0 recorded in Lemma 9.65; in either case the reduction does not disturb earlier choices.

Pressure and velocity gauges.

- $c(\tau)$ or $a_{z_0,r}(\tau)$: local pressure time gauge in (3.2) and (3.4); measurable in τ; used because pressure is defined modulo functions of time.
- $c_n(s)$: homogeneous velocity gauge at the minimal mesoscopic scale, Definition 9.66; bounded; either compactifies or assigns the descendant to $\mathcal{C}_{\mathrm{aff}}$ by Lemma 9.79; no pointwise derivative $\partial_s c_n$ is used unless that compactness has already been obtained.
- $\beta(s)\cdot y+\alpha(s)$: affine parasitic pressure paired with a spatially homogeneous velocity mode; quotient notation introduced in Definition 8.16 and used only in the gauged formulation of the affine/parasitic stratum.
- $L_n(x,s)=b_n(s)\cdot x+a_n(s)$: scaled affine pressure at the minimal mesoscopic scale, Lemma 9.74; affine in x, measurable in s; discarded in the non-affine quotient and assigned to $\mathcal{C}_{\mathrm{aff}}$ when it does not vanish.
- $a_{n,K}(t)$: local Seregin pressure time gauge on a compact cylinder K, used by the local pressure-gauge compactness theorem inherited from the setup paper and recorded in Proposition 1.21.

Local coordinate and pressure packages. Each row below is an interface for the profile named in its first column. It records only conclusions supplied by the cited definitions and theorems; in particular, covariance of the equation, compactness, pressure compatibility, and retention of velocity concentration are separate assertions.

TABLE 10. Local coordinate, topology, threshold, and pressure directory. The owner in the first column belongs to the fixed singular-profile ledger. Core and working regions, gauges, compactness inputs, and failure routes are local to that owner and to the coordinate package in the same row.

Package and local owner	**Domain and velocity map**	**Pressure representative and quotient**	**Topology and enabled local input**	**Output or named route**
Root centered chart for the profile selected from the fixed extraction at z_*.	The centered and physical representatives are related by (1.3) and (1.4). Local quantities are evaluated on fixed compact cylinders $Q_r(z_0)$ in the centered chart of the selected profile.	The setup pressure atlas supplies a representative on each cylinder modulo a time function. Overlap compatibility is Lemma 8.15; no affine quotient is taken before affine routing.	The setup extraction supplies local suitability, the local Seregin topology, and the root velocity thresholds η_0,η_v; see Theorem 1.7 and Definition 3.4.	If a setup lower state is selected, its named setup closure applies to this profile. Otherwise Proposition 1.5 routes the same fixed-ledger profile through the affine or retained row.
Ordinary local pressure chart for the current locally suitable profile.	The working cylinder is $Q_4=B_4\times(-16,0)$, the pressure split is made on Q_2, and harmonic interior estimates are read on the core Q_1. The velocity is unchanged.	With a time gauge $a_n(t)$, $p_n-a_n=p_n^{\mathrm{loc}}+h_n$, where p_n^{loc} is the cutoff Riesz part and h_n is spatially harmonic. The affine part of h_n is quotiented only in the affine/parasitic state.	Strong local L^3 convergence of velocity gives strong $L^{3/2}$ convergence of the local Riesz part. The pressure-inclusive threshold ε_* is used for local CKN activity, not to certify a retained velocity limit.	Lemmas 9.5 and 9.6 routes to neighboring velocity activity, $\mathcal{C}_{\mathrm{aff}}$, or a named pressure-loss alternative.
Exact centered or covariant observer of the current realized profile.	For $z=(\xi,\sigma)$, $\mathcal{R}_zU(y,s)=U(y+e^{s/2}\xi,\sigma+s)$. It maps $Q_r(0,0)$ bijectively to $\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma)$ with unit spacetime Jacobian; see Theorem 8.6.	Pressure is pulled back by the same map and remains defined modulo the local time gauge valid on the pulled-back cylinder. Affine pressure is available only after the affine quotient has been selected.	The map is an exact symmetry and is continuous in the local topology of $\mathfrak{X}_M$. It transports local equations and local integrals between the displayed paired windows.	For a sequence of observers, compactness and retained mass require the separate descendant contract below; covariance alone supplies neither conclusion.

Package and local owner	**Domain and velocity map**	**Pressure representative and quotient**	**Topology and enabled local input**	**Output or named route**
First-generation physical recentering of the selected setup terminal sequence.	For $\mathfrak{c} = (y_n, \tau_n)$, $V_n^{\mathfrak{c}}(y,s) = V_n(y + y_n, s + \tau_n)$. Every fixed compact core $K_{R,S}$ must have its physical image inside the fixed setup cylinder $\mathcal{Q}_{\mathrm{loc}}$; the compactness proof uses the corresponding enlarged working window $K_{2R,2S}$.	$P_n^{\mathfrak{c}} = P_n(y+y_n, s+\tau_n) - a_n^{\mathfrak{c}}(s)$, with the local setup-atlas time gauge valid on that physical image. No descendant or affine gauge is imported into this row.	Under the local-window condition, the setup bounds give strong local L^3 velocity compactness, weak local $L^{3/2}$ pressure compactness modulo time functions, and stability of the local energy inequality. A displayed $\liminf_n \mathcal{V}_1(V_n; z_n) > 0$ yields retention at the fixed descendant threshold η_*.	Lemma 9.8 gives the retained local limit. Failure of the local-window condition is routed by Lemma 9.10 to an exterior, diffuse, pressure/noncompact, or critical-tail state.
Covariant recentering of an already realized normalized descendant, including its diagonal lift.	The current owner is $(U, \Pi) \in \mathfrak{X}_M^{\#}$. Covariant sequences act by $\mathscr{G}_{g_n}$; a descendant lifted to the original terminal sequence uses $d_n = (g_n; \xi_n, \sigma_n)$ and the combined map of Definition 9.2.	Pressure is pulled back with the velocity and then placed in the compatible local time gauge from the descendant atlas. The affine quotient is used only if affine routing is selected.	Compact closure is tested in the normalized local Seregin topology. In its compact branch, a displayed $\liminf_n \mathcal{V}_1(\mathscr{G}_{g_n} U; 0) > 0$ and strong local L^3 convergence transfer retained velocity concentration; ancestry is supplied by Theorems 1.13 and 10.12.	Lemma 9.9 and Corollary 9.11 gives a retained descendant or routes lack of local/pressure compactness to the named exterior, diffuse, pressure/noncompact, or critical-tail state.
Canonical affine-normalized representative of the current pre-quotient profile.	On each compact cylinder, the exact symmetry is $S_c V(y,\tau) = V(y - 2c, \tau) - c$. Spatially homogeneous time-dependent modes are measured by the quotient seminorm $\mathrm{osc}_{3,\mathrm{sp}}$; non-affine activity is measured by Φ.	The transformed pressure is $S_c P(y,\tau) = P(y - 2c, \tau) + \frac{1}{2} c \cdot y$. Ordinary charts remain modulo time functions; affine spatial gauges are admitted only in the quotient state of Definition 8.14.	The map is continuous $\mathfrak{X}_M^{\#} \to \mathfrak{X}_{M+\lvert c\rvert}^{\#}$ for bounded c. The compact activity threshold η_K is produced only on a compact set already known to avoid the homogeneous quotient state.	Theorems 8.21 and 9.99 routes the current profile to $\mathcal{C}_{\mathrm{aff}}$ or produces its canonical representative with positive non-affine activity. Retained mass continues to be governed by its separate velocity and heredity interface.
Minimal mesoscopic blow-down of the current realized critical-tail or hidden-scale profile.	After the covariant center z_n and first-bad radius r_n, the ungauged field is $V_n(x,s) = (\mathcal{R}_{z_n} U_n)(r_n x, s)$; the quotient velocity is $W_n = V_n - c_n(s)$. Equations are tested on fixed compact core cylinders inside the selected blow-down window.	The actual pressure is $Q_n^{\mathrm{act}} = r_n^{-1} (\mathcal{R}_{z_n} \Pi_n)(r_n x, s)$, modulo time functions. The quotient pressure $Q_n^{\#} = Q_n^{\mathrm{act}} - L_n$ is used only in the affine topology. The local energy inequality always uses $(V_n, Q_n^{\mathrm{act}})$.	The first-bad threshold is ε_0. Velocity uses weak-* local L^∞/Young compactness; after the affine coefficient dichotomy the non-affine pressure converges strongly in local $L^{3/2}$, enabling the pressure–velocity product limit and the local energy passage.	Lemmas 9.77 and 9.78 routes a surviving affine coefficient to $\mathcal{C}_{\mathrm{aff}}$; otherwise Theorem 9.86 produces a realized descendant in $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, or $\mathcal{C}_{\mathrm{cohcrit}}$.

The threshold types in this directory are not interchangeable. The root quantities η_0, η_v belong to the fixed extraction, $\varepsilon_{\mathrm{sd}}$ is the local small-data threshold, η_* certifies retained descendant velocity, ε_* records pressure-inclusive unit CKN activity, and ε_0 selects a first-bad mesoscopic oscillation scale. An exact coordinate symmetry preserves the equation but supplies compactness or retained velocity concentration only when the corresponding row cites the required local-window, topology, lower-bound, realization, and heredity statements. Mildness and sequence-L^3 remain downstream profile properties produced by their named terminal theorems.

Table 11. Glossary of residual symbols.

Symbol	**Meaning**
$\mathcal{E}_r$	Scale-invariant parabolic CKN concentration.
D_r	Pressure quotient part of the CKN concentration, modulo time gauges.
$\mathcal{V}_r$	Scale-invariant velocity concentration used for retained nonzero descendants.
μ_n	Raw atlas defect measure for concentration-set and compactness arguments.
$\mathcal{A}_n(z,r)$	Spatial/log-tail averaged affine-quotient oscillation.
$\mathcal{H}_\infty$	Full covariant spacetime asymptotic hull.
$\mathcal{C}_{\mathrm{ax}}$	Axisymmetric bounded-circulation residual class.
$\mathcal{C}_{\mathrm{rot}}$	Rotational relative-equilibrium residual class.
$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$	Stationary-hull residual class with retained local velocity concentration.
$\mathcal{C}_{\mathrm{aff}}$	Affine/parasitic pressure residual quotient.
$\mathcal{C}_{\mathrm{log\,diff}}$	Log-diffuse critical-tail residual alternative.
$\mathcal{C}_{\mathrm{Young}}$	Young/defect critical-tail residual alternative, including non-Dirac Young measures and nonzero retained critical-tail pressure/viscous defect coordinates.
$\mathcal{C}_{\mathrm{homcrit}}$	Coherent homogeneous critical-tail alternative.
$\mathcal{C}_{\mathrm{logper}}$	Coherent log-periodic critical-tail alternative.
$\mathcal{C}_{\mathrm{ap}}$	Coherent aperiodic/minimal critical-tail alternative.
$\mathcal{C}_{\mathrm{gen}}$	Generic terminal residual class.
Retained descendant	Descendant with inherited positive unit velocity concentration.
Normalized support profile / support profile	Normalized defect support profile, not necessarily retained at the original threshold.
First-generation recentering	Recentered window of the original setup terminal sequence; compact only under the local-window condition.
Ancient-profile recentering	Covariant recentering of an already extracted bounded ancient profile; compact by global ancient boundedness.
Local-window condition	Requirement that every fixed recentered cylinder undoes into the original local Type I setup cylinder.
Exterior/diffuse terminal branch	Positive residual mass outside active neighborhoods, below unit velocity thresholds after the reduction step.
Pressure/noncompact defect	Failure of local Seregin compactness or pressure gauge compactness under a selected recentering.
Critical-tail defect	Nonzero normalized annular/logarithmic residual profile visible only after critical-tail blow-down.

4. Exclusion of the axisymmetric bounded-circulation residual class

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation. Related axisymmetric Liouville theorems for Navier–Stokes, in different settings, are proved in [8, 7].

Throughout the section we write cylindrical coordinates as (ρ, θ, Z), and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla) V + \nabla P = 0, \qquad \nabla \cdot V = 0. \tag{4.1}$$

For an axisymmetric solution define the centered angular circulation

$$\Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau). \tag{4.2}$$

This is the same scale-invariant quantity as the physical circulation rU_θ. If the physical representative is

$$U(x,t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \qquad r = \sqrt{-t}\, \rho, \quad z = \sqrt{-t}\, Z.$$

Lemma 4.1 (Centered circulation equation)**.** *Let* (V, P) *be a smooth bounded axisymmetric solution of* (4.1)*. Then* $\Gamma = \rho V_\theta$ *satisfies*

$$\partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma \tag{4.3}$$

on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$*. Moreover*

$$\Gamma(0, Z, \tau) = 0 \qquad \textit{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}. \tag{4.4}$$

Proof. The angular component of (4.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by ρ, using

$$\rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma,$$

$$\rho \left[\frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta\right] = \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) \Gamma,$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho}\right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (4.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in (Z, τ). Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (4.4). ∎

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

Lemma 4.2 (One-dimensional killed radial absorption)**.** *Fix $R > 1$, $M < \infty$, and a compact interval $K_\rho = [\delta, R-\delta] \Subset (0, R)$. Let ρ_t be any weak solution on $(0, R)$, killed at $\{0, R\}$, of*

$$d\rho_t = -\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) dt + \sqrt{2}\, dW_t, \qquad |\beta_t| \le M,$$

where β_t may be adapted. If $\tau_\partial := \inf\{t : \rho_t \in \{0, R\}\}$, then

$$\lim_{T\to\infty} \sup_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial > T) = 0.$$

More precisely, there are $T_0 = T_0(R, M, \delta) > 0$ and $\theta = \theta(R, M, \delta) > 0$ such that

$$\inf_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial \le T_0) \ge \theta.$$

Proof. Choose

$$\delta_0 := \min\{\delta/2,\ [4(M+2)]^{-1}\}.$$

We first show that every starting point in K_ρ reaches the left boundary layer $(0, \delta_0]$ in a fixed time with a probability bounded from below. While $\rho_t \ge \delta_0$, the drift satisfies

$$-\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) \le M.$$

Fix $T_1 > 0$ and the deterministic linear path

$$\ell(t) := -\frac{R + MT_1 + 2\delta_0}{T_1}\, t, \qquad 0 \le t \le T_1.$$

On the Brownian tube event

$$E_1 := \left\{ \sup_{0\le t\le T_1} \left|\sqrt{2}\, W_t - \ell(t)\right| \le \frac{\delta_0}{4} \right\},$$

the integral form of the SDE gives, for every $\rho_0 \in K_\rho$, that ρ_t reaches $(0, \delta_0]$ before time T_1, unless it has already hit 0. The probability of E_1 is a strictly positive number

$$p_1 = p_1(R, M, \delta) > 0.$$

For completeness, this positivity follows by partitioning $[0, T_1]$ into finitely many subintervals and using the one-dimensional Dirichlet heat kernel lower bound in the strip of width $\delta_0/4$ around the line ℓ.

It remains to absorb the process from the layer $(0, \delta_0]$. As long as the process stays in $(0, 2\delta_0]$,

$$-\left(\frac{1}{\rho_t} + \frac{\rho_t}{2} + \beta_t\right) \le M - \frac{1}{2\delta_0} \le -2.$$

Let $T_2 := \delta_0$. On the event

$$E_2 := \left\{ \sup_{0\le t\le T_2} \left(\sqrt{2}\, W_t - \frac{t}{4}\right) \le \frac{\delta_0}{4} \right\},$$

which has probability $p_2 = p_2(M, \delta) > 0$, the same integral estimate gives

$$\rho_t \le \rho_0 - 2t + \frac{\delta_0}{4} + \frac{t}{4} \qquad \text{until hitting } 0.$$

Thus every initial point $\rho_0 \le \delta_0$ hits 0 before T_2, on E_2. Again $p_2 > 0$ is an explicit finite-time heat-kernel barrier estimate for Brownian motion below the affine boundary $\delta_0/4 + t/4$.

Combining the two blocks and using the strong Markov property gives

$$\inf_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial \le T_1 + T_2) \ge p_1 p_2 =: \theta > 0.$$

With $T_0 := T_1 + T_2$, iteration gives

$$\sup_{\rho_0 \in K_\rho} \mathbb{P}_{\rho_0}(\tau_\partial > kT_0) \leq (1-\theta)^k,$$

and the asserted decay follows. The constants are independent of the axial coordinate and of any truncation in that coordinate. □

Lemma 4.3 (Axis-absorption Liouville theorem)**.** *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0,\infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

Let $G \in C^{2,1}((0,\infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve

$$\partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho}\partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2}\right)\partial_\rho G - \left(B_Z + \frac{Z}{2}\right)\partial_Z G \tag{4.5}$$

for all $\tau \in \mathbb{R}$, and assume

$$\lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \qquad \text{locally uniformly in } (Z, \tau). \tag{4.6}$$

Then $G \equiv 0$.

Proof. Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$h_R(\rho) := \frac{\displaystyle\int_0^{\min\{\rho,R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\displaystyle\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}. \tag{4.7}$$

Then $0 \leq h_R \leq 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0, \qquad h_R' > 0. \tag{4.8}$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho}\partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2}\right)\partial_\rho f - \left(B_Z + \frac{Z}{2}\right)\partial_Z f.$$

Since $B_\rho \geq -M_B$, (4.8) gives

$$\mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho\right) h_R' \leq h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho}\right) h_R' = 0. \tag{4.9}$$

Thus $(\partial_\tau - \mathcal{L}_\tau) h_R \geq 0$.

We now compare G to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the Dirichlet strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \qquad q_{s,R} = 0 \text{ on } \rho = 0, R, \qquad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed t,

$$\lim_{s \to -\infty} \sup_{(\rho,Z) \in K} q_{s,R}(\rho, Z, t) = 0. \tag{4.10}$$

To prove (4.10), fix $K \Subset \Omega_R$, and choose $\delta > 0$ such that

$$K \subset [\delta, R - \delta] \times \mathbb{R}.$$

We use a one-dimensional absorption estimate for the ρ-component. Let ρ_τ be any weak solution of

$$d\rho_\tau = -\left(\frac{1}{\rho_\tau} + \frac{\rho_\tau}{2} + \beta_\tau\right) d\tau + \sqrt{2}\, dW_\tau, \qquad |\beta_\tau| \leq M_B,$$

killed when it hits 0 or R. The coefficient β_τ may be any adapted process; in the application it is $B_\rho(\rho_\tau, Z_\tau, \tau)$. By Lemma 4.2, there are $T = T(R, M_B, \delta) > 0$ and $\theta = \theta(R, M_B, \delta) > 0$ such that

$$\inf_{\rho_0 \in [\delta, R-\delta]} \mathbb{P}_{\rho_0}\{\rho_\tau \text{ hits 0 or } R \text{ before time } T\} \geq \theta.$$

The parabolic maximum principle and the Feynman–Kac representation for the killed equation give

$$0 \leq q_{s,R}(\rho, Z, t) \leq \mathbb{P}_\rho\{\rho_\tau \text{ survives on } [s,t]\}.$$

Iterating the block estimate yields

$$\sup_{(\rho,Z)\in K} q_{s,R}(\rho, Z, t) \leq (1-\theta)^{\lfloor (t-s)/T \rfloor},$$

which tends to 0 as $s \to -\infty$. This proves (4.10) directly on the infinite strip.

Truncated Feynman–Kac justification. The displayed killed-kernel and survival-probability statements are first established on the bounded box

$$\Omega_{\varepsilon,R,L} := (\varepsilon, R) \times (-L, L) \qquad (0 < \varepsilon < R < \infty,\ L > 0)$$

with Dirichlet boundary conditions on the entire boundary of $\Omega_{\varepsilon,R,L}$, where the standard killed parabolic theory provides a Feynman–Kac representation of the strip kernel $q_s^{(\varepsilon,R,L)}(\rho, Z, t)$ by the killed diffusion (ρ_τ, Z_τ) with drift $(-(\rho^{-1} + \rho/2 + \beta_\tau), -(Z/2 + \gamma_\tau))$ and additive Brownian noise. The block survival estimate of Lemma 4.2 applies because the one-dimensional radial absorption in ρ does not depend on the Z-component or on ε or L once $\delta \leq \rho \leq R - \delta$; hence

$$\sup_{(\rho,Z)\in K} q_s^{(\varepsilon,R,L)}(\rho, Z, t) \leq (1-\theta)^{\lfloor (t-s)/T \rfloor}$$

holds uniformly in ε and L for any compact $K \Subset (\delta, R-\delta) \times [-L_0, L_0]$ and all $\varepsilon < \delta$, $L > L_0$. The truncated comparison $(W_R^\pm)_+ \leq q_s^{(\varepsilon,R,L)}$ on $\Omega_{\varepsilon,R,L}$ then follows from the parabolic maximum principle applied to the truncated problem (the $\rho = \varepsilon$ boundary contributes nonpositively because both G/M_G and h_R are bounded by 1, so the truncated boundary corrector $\rho_{\rm tr}(\tau) := 2\sup\{|G/M_G| + h_R\} \cdot \mathbf{1}_{\rho=\varepsilon \text{ or } |Z|=L}$ is killed by the same Dirichlet kernel and tends to zero on every fixed compact K as $L \to \infty$, $\varepsilon \downarrow 0$). Sending $L \to \infty$ recovers the half-strip $(\varepsilon, R) \times \mathbb{R}$ by monotone convergence of the killed semigroup; sending $\varepsilon \downarrow 0$ recovers $(0, R) \times \mathbb{R}$ using $h_R(0) = 0$ and the axis condition (4.6) on G; finally sending $R \to \infty$ gives the global statement (4.11) and the conclusion $G \equiv 0$ as in the next paragraph.

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \qquad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (4.5) and (4.9),

$$(\partial_\tau - \mathcal{L}_\tau) W_R^\pm \leq 0 \qquad \text{in } \Omega_R \times (s,t).$$

On $\rho = 0$, (4.6) and $h_R(0) = 0$ give $W_R^\pm \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^\pm \leq 0$. The parabolic comparison principle with the Dirichlet strip kernel therefore yields, for $\tau = t$,

$$(W_R^\pm)_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \qquad ((\rho, Z) \in \Omega_R).$$

Letting $s \to -\infty$ and using (4.10) gives $W_R^\pm \leq 0$ on compact subsets of Ω_R. Since the compact subset is arbitrary,

$$|G(\rho, Z, t)| \leq M_G h_R(\rho) \qquad (0 < \rho < R, Z \in \mathbb{R},\ t \in \mathbb{R}). \tag{4.11}$$

Finally, for each fixed $\rho < \infty$, the denominator in (4.7) tends to infinity super-exponentially as $R \to \infty$, while the numerator remains fixed. Hence $h_R(\rho) \to 0$. Letting $R \to \infty$ in (4.11) gives

$G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. $\square$

Theorem 4.4 (Exclusion of the axisymmetric bounded-circulation residual class)**.** *Let $\mathcal{C}_{\rm ax}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$\mathcal{C}_{\rm ax}(\mathcal{S}; I, J) = \varnothing. \tag{4.12}$$

Proof. Assume, for contradiction, that $V \in \mathcal{C}_{\rm ax}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\rm ax}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since V is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By Lemma 4.1, Γ satisfies (4.3) and vanishes at the axis. Applying Lemma 4.3 with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such V exists. $\square$

Remark 4.5 (Why this does not assert the full bounded-swirl Liouville theorem)*.* The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho\partial_\rho/2$ and $-Z\partial_Z/2$ in the scalar equation for Γ. It is therefore a Type I Seregin limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

5. The rotational relative-equilibrium residual class

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in Sections 9 and 10.

Throughout this section the centered equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla) V + \nabla P = 0, \qquad \nabla \cdot V = 0. \tag{5.1}$$

For $\Omega \in \mathfrak{so}(3)$, set

$$Q(\tau) := \exp(\tau\Omega) \in SO(3). \tag{5.2}$$

Definition 5.1 (Rotational relative-equilibrium profile)**.** A bounded smooth ancient solution (V, P) of (5.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field W, and a pressure profile Π, such that

$$V(y, \tau) = Q(\tau) W(Q(\tau)^T y), \qquad P(y, \tau) = \Pi(Q(\tau)^T y) \tag{5.3}$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\rm rot}$ consists of such residual Seregin profiles, after the axisymmetric bounded-circulation class and the previously closed alternatives in the setup paper have been removed.

Lemma 5.2 (Co-rotating stationary equation)**.** *Let (V, P) be a rotational relative equilibrium. Then the co-rotating profile (W, Π) satisfies*

$$-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W + \nabla \Pi = 0, \qquad \nabla \cdot W = 0. \tag{5.4}$$

Equivalently,

$$\mathbb{P}\Big[-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W \Big] = 0, \tag{5.5}$$

where $\mathbb{P}$ is the whole-space Leray projector.

Proof. Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T \Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y, \tau) = Q(\tau)\Big(\Omega W(z) - (\Omega z) \cdot \nabla W(z)\Big).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (5.1), followed by multiplication by $Q(\tau)^T$, gives (5.4). Applying the Leray projector gives (5.5). □

5.1. **The Bernoulli identity and the local Coriolis flux.** Write

$$F(y) := \frac{1}{2}y - \Omega y, \qquad A(y) := W(y) + F(y), \qquad \omega := \nabla \times W. \tag{5.6}$$

Since Ω is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$\Omega y = \varpi \times y. \tag{5.7}$$

Then

$$\nabla \times F = -2\varpi. \tag{5.8}$$

Lemma 5.3 (Coriolis Bernoulli identity)**.** *Let (W, Π) be a smooth solution of (5.4). Define*

$$B := \frac{1}{2}|W|^2 + \Pi + F \cdot W. \tag{5.9}$$

Then

$$\nabla B + \nabla \times \omega - A \times \omega = 0, \tag{5.10}$$

and consequently

$$-\Delta B + A \cdot \nabla B = -|\omega|^2 - \omega \cdot (\nabla \times F) = -|\omega|^2 + 2\varpi \cdot \omega. \tag{5.11}$$

Proof. The vector identity

$$(W \cdot \nabla)W = \nabla\left(\frac{1}{2}|W|^2\right) - W \times \omega$$

and the identity

$$(F \cdot \nabla)W + (\nabla F)^T W = \nabla(F \cdot W) - F \times \omega$$

apply because $(\nabla F)^T W = \frac{1}{2}W + \Omega W$. Thus (5.4) can be rewritten as

$$-\Delta W + \nabla\left(\frac{1}{2}|W|^2 + \Pi + F \cdot W\right) - (W + F) \times \omega = 0.$$

Since $\nabla \cdot W = 0$, one has $-\Delta W = \nabla \times \omega$, which gives (5.10).

Taking divergence in (5.10) gives

$$\Delta B = \nabla \cdot (A \times \omega),$$

because the divergence of a curl vanishes. Using

$$\nabla\cdot(A\times\omega)=\omega\cdot(\nabla\times A)-A\cdot(\nabla\times\omega),\qquad \nabla\times A=\omega+\nabla\times F,$$

we obtain

$$\Delta B=|\omega|^2+\omega\cdot(\nabla\times F)-A\cdot(\nabla\times\omega).$$

On the other hand, by (5.10),

$$A\cdot\nabla B=A\cdot(A\times\omega)-A\cdot(\nabla\times\omega)=-A\cdot(\nabla\times\omega).$$

Combining the last two identities gives (5.11). Finally, (5.8) follows from (5.7). ☐

Lemma 5.4 (Localized Coriolis flux identity)**.** *Let (W,Π) be as in Lemma 5.3, and let $\chi\in C_c^\infty(\mathbb{R}^3)$. Then*

$$\int_{\mathbb{R}^3}\chi\left(-\Delta B+A\cdot\nabla B+|\omega|^2\right)dy=-2\int_{\mathbb{R}^3}(W\times\varpi)\cdot\nabla\chi\,dy. \tag{5.12}$$

In particular the Coriolis contribution is supported only where the cutoff varies.

Proof. By (5.11),

$$-\Delta B+A\cdot\nabla B+|\omega|^2=2\varpi\cdot\omega.$$

Because ϖ is constant and $\omega=\nabla\times W$,

$$\varpi\cdot\omega=\nabla\cdot(W\times\varpi).$$

Multiplying by χ, integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2\int\chi\,\varpi\cdot\omega=2\int\chi\,\nabla\cdot(W\times\varpi)=-2\int(W\times\varpi)\cdot\nabla\chi,$$

which proves (5.12). ☐

Lemma 5.5 (Non-small Coriolis flux creates retained local velocity concentration)**.** *Let $\chi\in C_c^\infty(\mathbb{R}^3)$, let $A_\chi:=\operatorname{supp}\nabla\chi$, and put*

$$C_\chi:=\int_{A_\chi}|\nabla\chi|^{3/2}\,dy.$$

Fix $T\in(0,1]$, and let

$$\mathcal{T}_{\chi,T}:=\{(y,\tau):-T<\tau<0,\ Q(\tau)^Ty\in A_\chi\}$$

be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi,T}$ by finitely many unit terminal cylinders

$$\mathcal{T}_{\chi,T}\subset\bigcup_{m=1}^{N_{\chi,T}}Q_1(z_m).$$

The number $N_{\chi,T}$ depends only on the compact annulus A_χ, the time length T, and Ω, and the cover is chosen with bounded overlap inside the normalized terminal domain under consideration. Let $\varepsilon_{\rm sd}>0$ be the raw unit-window threshold used in Definition 9.3. If $\varpi\neq0$ and

$$\left|2\int_{\mathbb{R}^3}(W\times\varpi)\cdot\nabla\chi\,dy\right|\ge2|\varpi|\,C_\chi^{2/3}\left(\frac{N_{\chi,T}\varepsilon_{\rm sd}}{T}\right)^{1/3}, \tag{5.13}$$

then at least one cylinder $Q_1(z_m)$ is raw unit-window active at threshold $\varepsilon_{\rm sd}$ for the original centered solution V.

Proof. Write
$$\delta_\chi := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla\chi \, dy \right|.$$
Hölder's inequality gives
$$\begin{aligned} \delta_\chi &\le 2|\varpi| \int_{A_\chi} |W| \, |\nabla\chi| \, dy \\ &\le 2|\varpi| \left(\int_{A_\chi} |W|^3 \, dy \right)^{1/3} C_\chi^{2/3}. \end{aligned}$$
Thus (5.13) implies
$$\int_{A_\chi} |W|^3 \, dy \ge \frac{N_{\chi,T}\varepsilon_{\mathrm{sd}}}{T}.$$
Since $V(y,\tau) = Q(\tau)W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,
$$\begin{aligned} \iint_{\mathcal{T}_{\chi,T}} |V(y,\tau)|^3 \, dy \, d\tau &= \int_{-T}^{0} \int_{Q(\tau)A_\chi} |W(Q(\tau)^T y)|^3 \, dy \, d\tau \\ &= T \int_{A_\chi} |W(z)|^3 \, dz \ge N_{\chi,T}\varepsilon_{\mathrm{sd}}. \end{aligned}$$
Because the tube is covered by $N_{\chi,T}$ unit terminal cylinders with bounded overlap, one of them satisfies
$$\iint_{Q_1(z_m)} |V|^3 \, dy \, d\tau \ge \varepsilon_{\mathrm{sd}}.$$
The pressure contribution to the local unit defect measure is nonnegative after the allowed pressure gauge. Hence
$$\mu(Q_1(z_m)) \ge \varepsilon_{\mathrm{sd}}.$$
Thus the corresponding recentering sequence is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ in the sense of Definition 9.3. The retained nonzero descendant later obtained from this observer is certified by the displayed unit velocity lower bound, while the atlas pressure term is kept only in the raw defect measure. □

Lemma 5.6 (Local Coriolis-flux dichotomy)**.** *Fix $\chi \in C_c^\infty(\mathbb{R}^3)$ and $T \in (0,1]$. Along every terminal extraction of a rotational relative equilibrium, the following priority alternatives are applied:*

- *(i) $\varpi \neq 0$, the flux across $A_\chi = \operatorname{supp} \nabla\chi$ satisfies the threshold (5.13), and the swept annular tube contains a unit-scale velocity concentration;*
- *(ii) either $\varpi = 0$, or the threshold (5.13) fails; then the flux is a bounded compactly supported functional of W on A_χ, passes to terminal limits by local L^3 convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of Theorem 9.24.*

Proof. Alternative *(i)* is precisely Lemma 5.5. If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (5.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional
$$W \longmapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla\chi \, dy$$

depends only on W on the compact annular set A_χ. Under the local Seregin compactness supplied by the admissible setup hypotheses, velocities converge strongly in L^3_{loc}; since $\nabla\chi \in L^{3/2}$ and ϖ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla\chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla\chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (5.12). It is not a new location of local velocity concentration. Every actual location of compact velocity L^3 concentration is still one of the terminal concentration-set alternatives listed in Theorem 9.24. □

Remark 5.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in Lemma 5.4 vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for B. It gives only the local dichotomy in Lemma 5.6: either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space L^p bound, Gaussian energy estimate, or global pressure normalization is used.

5.2. **Reduction of the rotational relative-equilibrium class.**

Proposition 5.8 (Rotational residual reduction)**.** *Every rotational relative-equilibrium residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence satisfying the hypotheses of the terminal stratification assembly theorem.*

Proof. Let $V \in \mathcal{C}_{\mathrm{rot}}(\mathcal{S}; I, J)$. By Definition 5.1 and Lemma 5.2, there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile (W, Π) satisfying the Coriolis–Leray equation (5.4). The centered representative V is a normalized Seregin ancient limit, so by Theorem 1.7 it carries a fixed positive compact velocity L^3 concentration.

If $\Omega = 0$, then V is stationary in the centered frame. It is reduced to the stationary setup class if it lies in the already closed stationary alternative, and otherwise to the stationary-hull residual branch handled by Proposition 6.12. We therefore assume $\Omega \neq 0$.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy Lemma 5.6. A superthreshold Coriolis flux produces a unit-scale velocity concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of W, continuous under the local convergence supplied by the admissible setup compactness hypotheses, and hence remains inside the localized limiting equation; it carries no independent local velocity mass and creates no additional terminal alternative.

We now apply the paired terminal concentration-set/remainder decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by Theorem 9.27. By the preceding paragraph, the Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained velocity concentration profiles or in a single retained velocity concentration terminal profile, together with any residual diffuse, noncompact, or critical-tail remainder selected by the terminal extraction procedure. This gives a terminal input realized from the original blow-up sequence for the terminal stratification theorem; no R2 closure is used in this reduction step. □

Proposition 5.9 (Already-closed integrable subcase)**.** *Let V be a rotational relative equilibrium with co-rotating profile W. If*

$$W \in L^3(\mathbb{R}^3), \tag{5.14}$$

then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L^\infty_t L^3_x$ on its ancient time interval.

Proof. Rotations and the Navier–Stokes critical scaling preserve the L^3 norm. Thus $W \in L^3$ implies

$$\sup_{\tau \in \mathbb{R}} \|V(\cdot,\tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L^\infty_t L^3_x$. The endpoint L^3 class is one of the already closed alternatives used in the setup paper, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [5] and the stationary Nečas–Růžička–Šverák Liouville theorem [9]. Therefore $V \equiv 0$. A zero profile has zero local velocity concentration, so no such profile can belong to the residual nonvanishing class. □

6. The degenerate stationary-hull residual class

The third refined residual class consists of Type I ancient limits whose time-translation hull contains a stationary profile with retained local velocity concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\mathrm{ax}}$ and $\mathcal{C}_{\mathrm{rot}}$.

The retained-velocity concentration condition is essential. The mere presence of a stationary-family element in the hull of a residual ancient solution is insufficient. A stationary element may occur as a limit without retained local velocity concentration and therefore need not retain the local velocity concentration inherited from the original singular point. Such a hull element is not placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-velocity occurrence*: the stationary-family profile must itself occur as a Seregin limit carrying a positive local velocity concentration.

Throughout this section the centered equation is

$$\partial_\tau V - \Delta V + \frac{1}{2} y \cdot \nabla V + \frac{1}{2} V + (V \cdot \nabla)V + \nabla P = 0, \qquad \nabla \cdot V = 0. \tag{6.1}$$

Its projected form is

$$\partial_\tau V = \mathcal{F}(V), \qquad \mathcal{F}(W) := \mathbb{P}\left[\Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W\right], \tag{6.2}$$

where $\mathbb{P}$ is the whole-space Leray projector.

6.1. Stationary families and retained-velocity hull limits.

Definition 6.1 (Admissible stationary phase space)**.** An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

(i) $\mathfrak{B} \hookrightarrow C^{2,\alpha}_{\mathrm{loc}}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0,1)$;
(ii) the map $\mathcal{F}$ in (6.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{Y}$;
(iii) $\mathcal{F} : \mathcal{U} \to \mathfrak{Y}$ is C^1.

Definition 6.2 (Nontrivial stationary family)**.** Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that W lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a C^1 curve

$$a \mapsto W_a \in \mathcal{U}, \qquad |a| < \varepsilon,$$

such that

$$W_0 = W, \qquad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon), \tag{6.3}$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$\dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\} \tag{6.4}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

Definition 6.3 (Retained stationary-hull velocity concentration)**.** Let (W, Π) be a stationary centered profile. We say that (W, Π) is retained-stationary concentrating if it has retained velocity concentration in the normalized unit sense: there is a recentering/rescaling realized from the original blow-up sequence $\mathscr{T}$ such that

$$\mathcal{V}_1\big(\mathscr{T} W; 0\big) \geq \eta_*. \tag{6.5}$$

Equivalently, before the final unit normalization one may require $\mathcal{V}_r(W; z_0) \geq \eta_*$ for some parabolic cylinder $Q_r(z_0)$. Unnormalized large-cylinder mass is not a retained velocity concentration criterion. It has stationary-hull CKN activity if the same realized normalization also satisfies

$$\mathcal{E}_1(\mathscr{T} W, \mathscr{T}\Pi; 0) \geq \varepsilon_*.$$

Only the velocity condition (6.5) certifies that the stationary-hull profile is nonzero. The accompanying CKN activity condition is used only for CKN smallness and pressure estimates.

Definition 6.4 (Degenerate stationary-hull residual class with retained velocity concentration)**.** Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local velocity concentration

$$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\mathrm{ax}}$ and $\mathcal{C}_{\mathrm{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile (W, Π) such that

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi) \tag{6.6}$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

(i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
(ii) W lies on an admissible nontrivial stationary family in the sense of Definition 6.2;
(iii) (W, Π) is retained-stationary concentrating in the sense of Definition 6.3;
(iv) (W, Π) is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained velocity concentration as the degenerate stationary-hull alternative. The hull cases without retained local velocity concentration are assigned to the generic terminal residual class, where they are handled by the terminal concentration-set decomposition rather than by a stationary-profile exclusion.

Remark 6.5 (Why retained velocity concentration is necessary). The condition (6.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local velocity concentration on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

6.2. **Hull limits are again Seregin limits.** The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

Lemma 6.6 (Scale reselection for centered hull limits)**.** *Let (V, P) be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution (u, p) at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \to (W, \Pi)$$

locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, (W, Π) is itself a centered Type I Seregin limit of (u, p) at z_*. If the convergence to (W, Π) retains a positive local velocity lower bound, then (W, Π) is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.

Proof. Let $\lambda_n \downarrow 0$ be a sequence of scales producing (V, P). In centered variables, a time translation by s is a change of the physical blow-up scale by a factor depending only on s. With the convention used in (6.1), write this adjusted scale as

$$\lambda_{n,s} := e^{-s/2}\lambda_n. \tag{6.7}$$

Indeed, writing $t = -e^{-\tau}$, a shift $\tau \mapsto \tau + s$ corresponds to multiplying the physical length scale by $e^{-s/2}$, so that

$$V_{\lambda e^{-s/2}}(y, \tau) = V_\lambda(y, \tau + s)$$

with the corresponding pressure gauge. Thus the rescaled fields built from (u, p) at scale $\lambda_{n,s}$ are the same as the fields at scale λ_n, translated by s in centered time.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each k, first require

$$\lambda_{n(k)} \leq k^{-1} e^{s_k/2}$$

when $s_k < 0$, and otherwise require simply $\lambda_{n(k)} \leq k^{-1}$. Then $\lambda_{n(k),s_k} = e^{-s_k/2}\lambda_{n(k)} \to 0$. Increasing $n(k)$ if needed, arrange that the rescaled fields at scale $\lambda_{n(k)}$, shifted by s_k, are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on K_k in the local Seregin topology. This is possible because the original sequence converges locally to (V, P).

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to (W, Π) on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k),s_k}$ therefore converges locally to (W, Π). Hence (W, Π) is again a centered Type I Seregin limit at z_*. The assertion about nonvanishing follows from strong local L^3 convergence of the velocity and the retained velocity lower bound; pressure compactness is used only to keep the limiting pressure gauge admissible. ☐

6.3. **Elementary consequences of stationary-family degeneracy.** The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is constrained to a stationary family, then it is stationary.

Lemma 6.7 (Stationary family implies linearized degeneracy)**.** *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family satisfying (6.3). Then*

$$D\mathcal{F}(W_0)\dot{W}_0 = 0. \tag{6.8}$$

In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.

Proof. Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is C^1 on $\mathfrak{B}$, this gives (6.8). The final statement is the nontriviality condition in Definition 6.2. □

Lemma 6.8 (Trajectories constrained to a stationary family are stationary)**.** *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (6.2) has the form*

$$V(\tau) = W_{a(\tau)} \tag{6.9}$$

for a C^1 function $a(\tau)$. Then a is constant, and hence V is stationary.

Proof. Since $\mathcal{F}(W_a) = 0$ for every a, equation (6.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus a is constant on the interval. □

Corollary 6.9 (Exact stationary-family subcase)**.** *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary L^3 Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

Proof. Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Lemma 6.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence a is constant and V is a single stationary centered profile. If that stationary profile satisfies the stationary L^3 Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. □

6.4. **Stationary-hull profiles which occur as Seregin limits.** The preceding lemma removes the possible internal dynamics along a stationary family: exact stationary-family motion is stationary by Lemma 6.8. The remaining question is attainability by blow-up: whether a degenerate stationary-family profile can occur as a Seregin limit of a finite-energy suitable weak solution while retaining local velocity concentration. The following statement gives the precise Seregin-limit occurrence exclusion required to close the degenerate stationary-hull class with retained local velocity concentration. Its proof is deferred until the terminal stratification assembly theorem has been established.

Proposition 6.10 (R3 degenerate stationary-family reduction target)**.** *Fix an admissible stationary phase space $\mathfrak{B}$. Let (W, Π) be a stationary centered profile satisfying:*

- *(i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and W lies on an admissible nontrivial stationary family;*
- *(ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;*
- *(iii) (W, Π) is retained-stationary concentrating in the sense of Definition 6.3;*
- *(iv) (W, Π) is not in the small, stationary L^3, tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.*

Then no such (W, Π) *exists after the terminal stratification assembly is inserted.*

Proof deferred. The proof is given after the terminal stratification assembly theorem in Corollary 10.69. The present statement records the exact R3 occurrence input that must be proved; no R3 exclusion is used before the terminal assembly theorem is available. □

Remark 6.11 (Interpretation of the Seregin limit occurrence theorem). Proposition 6.10 excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local velocity concentration after all previously closed classes have been removed. This is the formulation required for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

6.5. Reduction of the degenerate stationary-hull concentration class.

Proposition 6.12 (R3 stationary-hull reduction)**.** *Every retained stationary-hull residual profile either belongs to one of the stationary, tight, or structured classes already excluded by the setup paper, or it produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of the terminal stratification assembly theorem.*

Proof. Let $V \in \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$. By Definition 6.4, there is a stationary profile (W, Π) in the centered time-translation hull of (V, P), and (W, Π) is retained-stationary concentrating. The profile W belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$, it is not in any of the previously closed lower classes.

By Lemma 6.6, the hull profile (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local velocity concentration is retained, (W, Π) satisfies every condition in Proposition 6.10, except that the actual contradiction is deferred to the terminal stratification assembly theorem. Thus the R3 branch has been reduced to a stationary terminal concentration profile realized from the original blow-up sequence; no R3 closure is used in this reduction step. □

Proposition 6.13 (Already-closed integrable stationary-family subcase)**.** *Let* (W, Π) *be a stationary centered profile lying on an admissible stationary family. If*

$$W \in L^3(\mathbb{R}^3), \tag{6.10}$$

then $W \equiv 0$. *Hence no retained-velocity hull profile satisfying* (6.10) *can occur in* $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$.

Proof. The stationary physical representative associated with W is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary L^3 class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [9], equivalently by the endpoint ancient Liouville theorem already recorded in the setup paper and cited below as Theorem 10.9. Hence $W \equiv 0$. A zero profile has zero local velocity concentration, so it cannot be retained-stationary concentrating. □

Remark 6.14 (Closure after terminal stratification assembly). The preceding argument reduces the stationary-hull class with retained local velocity concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. Proposition 6.10 is supplied only after the terminal stratification assembly theorem has been proved.

Remark 6.15 (Residual decomposition statement). For the residual decomposition, the third line is reduced as follows and closed only after the terminal stratification assembly theorem:

$$\text{stationary-hull residual with retained velocity concentration}$$
$$\xrightarrow{\text{stationary-family Seregin limit reduction}} \text{terminal stationary concentration profile}$$
$$\xrightarrow{\text{terminal closure}} \varnothing.$$

The precise class is the *degenerate stationary-hull residual class with retained local velocity concentration.*

7. Reduction of degenerate stationary-hull profiles with retained velocity concentration

We record the terminal concentration input needed for the degenerate stationary-hull class. The actual closure is deferred until after the terminal stratification theorem. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local velocity concentration, it is a stationary terminal concentration profile. Such profiles are later excluded by the same terminal stratification assembly and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$\partial_\tau U - \Delta U + \frac{1}{2} y \cdot \nabla U + \frac{1}{2} U + (U \cdot \nabla)U + \nabla \Pi = 0, \qquad \nabla \cdot U = 0 \tag{7.1}$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

7.1. **Deferred terminal stratification corollary for R3.** We record the following specialization of the terminal stratification assembly theorem. It is stated here exactly in the form needed for the stationary-hull closure with retained local velocity concentration as a *reduction target*; the substantive proof is the terminal stratification proof later in the paper. Following the recommendation that only one actual R3 closure theorem appear after terminal assembly, this statement is recorded as a proposition rather than a theorem.

Proposition 7.1 (R3 reduction target). *Let (U, Π) be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local velocity concentration, i.e. for some admissible recentering/rescaling $\mathscr{T}_{z_0,r}$,*

$$\mathcal{V}_1\big(\mathscr{T}_{z_0,r} U; 0\big) \geq \eta_*. \tag{7.2}$$

Assume that (U, Π) is outside the previously closed small-amplitude, stationary L^3, tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.

- *(i) Every terminal extraction of (U, Π) has the paired decomposition of Theorem 9.24: a retained velocity concentration set together with a residual remainder which is locally vanishing, diffuse, or noncompact after neighborhoods of the concentration set are removed.*
- *(ii) Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact velocity concentration (7.2).*
- *(iii) Mixed compact–diffuse residual profiles are impossible in the refined residual class, while compact–noncompact configurations are routed by the terminal recentering protocol to the diffuse, pressure/noncompact, or critical-tail branches. Infinite chains of retained concentrating descendants are impossible as well.*
- *(iv) Finite separated families of retained velocity concentration profiles are impossible.*

(v) *Any remaining single retained velocity concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*

(vi) *Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \to -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty. \tag{7.3}$$

(vii) *Retained terminal concentration profiles are either affine/parasitic lower-stratum profiles or have parasitic-free finite-shift mild physical pullbacks. Therefore the Albritton–Barker endpoint Liouville theorem applies to any normalized residual profile satisfying* (7.3).

Proof deferred. This is the R3 specialization of Theorem 10.66. It is stated here only as a reduction target for the stationary-hull section; its proof is the terminal stratification proof in Sections 9 and 10. □

Remark 7.2 (Local nature of the theorem). The theorem above is a local velocity concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [4] together with the imported setup results: local positive concentration and Type I Seregin compactness. It then uses paired terminal profile decomposition, sole-source exclusions, critical-tail compactification with realized critical-tail rigidity, recurrence for concentration chains, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For reference, we record the stationary specialization of the atomic sequence-L^3 lemma used after the terminal stratification assembly theorem.

Lemma 7.3 (Indecomposable local profiles have backward sequence-L^3 control). *Let* (U, Π) *be a bounded smooth ancient solution of* (7.1) *which belongs to the normalized parasitic-free compact state class* $\mathfrak{X}_M^{\#}$ *imported from the setup construction. Suppose that* (U, Π) *has retained local velocity concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration in the parabolic CKN sense, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence* $\tau_k \to -\infty$ *such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty. \tag{7.4}$$

Proof. This is the stationary-profile specialization of Lemma 10.6. The hypotheses listed here are the terminal indecomposability hypotheses used there: retained unit velocity concentration, no retained concentrating tail limit, no diffuse exterior concentration, and no escaping recentering with local compactness or pressure-gauge failure. Applying Lemma 10.6 gives (7.4). No separate exterior activity level is introduced in this stationary specialization; the only exterior activity selection is the velocity quantity m_n^{vel} in Lemma 10.6, and pressure-inclusive CKN activity is not used to certify nontriviality of a tail profile. □

7.2. **Deferred stationary concentration statement.**

Remark 7.4 (Deferred R3 stationary consequence). Let (W, Π) be a stationary solution of the centered equation,

$$-\Delta W + \frac{1}{2} y \cdot \nabla W + \frac{1}{2} W + (W \cdot \nabla) W + \nabla \Pi = 0, \qquad \nabla \cdot W = 0. \tag{7.5}$$

Assume that:

(i) (W, Π) is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
(ii) (W, Π) has retained local velocity concentration in the sense of (7.2);
(iii) (W, Π) is outside all previously closed lower classes listed in Proposition 7.1.

Then no such (W, Π) exists. This statement is the stationary case of the R3 closure after Theorem 10.66; it follows from Corollary 10.69 and is recorded here only to name the exact stationary concentration statement used by the stationary-hull reduction.

7.3. **Deferred degenerate stationary-hull closure statement.**

Remark 7.5 (Deferred R3 degenerate-family consequence)*.* Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile (W, Π) satisfying all of the following conditions:
(i) $W \in \mathfrak{B}$, W solves the stationary centered equation (7.5), and W lies on an admissible nontrivial stationary family;
(ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
(iii) (W, Π) has retained local velocity concentration;
(iv) (W, Π) is outside the previously closed lower classes.

This statement is recorded as a deferred R3 consequence; it follows from the post-terminal R3 closure Corollary 10.69. The stationary-family condition in (i) is additional structure; once the terminal stratification assembly theorem is available, the stationary concentration argument applies to these profiles in particular.

Proposition 7.6 (Stationary-hull reduction to terminal stratification)**.** *Let $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local velocity concentration. Every element of this class produces a stationary terminal concentration profile realized from the original blow-up sequence and satisfying the hypotheses of Theorem 10.66.*

Proof. Let $V \in \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local velocity concentration, the centered translation hull of (V, P) contains a stationary profile (W, Π) with retained local velocity concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus (W, Π) satisfies the hypotheses of the terminal stratification assembly theorem in the stationary case. The actual emptiness conclusion is recorded after that terminal stratification assembly theorem as Corollary 10.69. $\square$

Remark 7.7 (How this enters the residual decomposition)*.* The third residual line is reduced as follows and closed only after the terminal stratification theorem:

$$\begin{gathered}\text{degenerate stationary-hull residual with retained velocity concentration}\\ \Longrightarrow \text{stationary profile with retained velocity concentration} \Longrightarrow \varnothing.\end{gathered}$$

The second implication is proved by the terminal stratification assembly: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence-L^3 control and are ruled out by the endpoint ancient Liouville theorem.

8. Tail geometry of the generic terminal residual

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under ordinary fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

Definition 8.1 (Covariant translations)**.** For $a \in \mathbb{R}^3$, define

$$(T_aV)(y,\tau) := V(y + e^{\tau/2}a, \tau), \qquad (T_aP)(y,\tau) := P(y + e^{\tau/2}a, \tau).$$

Lemma 8.2 (Covariant translation invariance)**.** *If (V, P) solves (1.2), then (T_aV, T_aP) also solves (1.2).*

Proof. Set $z = y + e^{\tau/2}a$. Then

$$\partial_\tau (T_aV)(y,\tau) = (\partial_\tau V)(z,\tau) + \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z,\tau).$$

Moreover,

$$\frac{1}{2}y \cdot \nabla (T_aV)(y,\tau) = \frac{1}{2}z \cdot \nabla V(z,\tau) - \frac{1}{2}e^{\tau/2}a \cdot \nabla V(z,\tau).$$

The two additional transport terms cancel in $\partial_\tau T_aV + \frac{1}{2}y \cdot \nabla T_aV$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into (1.2) proves the claim. $\square$

Remark 8.3 (Fixed translates create a spurious drift)*.* If one sets $V_n(y,\tau) = V(y + x_n, \tau)$ with fixed x_n, then the centered equation acquires the additional transport term

$$-\frac{1}{2}x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

Definition 8.4 (Spacetime covariant action)**.** For $s \in \mathbb{R}$, define the centered time translation

$$(\Theta_sV)(y,\tau) := V(y,\tau+s), \qquad (\Theta_sP)(y,\tau) := P(y,\tau+s).$$

For a tail center $x \in \mathbb{R}^3$ observed at centered time σ, define the covariant tail recentering

$$\mathscr{T}_{(x,\sigma)}V(y,s) := V(y + e^{s/2}x, \sigma + s), \qquad \mathscr{T}_{(x,\sigma)}P(y,s) := P(y + e^{s/2}x, \sigma + s).$$

Equivalently,

$$\mathscr{T}_{(x,\sigma)} = \Theta_\sigma T_{e^{-\sigma/2}x}.$$

Thus $\mathscr{T}_{(x,\sigma)}$ is a composition of exact symmetries of the centered equation. Ordinary parabolic recenterings are used below only for local terminal extraction before one has passed to a centered ancient profile; all escaping tail limits of centered profiles use $\mathscr{T}_{(x,\sigma)}$.

Lemma 8.5 (Semidirect covariance of the centered action)**.** *The centered time translations and spatial covariant translations satisfy*

$$\Theta_sT_a\Theta_{-s} = T_{e^{s/2}a}, \qquad s \in \mathbb{R},\ a \in \mathbb{R}^3.$$

Equivalently, if $G = \mathbb{R}^3 \rtimes \mathbb{R}$ acts by

$$\mathscr{G}_{(a,s)} := \Theta_sT_a,$$

then the group law in this convention is

$$(a,s)\cdot(b,t) = (e^{-t/2}a + b,\ s+t), \qquad \mathscr{G}_{(a,s)}\mathscr{G}_{(b,t)} = \mathscr{G}_{(a,s)\cdot(b,t)}. \tag{8.1}$$

In particular the subgroup $\{T_a : a \in \mathbb{R}^3\}$ is normal in the full spacetime covariant action, and the subspace of observables invariant under all spatial covariant translations is invariant under every Θ_s.

Proof. For a profile V,

$$(\Theta_s T_a \Theta_{-s} V)(y,\tau) = V(y + e^{(\tau+s)/2}a, \tau) = (T_{e^{s/2}a} V)(y,\tau).$$

The pressure transforms identically. The asserted invariance of the spatially invariant observable subspace is the same identity written on the induced L^2-representation. The group law (8.1) follows from

$$\Theta_s T_a \Theta_t T_b = \Theta_{s+t} T_{e^{-t/2}a+b}.$$

□

Theorem 8.6 (Covariant observer calculus)**.** *Let $Z := \mathbb{R}^3 \times \mathbb{R}$ denote the centered observer space, with points $z = (\xi, \sigma)$. For $a \in \mathbb{R}^3$ and $\sigma \in \mathbb{R}$,*

$$(T_a U)(y,\tau) = U(y + e^{\tau/2}a, \tau), \qquad (\Theta_\sigma U)(y,\tau) = U(y, \tau+\sigma).$$

Then

$$\Theta_\sigma T_a = T_{e^{\sigma/2}a}\Theta_\sigma, \qquad \Theta_\sigma T_a \Theta_{-\sigma} = T_{e^{\sigma/2}a}.$$

For $z = (\xi,\sigma)$, define the covariant observer

$$\mathcal{R}_z U := \Theta_\sigma T_{e^{-\sigma/2}\xi} U.$$

Equivalently,

$$(\mathcal{R}_{(\xi,\sigma)} U)(y,s) = U(y + e^{s/2}\xi, \sigma + s).$$

The covariant tube pulled back from $Q_r(0,0)$ is

$$\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma) := \left\{(Y,\tau) : |\tau - \sigma| < r^2,\ |Y - e^{(\tau-\sigma)/2}\xi| < r\right\}.$$

The map

$$(y,s) \mapsto (Y,\tau) = (y + e^{s/2}\xi, \sigma + s)$$

is a bijection from $Q_r(0,0)$ onto $\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma)$ with spacetime Jacobian one. The same formulas apply to the pressure, modulo the admissible time-dependent gauges.

Proof. The semidirect identities are Lemma 8.5; written without conjugation,

$$(\Theta_\sigma T_a U)(y,\tau) = U(y + e^{(\tau+\sigma)/2}a, \tau+\sigma) = (T_{e^{\sigma/2}a}\Theta_\sigma U)(y,\tau).$$

Taking $a = e^{-\sigma/2}\xi$ gives

$$(\Theta_\sigma T_{e^{-\sigma/2}\xi} U)(y,s) = U(y + e^{(s+\sigma)/2}e^{-\sigma/2}\xi, \sigma+s) = U(y + e^{s/2}\xi, \sigma+s).$$

If $Y = y + e^{s/2}\xi$ and $\tau = \sigma + s$, then

$$y = Y - e^{(\tau-\sigma)/2}\xi, \qquad s = \tau - \sigma,$$

so $|s| < r^2$ and $|y| < r$ are equivalent to the defining inequalities for $\mathcal{Q}_r^{\mathrm{cov}}(\xi,\sigma)$. The transformation is a time-dependent spatial translation, hence has Jacobian one. □

Definition 8.7 (Terminal profile class and local topology)**.** For $M < \infty$, let $\mathfrak{X}_M$ be the class of pairs (U,Π), modulo addition of arbitrary functions of time to Π, such that:

(i) U is a smooth bounded ancient solution of (1.2) on $\mathbb{R}^3 \times \mathbb{R}$, with $\|U\|_{L^\infty} \le M$;
(ii) (U,Π) is locally suitable on every compact terminal cylinder;
(iii) on every compact cylinder the pressure has an admissible $L^{3/2}$-representative after subtracting a time-dependent gauge.

The topology on $\mathfrak{X}_M$ is

$$U_n \to U \quad \text{in } C^\infty_{\text{loc}}, \qquad \Pi_n - c_n(\tau) \rightharpoonup \Pi \quad \text{in } L^{3/2}_{\text{loc}}$$

for suitable time-dependent gauges c_n. All profile limits, tail descendants, concentrating descendant hulls, and recurrent-core limits below are taken in this topology.

Definition 8.8 (Observer oscillation and spatial oscillation)**.** Let $Q = B \times I$ be a bounded covariant terminal cylinder. For a velocity field U, define the constant-in-time oscillation functional

$$\operatorname{osc}_{3,\text{ct}}(U;Q) := \inf_{c\in\mathbb{R}^3} \iint_Q |U-c|^3 \, dy \, d\tau.$$

Define also the spatial oscillation modulo time-dependent homogeneous gauges

$$\operatorname{osc}_{3,\text{sp}}(U;Q) := \inf_{c\in L^3(I;\mathbb{R}^3)} \iint_Q |U(y,\tau) - c(\tau)|^3 \, dy \, d\tau.$$

Thus $\operatorname{osc}_{3,\text{ct}}$ detects nonconstant time-dependent homogeneous modes, while $\operatorname{osc}_{3,\text{sp}}$ is the quotient seminorm used to remove spatially homogeneous parasitic modes. A covariant observer $z \in Z$ is η-oscillatory for U if

$$\operatorname{osc}_{3,\text{ct}}\big(U; \mathcal{Q}_1^{\text{cov}}(z)\big) \ge \eta.$$

Fix once and for all a proper covariant metric d_Z on Z, and choose a separated covariant unit-net $\{z_k\}_{k\in\mathbb{N}} \subset Z$ such that

$$Z \subset \bigcup_{k\in\mathbb{N}} B_Z(z_k, 1/10), \qquad \#\{k : z_k \in B_Z(z,2)\} \le N_0$$

for every $z \in Z$. The associated covariant unit cylinders therefore have bounded overlap with constant depending only on N_0. The oscillation measure on observer-center sets $E \subset Z$ is the atomic measure on this fixed net,

$$\mu_U^{\text{osc}}(E) := \sum_{z_k\in E} \operatorname{osc}_{3,\text{ct}}\big(U; \mathcal{Q}_1^{\text{cov}}(z_k)\big).$$

For a sequence U_n we write μ_n^{osc} for this measure. The fixed packing constant N_0 is used below in the moving-window estimate for escaping observer regions. The concentration-set extraction uses $\operatorname{osc}_{3,\text{ct}}$, because after parasitic-free normalization it detects genuine non-affine oscillation. Mesoscopic cylinders use $\operatorname{osc}_{3,\text{sp}}$, equivalently the functional $\mathcal{A}_n(z,r)$ of Definition 9.60, because the minimal-scale quotient must ignore time-dependent homogeneous gauges. Thus the concentration-set construction and the mesoscopic affine quotient are kept distinct.

Lemma 8.9 (Local compactness and continuity of the profile transformations)**.** *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a sequence with the local suitability and pressure-gauge bounds supplied by the setup paper on every fixed compact cylinder. Then every subsequence has a further subsequence converging in the topology of $\mathfrak{X}_M$. Moreover the actions T_a, Θ_s, the covariant tail recenterings $\mathscr{T}_{(x,\sigma)}$ are continuous on $\mathfrak{X}_M$. The affine maps satisfy the bound-tracking continuity*

$$S_c : \mathfrak{X}_M \longrightarrow \mathfrak{X}_{M+|c|},$$

again with the pressure interpreted modulo the corresponding transformed time gauges.

Proof. On each compact cylinder K, the setup local suitability, dissipation, strong local L^3-velocity compactness, and pressure-gauge compactness give a Seregin subsequential limit in the local topology of $\mathfrak{X}_M$. Diagonalizing over the countable compact exhaustion gives the asserted subsequence. With the setup pressure-atlas bounds and local suitability fixed, interior parabolic estimates for (1.2) upgrade the velocity convergence to C^∞_{loc}; the pressure representatives converge weakly in $L^{3/2}_{\mathrm{loc}}$ after the local gauges. The formulas defining T_a, Θ_s, $\mathscr{T}_{(x,\sigma)}$, and S_c are smooth changes of variables and affine pressure changes on each fixed compact cylinder. The first three preserve the L^∞ bound M. The affine map S_c subtracts the constant vector c, so its velocity bound is at most $M+|c|$. All four maps preserve the local topology and transform admissible gauges into admissible gauges. No global spatial norm or global pressure representative is used. □

Definition 8.10 (Compact normalized descendant profile class)**.** Let $\mathfrak{X}^{\#}_M \subset \mathfrak{X}_M$ denote the sequential closure, in the local topology of $\mathfrak{X}_M$, of all normalized Seregin profiles from the setup paper and all profiles obtained from them by the limiting operations used in this paper: centered time translations, covariant translations, covariant tail recenterings, terminal profile limits, concentrating descendant limits, recurrent-core limits, and the canonical affine normalization whenever the affine parameter is bounded. The closure is always taken with the local pressure gauges supplied by the normalized Seregin compactness theorem.

Lemma 8.11 (Compactness and continuity on the normalized profile class)**.** *The space $\mathfrak{X}^{\#}_M$ is compact in the local topology inherited from $\mathfrak{X}_M$. The maps T_a, Θ_s, and the covariant recenterings used to form descendants act continuously on $\mathfrak{X}^{\#}_M$. The affine map satisfies the bound-tracking continuity*

$$S_c : \mathfrak{X}^{\#}_M \longrightarrow \mathfrak{X}^{\#}_{M+|c|}.$$

Proof. By definition, every sequence in $\mathfrak{X}^{\#}_M$ is a diagonal limit of normalized Seregin descendants with the same local L^∞, local suitability, pressure-gauge, and interior smooth bounds on each fixed compact cylinder. Applying Lemma 8.9 on an exhaustion of compact cylinders gives a further subsequential limit in $\mathfrak{X}_M$. The closure defining $\mathfrak{X}^{\#}_M$ contains that limit, so $\mathfrak{X}^{\#}_M$ is sequentially compact; the local topology is metrizable on bounded sets by the standard countable exhaustion, hence compact.

Continuity of the limiting operations is the continuity already proved in Lemma 8.9, restricted to the closed normalized profile class. The affine map changes only the velocity bound and the explicit affine pressure gauge, so it maps $\mathfrak{X}^{\#}_M$ continuously into $\mathfrak{X}^{\#}_{M+|c|}$. □

Definition 8.12 (Affine homogeneous symmetry)**.** For $c \in \mathbb{R}^3$, define

$$(S_cV)(y,\tau) := V(y-2c,\tau) - c, \qquad (S_cP)(y,\tau) := P(y-2c,\tau) + \frac{1}{2}c\cdot y.$$

Lemma 8.13 (Affine renormalized Galilean symmetry)**.** *If (V,P) solves (1.2), then (S_cV, S_cP) also solves (1.2).*

Proof. Set $U(y,\tau) = V(y-2c,\tau)-c$, $\Pi(y,\tau) = P(y-2c,\tau)+\frac{1}{2}c\cdot y$, and $z = y-2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z,\tau), \qquad \Delta U = (\Delta V)(z,\tau), \qquad \nabla U = (\nabla V)(z,\tau).$$

The nonlinear term is

$$(U\cdot\nabla)U = (V-c)\cdot\nabla V = (V\cdot\nabla)V - c\cdot\nabla V,$$

where all quantities on the right are evaluated at (z,τ). The drift term satisfies

$$\frac{1}{2}y\cdot\nabla U = \frac{1}{2}(z+2c)\cdot\nabla V = \frac{1}{2}z\cdot\nabla V + c\cdot\nabla V.$$

The two $c\cdot\nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla\Pi = \frac{1}{2}(V-c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point (z,τ) gives the equation for (U,Π). The divergence condition is unchanged by translation and subtracting a constant vector. □

Definition 8.14 (Pressure gauges and affine quotient)**.** On every ordinary local cylinder Q, pressure representatives are identified modulo time functions:

$$P_Q \sim P_Q + c_Q(\tau).$$

At the affine/parasitic quotient level, and only there, one also identifies pressure representatives modulo the affine spatial functions paired with spatially homogeneous parasitic velocities:

$$P_Q \sim P_Q + b_Q(\tau)\cdot y + c_Q(\tau).$$

The affine term is allowed exactly when it corresponds to a removable homogeneous mode in the velocity equation, as in Definition 8.16. Generic terminal residual branches are always considered after this quotient has either been fixed by the canonical parasitic-free representative or assigned to $\mathcal{C}_{\mathrm{aff}}$.

Lemma 8.15 (Pressure atlas compatibility)**.** *Let Q, Q' be overlapping local cylinders in a normalized Seregin extraction. Before affine quotienting, admissible pressure representatives satisfy*

$$P_Q - P_{Q'} = c_{Q,Q'}(\tau)$$

on $Q\cap Q'$. At the affine/parasitic quotient level they satisfy

$$P_Q - P_{Q'} = b_{Q,Q'}(\tau)\cdot y + c_{Q,Q'}(\tau)$$

with the affine term paired with the corresponding removable homogeneous velocity mode. The velocity equation is unchanged after the time-gauge identification, and after the affine identification it is interpreted in the quotient equation with the parasitic correction removed or assigned to $\mathcal{C}_{\mathrm{aff}}$.

Proof. The time-gauge statement is the setup pressure atlas compatibility imported in Theorem 1.3. The affine statement follows by substituting a spatially homogeneous velocity $b(\tau)$ into the centered equation: the only pressure gradient it can create is $-(\partial_\tau b + \frac{1}{2}b)$, hence an affine pressure in y, modulo a time function. This is the quotient convention in Definition 8.14. All terminal limiting operations below choose their pressure gauges before taking compactness limits, and if the affine term is nontrivial the profile is assigned to $\mathcal{C}_{\mathrm{aff}}$. □

Definition 8.16 (Affine/parasitic lower stratum and residual normalization)**.** The affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$ consists of the spatially homogeneous modes which are invisible to the spatial quotient seminorm $\mathrm{osc}_{3,\mathrm{sp}}$. It has two equivalent realizations.

(i) A genuine centered profile belongs to $\mathcal{C}_{\mathrm{aff}}$ if, after one affine homogeneous symmetry S_c, its velocity is spatially constant. In the parasitic-free representative this is the zero profile.

(ii) A quotient limit belongs to $\mathcal{C}_{\mathrm{aff}}$ if it is represented by a locally bounded spatially homogeneous velocity $b(\tau)$. In the weak centered equation this mode is paired with the affine pressure functional

$$\big(-\partial_\tau b(\tau) - \tfrac{1}{2}b(\tau)\big)\cdot y$$

in $\mathcal{D}'_\tau\mathcal{D}'_y$. If the mode is realized as a smooth suitable profile, this affine pressure is an ordinary local pressure; if it arises only as a measurable homogeneous gauge $c_n(\tau)$, it is used only in the quotient weak formulation and never as an independent $L^{3/2}_{\mathrm{loc}}$ pressure.

In the ordered residual decomposition this stratum is part of the previously closed lower classes, not part of $\mathcal{C}_{\mathrm{gen}}$. Equivalently, every generic residual profile is considered after the canonical affine normalization which removes its spatially homogeneous mild mode. Retained non-affine oscillation in the residual class is therefore always measured after this normalization.

Remark 8.17 (How affine constants are used). When a compact recurrent tail core is shown below to consist only of spatially homogeneous parasitic modes, the contradiction is not that an unnormalized nonzero constant has zero local L^3-mass. The contradiction is that these modes have already been assigned to the closed lower stratum in Definition 8.16; after the canonical affine normalization they are the zero profile and cannot represent a generic residual profile carrying retained velocity concentration or non-affine oscillation.

Lemma 8.18 (Homogeneous gauges generate only the affine/parasitic stratum). *Let $c_n : I \to \mathbb{R}^3$ be bounded in $L^\infty(I)$, and let*

$$C_n(y,\tau) := c_n(\tau).$$

Every local weak limit, time translate, or normalized noncompactness limit of (C_n) has vanishing spatial quotient oscillation on every compact cylinder and belongs to $\mathcal{C}_{\mathrm{aff}}$. Conversely, after the canonical parasitic-free normalization, no nonzero spatially homogeneous mode remains in the residual profile class.

Proof. For every compact cylinder Q,

$$\inf_{c(\tau)} \iint_Q |C_n(y,\tau) - c(\tau)|^3 \, dy \, d\tau = 0,$$

with the choice $c(\tau) = c_n(\tau)$. The same identity is stable under local weak limits and time recenterings. Hence every such limit has zero spatial quotient oscillation on every compact cylinder and is, by Definition 8.16, an affine/parasitic quotient mode. If a homogeneous mode is realized as a smooth centered solution, substituting $U = b(\tau)$ into (1.2) gives

$$\partial_\tau b + \frac{1}{2} b + \nabla P = 0,$$

so the pressure is affine in y up to a time gauge, exactly as in the definition of $\mathcal{C}_{\mathrm{aff}}$. If b is only a bounded measurable gauge, the same identity is understood distributionally and only in the quotient formulation. The residual representative is defined after removal of this quotient direction; therefore a nonzero retained effect of b is assigned to the closed lower stratum, while the parasitic-free residual profile has no such mode. $\square$

Theorem 8.19 (Closed non-affine oscillation functional). *Let $Q \Subset \mathbb{R}^3 \times \mathbb{R}$ be a bounded connected parabolic cylinder. For $U \in \mathfrak{X}_M$, define*

$$d_Q(U) := \mathrm{osc}_{3,\mathrm{ct}}(U;Q)^{1/3}.$$

Then

$$|d_Q(U) - d_Q(V)| \le \|U - V\|_{L^3(Q)}.$$

In particular d_Q is continuous under the local topology of $\mathfrak{X}_M$. Moreover,

$$d_Q(U) = 0 \iff U \text{ is spatially and temporally constant on } Q.$$

Fix once and for all a nested exhaustion

$$Q_1 \Subset Q_2 \Subset \cdots, \qquad \bigcup_{j=1}^\infty Q_j = \mathbb{R}^3 \times \mathbb{R},$$

by bounded connected parabolic cylinders. Define

$$\Phi(U) := \sum_{j=1}^\infty 2^{-j} \min\{1, d_{Q_j}(U)\}.$$

Then Φ is continuous on $\mathfrak{X}_M$, and

$$\Phi(U) = 0 \iff U \text{ is globally constant.}$$

Consequently, for every $\eta > 0$, the non-affine oscillation set

$$\mathfrak{A}_\eta := \{(U, \Pi) \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

is closed in the local profile topology.

Proof. For every $c \in \mathbb{R}^3$, the triangle inequality in $L^3(Q)$ gives

$$\|U - c\|_{L^3(Q)} \leq \|U - V\|_{L^3(Q)} + \|V - c\|_{L^3(Q)}.$$

Taking the infimum over c gives

$$d_Q(U) \leq \|U - V\|_{L^3(Q)} + d_Q(V).$$

Interchanging U and V proves the Lipschitz estimate. Local smooth convergence implies local L^3-convergence, so d_Q is continuous in the topology of $\mathfrak{X}_M$.

If $d_Q(U) = 0$, choose $c_k \in \mathbb{R}^3$ such that $\|U - c_k\|_{L^3(Q)} \to 0$. The sequence c_k is bounded; otherwise

$$\|U - c_k\|_{L^3(Q)} \geq |c_k|\,|Q|^{1/3} - \|U\|_{L^3(Q)} \to \infty,$$

which is impossible. Passing to a subsequence, $c_k \to c$, and hence $U = c$ almost everywhere on Q. Since U is smooth, $U \equiv c$ pointwise on Q. The converse is immediate.

Each summand in the definition of Φ is continuous and bounded by 2^{-j}. The Weierstrass test gives uniform convergence of the defining series, hence continuity of Φ. If $\Phi(U) = 0$, then $d_{Q_j}(U) = 0$ for every j, so U is constant on every Q_j. The cylinders are nested and overlap, so the constants agree and U is globally constant. The converse is again immediate. Closedness of $\mathfrak{A}_\eta$ follows from continuity of Φ. □

Corollary 8.20 (Compact normalized non-affine oscillation sets)**.** *For every $\eta > 0$,*

$$\mathfrak{A}^{\#}_\eta := \{(U, \Pi) \in \mathfrak{X}^{\#}_M : \Phi(U) \geq \eta\}$$

is compact in the normalized profile topology.

Proof. By Lemma 8.11, $\mathfrak{X}^{\#}_M$ is compact. By Theorem 8.19, Φ is continuous in the local topology. Therefore $\mathfrak{A}^{\#}_\eta$ is a closed subset of a compact space. □

Theorem 8.21 (Affine normalization dichotomy)**.** *Let S_c be the affine homogeneous symmetry of Definition 8.12, and suppose the ordered lower strata include all constant centered profiles as in Definition 8.16. For every $(U, \Pi) \in \mathfrak{X}_M$, exactly one of the following alternatives holds:*

- (i) $\mathrm{osc}_{3,\mathrm{sp}}(U; Q) = 0$ *on every compact cylinder Q. Then U is a spatially homogeneous quotient mode, and (U, Π) belongs to the affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$.*
- (ii) *For the parasitic-free representative, after an admissible affine symmetry S_c, one has $\Phi(S_cU) > 0$. Equivalently, for some compact cylinder Q,* $\mathrm{osc}_{3,\mathrm{ct}}(S_cU; Q) > 0$.

In the retained normalized residual space, alternative (i) *has already been removed by the canonical affine/parasitic normalization, so only alternative* (ii) *can remain. Moreover, if $K \subset \mathfrak{X}_M$ is compact and*

$$K \cap \{(U, \Pi) : \Phi(U) = 0\} = \varnothing,$$

then there exists $\eta_K > 0$ such that

$$\Phi(U) \geq \eta_K \qquad \textit{for every } (U, \Pi) \in K.$$

Proof. If $\mathrm{osc}_{3,\mathrm{sp}}(U; Q) = 0$ on every compact cylinder, then on each time slice U is spatially constant in the distributional sense. The local constants are compatible on overlapping cylinders, so $U(y, \tau) = b(\tau)$ locally in time for a measurable bounded function b. By Definition 8.16, this is precisely the affine/parasitic quotient stratum; if it is realized by a smooth profile, the corresponding affine pressure is the one described there, and otherwise it is used only in the quotient weak formulation.

If the spatial oscillation does not vanish identically, U is not a spatially homogeneous quotient mode. After choosing the canonical parasitic-free representative and applying any admissible affine symmetry S_c, the resulting velocity is not globally constant. Hence, by Theorem 8.19, $\Phi(S_cU) > 0$. Conversely, if $\Phi(S_cU) = 0$, then S_cU is globally constant, and the original profile is in $\mathcal{C}_{\mathrm{aff}}$. This proves the dichotomy and the retained normalized statement. For the compact statement, Φ is continuous and strictly positive on K, so it attains a positive minimum

$$\eta_K := \min_{(U,\Pi)\in K} \Phi(U) > 0.$$

$\square$

Definition 8.22 (Escaping covariant hull)**.** Let V be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{H}_\infty(V) := \left\{W :\ \exists (a_n, s_n) \in G,\ |(a_n, s_n)|_{\mathrm{cov}} \to \infty, \mathscr{G}_{(a_n,s_n)}V \to W \text{ in } C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})\right\}.$$

Here $G = \mathbb{R}^3 \rtimes \mathbb{R}$ is the covariant spacetime group in Lemma 8.5, and any proper left-invariant metric may be used for $|(a, s)|_{\mathrm{cov}}$. Thus the recurrent tail hull is invariant under the full covariant action, not merely under ordinary spatial translations.

Theorem 8.23 (Compactness of the normalized escaping covariant hull)**.** *Let $(V, P) \in \mathfrak{X}^\#_M$ be represented after canonical affine/parasitic normalization. Assume that every escaping covariant sequence for (V, P) has a compact subsequence in the normalized local topology with the setup pressure atlas fixed; if this assumption fails, the sequence is assigned instead to the local-compactness or pressure-gauge-loss terminal alternative. In the compactness branch, $\mathcal{H}_\infty(V)$ is nonempty and compact in $C^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{H}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \le M$.*

Proof. For any escaping covariant sequence (a_n, s_n), the transformed solutions $\mathscr{G}_{(a_n,s_n)}V$ solve the same equation and have the same L^∞ bound by Lemmas 8.2 and 8.5. The L^∞ bound alone is not used to obtain a time modulus or derivative bound. By the compactness-branch assumption, the transformed profiles have a subsequence converging in the normalized pressure-atlas topology, hence locally smoothly in velocity. Passing to the limit in the equation gives another bounded ancient solution. Nonemptiness follows by applying this compactness branch to any escaping sequence in G.

It remains to record closedness of the escaping hull. Suppose

$$W_j \in \mathcal{H}_\infty(V), \qquad W_j \to W \quad \text{in } C^\infty_{\mathrm{loc}}.$$

For each j, choose an escaping sequence $g_{j,k} \in G$ with $\mathscr{G}_{g_{j,k}}V \to W_j$. Choose $k(j)$ so large that

$$|g_{j,k(j)}|_{\mathrm{cov}} \ge j, \qquad d_j\Big(\mathscr{G}_{g_{j,k(j)}}V, W_j\Big) \le j^{-1},$$

where d_j is a metric for C^j-convergence on $\overline{B_j} \times [-j, j]$. Then

$$\mathscr{G}_{g_{j,k(j)}}V \to W \quad \text{in } C^\infty_{\mathrm{loc}}, \qquad |g_{j,k(j)}|_{\mathrm{cov}} \to \infty.$$

Hence $W \in \mathcal{H}_\infty(V)$. This proves closedness.

To see sequential precompactness of the whole hull, let $W_j \in \mathcal{H}_\infty(V)$ be arbitrary. For each j, choose an escaping sequence $g_{j,k}$ with $\mathscr{G}_{g_{j,k}}V \to W_j$, and choose $k(j)$ so that

$$|g_{j,k(j)}|_{\mathrm{cov}} \ge j, \qquad d_j\Big(\mathscr{G}_{g_{j,k(j)}}V, W_j\Big) \le j^{-1}.$$

The sequence $\mathscr{G}_{g_{j,k(j)}}V$ is an escaping transformed sequence in the compactness branch, so it has a locally smooth convergent subsequence. Along that subsequence, W_j converges to the same limit. The closedness argument places the limit in $\mathcal{H}_\infty(V)$. Thus every sequence in the hull has a convergent subsequence with limit still in the hull; metrizability of the local smooth topology gives compactness. $\square$

Lemma 8.24 (Full covariant invariance of the escaping covariant hull)**.** *For every $g \in G$,*

$$\mathscr{G}_g\mathcal{H}_\infty(V) = \mathcal{H}_\infty(V).$$

In particular, if $W \in \mathcal{H}_\infty(V)$, then both T_aW and Θ_sW belong to $\mathcal{H}_\infty(V)$ for all $a \in \mathbb{R}^3$ and $s \in \mathbb{R}$.

Proof. Let $W = \lim_n \mathscr{G}_{g_n}V$, with $|g_n|_{\mathrm{cov}} \to \infty$. Then

$$\mathscr{G}_gW = \lim_n \mathscr{G}_g\mathscr{G}_{g_n}V = \lim_n \mathscr{G}_{gg_n}V.$$

Since the metric on G is proper and left-invariant, $|gg_n|_{\mathrm{cov}} \to \infty$. Hence $\mathscr{G}_gW \in \mathcal{H}_\infty(V)$. Applying the same argument to g^{-1} gives equality. $\square$

Definition 8.25 (Tail alternatives)**.** Let $\mathcal{C}$ denote the set of constant vector fields. Define the following classes only in the compact normalized hull branch, after the local-compactness and pressure-gauge-loss alternatives identified in Theorem 8.23:

$$\mathcal{C}_{\mathrm{gen}_{\text{tail-const}}} := \{V \in \mathcal{C}_{\mathrm{gen}} : \mathcal{H}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\mathrm{gen}_{\text{tail-prof}}} := \{V \in \mathcal{C}_{\mathrm{gen}} : \mathcal{H}_\infty(V) \text{ contains a nonconstant profile}\}.$$

Proposition 8.26 (Tail dichotomy and tail-zero normalization)**.** *Every $V \in \mathcal{C}_{\mathrm{gen}}$ belongs to $\mathcal{C}_{\mathrm{gen}_{\text{tail-const}}} \cup \mathcal{C}_{\mathrm{gen}_{\text{tail-prof}}}$. If $\mathcal{H}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty}V$$

solves (1.2) and satisfies

$$\mathcal{H}_\infty(U) = \{0\}.$$

Proof. The dichotomy is the definition. Since T_a and S_c commute,

$$T_a(S_{C_\infty}V) = S_{C_\infty}(T_aV).$$

Thus every far-field limit of $S_{C_\infty}V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by Lemma 8.13. $\square$

Definition 8.27 (Tail-minimal recurrent core)**.** A nonempty compact set $\mathcal{M} \subset \mathcal{H}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations T_a, invariant under all time translations

$$(\Theta_sW)(y,\tau) := W(y,\tau+s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

Lemma 8.28 (Pressure gradients in invariant local averages)**.** *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations T_a and the time translations Θ_s. Let μ be a Borel probability measure on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let P_W be any smooth local pressure near $(0,0)$ satisfying (1.2). Put*

$$u_i(W) := W_i(0,0), \qquad p_i(W) := \partial_iP_W(0,0), \qquad b_{ij}(W) := u_i(W)u_j(W).$$

Let Π_0 denote the orthogonal projection in $L^2(\mathcal{M},\mu)$ onto the closed subspace of functions invariant under all spatial covariant translations T_a, and set

$$u_i^0 := \Pi_0 u_i, \qquad p_i^0 := \Pi_0 p_i.$$

Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies

$$\int_{\mathcal{M}} u_i(W)p_i(W)\,d\mu(W) = \int_{\mathcal{M}} u_i^0(W)p_i^0(W)\,d\mu(W). \tag{8.2}$$

Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:

$$\int_{\mathcal{M}} u_i(W)\big(p_i(W)-p_i^0(W)\big)\,d\mu(W) = 0. \tag{8.3}$$

Proof. The additive pressure gauge does not enter, because only ∇P_W appears. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y\cdot\nabla W) - \partial_\tau W - (W\cdot\nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. The local pressure reconstruction step in the imported setup results supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let U_a be the unitary representation of spatial covariant translations on $L^2(\mathcal{M},\mu)$,

$$(U_aF)(W) := F(T_aW).$$

Let D_1, D_2, D_3 be its skew-adjoint infinitesimal generators. For a smooth local observable F, D_kF is obtained by differentiating $F(T_{he_k}W)$ at $h=0$. In particular, at $\tau=0$,

$$D_ku_i(W) = \partial_kW_i(0,0).$$

Hence

$$D_iu_i = 0 \qquad \text{in } L^2(\mathcal{M},\mu), \tag{8.4}$$

because each W is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$-\Delta_y P_W = \partial_i\partial_j(W_iW_j). \tag{8.5}$$

Indeed, the divergence of $\partial_\tau W$, ΔW, and W is zero; the divergence of $y\cdot\nabla W$ is $\partial_i(y_j\partial_jW_i) = \partial_jW_j + y_j\partial_j\partial_iW_i = 0$; and $(W\cdot\nabla)W_i = \partial_j(W_jW_i)$. Applying the infinitesimal translation generators at $(0,0)$ gives

$$D_ip_i = -D_iD_jb_{ij}. \tag{8.6}$$

The pressure gradient is curl-free, hence

$$D_kp_i = D_ip_k. \tag{8.7}$$

Let

$$A := -\sum_{k=1}^{3} D_k^2.$$

By the spectral theorem for the commuting unitary group U_a, A is self-adjoint and nonnegative, and Π_0 is the spectral projection of A at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I-\Pi_0)p_i.$$

Equations (8.6) and (8.7) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1}(I - \Pi_0) D_i D_j b_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k(\varepsilon + A)^{-1}(I - \Pi_0) D_i D_j b_{ij}, \qquad \varepsilon > 0,$$

the joint spectral resolution of D_1, D_2, D_3 gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M}, \mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is precisely the multiplier convergence

$$\frac{\xi_k \xi_i \xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k \xi_i \xi_j}{|\xi|^2} \qquad (\xi \neq 0),$$

together with $(I - \Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (8.7) says that the nonzero spatial spectral part of p is parallel to ξ, and (8.6) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of D_k and (8.4) give

$$\begin{aligned}\int_{\mathcal{M}} u_k q_{k,\varepsilon}\, d\mu &= \int_{\mathcal{M}} u_k D_k(\varepsilon + A)^{-1}(I - \Pi_0) D_i D_j b_{ij}\, d\mu \\ &= -\int_{\mathcal{M}} (D_k u_k)(\varepsilon + A)^{-1}(I - \Pi_0) D_i D_j b_{ij}\, d\mu = 0.\end{aligned}$$

Passing to the limit in L^2 yields

$$\int_{\mathcal{M}} u_k q_k\, d\mu = 0,$$

which is (8.3). Since p_i^0 lies in the range of Π_0 and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0) p_i^0\, d\mu = 0.$$

Combining this with (8.3) proves (8.2). □

Theorem 8.29 (Rigidity of recurrent profile tail cores)**.** *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the normalized Seregin setup in the companion setup paper, with the affine zero mode interpreted by the affine symmetry Lemma 8.13. Equivalently, after applying one fixed affine normalization S_c, the spatially homogeneous part of the core satisfies the bounded mild centered formulation recorded in the imported setup results. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y, \tau) \equiv m \qquad \textit{for every } W \in \mathcal{M}.$$

After applying the affine symmetry S_m, the core is reduced to the zero solution.

Proof. The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure μ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W)\, d\mu(W)$$

for local observables. Let P_W be a local pressure associated to W. As in Lemma 8.28, set

$$u_i(W) := W_i(0,0), \qquad p_i(W) := \partial_i P_W(0,0),$$

and let Π_0 be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \qquad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0,0)\rangle = 0, \qquad \langle \partial_j F(W)(0,0)\rangle = 0$$

for every smooth local observable F. These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate (1.2) at $(0,0)$. The drift term vanishes at $y=0$, which gives

$$\partial_\tau W(0,0) - \Delta W(0,0) + \frac{1}{2}W(0,0) + (W(0,0)\cdot\nabla)W(0,0) + \nabla P_W(0,0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under Π_0: the Laplacian is $D_jD_ju_i$, and the nonlinear term is $D_j(u_ju_i)$ because $D_ju_j=0$. Time translations do not literally commute with the covariant spatial translations; by Lemma 8.5 they normalize them:

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}.$$

Hence the subspace of observables invariant under all T_a is invariant under Θ_s, and the orthogonal projection Π_0 commutes with the time-translation representation. Therefore $\Pi_0\partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$\partial_\tau u_i^0 + \frac{1}{2}u_i^0 + p_i^0 = 0 \qquad \text{in } L^2(\mathcal{M},\mu). \tag{8.8}$$

Taking the inner product of (8.8) with u_i^0, summing in i, and using invariance under time translations,

$$\left\langle u_i^0\partial_\tau u_i^0\right\rangle = \frac{1}{2}\left\langle \partial_\tau |u^0|^2\right\rangle = 0,$$

gives

$$\left\langle u_i^0 p_i^0\right\rangle = -\frac{1}{2}\left\langle |u^0|^2\right\rangle. \tag{8.9}$$

Testing the equation against W, evaluating at $(0,0)$, and averaging, the time term is

$$\langle W\cdot\partial_\tau W\rangle = \frac{1}{2}\left\langle \partial_\tau |W|^2\right\rangle = 0.$$

The diffusion term gives

$$-W\cdot\Delta W = |\nabla W|^2 - \operatorname{div}(W\cdot\nabla W),$$

and therefore

$$-\langle W\cdot\Delta W\rangle = \left\langle |\nabla W|^2\right\rangle.$$

The nonlinear term is a divergence:

$$W\cdot(W\cdot\nabla)W = \frac{1}{2}W\cdot\nabla|W|^2 = \frac{1}{2}\operatorname{div}(|W|^2W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by Lemma 8.28, and the remaining spatial zero mode is evaluated by (8.9). Hence

$$\langle u_ip_i\rangle = \left\langle u_i^0p_i^0\right\rangle = -\frac{1}{2}\left\langle |u^0|^2\right\rangle.$$

Thus

$$\left\langle |\nabla W|^2\right\rangle + \frac{1}{2}\left\langle |W|^2\right\rangle - \frac{1}{2}\left\langle |u^0|^2\right\rangle = 0. \tag{8.10}$$

Both terms are nonnegative. The second nonnegative quantity is

$$\left\langle |u|^2\right\rangle - \left\langle |u^0|^2\right\rangle = \left\langle |u-u^0|^2\right\rangle,$$

because Π_0 is an orthogonal projection in $L^2(\mathcal{M},\mu)$. Therefore

$$\langle |\nabla W|^2 \rangle = 0, \qquad \langle |u-u^0|^2 \rangle = 0.$$

Since μ has full support and the observables are continuous in the local smooth topology, $\nabla W(0,0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0,0)$ to any (y,τ), so $\nabla W(y,\tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y,\tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the normalization in the setup paper quoted in the statement, there is one fixed affine normalization S_c for which the spatially homogeneous representatives satisfy the bounded mild centered formulation recorded in the imported setup results. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in the mild formulation vanishes, and the Ornstein–Uhlenbeck–Stokes semigroup acts on constants by

$$S(\tau-\sigma)B(\sigma) = e^{-(\tau-\sigma)/2}B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau-\sigma)/2}B(\sigma) \qquad (\sigma < \tau).$$

Letting $\sigma \to -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every τ. Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y,\tau) \equiv m \qquad (W \in \mathcal{M}).$$

The affine symmetry S_m then maps the core to the zero solution. □

9. Terminal concentration profiles and separated centers

The terminal part of the proof is local in parabolic cylinders. It does not require a global L^3 bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

Definition 9.1 (Parabolic recentering sequence)**.** Let (V_n,P_n) be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3\times\mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathfrak{c} = (z_n)_{n\ge 1}, \qquad z_n = (y_n,\tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \Big(|y_n - y_n'| + |\tau_n - \tau_n'|^{1/2} \Big) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathfrak{c}$ are denoted

$$V_n^{\mathfrak{c}}(y,s) := V_n(y+y_n, s+\tau_n), \qquad P_n^{\mathfrak{c}}(y,s) := P_n(y+y_n,s+\tau_n) - a_n^{\mathfrak{c}}(s),$$

where $a_n^{\mathfrak{c}}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables (y,s).

Definition 9.2 (Combined parabolic-covariant descendant recentering)**.** Let $g_n = (y_n,\tau_n)$ be an ordinary parabolic recentering sequence for a terminal extraction (V_n,P_n), and let $\zeta_n = (\xi_n,\sigma_n)$ be a covariant tail recentering sequence in the centered limit generated by g_n. The combined parabolic-covariant descendant recentering is denoted

$$d_n = (g_n;\xi_n,\sigma_n)$$

and acts on the original terminal sequence by

$$V_n^d(y,s) := V_n(y + y_n + e^{s/2}\xi_n,\ s + \tau_n + \sigma_n),$$

with the pressure pulled back by the same formula and then adjusted by an admissible time-dependent gauge. Its relative distance from g_n is

$$|\xi_n| + |\sigma_n|^{1/2}.$$

Thus d_n is separated from g_n exactly when this relative distance tends to infinity. This definition is used only to lift descendants of a centered profile back to the original terminal extraction; ordinary first-generation terminal recenterings remain the ordinary parabolic recenterings of Definition 9.1.

Definition 9.3 (Raw atlas defect measure and raw unit-window active recenterings)**.** After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \Big(|V_n|^3 + |P_n - a_n(\tau)|^{3/2}\Big)\, dy\, d\tau, \qquad d\nu_n := |\nabla V_n|^2\, dy\, d\tau.$$

The measure μ_n is the atlas-dependent Radon measure from Definition 3.1. It is raw, gauge-dependent, and used on fixed unit observer cylinders for local compactness and concentration-set extraction. It is not a scale-invariant CKN measure. Scale-invariant threshold and regularity statements at arbitrary radius are written with $\mathcal{V}_r$, D_r, or $\mathcal{E}_r$. The raw mass condition below is only a selection criterion for observer windows. It never certifies that a limiting profile is nonzero: retained nonzero profiles are certified by $\mathcal{V}_1$ or by the explicitly stated non-affine oscillation functional. If the selected raw mass is pressure-only, it is treated by Lemma 9.5. The local mass of a recentering sequence $\mathfrak{c} = (z_n)$ is

$$m(\mathfrak{c}) := \limsup_{n\to\infty} \mu_n(Q_1(z_n)).$$

For a combined parabolic-covariant descendant recentering $d_n = (g_n; \xi_n, \sigma_n)$, the same notation means the mass of the pulled back defect measure in the combined coordinates:

$$m(\mathfrak{d}) := \limsup_{n\to\infty} \mu_n^{\mathfrak{d}}(Q_1(0,0)),$$

where $\mu_n^{\mathfrak{d}}$ is obtained from μ_n by the coordinate change in Definition 9.2 and by the admissible pressure gauges used in that recentered frame. Thus first-generation recentering sequences and lifted covariant descendants use the same local unit-cylinder criterion, expressed in their own local coordinates. For a fixed raw unit-window threshold $\varepsilon_{\mathrm{sd}} > 0$, the recentering sequence is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ if $m(\mathfrak{c}) \geq \varepsilon_{\mathrm{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n\to\infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

Lemma 9.4 (Gauge compatibility and retained-velocity semicontinuity)**.** *Let $(U_n, \Pi_n) \to (U, \Pi)$ in the topology of $\mathfrak{X}_M$ on a fixed compact cylinder containing $Q_1(0,0)$. After choosing the pressure gauges used in that convergence, the following hold.*

(i) *The pressure gauges are compatible with centered time shifts, covariant translations, covariant tail recenterings, affine normalization, and diagonal extraction of descendant profiles: applying any of these operations only changes the gauge by another admissible function of time.*

(ii) *The local CKN quantity is lower semicontinuous in the only direction used for pressure-inclusive limits:*

$$\mathcal{E}_1(U,\Pi;0) \le \liminf_{n\to\infty} \mathcal{E}_1(U_n,\Pi_n;0).$$

Consequently, if the limit profile has $\mathcal{E}_1(U,\Pi;0) \ge \eta$, then all sufficiently far approximating profiles have CKN mass at least any prescribed $\eta' < \eta$.

(iii) *If retained nontriviality is carried by the velocity part, then it is closed under the same convergence:*

$$\liminf_{n\to\infty} \mathcal{V}_1(U_n;0) \ge \eta \quad \Longrightarrow \quad \mathcal{V}_1(U;0) \ge \eta.$$

In applications where the lower bound is first obtained for velocity alone and then converted into a CKN lower bound, the descendant threshold is chosen below the resulting positive constant as in Definition 3.4.

Proof. Time shifts and covariant translations compose the pressure with a smooth local change of variables; affine normalization adds the explicit linear pressure $\frac{1}{2}c \cdot y$; and every local pressure is defined only up to a time-dependent function. Thus admissible gauges remain admissible under all operations listed in (i). For (ii), strong local convergence of U_n gives

$$\iint_{Q_1} |U_n|^3 \to \iint_{Q_1} |U|^3.$$

For the pressure part, pass to the quotient space

$$L^{3/2}(Q_1)/L^{3/2}((-1,0);\{\text{spatial constants}\}).$$

The quotient norm is convex and weakly lower semicontinuous; equivalently, after choosing any admissible atlas gauges realizing the local convergence,

$$D_1(\Pi;0) \le \liminf_{n\to\infty} D_1(\Pi_n;0).$$

Adding this to the strong L^3 convergence of the velocity gives the claimed lower semicontinuity direction for $\mathcal{E}_1$. The velocity-only assertion follows directly from strong $L^3(Q_1)$ convergence. The statement is entirely local on Q_1. □

Lemma 9.5 (No pressure-only retained active limit)**.** *Let $(U_n,\Pi_n) \to (U,\Pi)$ in the local Seregin topology on $Q_2(0,0)$, after passing to compatible pressure gauges whenever such gauges are available. Assume that, on the cylinders where pressure compactness is invoked, the pressure representatives are the normalized whole-space Riesz pressures modulo functions of time and modulo the affine/parasitic quotient of Definition 8.14. If $U = 0$ on $Q_2(0,0)$, then*

$$D_1(\Pi;0) = 0.$$

Consequently, suppose

$$\liminf_{n\to\infty} \mathcal{E}_1^{\mathfrak{a}}(U_n,\Pi_n;0) \ge \eta > 0 \qquad \textit{but} \qquad \mathcal{V}_1(U;0) = 0.$$

Then the velocity part of the approximating concentration vanishes in the limit, and the remaining lower bound is pressure-only. At least one of the following ordered alternatives holds; when more than one applies, the proof uses the first applicable alternative:

(i) *the pressure gauges or pressure masses are not strongly compact in the local $L^{3/2}$ quotient topology on $Q_2(0,0)$, so the recentering belongs to the pressure-loss part of the noncompact terminal alternative;*

(ii) *the limiting pressure defect is affine in space modulo functions of time and is assigned to the affine/parasitic quotient stratum $\mathcal{C}_{\mathrm{aff}}$;*

(iii) *the local Riesz part on a slightly larger cylinder carries the pressure lower bound, and hence yields a positive velocity lower bound on a finite family of neighboring unit cylinders.*

Consequently no pressure-only lower bound is used below to certify a retained nonzero terminal profile. Retained descendants are certified by $\mathcal{V}_1$, or by the non-affine oscillation functional Φ, and pressure-inclusive $\mathcal{E}_1$ is used only after one of the three alternatives above has been resolved.

Proof. If $U = 0$ on Q_2, then the limiting centered equation gives, in distributions on Q_2,

$$\nabla\Pi = -\partial_\tau U + \Delta U - \frac{1}{2}y \cdot \nabla U - \frac{1}{2}U - (U \cdot \nabla)U = 0.$$

Thus $\Pi = c(\tau)$ on Q_2, and its quotient pressure concentration on Q_1 vanishes:

$$D_1(\Pi; 0) = 0.$$

If the gauges are not compact, or if the pressure representatives are only weakly compact and retain a nonzero $L^{3/2}$ defect in the local quotient, this is alternative (i) by definition of the pressure-loss part of the noncompact terminal alternative. Assume therefore that, after subtracting the chosen time gauges, the pressures are strongly compact in $L^{3/2}_{\mathrm{loc}}$ modulo spatial constants. Since $\mathcal{V}_1(U; 0) = 0$ and $U_n \to U$ strongly in $L^3(Q_1)$, the velocity contribution to $\mathcal{E}_1^{\mathfrak{a}}(U_n, \Pi_n; 0)$ tends to zero. Any remaining positive lower bound is therefore represented by the strong $L^{3/2}$ quotient pressure limit modulo gauges.

If $U = 0$ on Q_2, the first paragraph shows that this quotient pressure limit has zero D_1-mass, a contradiction. Hence a genuine pressure lower bound with compact pressure gauges can only survive through either an affine/parasitic pressure mode, through a local Riesz part generated by velocity products on the finite enlargement used by the pressure representation, or through failure of the chosen normalized whole-space pressure representative to remain compact in the local quotient.

On Q_2, the pressure satisfies the local decomposition

$$\Pi_n - a_n(\tau) = \mathcal{R}_i\mathcal{R}_j(U_{n,i}U_{n,j}) + h_n,$$

where h_n is harmonic in the spatial variables on the smaller cylinder, after the standard cutoff used in the setup pressure atlas. If the velocity products on the finite enlargement used by this cutoff decomposition carry a positive $L^{3/2}$ lower bound, then a finite unit-cylinder cover gives (iii). Otherwise the Riesz source tends to zero strongly in $L^{3/2}$ on that enlargement, and the Riesz part of the pressure converges strongly to zero in $L^{3/2}(Q_1)$. Any remaining nonzero pressure defect is therefore harmonic in space on the unit cylinder. Its affine part is the parasitic pressure mode allowed in $\mathcal{C}_{\mathrm{aff}}$, giving (ii). If the harmonic representative is not affine after quotienting by time functions, then it is not an admissible compact pressure-only remnant under the normalized whole-space Riesz representation modulo time and affine gauges assumed in the lemma. It is therefore recorded as pressure-gauge or pressure-representation failure, namely alternative (i). This proves that a retained terminal profile is never certified by pressure alone. □

Lemma 9.6 (Local pressure decomposition and pressure-only alternatives)**.** *Let (u_n, p_n) be locally suitable on a parabolic cylinder $Q_4 = B_4 \times (-16, 0)$ with $\|u_n\|_{L^\infty(Q_4)} \le M$ and pressure gauges $a_n(t)$ compatible with Definition 8.14. Fix a compactly supported cutoff $\chi \in C_c^\infty(B_3)$ with $\chi \equiv 1$ on B_2. Then on B_2*

$$p_n - a_n(t) = p_n^{\mathrm{loc}} + h_n, \qquad p_n^{\mathrm{loc}} := \mathcal{R}_i\mathcal{R}_j(\chi u_{n,i}u_{n,j}), \tag{9.1}$$

where h_n is harmonic in the spatial variables on B_2 for almost every $t \in (-4, 0)$ and depends only on the values of $(u_n, p_n - a_n)$ on $Q_4 \setminus Q_2$ (the boundary data of the cutoff representation). The following four properties hold.

(i) Local Riesz strong continuity. *If $u_n \to 0$ strongly in $L^3(Q_4)$, then $p_n^{\mathrm{loc}} \to 0$ strongly in $L^{3/2}(Q_2)$, with $\|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)} \le C\|u_n\|^2_{L^3(Q_4)}$.*

(ii) Harmonic moment compactness. *For every multi-index α with $|\alpha| \le K$ and every Schwartz test function ψ compactly supported in B_1,*

$$t \mapsto \int (\partial^\alpha h_n)(x,t)\,\psi(x)\,dx$$

is bounded uniformly in n by a constant depending only on $\|u_n\|_{L^3(Q_4)}$, $\|p_n - a_n\|_{L^{3/2}(Q_4)}$, K, and ψ.

(iii) Affine quotient record. *The quotient of h_n by spatial affine functions*

$$h_n(x,t) \sim h_n(x,t) + (\alpha_n(t) + \beta_n(t)\cdot x)$$

is bounded in any local norm by the same data; the quotient $[h_n] \in D_1$ is the affine quotient of Definition 8.14. If the affine part of h_n is the only carrier of a positive pressure lower bound, the corresponding recentering belongs to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.

(iv) Non-affine harmonic record. *If $[h_n]$ has a nonzero limit $[h]$ in the quotient D_1, then either $[h]$ is purely affine (case* (ii) *of Lemma 9.5), or the non-affine harmonic remainder is not used as retained activity. If a nonzero local Riesz part is present on the finite enlargement, it gives neighboring velocity activity by* (i)*. If no such Riesz part is present, the non-affine harmonic remainder is recorded as failure of the normalized whole-space pressure representative, modulo time and affine gauges, to be compact in the local quotient; this is the pressure/noncompact alternative.*

The conclusion of (iv) *is the precise mechanism behind the reduction of the pressure-only case: only the local Riesz component is allowed to produce neighboring velocity activity. A genuinely non-affine harmonic quotient is not converted into a velocity source by itself; outside the affine quotient it is treated as pressure-representation or pressure-compactness failure.*

Proof. The decomposition (9.1) is the standard local pressure cutoff: writing

$$-\Delta(p_n - a_n) = \partial_i\partial_j(u_{n,i}u_{n,j})$$

on Q_4 and applying the Riesz operators $\mathcal{R}_i\mathcal{R}_j$ to the cutoff right-hand side $\partial_i\partial_j(\chi u_{n,i}u_{n,j})$ gives a local representative of the velocity-product part of the pressure, and the difference $h_n = p_n - a_n - p_n^{\mathrm{loc}}$ is then a harmonic function in space on B_2 at almost every time. This is the same construction used in the setup paper for the local pressure atlas (Theorem 1.3 (3)).

(i) The boundedness of Riesz transforms on $L^{3/2}$ gives

$$\|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)} \le C\|\chi u_{n,i}u_{n,j}\|_{L^{3/2}(Q_4)} \le C\|u_n\|^2_{L^3(Q_4)}.$$

Strong L^3-vanishing of u_n on Q_4 thus forces strong $L^{3/2}$-vanishing of p_n^{loc} on Q_2.

(ii) The standard interior estimates for harmonic functions in B_2 bound every spatial derivative of h_n on B_1 by an $L^{3/2}$-norm of h_n on B_2, which in turn is bounded by $\|p_n - a_n\|_{L^{3/2}(Q_4)} + \|p_n^{\mathrm{loc}}\|_{L^{3/2}(Q_2)}$. The displayed bound on time-integrated moments follows.

(iii) The affine subspace of harmonic functions on B_2 is finite-dimensional; the projection onto it is bounded for any local norm. The quotient $[h_n]$ is therefore bounded. By the affine pressure quotient fixed in Definition 8.14, the affine part of h_n is the parasitic pressure mode allowed in $\mathcal{C}_{\mathrm{aff}}$; a recentering with positive lower bound carried only by this mode is the affine/parasitic recentering and is recorded in $\mathcal{C}_{\mathrm{aff}}$.

(iv) Assume the limit $[h]$ is non-affine in the quotient D_1. A harmonic function may be non-affine while still having no local Poisson source in B_2; therefore this fact alone is not used to produce a velocity lower bound. If the localized Riesz component p_n^{loc} has a nonzero lower bound on the finite enlargement entering the cutoff, (i) and boundedness of the Riesz transforms give a neighboring unit cylinder with positive velocity-product mass, hence the velocity alternative of Lemma 9.5. If $p_n^{\mathrm{loc}} \to 0$ strongly and $[h]$ is non-affine, then the surviving pressure lower bound is neither affine/parasitic nor represented by the normalized whole-space Riesz pressure modulo the allowed gauges. It is therefore assigned to the pressure-representation or pressure-compactness failure alternative, not to retained velocity concentration. □

Definition 9.7 (Local-window condition for first-generation recenterings)**.** Let (V_n, P_n) be a first-generation terminal sequence obtained by undoing the setup paper local Type I rescaling on a fixed physical cylinder $\mathcal{Q}_{\rm loc}$. A parabolic recentering sequence $\mathfrak{c} = (z_n)$, $z_n = (y_n, \tau_n)$, satisfies the *local-window condition* if, for every compact cylinder

$$K_{R,S} := B_R(0) \times [-S, S] \Subset \mathbb{R}^3 \times \mathbb{R},$$

the cylinder obtained by writing $K_{R,S}$ in the $\mathfrak{c}$-centered coordinates and then undoing the setup rescaling is contained in $\mathcal{Q}_{\rm loc}$ for all sufficiently large n. Equivalently, every fixed compact cylinder in the recentered variables lies in the part of the first-generation extraction where the setup pressure atlas, local energy inequality, and endpoint Type I bounds are available.

Lemma 9.8 (Local-window compactness for first-generation recenterings)**.** *Let (V_n, P_n) be a first-generation admissible terminal sequence from the setup extraction, and let $\mathfrak{c} = (z_n)$ satisfy the local-window condition of Definition 9.7. If*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) > 0,$$

then the recentered sequence admits a subsequence and admissible pressure gauges for which

$$(V_n^{\mathfrak{c}}, P_n^{\mathfrak{c}}) \longrightarrow (U, \Pi)$$

locally in the Seregin topology. The limit is locally suitable, pressure normalized, bounded on compact terminal windows by the imported setup bounds, and nonzero:

$$\mathcal{V}_1(U; 0) > 0$$

after lowering the velocity threshold once.

Proof. Fix $R, S < \infty$. By the local-window condition, the physical image of $K_{2R,2S}$ in the $\mathfrak{c}$-frame lies inside the setup local Type I cylinder for all large n. The imported setup results therefore gives the local energy, dissipation, velocity L^3, and pressure-gauge bounds on this physical image, uniformly in n, after the pressure gauges supplied by the setup atlas. Rescaling back to the $\mathfrak{c}$-coordinates gives the standard Seregin compactness bounds on $K_{R,S}$: strong local L^3 compactness for the velocity, weak local $L^{3/2}$ compactness for pressure modulo time functions, and stability of the local energy inequality.

Diagonalizing over $R, S \in \mathbb{N}$ gives a locally suitable terminal limit. The retained lower bound is carried by the velocity part, so strong $L^3_{\rm loc}$ convergence gives a positive $\mathcal{V}_1$ lower bound in the limit after the fixed threshold loss. No compactness is asserted outside the local-window region. □

Lemma 9.9 (Covariant compactness or pressure-atlas loss for descendants)**.** *Let $(U, \Pi) \in \mathfrak{X}_M^{\#}$ be represented after canonical affine/parasitic normalization. For every covariant recentering sequence $g_n \in G$, either the transforms*

$$\mathscr{G}_{g_n}(U, \Pi)$$

have no compact closure in the normalized local topology with the setup pressure atlas fixed, in which case the sequence is assigned to the local-compactness or pressure-gauge-loss terminal alternative, or they admit a locally compact pressure-gauge subsequence in the Seregin topology. In the compactness branch, if

$$\liminf_{n\to\infty} \mathcal{V}_1(\mathscr{G}_{g_n} U; 0) > 0,$$

then the limiting descendant is nonzero.

Proof. The covariant action is an exact symmetry of the centered equation by Lemma 8.5. Hence the transformed velocities remain bounded by $\|U\|_{L^\infty}$ and solve the same centered equation on all of $\mathbb{R}^3 \times \mathbb{R}$. This boundedness alone is not used as a compactness theorem. If the setup pressure-atlas representatives and local compactness bounds fail to have a compact subsequence, this is precisely the noncompact/pressure-gauge terminal alternative. Otherwise the transformed profiles lie in a compact subset of $\mathfrak{X}_M^{\#}$, so a diagonal subsequence converges in the normalized local topology to a bounded ancient suitable descendant. Strong local L^3 convergence transfers any velocity lower bound to the limit. □

Lemma 9.10 (Escaping first-generation recenterings are assigned to terminal alternatives)**.** *Let (V_n, P_n) be a first-generation terminal sequence and let $\mathfrak{c} = (z_n)$ carry retained unit velocity concentration:*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) > 0.$$

If $\mathfrak{c}$ fails the local-window condition, then it is not handled by local compactness. Instead it is assigned, according to the compact-window-to-expanding-region protocol of Theorem 9.17, to one of the terminal exterior alternatives: an exterior retained velocity profile, a diffuse defect, a pressure/noncompact defect, or a critical-tail defect.

Proof. Failure of the local-window condition means that some fixed recentered compact cylinder leaves the physical setup cylinder after undoing the first-generation scaling. The imported local Seregin compactness theorem is therefore not available on that cylinder, so the recentering cannot be treated by Lemma 9.8. The retained unit velocity concentration gives a positive residual measure in the corresponding selected expanding terminal region. Applying Theorem 9.17 to that region selects one of the protocol's priority alternatives: a selected unit observer with velocity concentration, a residual measure escaping to the observer boundary, a failure of local compactness or pressure-gauge compactness, or a nonzero normalized critical-tail scale. These are reduction alternatives, not compactness conclusions. □

Corollary 9.11 (Local compactness or alternative selection for retained recenterings)**.** *Let (V_n, P_n) be a terminal sequence obtained either as a first-generation setup sequence or as a descendant of an already extracted bounded ancient profile. Any recentering sequence carrying retained unit velocity concentration is of exactly one of the following types.*

(i) *It is first-generation and satisfies the local-window condition; then Lemma 9.8 gives a nonzero locally compact Seregin limit.*
(ii) *It acts on an already extracted normalized ancient profile and its pressure-atlas orbit has compact closure; then Lemma 9.9 gives a nonzero locally compact descendant.*
(iii) *It is either first-generation and fails the local-window condition, or it is an ancient-profile recentering whose pressure-atlas orbit has no compact closure; then Lemmas 9.9 and 9.10 assigns it to an exterior, diffuse, pressure/noncompact, or critical-tail terminal branch.*

Thus retained unit velocity concentration under an arbitrary recentering is never used by itself as a compactness hypothesis.

Proof. First-generation recenterings either satisfy or fail the local-window condition. The two cases are handled by Lemmas 9.8 and 9.10. Recenterings of extracted ancient profiles act on a bounded ancient profile defined on the full centered ancient space. They are handled by Lemma 9.9: compact pressure-atlas orbits give locally compact descendants, while noncompact or pressure-gauge-loss orbits are assigned to the corresponding terminal alternative. These cases exhaust the recenterings used in the terminal extraction arguments. □

Theorem 9.12 (Local concentration-set extraction). *Let $Z = \mathbb{R}^3 \times \mathbb{R}$ be the centered covariant observer space, and let $K_m \Subset Z$, $m \geq 1$, be a compact exhaustion. Let μ_n be the raw atlas defect measures of a terminal sequence, with local finite mass on compact observer windows:*

$$\sup_n \mu_n(K_m^+) < \infty \qquad (m \geq 1),$$

where K_m^+ is a compact enlargement such that every covariant unit cylinder $\mathcal{Q}_1^{\mathrm{cov}}(z)$, $z \in K_m$, lies in K_m^+. The compact-window concentration sets are

$$\widetilde{A}_{n,m}^{\eta} := \{z \in K_m : \mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta\}.$$

For the Hausdorff–Fell extraction we use the equivalent smoothed version below; thresholds are adjusted only by fixed cutoff constants. Fix $\chi \in C_c^\infty(Q_1(0,0))$, $0 \leq \chi \leq 1$, with $\chi \equiv 1$ on $Q_{1/2}(0,0)$. For $z \in Z$, put

$$\chi_z(Y,\tau) := \chi\big(\mathcal{R}_z^{-1}(Y,\tau)\big), \qquad M_n(z) := \int \chi_z \, d\mu_n.$$

For $\eta > 0$, define the compact-window concentration set

$$A_{n,m}^{\eta} := \{z \in K_m : M_n(z) \geq \eta\}.$$

After passing to a subsequence, for every m and every rational $\eta > 0$, the compact sets $A_{n,m}^{\eta}$ converge in the Hausdorff–Fell topology to compact sets $A_m^{\eta} \subset K_m$. Moreover, for every open neighborhood $O \subset K_m$ of A_m^{η},

$$\limsup_{n\to\infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta.$$

Thus outside every neighborhood of the concentration set A_m^{η}, no η-concentrating covariant unit tube remains in the compact observer window K_m. The concentration set is the closed locus A_m^{η}, not a finite or countable enumeration of tubes.

Proof. For fixed n, the map $z \mapsto \chi_z$ is continuous in C_c on K_m, and all supports are contained in the compact enlargement K_m^+. Since μ_n is a Radon measure finite on K_m^+, M_n is continuous on K_m. Therefore $A_{n,m}^{\eta}$ is compact.

The hyperspace of compact subsets of the compact metric space K_m is compact in the Hausdorff–Fell topology. Hence, for each fixed pair (m,η), a subsequence exists along which $A_{n,m}^{\eta}$ converges. A diagonal argument over $m \in \mathbb{N}$ and $\eta \in \mathbb{Q}_+$ gives simultaneous convergence.

Let $O \subset K_m$ be an open neighborhood of A_m^{η}. If the stated residual bound failed, there would exist $\delta > 0$, a subsequence n_k, and points $z_{n_k} \in K_m \setminus O$ such that

$$M_{n_k}(z_{n_k}) \geq \eta + \delta.$$

After replacing O by a slightly smaller open neighborhood of A_m^{η}, we may assume $K_m \setminus O$ is compact. Passing to a further subsequence,

$$z_{n_k} \to z_* \in K_m \setminus O.$$

For all large k,

$$z_{n_k} \in A_{n_k,m}^{\eta+\delta/2}.$$

Hausdorff–Fell convergence gives $z_* \in A_m^{\eta+\delta/2}$. Since $A_m^{\eta+\delta/2} \subset A_m^{\eta} \subset O$, this contradicts $z_* \in K_m \setminus O$. The residual estimate follows. All bounds used here are local on K_m^+; no global CKN budget or countability of concentrating profiles is assumed. □

Lemma 9.13 (Closed-active-locus reduction)**.** *Let $A_m^\eta \subset K_m$ be the compact concentration loci produced by Theorem 9.12 for a rational threshold $\eta \in (0, \varepsilon_{\mathrm{CKN}}/2)$ and a compact observer window $K_m \Subset Z$. Then A_m^η is a compact subset of K_m for which the following reduction holds. For every open neighborhood $O \Supset A_m^\eta$ in K_m there is $N(O) \in \mathbb{N}$ such that for all $n \geq N(O)$ the following hold simultaneously.*

(1) Sole carrier of compact retained concentration. *Every covariant unit cylinder $\mathcal{Q}_1^{\mathrm{cov}}(z) \subset K_m$ with center $z \in K_m \setminus O$ carries less than η defect mass: $M_n(z) < \eta$.*
(2) Locality of the residual. *The residual measure $\mu_n \lfloor (K_m \setminus O)$ satisfies $\sup_{z \in K_m \setminus O} M_n(z) < \eta$; equivalently, no covariant unit cylinder centered in $K_m \setminus O$ is η-concentrating.*
(3) No additional compact concentration locus. *Any further Hausdorff–Fell limit point of compact η-concentration sets in K_m is contained in A_m^η; the compact concentration locus is the unique maximal closed locus of compact retained concentration in K_m.*

Consequently, the only candidate compact carriers of retained terminal concentration on K_m at threshold η are the points of A_m^η, and the trichotomy alternatives of Definition 9.15 (vanishing, diffuse, or noncompact residual) describe the entirety of the residual after a ρ-neighborhood of A_m^η is removed.

Proof. Compactness of A_m^η is recorded in Theorem 9.12. Statement (1)–(2) is the displayed residual estimate of that theorem applied with neighborhood O of A_m^η. Statement (3) follows because every other compact η-concentration limit $A' \subset K_m$ along the same extraction is, by Hausdorff–Fell limit-point characterization, contained in A_m^η: otherwise a point $z' \in A' \setminus A_m^\eta$ would carry a covariant unit defect mass $\geq \eta$ and could not be evicted from $K_m \setminus O$ for O any open neighborhood of A_m^η, contradicting (1). The trichotomy then exhausts the residual on $K_m \setminus U_\rho(A_m^\eta)$ by definition of Definition 9.15. ☐

Lemma 9.14 (Closed active loci reduce to single, finite separated, or chain alternatives)**.** *Let $A_m^\eta \subset K_m$ be a nonempty compact active locus from Theorem 9.12. For each $z \in A_m^\eta$, choose centers $z_n \in A_{n,m}^{\eta'}$, with $\eta' < \eta$, converging to z in the Hausdorff–Fell realization and extract the corresponding locally compact retained velocity profile whenever the local-window/ancient-profile compactness hypotheses of Corollary 9.11 hold. Declare two realized centers preliminarily related if their realizing recentering sequences have bounded covariant parabolic distance and converge, after the admissible pressure gauges and after passing to a common subsequence, to the same retained terminal profile class. The non-separated local profile equivalence relation is the transitive closure of this preliminary relation. Then the compact active locus contributes to the terminal assembly in exactly one of the following priority ways.*

(i) *All realized centers in A_m^η lie in one equivalence class; the compact active locus represents a single retained profile class.*
(ii) *There are at least two separated equivalence classes; choosing one realizing sequence from each of finitely many such classes gives a finite separated family of retained velocity concentration profiles in the sense of Definition 9.115.*
(iii) *After every finite separated subfamily has been selected, a further retained active descendant remains separated from the selected family. Iterating the realized descendant construction and using Lemma 10.11 and Theorem 10.12 produces an infinite retained concentration chain.*

Thus a continuum active set is not a separate terminal alternative: after quotienting by non-separated local profile equivalence, it enters the terminal assembly as a single profile class, a finite separated family, or an infinite retained chain.

Proof. The relation is an equivalence relation by construction. If all realized centers are equivalent, the active locus has one non-separated local profile class, proving (i).

If not, choose two inequivalent realized center sequences. Their covariant distance cannot stay bounded along a subsequence that realizes the same local profile class, because then they would be related by the preliminary relation. Thus, after passing to the realizing subsequences defining the inequivalent classes, they are separated in the sense of Definition 9.115. The same argument selects any finite number of separated inequivalent classes, giving (ii) whenever the process stops after finitely many selections.

If the process does not stop, construct inductively a sequence of retained active descendants, each separated from the finite family already chosen. The diagonal lifting lemma realizes each descendant as a recentering of the original terminal extraction, and descendant heredity keeps the profile in the same normalized retained residual class unless it has already fallen into a closed lower stratum. Removing the closed lower strata, the inductive sequence is an infinite retained concentration chain. This is (iii). □

Definition 9.15 (Residual measure after concentration-set removal)**.** Let $U_\rho(A^\eta_{n,m})$ be the ρ-neighborhood of the concentration set in the compact observer window K_m. The residual measure after deleting positive-threshold concentration neighborhoods is

$$\mu^{\mathrm{rem}}_{n,m,\eta,\rho} := \mu_n \lfloor \big(K_m \setminus U_\rho(A^\eta_{n,m})\big)\,.$$

The terminal trichotomy uses this restricted measure: vanishing means $\mu^{\mathrm{rem}}_{n,m,\eta,\rho}(K_m) \to 0$ along the selected extraction; diffuse defect means positive residual mass remains while no unit cylinder exceeds the concentration threshold after deletion; noncompact defect means that an escaping residual recentering loses local compactness or pressure-gauge control under its own normalization.

Definition 9.16 (Canonical countable family of expanding terminal regions)**.** The *canonical countable family* $\mathcal{E} \subset 2^Z$ is the closure under finite intersection of the family of regions of one of the following elementary types:

(i) exterior balls $\{|y| > N\}$ with $N \in \mathbb{N}$;
(ii) dyadic spatial annuli $\{2^k < |y| < 2^{k+1}\}$ with $k \in \mathbb{Z}$;
(iii) time slabs $\{|\tau - q| < \ell\}$ with $q \in \mathbb{Q}$ and $\ell \in \mathbb{Q}_{>0}$;
(iv) logarithmic annuli $\{2^k < |y - y_0| < 2^{k+1}\}$ with $y_0 \in \mathbb{Q}^3$ and $k \in \mathbb{Z}$,

together with their images under finitely many covariant observer shifts T_a with $a \in \mathbb{Q}^3 \rtimes \mathbb{Q}$ and parabolic rescalings Θ_s with $s \in \mathbb{Q}$. Equivalently, $\mathcal{E}$ is the set of all finite intersections of the rationally parametrized images of (i)–(iv). This family is countable.

Every concrete exterior or tail window used in the residual proof is of one of the following two forms: (a) an element of $\mathcal{E}$ up to covariant action; (b) a multiplicative or additive scale-shift of an element of $\mathcal{E}$ by an index-dependent parameter $R_n \to \infty$ or $R_n \downarrow 0$. In case (a) the inclusion in $\mathcal{E}$ is direct; in case (b) the relevant scale-shifted region is one of the canonical dyadic annuli or logarithmic annuli of $\mathcal{E}$ after substitution $R_n \mapsto 2^{k(n)}$ for an integer index $k(n)$ chosen within a bounded distance from $\log_2 R_n$, at the cost of an absolute multiplicative loss in the captured mass.

For convenience we write $E^{(j)}_n = g^{(j)}_n \cdot E^{(j)}$, where $E^{(j)} \in \mathcal{E}$ and $g^{(j)}_n$ is the index-dependent covariant shift or scale change selecting the j-th window at time n.

Theorem 9.17 (Compact-window to expanding-region terminal protocol)**.** *Let $K_m \uparrow Z$ be the compact observer exhaustion used in Theorem 9.12, and let μ_n be the raw terminal defect measures. Let $\mathcal{E}$ be the canonical countable family of expanding terminal regions of Definition 9.16, indexed by $j \in \mathbb{N}$, and write $E^{(j)}_n \subset Z$ for its members along the extraction. After one diagonal subsequence, for every j, every rational $\eta > 0$, and every rational $\rho > 0$, and for every CKN threshold $\eta \le \varepsilon_{\mathrm{CKN}}/2$, exactly one of the following* priority alternatives *occurs in the order listed* *for the concentration-deleted measures*

$$\mu_n \lfloor \big(E_n^{(j)} \setminus U_\rho(A_{n,m}^\eta)\big).$$

(a) Expanding-region vanishing. *For every m,*

$$\mu_n\big(E_n^{(j)} \cap K_m \setminus U_\rho(A_{n,m}^\eta)\big) \longrightarrow 0,$$

and the remaining mass outside K_m tends to zero as $m \to \infty$ along the selected protocol.

(b) Compact subthreshold regular residual. *A positive amount of residual mass remains in some fixed compact observer window K_m after deleting $U_\rho(A_{n,m}^\eta)$, every unit cylinder inside the residual region has CKN activity $\mathcal{E}_1 < \eta$, pressure gauges are compact, and no escape outside K_m takes place.*

(c) Boundary diffuse defect. *A positive amount of residual mass leaves every fixed K_m; normalized restrictions converge on $\overline{Z}$ to a nonzero probability measure supported on $\overline{Z} \setminus Z$.*

(d) Noncompact or pressure-gauge defect. *There exists a recentering sequence $z_n \in E_n^{(j)}$ carrying positive velocity or atlas CKN mass for which the local Seregin compactness or pressure-gauge compactness required in Definition 9.3 fails.*

(e) Critical-tail defect. *No unit observer carries a retained velocity threshold and no compactness failure occurs, but there are scales $R_n \to \infty$ or $R_n \downarrow 0$, according to the observer convention, for which the normalized residual measure on the associated annular/logarithmic windows has nonzero limiting mass. This is the critical-tail compactification input of Definition 9.39.*

The priority order is: (a) *preempts the others; failing* (a)*, the alternatives* (b)–(e) *are intrinsically distinct after deleting unit cylinders above the CKN threshold, with* (b) *being the only compact-window alternative in which no unit cylinder has CKN activity above η. By Lemma 9.18, alternative* (b) *carries no retained terminal velocity concentration and is removed from the residual decomposition before any later use; the terminal stratification theorem Theorem 10.66 therefore lists only the alternatives* (c)–(e) *together with the four compact retained alternatives proved later.*

Proof. The extraction in Theorem 9.12 is already diagonal over the countable pairs (m, η). Refine it over the countable selected regions $E_n^{(j)}$ and rational deletion radii ρ; since $\mathcal{E}$ is itself countable by Definition 9.16, the resulting diagonal extraction is a countable refinement. If alternative (a) fails, then for some j, η, ρ either a positive amount of mass remains in some fixed compact window after deleting $U_\rho(A_{n,m}^\eta)$, or positive mass escapes every fixed compact window.

In the first case, if a unit observer inside the residual set has velocity activity above the velocity threshold, it is an active velocity recentering and its observer cylinder is raw-measure active by Lemma 9.21; but every raw-measure active observer at threshold η has been deleted by construction, so this case does not occur after deletion. If the local compactness or pressure gauges fail along the selected observer sequence, this is (d). If compactness holds but the lower bound is pressure-only, Lemma 9.5 places it either in (d), to the affine/parasitic quotient, or to a neighboring velocity observer. The remaining sub-alternative is (b): every unit cylinder in $K_m \setminus U_\rho(A_{n,m}^\eta)$ has CKN activity $\mathcal{E}_1 < \eta$, pressure gauges are compact, and the residual mass is contained in K_m for some fixed m (no escape).

In the second case, normalize the residual restrictions outside K_m and use weak compactness of probability measures on $\overline{Z}$. The limit has no mass on any compact subset of Z, hence is supported on the boundary and gives (c). Finally suppose that the residual mass is invisible at every fixed unit observer but is not lost in the compactification. Cover the selected tail region by dyadic annular/logarithmic windows and normalize the residual restriction in that region to a probability measure. Apply Lemma 9.19. If a fixed-width logarithmic window captures a positive amount of the normalized mass after a suitable logarithmic recentering, then that window gives a fixed multiplicative annulus at macroscopic scales R_n, hence the macroscopic critical-tail input of Definition 9.39. If no such window exists, the log-radius mass is diffuse along the logarithmic axis; its weak-$*$ limit on the logarithmic compactification is recorded as the log-diffuse critical-tail coordinate $\Lambda \neq 0$. Both subcases are alternative (e). This selection is local in the chosen windows and uses no global mass budget. □

Lemma 9.18 (Compact subthreshold regular residual cannot carry retained velocity concentration)**.** *Let the residual measure $\mu^{\mathrm{rem}}_{n,m,\eta,\rho}$ of Definition 9.15 fall into alternative* (b) *of Theorem 9.17 at the threshold $\eta \leq \varepsilon_{\mathrm{CKN}}/2$, where $\varepsilon_{\mathrm{CKN}}$ is the CKN smallness constant of Theorem 1.3 (1). Then the corresponding portion of the residual measure is supported on a relatively compact set of $\varepsilon_{\mathrm{CKN}}$-regular points of the underlying suitable solutions, and the following two conclusions hold.*

(i) Local regularity. *The compact-window residual mass is carried by velocity fields which are uniformly bounded and locally smooth on a slight enlargement of the selected residual region; in particular it cannot accumulate retained terminal velocity concentration at any compact observer.*

(ii) Disjointness from singular cores. *After alternative* (b) *has been recorded and removed, no surviving residual measure with retained velocity concentration intersects the same compact region. Equivalently, alternative* (b) *cannot coexist with a retained singular concentration core inside the same compact observer window after the CKN threshold has been chosen below $\varepsilon_{\mathrm{CKN}}$.*

Proof. Fix the compact window K_m and the deletion radius ρ provided by the alternative. Let $R_m = \mathrm{dist}(K_m, \partial K_{m+1}) > 0$. Cover K_m by a finite collection of unit covariant cylinders $\mathcal{Q}^{\mathrm{cov}}_1(z_i)$, $i = 1, \ldots, N_m$, arranged so that the slightly enlarged cylinders $\mathcal{Q}^{\mathrm{cov}}_{1+\delta}(z_i)$ with $\delta \in (0, R_m/4)$ still lie in $K_{m+1} \setminus U_{\rho/2}(A^{\eta}_{n,m})$ for all sufficiently large n, using that the CKN-threshold locus has been deleted at radius ρ and the cylinders are chosen at distance $\rho/2$ from it.

By the compact subthreshold hypothesis, every such cylinder satisfies

$$\mathcal{E}_1(U_n, \Pi_n; z_i) < \eta \leq \varepsilon_{\mathrm{CKN}}/2.$$

The pressure gauge attached to (U_n, Π_n) on K_{m+1} is compact in the relevant atlas (Definition 9.3). By the imported ε-regularity criterion of Caffarelli–Kohn–Nirenberg, recorded in Theorem 1.3 (1), each cylinder $\mathcal{Q}^{\mathrm{cov}}_1(z_i)$ is $\varepsilon_{\mathrm{CKN}}$-regular for the underlying suitable solution (U_n, Π_n) along the diagonal extraction: the velocity is bounded by an absolute constant on $\mathcal{Q}^{\mathrm{cov}}_{1-\delta}(z_i)$ and is locally smooth there. Taking the finite union over i and over the at most countably many windows in alternative (b) gives a uniformly bounded smooth representative of the velocity on a slight enlargement of the residual support, proving (i).

For (ii), a retained singular concentration core inside the same compact window has, by definition of the CKN threshold below $\varepsilon_{\mathrm{CKN}}$, at least one unit cylinder with CKN activity exceeding η. All such cylinders are removed by passing to the concentration-deleted measure at radius ρ. The compact subthreshold alternative is recorded only on the complement of these cylinders. Hence the compact subthreshold residual and any retained singular core are disjointly supported on the compact window after deletion. In particular the compact subthreshold residual carries no retained terminal velocity concentration, and is removed from the residual decomposition before the terminal stratification theorem Theorem 10.66 is applied. □

Lemma 9.19 (Annular tight/log-escape dichotomy)**.** *Let λ_n be normalized residual probability measures on dyadic annuli or logarithmic windows in escaping tail regions, with $\lambda_n(\mathbb{R}^3 \setminus \{0\}) = 1$*

and supports tending to infinity or to zero in the radial sense. Write

$$\Lambda_n := (\log|\cdot|)_*\lambda_n.$$

Then, after passing to a subsequence, the following priority alternatives exhaust the possibilities.

(1) Macroscopic annular tightness. *There exist macroscopic scales R_n (with $R_n \to \infty$ or $R_n \downarrow 0$ according to the observer convention) and constants $0 < a < b < \infty$ and $\delta > 0$ such that*

$$\lambda_n\big(\{aR_n < |x| < bR_n\}\big) \geq \delta \qquad \textit{for all sufficiently large } n.$$

(2) Logarithmic escape. *The pushforwards Λ_n have no positive mass concentration in any fixed-width logarithmic window after arbitrary recentering:*

$$\lim_{n\to\infty} \sup_{\ell\in\mathbb{R}} \Lambda_n([\ell - L, \ell + L]) = 0 \qquad \textit{for every } L < \infty.$$

In this case every weak-$*$ limit of Λ_n on the logarithmic compactification $\overline{\mathbb{R}}_{\log} = [-\infty, +\infty]$ is supported at the logarithmic boundary and gives the log-diffuse coordinate $\Lambda \neq 0$.

Proof. For $L > 0$, define the logarithmic concentration function

$$Q_n(L) := \sup_{\ell\in\mathbb{R}} \Lambda_n([\ell - L, \ell + L]).$$

If for some $L < \infty$,

$$\limsup_{n\to\infty} Q_n(L) > 0,$$

then, after passing to a subsequence, there are logarithmic centers ℓ_n and $\delta > 0$ such that

$$\Lambda_n([\ell_n - L, \ell_n + L]) \geq \delta.$$

With $R_n = e^{\ell_n}$, this is exactly

$$\lambda_n\big(\{e^{-L}R_n \leq |x| \leq e^{L}R_n\}\big) \geq \delta$$

for all sufficiently large n. Since the original measures live on escaping tail regions, the selected R_n tends to infinity or to zero according to the observer convention after passing to a further subsequence. Thus alternative (1) holds with $a = e^{-L}$ and $b = e^{L}$.

If no such L exists, then $Q_n(L) \to 0$ for every rational $L > 0$, hence for every finite L by monotonicity. This is the stated logarithmic diffuseness condition. Compactness of probability measures on $\overline{\mathbb{R}}_{\log}$ gives a weak-$*$ limit point. The vanishing of $Q_n(L)$ for every finite L rules out mass at any finite logarithmic point, because a neighborhood of such a point contains some fixed interval $[\ell - L, \ell + L]$. Since each Λ_n has total mass one, the limiting measure is nonzero and is supported on $\{-\infty, +\infty\}$. This is the log-diffuse alternative. ∎

Remark 9.20 (How the dichotomy enters the protocol)*.* The dichotomy Lemma 9.19 is invoked exactly at the last step of the proof of Theorem 9.17: the normalized residual measure on the selected dyadic annular window is either macroscopically annular tight, in which case it gives the macroscopic critical-tail input of Definition 9.39, or it has nonzero logarithmic escape, in which case it yields the log-defect measure Λ used in the critical-tail compactification of Definition 9.39. In both cases alternative (e) of the protocol is realized.

Lemma 9.21 (Oscillatory observers are raw-measure active)**.** *Let μ_n be the raw atlas defect measure used in Theorem 9.12, so that μ_n dominates the velocity measure $|U_n|^3\,dy\,d\tau$. If a covariant unit observer z satisfies*

$$\operatorname{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta,$$

then the same observer is raw-measure active at threshold η:

$$\mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)) \geq \eta.$$

Consequently, after the raw atlas concentration sets of Theorem 9.12 are removed at every positive threshold, no positive constant-in-time oscillatory unit observer remains outside those sets.

Proof. By definition of the constant-in-time oscillation,

$$\operatorname{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) = \inf_{c\in\mathbb{R}^3} \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n - c|^3 \le \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n|^3.$$

The raw atlas defect measure contains the velocity contribution, hence

$$\iint_{\mathcal{Q}_1^{\mathrm{cov}}(z)} |U_n|^3 \le \mu_n(\mathcal{Q}_1^{\mathrm{cov}}(z)).$$

Thus oscillation implies raw-measure activity with the same threshold. The last statement follows by applying the local concentration-set extraction theorem to all rational positive thresholds and then lowering thresholds by harmless fixed cover and cutoff constants when the smoothed bump convention is used. □

Lemma 9.22 (Spatial quotient defect measures are locally CKN-controlled)**.** *On the same bounded-overlap covariant unit-cylinder cover, define the spatial quotient oscillation measure*

$$\nu_U^{\mathrm{sp}}(E) := \sum_{z_k\in E} \operatorname{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big).$$

Then ν_U^{sp} is locally finite on compact observer windows and is dominated, up to the fixed cover multiplicity, by the local velocity part of the defect measure. In particular, every compactness argument in Theorem 9.29 applies without change to ν_U^{sp} after concentration-set removal.

Proof. For each unit covariant cylinder,

$$\operatorname{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big) \le \iint_{\mathcal{Q}_1^{\mathrm{cov}}(z_k)} |U|^3.$$

Summing over a bounded-overlap cover gives local finiteness and domination by the local velocity contribution, hence by the raw atlas defect measure. The proof of Theorem 9.29 uses only positivity, local finiteness, bounded-overlap domination, and weak compactness of normalized restrictions on the observer compactification. These properties hold for ν_U^{sp} exactly as they hold for the constant-in-time oscillation measure μ_U^{osc}. In particular,

$$\nu_U^{\mathrm{sp}}(\{z_k\}) = \operatorname{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big) \le \operatorname{osc}_{3,\mathrm{ct}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big).$$

Hence, on escaping observer windows where all positive-threshold unit $\operatorname{osc}_{3,\mathrm{ct}}$-observers have been removed, the atomic masses of ν_U^{sp} tend to zero, so the finite atomic window-selection lemma applies to ν_U^{sp} exactly as it does to μ_U^{osc}. □

Lemma 9.23 (Separation of recentered cylinders)**.** *Let $\mathfrak{c}_i = (z_{i,n})$ and $\mathfrak{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathfrak{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity L^3 concentration assigned to $\mathfrak{c}_j$ is visible on any fixed compact cylinder of the $\mathfrak{c}_i$-frame.*

Proof. In the $\mathfrak{c}_i$-coordinates the center of $Q_R(z_{j,n})$ is

$$\big(y_{j,n} - y_{i,n},\ \tau_{j,n} - \tau_{i,n}\big).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder K, one has

$$K \cap \big(Q_R(z_{j,n}) - z_{i,n}\big) = \varnothing$$

for all sufficiently large n. This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. $\square$

Theorem 9.24 (Paired terminal concentration-profile decomposition)**.** *Every terminal concentration sequence admits a paired decomposition*

$$\textit{retained velocity concentration set} \quad + \quad \textit{residual remainder.}$$

More precisely, after passing to a subsequence, the compact observer-window concentration sets A_m^η of Theorem 9.12 exist simultaneously for every m and every rational $\eta > 0$. Every observer sequence that remains in these sets at a fixed positive velocity threshold and has no loss of local compactness generates a bounded retained terminal profile after local pressure gauges. Outside every neighborhood of the concentration set in each fixed compact observer window, the residual raw unit-window mass is below the chosen threshold in the precise concentration-deleted sense of $\mu^{\mathrm{rem}}_{n,m,\eta,\rho}$ from Definition 9.15. Applying the compact-window to expanding-region protocol Theorem 9.17, the compact subthreshold regular residual is removed first by Lemma 9.18. The surviving residual remainder is then of one of the following priority types:

$$\mathcal{R}_{\mathrm{van}}, \qquad \mathcal{R}_{\mathrm{diff}}, \qquad \mathcal{R}_{\mathrm{noncomp}}, \qquad \mathcal{R}_{\mathrm{crit}}.$$

Here $\mathcal{R}_{\mathrm{van}}$ means the concentration-deleted residual measures vanish on compact observer windows and then on the selected expanding regions; $\mathcal{R}_{\mathrm{diff}}$ means positive concentration-deleted residual mass persists at the boundary of the observer compactification while every fixed unit cylinder is below threshold; and $\mathcal{R}_{\mathrm{noncomp}}$ means that some escaping residual recentering loses the local Seregin compactness or pressure-gauge control; and $\mathcal{R}_{\mathrm{crit}}$ means that residual mass is first visible only after annular/logarithmic critical-tail normalization. Thus mixed configurations such as one compact concentration profile plus diffuse exterior remainder are part of the decomposition; they are not silently identified with the pure diffuse stratum.

Proof. Apply the local concentration-set extraction theorem, Theorem 9.12, on the compact observer exhaustion $\{K_m\}$. It produces closed concentration sets A_m^η and the residual estimate

$$\limsup_{n\to\infty} \sup_{z\in K_m\setminus O} M_n(z) \le \eta$$

outside every neighborhood O of A_m^η. This is the precise replacement for a maximal finite or countable tube family: concentrating centers are recorded by closed loci, and the residual is defined by restricting away from their neighborhoods.

If an observer sequence stays in a positive-threshold concentration set and has no loss of the local compactness specified in Definition 9.3, then Corollary 9.11 gives a bounded retained terminal profile: the diagonal compactness is supplied there either by the first-generation local-window compactness lemma or by covariant compactness for bounded ancient profiles. If such an observer sequence loses local dissipation or pressure-gauge control, the corresponding remainder is $\mathcal{R}_{\mathrm{noncomp}}$.

It remains to classify what is outside the concentration sets. Apply the compact-window-to-expanding-region protocol Theorem 9.17 to the selected exterior and tail regions used in the terminal extraction. Its vanishing alternative is $\mathcal{R}_{\mathrm{van}}$. Its boundary-measure alternative is $\mathcal{R}_{\mathrm{diff}}$. Its compactness or pressure-gauge failure alternative is $\mathcal{R}_{\mathrm{noncomp}}$, while its critical-scale alternative is recorded for the later critical-tail branch. These residual alternatives are mutually exclusive once the concentration sets and selected protocol are fixed, and they exhaust the residual possibilities. No step asserts that a diffuse or exterior remainder cannot coexist with compact concentration profiles. $\square$

Selected terminal-output directory. Fix the terminal sequence carried by the singular-profile ledger and the single diagonal subsequence chosen in Theorems 9.12 and 9.17. For fixed selected indices j, m, η, ρ, the active locus $A^{\eta}_{n,m}$ and the complementary restriction

$$\mu_n \lfloor \big(E_n^{(j)} \setminus U_\rho(A^{\eta}_{n,m})\big)$$

use the same raw atlas measure, threshold, deletion radius, terminal region, and diagonal subsequence. The compact subthreshold regular part is removed by Lemma 9.18 before the surviving outputs below are used.

TABLE 12. Selected-sequence terminal outputs. Each row records data produced from the fixed terminal sequence and its named local windows. The final column contains downstream routes, not premises of the paired decomposition.

Selected output	**Exact local certification**	**Object produced at this stage**	**Downstream route only**
Compact retained profile.	A selected observer sequence carries a positive unit velocity threshold and lies in the compact branch of Corollary 9.11.	The applicable compactness result in Lemmas 9.8 and 9.9 produces the bounded suitable local profile with the pressure representatives attached to its charts. Velocity retention follows from Lemma 9.4; for a lifted descendant, ancestry and heredity are supplied by Theorems 1.13 and 10.12.	Lemma 9.14 passes the realized active locus to the single-profile, separated-family, or retained-chain analysis.
Vanishing residual.	The concentration-deleted restrictions vanish on compact observer windows and then on the selected expanding regions, as in alternative (a) of Theorem 9.17.	No residual profile is produced.	The later sole-carrier argument in Lemma 9.26 and Theorem 9.27 uses this output only for the retained compact obstruction of the selected extraction.
Diffuse residual measure.	Positive concentration-deleted mass leaves each fixed compact observer window, while the selected unit windows remain below the retained threshold; this is alternative (c) of Theorem 9.17.	The normalized restrictions have a weak-* probability-measure limit on the fixed observer compactification $\overline{Z}$. At this stage the output is measure data, not an ancient solution.	The selected measure data are consumed by Theorems 9.29 and 9.38. Compact–diffuse coexistence remains recorded until Theorem 9.113 is available.
Pressure or compactness loss.	A selected residual recentering loses the local Seregin compactness or the pressure-gauge control required on its own windows; this is alternative (d) of Theorem 9.17.	The output is the named noncompact or pressure-loss alternative. Its pressure representative and time gauge belong to the selected recentered chart; no retained profile is asserted.	Lemma 9.114 passes the selected failure to the named pressure, affine, diffuse, or critical-tail analysis.
Critical-tail window.	No selected unit observer carries retained velocity concentration, local compactness has not failed, and positive residual mass first appears after the selected annular or logarithmic normalization; this is alternative (e) of Theorem 9.17.	The output is a normalized selected window or defect datum. No Young, log-diffuse, or coherent profile has yet been produced.	Theorem 9.91 consumes this window and produces the later named critical-tail routes.

Only the first row produces a retained profile, and it does so through the displayed velocity and compactness contracts. The other rows record vanishing, measure data, compactness or pressure loss, or a selected scale window. Each pressure-dependent row uses the representative valid on its own local chart. Mildness, sequence-L^3, finite-shift mildness, the setup residual theorem, and final assembly are downstream of this directory.

Corollary 9.25 (Persistent local velocity concentration forces retained velocity concentration)**.** *If a terminal sequence satisfies*

$$\liminf_{n\to\infty} \iint_{Q_1(0,0)} |V_n|^3 \, dy \, d\tau \geq \eta_0 > 0$$

with $\eta_0 \geq \varepsilon_{\mathrm{sd}}$, then the terminal decomposition contains a local concentration profile unless the distinguished observer sequence has loss of local compactness. In the generic residual after local vanishing and local compactness failure are excluded, the persistent compact velocity concentration is therefore carried by the retained velocity concentration set of Theorem 9.24; the locus may contain one, finitely many, infinitely many, or continuum many concentrating observer classes, and the residual remainder may still be diffuse until the mixed compact–diffuse stratum is excluded.

Proof. The raw atlas defect measure μ_n dominates the velocity contribution. Hence the origin recentering sequence has μ-mass at least η_0, and is raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$ unless it has loss of local compactness. If it has no loss of local compactness, concentration-set extraction places its observer class in a positive-threshold concentration set A^{η}_m of Theorem 9.24. □

Lemma 9.26 (Local vanishing and exterior alternatives do not carry compact concentration)**.** *Let a terminal extraction inherit the positive compact velocity L^3 concentration supplied by the imported setup results. Then the retained compact velocity concentration cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

Proof. The assertion is local. By the local no-escape and concentration input in the imported setup results, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. The setup paper carries this through the Type I Seregin extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder K in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since μ_n dominates $|V_n|^3 \, dy \, d\tau$, choosing $\varepsilon_{\mathrm{sd}} \leq \eta_K$ makes that unit cylinder raw unit-window active at threshold $\varepsilon_{\mathrm{sd}}$. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of raw unit-window active recentering sequences means that the residual profile has raw unit-window mass below $\varepsilon_{\mathrm{sd}}$ on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By Lemma 9.23, it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact lower-bound threshold from the imported setup results, the positive concentration input in that setting would select a unit terminal cylinder with mass at least the velocity threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity concentration. □

Theorem 9.27 (Local exclusion of noncompact and nonconcentrating terminal alternatives)**.** *For terminal sequences generated by the Seregin setup theorem in the companion setup paper, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity concentration.*

Proof. Retained recentering sequences with loss of local compactness are first routed by Corollary 9.11 to one of the terminal branches. If the routed branch is locally compact, it is handled by the retained-profile protocol. If it is diffuse, pressure/noncompact, or critical-tail, it is excluded only after applying the corresponding branch closure theorem; thus the routing lemma is not itself used as a contradiction. Local vanishing, exterior escape, and diffuse exterior concentration are excluded by Lemma 9.26. The proof uses only the positive concentration input and the local compactness hypotheses in the imported setup results, and fixed compact terminal cylinders. □

Definition 9.28 (Observer compactification)**.** We fix a metrizable compactification $\overline{Z}$ of the observer space $Z = \mathbb{R}^3 \times \mathbb{R}$ with the following properties.

(i) Compact observer balls of finite covariant radius embed continuously.
(ii) The covariant observer maps generated by T_a and Θ_s extend continuously to $\overline{Z}$.
(iii) Unit-cylinder concentration evaluation maps are upper semicontinuous after the fixed smoothing convention of Theorem 9.12.
(iv) The normalized restrictions of locally finite defect measures are tight as probability measures on $\overline{Z}$.

Such a compactification is obtained by embedding Z into the product of compact intervals generated by a countable separating family of bounded continuous observer functions, including the smoothed unit-cylinder evaluation functions and the coordinate cutoffs for a compact exhaustion, and then taking the closure.

Theorem 9.29 (Diffuse-defect compactness)**.** *Let μ_n^{osc} be the residual constant-in-time oscillation measures of Definition 8.8, after removing neighborhoods of all positive-threshold concentration sets from Theorem 9.12. By Lemma 9.21, this removal also removes all positive-threshold oscillatory unit observers on compact windows. Assume there are observer regions $E_n \subset Z$, escaping every compact subset of Z, and a number $m_0 > 0$ such that*

$$0 < m_0 \le \mu_n^{\mathrm{osc}}(E_n) < \infty,$$

while no unit observer in E_n carries a positive oscillation threshold:

$$\sup_{z \in E_n} \mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) \longrightarrow 0.$$

Let $\overline{Z}$ be the compact metrizable observer compactification of Definition 9.28, and define probability measures

$$\lambda_n := \frac{\mu_n^{\mathrm{osc}} \lfloor E_n}{\mu_n^{\mathrm{osc}}(E_n)} \qquad \text{on } \overline{Z}.$$

After passing to a subsequence,

$$\lambda_n \rightharpoonup \lambda$$

weakly as probability measures on $\overline{Z}$. The limit satisfies

$$\lambda(\overline{Z}) = 1, \qquad \lambda(B) = 0 \quad \text{for every } B \Subset Z.$$

Thus the limit is a nonzero diffuse-defect profile supported at the boundary of the observer compactification, not a local concentration profile.

Proof. The theorem is stated directly in the finite-mass normalization regime needed to define λ_n. If one starts only from a positive lower bound on an escaping oscillation region, Lemma 9.30 provides an escaping finite-mass net-cell subset to which the present argument applies.

The measures λ_n are probability measures on the compact metric space $\overline{Z}$, so weak compactness gives a convergent subsequence. Testing against the constant function 1 preserves total mass and gives $\lambda(\overline{Z}) = 1$.

Let $B \Subset Z$. Since E_n escapes every compact subset of Z, one has $B \cap E_n = \varnothing$ for all sufficiently large n. Hence $\lambda_n(B) = 0$ eventually, and the portmanteau theorem gives $\lambda(B) = 0$. The additional unit-scale diffuseness assumption gives the stronger moving-window estimate

$$\sup_{z\in Z} \lambda_n\big(B_Z(z,1)\big) \le \frac{N_0}{m_0} \sup_{\zeta\in E_n\cap\{z_k\}} \mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(\zeta)) \longrightarrow 0.$$

Indeed, λ_n is atomic on the fixed observer net of Definition 8.8, and every unit ball for d_Z contains at most N_0 net points. Therefore

$$\begin{aligned}\lambda_n\big(B_Z(z,1)\big) &= \frac{1}{\mu_n^{\mathrm{osc}}(E_n)} \sum_{z_k\in E_n\cap B_Z(z,1)} \mathrm{osc}_{3,\mathrm{ct}}\big(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z_k)\big) \\ &\le \frac{N_0}{m_0} \sup_{\zeta\in E_n\cap\{z_k\}} \mathrm{osc}_{3,\mathrm{ct}}\big(U_n; \mathcal{Q}_1^{\mathrm{cov}}(\zeta)\big).\end{aligned}$$

Equivalently, after replacing E_n by its net-cell representative, every unit observer ball contains at most N_0 contributing atoms of μ_n^{osc}. Therefore no positive unit-scale local oscillation profile is hidden inside the escaping windows. The proof is local on the chosen observer windows and uses no global critical-mass or energy budget. □

Lemma 9.30 (Finite diffuse-window selection)**.** *Let μ_n^{osc} be the observer oscillation measures from Definition 8.8. Suppose $E_n \subset Z$ escapes every compact subset of Z and*

$$\liminf_{n\to\infty} \mu_n^{\mathrm{osc}}(E_n) > 0, \qquad \sup_{z\in E_n} \mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) \to 0.$$

Then for every $m_0 > 0$ below the surviving mass there is, after passing to a subsequence, an escaping net-cell subset $E_n' \subset E_n \cap \{z_k\}_{k\in\mathbb{N}}$ such that

$$m_0 \le \mu_n^{\mathrm{osc}}(E_n') \le 2m_0$$

for all large n.

Proof. Since m_0 is below the surviving mass, after passing to a subsequence we may assume

$$\mu_n^{\mathrm{osc}}(E_n) \ge m_0$$

for every n. Since the unit-scale oscillation threshold on E_n tends to zero, after passing to a further subsequence we may also assume

$$\sup_{z\in E_n} \mathrm{osc}_{3,\mathrm{ct}}(U_n; \mathcal{Q}_1^{\mathrm{cov}}(z)) \le m_0$$

for every n. Replace E_n by its net-cell representative $E_n \cap \{z_k\}_{k\in\mathbb{N}}$, which does not change $\mu_n^{\mathrm{osc}}(E_n)$ because the measure is atomic on the fixed observer net.

Enumerate the net points in E_n arbitrarily and add them one by one until the accumulated μ_n^{osc}-mass first reaches or exceeds m_0. The partial sum at the preceding step is $< m_0$, and the last added atom has mass at most m_0, so the resulting finite subset $E_n' \subset E_n$ satisfies

$$m_0 \le \mu_n^{\mathrm{osc}}(E_n') \le 2m_0.$$

Because E_n escapes every compact subset of Z, every subset $E_n' \subset E_n$ also escapes every compact subset. Thus the selected finite windows retain positive normalized oscillation mass while avoiding any division by an infinite total mass. □

Lemma 9.31 (Finite window selection for atomic observer defects)**.** *Let γ_n be a nonnegative atomic measure on the fixed observer net $\{z_k\}_{k\in\mathbb{N}}$. Suppose $E_n \subset Z$ escapes every compact subset of Z,*

$$\liminf_{n\to\infty} \gamma_n(E_n) > 0, \qquad \sup_{z_k\in E_n} \gamma_n(\{z_k\}) \to 0.$$

Then, for every $m_0 > 0$ below the surviving mass, after passing to a subsequence there is an escaping finite net-cell subset $E_n' \subset E_n \cap \{z_k\}_{k\in\mathbb{N}}$ such that

$$m_0 \le \gamma_n(E_n') \le 2m_0$$

for all large n.

Proof. Since m_0 lies below the surviving mass, after passing to a subsequence we may assume $\gamma_n(E_n) \ge m_0$ for every n. Since the atomic masses on E_n tend to zero, after passing to a further subsequence we may also assume

$$\sup_{z_k\in E_n} \gamma_n(\{z_k\}) \le m_0$$

for every n. Replacing E_n by $E_n \cap \{z_k\}_{k\in\mathbb{N}}$ does not change $\gamma_n(E_n)$ because γ_n is supported on the fixed observer net.

Enumerate the atoms of E_n arbitrarily and add them one by one until the accumulated γ_n-mass first reaches or exceeds m_0. The partial sum at the previous step is $< m_0$, and the last added atom has mass at most m_0, so the resulting finite subset $E_n' \subset E_n$ satisfies

$$m_0 \le \gamma_n(E_n') \le 2m_0.$$

Because E_n escapes every compact subset of Z, every subset $E_n' \subset E_n$ also escapes every compact subset. □

Lemma 9.32 (Support transfer for the diffuse-defect compactification)**.** *Let $X_{\rm def}$ be the compact metrizable diffuse-defect space of Theorem 9.34, and let*

$$\lambda_n = \frac{\mu_n^{\rm osc}\lfloor E_n}{\mu_n^{\rm osc}(E_n)}$$

be the normalized realized diffuse-defect measures from Theorem 9.29, viewed in $X_{\rm def}$. If $\lambda_n \rightharpoonup \lambda$ and $A \subset X_{\rm def}$ is a closed union of already excluded diffuse alternatives, then support of λ in A implies that no diffuse defect mass survives outside A. Conversely, surviving normalized diffuse mass outside A produces a sequence-realized support profile in $X_{\rm def} \setminus A$.

Proof. This is Lemma 1.14 with $X = X_{\rm def}$, $\mu_n = \mu_n^{\rm osc}$, and the escaping observer regions E_n. The construction theorem makes $X_{\rm def}$ compact metrizable and keeps the limit inside the closed sequence-realized subset, so the support point obtained outside A is a realized diffuse-defect profile. □

Definition 9.33 (Defect observables)**.** Let $M \ge 1$ be the velocity bound of the underlying admissible class. Fix the following six countable families of bounded continuous observables, each defined on diffuse-defect candidates produced by Theorem 9.29.

(1) *Observer test functions.* A countable dense family $\{\varphi_j\}_{j\in\mathbb{N}} \subset C(\overline{Z})$, separating probability measures on the compact metrizable observer compactification $\overline{Z}$ of Definition 9.28.
(2) *Smoothed unit-cylinder activity observables.* For each rational $z \in Z$ and each $\eta \in \mathbb{Q}_{>0}$, the smooth bounded functional $\Psi_{z,\eta}$ recording the convolved CKN activity $\mathcal{E}_1$ of a defect representative on the unit covariant cylinder $\mathcal{Q}_1^{\rm cov}(z)$, thresholded by a smooth cutoff at η. This family is countable.
(3) *Compact-exhaustion cutoff observables.* For each $m \in \mathbb{N}$, a smooth cutoff ζ_m supported in K_{m+1} and equal to 1 on K_m, recording the mass that the defect measure places on the compact exhaustion of Theorem 9.12.
(4) *Local velocity observables.* For each compact cylinder K_m and each finite list $F_1, \dots, F_N$ of polynomials of total degree $\le m$ in the variables $W, \nabla W, \dots, \nabla^m W$, the localized observable $\int F_j(W, \nabla W, \dots)\zeta_m \, dy \, d\tau$.

(5) *Pressure-gradient observables modulo gauges.* For each compactly supported divergence-free test field $\Psi \in C_c^\infty(K_m, \mathbb{R}^3)$ with rational coefficients in a countable dense subspace, the bounded observable
$$\mathcal{P}_\Psi(W, P) = \langle \nabla P, \Psi \rangle,$$
computed for representatives modulo the time-only and affine pressure gauges of Definition 8.14.

(6) *Log-annular observables.* For each rational ρ_0 and $A > 0$, the cutoffs $\chi_{\rho_0, A}$ of Definition 9.42 applied to the log-radial pushforward of any defect representative.

The combined family is countable; let $\{\mathcal{O}_k\}_{k\in\mathbb{N}}$ enumerate it after fixing a uniform bound $\|\mathcal{O}_k\|_\infty \le C_k$. Fix also a countable dense subgroup $G_\mathbb{Q} \subset G = \mathbb{R}^3 \rtimes \mathbb{R}$, and enlarge the observable family by all rational pullbacks $\mathcal{O}_k \circ q$, $q \in G_\mathbb{Q}$. The enlarged family is still countable and is again denoted by $\{\mathcal{O}_k\}_{k\in\mathbb{N}}$.

Theorem 9.34 (Construction of the diffuse-defect compactification)**.** *There exists a compact metrizable space $X_{\rm def}$, a continuous action $\sigma\colon G \times X_{\rm def} \to X_{\rm def}$ of $G = \mathbb{R}^3 \rtimes \mathbb{R}$, and a closed G-invariant subset*
$$X_{\rm def}^{\rm anc} \subset X_{\rm def},$$
such that every normalized diffuse-defect sequence generated by the original terminal sequence has a subsequential limit in $X_{\rm def}^{\rm anc}$.

Proof. Let $\{\mathcal{O}_k\}_{k\in\mathbb{N}}$ be the observable family of Definition 9.33 with uniform bound $\|\mathcal{O}_k\|_\infty \le C_k$. Choose an exhaustion
$$C_1 \Subset C_2 \Subset \cdots \Subset G$$
by compact sets with $e \in C_1$ and such that for every compact $H \Subset G$ and every m there exists an index $m(H, m)$ satisfying $C_m H \subset C_{m(H,m)}$.

For each diffuse-defect candidate ω and each pair (k, m), define the orbit map
$$\iota_{k,m}(\omega)(g) := \mathcal{O}_k(g \cdot \omega), \qquad g \in C_m.$$
By Lemma 8.9 and Theorem 8.6, the family $\{\iota_{k,m}(\omega) : \omega\}$ is uniformly bounded by C_k and equicontinuous on C_m. Arzelà–Ascoli therefore gives a compact metrizable closure
$$K_{k,m} := \overline{\{\iota_{k,m}(\omega) : \omega\}}^{C(C_m)} \subset C(C_m, [-C_k, C_k]).$$
Set
$$\mathcal{X}_0 := \prod_{k,m\in\mathbb{N}} K_{k,m}, \qquad \iota(\omega) := \big(\iota_{k,m}(\omega)\big)_{k,m},$$
and define
$$X_{\rm def} := \overline{\iota(\text{all defect candidates})} \subset \mathcal{X}_0.$$
Then $\mathcal{X}_0$ is compact metrizable as a countable product of compact metrizable spaces, so $X_{\rm def}$ is compact metrizable as a closed subset.

For every defect candidate the orbit maps satisfy the compatibility relation
$$\iota_{k,m'}(\omega)|_{C_m} = \iota_{k,m}(\omega) \qquad (m' \ge m).$$
Since the restriction maps $C(C_{m'}) \to C(C_m)$ are continuous, every point $x = (f_{k,m}) \in X_{\rm def}$ satisfies the same compatibility. Now fix $h \in G$. Given m, choose any index m' with $C_m h \subset C_{m'}$ and define
$$(\sigma_h x)_{k,m}(g) := f_{k,m'}(gh), \qquad g \in C_m.$$
Compatibility makes this independent of the choice of m'. For every defect candidate ω,
$$(\sigma_h \iota(\omega))_{k,m}(g) = \mathcal{O}_k(gh \cdot \omega) = \iota_{k,m}(h \cdot \omega)(g),$$

so $\sigma_h \iota(\omega) = \iota(h \cdot \omega)$. Hence σ_h maps the dense embedded set ι(all defect candidates) into itself and therefore preserves the closure $X_{\rm def}$.

To check joint continuity, fix a compact $H \Subset G$ and a coordinate (k, m). Choose m_H with $C_m H \subset C_{m_H}$. Then the map

$$(h, f) \longmapsto \big[g \mapsto f(gh)\big]$$

is continuous from $H \times C(C_{m_H})$ to $C(C_m)$ in the sup norm, because right translation on the compact set C_m depends continuously on $h \in H$. Therefore each coordinate of $(h, x) \mapsto \sigma_h x$ is continuous on $H \times X_{\rm def}$; since the product topology is generated by finitely many coordinates at a time, $(h, x) \mapsto \sigma_h x$ is jointly continuous on $G \times X_{\rm def}$. The group law holds on the dense embedded set ι(all defect candidates) and hence on all of $X_{\rm def}$ by continuity.

The sequence-realized class:

$$X_{\rm def}^{\rm anc} := \overline{\iota(\text{sequence-realized defect profiles})} \subset X_{\rm def},$$

where "sequence-realized" means realized from the original blow-up sequence in the sense of Definition 1.11. This gives a closed subset of the compact product, and the diagonal argument below verifies that this closure is still represented by limits of realized states rather than by an abstract compactification artifact.

It remains to check G-invariance of $X_{\rm def}^{\rm anc}$ and the diagonal-closure property. G-invariance follows because the covariant observer maps belong to the allowed realized operations. For the diagonal-closure property, suppose $\omega_j \in X_{\rm def}^{\rm anc}$ and $\omega_j \to \omega$ in $X_{\rm def}$. By definition of the closure each ω_j is a limit point of a sequence $\iota(\omega_j^{(n)})$ with $\omega_j^{(n)}$ realized from the original blow-up sequence. Choose indices $n = n(j)$ so large that $d(\iota(\omega_j), \iota(\omega_j^{(n(j))})) < 2^{-j}$. Then $\iota(\omega_j^{(n(j))}) \to \iota(\omega)$, and the $\omega_j^{(n(j))}$ are realized. Hence ω is in the closure of the realized states, i.e. $\omega \in X_{\rm def}^{\rm anc}$.

Finally, every normalized diffuse-defect sequence generated by the original terminal sequence is a sequence ω_n of realized states. Their images $\iota(\omega_n)$ lie in the compact metrizable space $X_{\rm def}$ and have a convergent subsequence; the limit lies in $X_{\rm def}^{\rm anc}$ by the closure argument above. □

Definition 9.35 (Diffuse-defect profile class)**.** The compact metrizable space $X_{\rm def} = \mathfrak{D}_M$ and its closed G-invariant sequence-realized subset $X_{\rm def}^{\rm anc} = \mathfrak{D}_M^{\rm anc}$ are those constructed in Theorem 9.34. The diffuse-defect mass functional is normalized to be identically one on $\mathfrak{D}_M$. The covariant observer group $G = \mathbb{R}^3 \rtimes \mathbb{R}$ acts on $\mathfrak{D}_M$ continuously through the action σ of Theorem 9.34.

Proposition 9.36 (Amenable averaging protocol for realized diffuse defects)**.** *Let $G = \mathbb{R}^3 \rtimes \mathbb{R}$ act continuously on the compact metrizable diffuse-defect compactification $X_{\rm def} = \mathfrak{D}_M$. Let ν_0 be a Borel probability measure supported on the closed sequence-realized class $\mathfrak{D}_M^{\rm anc}$. Then there exists a G-invariant Borel probability measure ν on $X_{\rm def}$, obtained as a weak-* limit of Følner averages*

$$\nu_\alpha := \frac{1}{|F_\alpha|} \int_{F_\alpha} g_\# \nu_0 \, dg,$$

where (F_α) is a left Følner net in G. Moreover $\nu(\mathfrak{D}_M^{\rm anc}) = 1$, so the invariant diffuse defect remains sequence-realized.

Proof. The semidirect group $G = \mathbb{R}^3 \rtimes \mathbb{R}$ is solvable and hence amenable; choose a left Følner net (F_α) for a left Haar measure. The action map $g \mapsto g_\# \nu_0$ is weak-* continuous because the action on $\mathfrak{D}_M$ is continuous. Therefore each ν_α is a Borel probability measure. Compactness of $X_{\rm def}$ implies weak-* compactness of $\mathcal{P}(X_{\rm def})$, so a subnet converges to a probability measure ν.

Let $h \in G$ and let $F \in C(X_{\rm def})$. Then

$$\int F \, d(h_\# \nu_\alpha) - \int F \, d\nu_\alpha = \frac{1}{|F_\alpha|} \int_{F_\alpha} \left[\int F(hg\omega) \, d\nu_0(\omega) - \int F(g\omega) \, d\nu_0(\omega) \right] dg.$$

The absolute value is bounded by

$$2\|F\|_{L^\infty}\frac{|hF_\alpha\triangle F_\alpha|}{|F_\alpha|},$$

which tends to zero by the Følner property. Passing to the weak-* limit gives $h_\#\nu=\nu$. Since $\mathfrak{D}_M^{\mathrm{anc}}$ is closed and G-invariant, each averaged measure ν_α is supported on $\mathfrak{D}_M^{\mathrm{anc}}$. Weak-* limits preserve support on closed sets, so $\nu(\mathfrak{D}_M^{\mathrm{anc}})=1$. □

Theorem 9.37 (Recurrent diffuse-defect measure)**.** *The covariant observer action of G is continuous on $\mathfrak{D}_M$. Every nonzero diffuse-defect profile $\lambda\in\mathfrak{D}_M$ generates a G-invariant Borel probability measure $\mathbb{P}$ on $\mathfrak{D}_M$. The normalized diffuse-defect mass remains one under this averaging.*

Proof. The continuity of the covariant observer action is the conclusion of the construction theorem Theorem 9.34. In that construction, the action is first defined on realized defect profiles by the covariant observer maps and then extended continuously to the compact product closure of the countable observable family. Apply Proposition 9.36 to $\nu_0=\delta_\lambda$. The resulting invariant probability measure $\mathbb{P}$ is supported on the closed sequence-realized diffuse-defect class. Since the diffuse-defect mass is identically one on $\mathfrak{D}_M$, its average against $\mathbb{P}$ is also one. □

Theorem 9.38 (Diffuse-defect trichotomy)**.** *Let $(U_n,\Pi_n)\subset\mathfrak{X}_M$ be a normalized terminal residual sequence with $\|U_n\|_{L^\infty}\le M$. Suppose that, after removing the concentration sets at every positive affine-normalized threshold, there are escaping observer windows E_n such that*

$$\mu_n^{\mathrm{osc}}(E_n)\ge m_0>0.$$

Then, after passing to a subsequence, the residual diffuse defect falls into one of the following ordered alternatives:

- D1. Regenerated concentration. *There exist covariant observers z_n and a threshold $\eta>0$ such that*
$$\mathrm{osc}_{3,\mathrm{ct}}(U_n;\mathcal{Q}_1^{\mathrm{cov}}(z_n))\ge\eta.$$
This produces a retained concentrating descendant.
- D2. Affine/parasitic collapse. *Every velocity tangent associated with the diffuse-defect hull is locally constant after the admissible affine normalization; the defect belongs to the affine/parasitic lower stratum of Definition 8.16.*
- D3. Critical diffuse tail. *There is a nonzero recurrent diffuse-defect profile supported on $\overline{Z}\setminus Z$, carrying normalized diffuse oscillation mass one, annihilating every fixed unit concentration observable, and not contained in the affine/parasitic lower stratum.*

Equivalently, residual diffuse defects are contained in the union of the regenerated concentration alternative, the affine/parasitic stratum, and the critical diffuse-tail class $\mathcal{C}_{\mathrm{critdiff}}$.

Proof. If D1 occurs, the asserted trichotomy holds. Assume D1 does not occur. Then, for every fixed $\eta>0$, all sufficiently large n satisfy

$$\sup_{z\in E_n}\mathrm{osc}_{3,\mathrm{ct}}(U_n;\mathcal{Q}_1^{\mathrm{cov}}(z))<\eta.$$

Since D1 does not occur, the atom size of μ_n^{osc} on E_n tends to zero. By Lemma 9.30, after passing to a subsequence we may replace E_n by an escaping net-cell subset $E_n'\subset E_n$ such that

$$m_0\le\mu_n^{\mathrm{osc}}(E_n')\le 2m_0.$$

Relabel E_n' as E_n. The hypotheses of Theorem 9.29 now apply to $\mu_n^{\mathrm{osc}}\lfloor E_n$, and hence produce a diffuse-defect profile λ with total mass one and zero mass on each compact subset of Z. By Theorem 9.37, the orbit closure of λ carries a G-invariant probability measure $\mathbb{P}$.

If every local velocity tangent in the support of $\mathbb{P}$, after the canonical affine normalization, is locally constant, then Definition 8.16 places the defect in the affine lower stratum. This is D2. If not, the invariant diffuse boundary profile has positive normalized spatial quotient oscillation mass, no retained unit-scale velocity concentration observable, and non-affine velocity content. This is precisely the critical diffuse-tail alternative D3. The alternatives are ordered by first extracting all regenerated concentration and then all affine/parasitic collapse; after those removals, the remaining nonzero diffuse profile is $\mathcal{C}_{\mathrm{critdiff}}$. □

Definition 9.39 (Critical-tail compactification)**.** The class $\mathcal{C}_{\mathrm{critdiff}}$ consists of recurrent diffuse-defect profiles produced by Theorem 9.38 which are supported on the boundary of the covariant observer compactification, carry normalized spatial quotient oscillation mass one, annihilate every fixed unit concentration observable, and are not affine/parasitic.

The critical-tail compactification $\mathfrak{T}_M$ is the compact profile class of tuples

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda; \widehat{\mathscr{P}}, \mathscr{D}, \mathfrak{q}_{\mathrm{aff}}, \mathfrak{v}_{\mathrm{vir}}, \mathfrak{s})$$

generated by such critical diffuse profiles after macroscopic blow-down. Here $\bar{W}$ is the affine-normalized barycentric macroscopic velocity, $\mathcal{Y}_{x,s}$ is the Young measure generated by the affine-normalized fields $W_n = V_n - c_n(s)$ and recording unresolved bounded macroscopic oscillation, with velocity support in $\overline{B_{M_{\mathrm{tail}}}}$ where $M_{\mathrm{tail}} = 2M$, and Λ is the log-scale diffuse measure recording mass escaping every fixed multiplicative annulus. The coordinates $\widehat{\mathscr{P}}$, $\mathscr{D}$, $\mathfrak{q}_{\mathrm{aff}}$, $\mathfrak{v}_{\mathrm{vir}}$, and $\mathfrak{s}$ record, respectively, the actual pressure-defect pairings, viscous-defect pairings, affine quotient data, optional virial-tight coordinates, and sequence-realized support data. When these extra coordinates are zero or irrelevant, we suppress them and write simply $(\bar{W}, \mathcal{Y}, \Lambda)$. In the coarse log-diffuse case $\Lambda \neq 0$, annular Young, pressure, viscous, and virial coordinates are inactive until one restricts to normalized log-window support profiles, where the annular compactification is applied to the selected window. The exact PDE and pressure/viscous Young balance are written instead for the ungauged blow-downs V_n and their ungauged Young measure, denoted $\mathcal{Y}^V$; outside $\mathcal{C}_{\mathrm{aff}}$, the homogeneous gauge bridge of Lemma 9.90 transfers Dirac/non-Dirac and hidden-scale alternatives between $\mathcal{Y}^V$ and the affine-normalized Young measure $\mathcal{Y}$. The coherent subspace $\mathfrak{T}_{\mathrm{coh}}$ is the special case

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}, \qquad \Lambda = 0, \qquad \widehat{\mathscr{P}} = 0, \qquad \mathscr{D} = 0,$$
$$\text{and no retained affine or pressure/noncompact defect.}$$

The purely log-diffuse subspace is denoted $\mathfrak{T}_{\log\mathrm{diff}}$, and the Young/defect tail subspace is denoted $\mathfrak{T}_{\mathrm{Young}}$. A sequence

$$\mathcal{T}_n = (\bar{W}_n, \mathcal{Y}_n, \Lambda_n; \widehat{\mathscr{P}}_n, \mathscr{D}_n, \mathfrak{q}_{\mathrm{aff},n}, \mathfrak{v}_{\mathrm{vir},n}, \mathfrak{s}_n)$$

with actual annular pressure representatives $\widehat{Q}_n^{\mathrm{act}}$, defined modulo time-only gauges, and affine-quotient representatives $Q_n^{\#}$, converges to $\mathcal{T}$, with limiting annular pressure representative Q, in $\mathfrak{T}_M$ if:

(i) $\bar{W}_n \rightharpoonup^* \bar{W}$ in L^∞_{loc} on macroscopic annular cylinders;

(ii) the affine-normalized Young measures generated by W_n converge against every bounded continuous velocity observable with compact support in the macroscopic variables: for every $\psi \in C_c^\infty$ supported in a macroscopic annular cylinder and every $F \in C_b(\mathbb{R}^3)$,

$$\iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{n,x,s}(\xi)\, dx\, ds \longrightarrow \iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds;$$

(iii) the log-scale defect measures Λ_n converge vaguely on the logarithmic annular compactification;

(iv) the actual pressure gradients $\nabla\widehat{Q}_n^{\mathrm{act}}$ converge annularly distributionally after the time-dependent gauges fixed in Definition 9.44, and for every pair (F,ψ) in the fixed countable C^2-test family of Definition 9.44 the scalar defect coordinates $\widehat{\mathscr{P}}_{n,F}[\psi]$ and $\mathscr{D}_{n,F}[\psi]$ converge;

(v) on the scale-critical virial-tight recurrent subhull of Definition 9.48, the local virial, dissipation, and pressure-flux coordinates on the fixed rational family of log-windows are recorded as bounded compactification data rather than asserted to be continuous for the weak velocity-pressure topology alone;

(vi) the affine quotient data, including the quotient pressures $Q_n^{\#}$ and any retained affine-force coordinates, converge in the finite-dimensional local quotient topology; and

(vii) when the profiles are sequence-realized, the limiting profile remains in the closed sequence-realized support class generated by the original terminal sequence; equivalently, the associated support data converge in the diffuse-defect state space $X_{\mathrm{def}}^{\mathrm{anc}}$.

For coherent elements, there is an ungauged representative (V,Q^{act}) satisfying on punctured annular cylinders $\{0<r_0<|x|<r_1<\infty\}\times I$

$$\partial_s V+\frac{1}{2}x\cdot\nabla V+\frac{1}{2}V+\nabla Q^{\mathrm{act}}=0,\qquad \nabla\cdot V=0$$

in distributions. The affine-normalized representative is $W=V-c(s)$. The pair $(W,Q)\in\mathcal{C}_{\mathrm{cohcrit}}$ is understood modulo the affine/parasitic pressure direction: equivalently, W satisfies the same transport equation only in the quotient by spatially homogeneous velocity modes and affine pressure forces. If one chooses to write a literal equation for W, the compensating affine pressure $(c'(s)+\frac{1}{2}c(s))\cdot x$ is allowed as a distribution in time. The coherent tail has nonzero spatial quotient oscillation and recurrence under the induced logarithmic dilation flow.

Theorem 9.40 (Compactness of the critical-tail compactification)**.** *For every sequence-realized family of critical-tail profiles, either an affine pressure descendant or a pressure/noncompact descendant is generated and routed to the corresponding lower stratum, or else, after those alternatives have been routed away, the pressure and viscous defect coordinates are uniformly bounded on the fixed countable test family and the family has a convergent subsequence in $\mathfrak{T}_M$. If all profiles in the retained bounded sequence are sequence-realized, then the limit is sequence-realized.*

Proof. First apply the affine-pressure and pressure/noncompact routing alternatives. A nonzero retained affine-force coordinate is assigned to $\mathcal{C}_{\mathrm{aff}}$. In the complementary retained branch the affine-pressure coefficient dichotomy gives $b_n\to 0$ on compact time intervals, while Lemma 9.74 gives $\|q_n^{\mathrm{na}}\|_{L^p(K)}\le C_{p,K,M}R_n^{-1}$ on every fixed annular cylinder. The retained alternative in Lemma 9.69 gives

$$R_n^{-2}\iint|\nabla V_n|^2\phi\le C_{M,K,\phi}.$$

Consequently, for every pair in the fixed countable test family, the viscous coordinates are bounded by this local-energy bound and the bounded Hessian of F. For the pressure coordinates, write $\widehat{Q}_n^{\mathrm{act}}=q_n^{\mathrm{na}}+b_n(s)\cdot x+a_n(s)$. The time-only gauge has zero spatial gradient; the affine part is bounded, and in fact vanishes in the retained branch, because $b_n\to 0$; and the non-affine part is bounded after integrating by parts, using the finite-p pressure estimate and the same local-energy bound. More explicitly, on a fixed annular cylinder K, choose $p=2$ in Lemma 9.74; then

$$\left|\iint q_n^{\mathrm{na}}\operatorname{div}_x(\psi\nabla_\xi F(V_n))\right|\le C\|q_n^{\mathrm{na}}\|_{L^2(K)}\big(1+\|\nabla V_n\|_{L^2(K)}\big)\le C_{F,\psi,K,M},$$

because $\|q_n^{\mathrm{na}}\|_{L^2(K)} = O(R_n^{-1})$ and $\|\nabla V_n\|_{L^2(K)} = O(R_n)$ in the retained local-energy branch. Thus all pressure and viscous defect coordinates on the fixed countable critical-tail test family are uniformly bounded. We therefore work in this retained bounded-coordinate branch.

The barycentric velocities are bounded in L^∞_{loc}, so Banach–Alaoglu gives local weak-$*$ subsequential convergence. The fundamental compactness theorem for Young measures gives convergence of the oscillation measures against bounded continuous velocity observables. The logarithmic annular compactification is metrizable on the countable family of windows used in the extraction, hence Prokhorov compactness gives a vaguely convergent subsequence of the log-scale measures. The pressure components are recorded only through distributional gradients after the time and affine gauges; local pressure compactness modulo these gauges gives distributional subsequential limits. The pressure-defect coordinates $\widehat{\mathscr{P}}_{n,F}[\psi]$ and viscous-defect coordinates $\mathscr{D}_{n,F}[\psi]$ are part of the defining topology from Definitions 9.39 and 9.44; on the fixed countable family they are uniformly bounded in the retained branch by the finite-n Young identity and the retained local-energy alternative in Lemma 9.69, so a diagonal extraction gives convergent subsequences for all of them simultaneously. For realized recurrent profiles lying in the scale-critical virial-tight subhull of Definition 9.48, the virial coordinates are likewise uniformly bounded on the countable rational family by Lemma 9.49, hence admit simultaneous limits as coordinate data of the recurrent virial hull. The affine quotient data are finite-dimensional on each fixed time window, and the quotient topology was chosen so that bounded sequences have subsequential limits. Diagonalizing over the countable annular exhaustion gives convergence in all components listed in Definition 9.39. For sequence-realized sequences, choose for each approximating profile a realizing subsequence from the original terminal sequence and diagonalize over the annular exhaustion and the countable observables in the topology above. The diagonal realizing sequence converges to the same limit, so the limiting critical-tail profile remains sequence-realized. □

Lemma 9.41 (Support transfer for the critical-tail compactification)**.** *Let $\mathfrak{T}_M$ be the compact metrizable critical-tail compactification and let*

$$\lambda_n = \frac{\mu_n \lfloor E_n}{\mu_n(E_n)}$$

be normalized realized critical-tail defect measures, where $\mu_n(E_n) > 0$. If $\lambda_n \rightharpoonup \lambda \in \mathcal{P}(\mathfrak{T}_M)$ and $A \subset \mathfrak{T}_M$ is closed, then support of λ in A removes all surviving critical-tail defect mass outside A. If positive normalized critical-tail mass survives outside A, then a sequence-realized support profile in $\mathfrak{T}_M \setminus A$ exists.

Proof. This is Lemma 1.14 with $X = \mathfrak{T}_M$. The compactness theorem above ensures that subsequential limits of realized critical-tail profiles remain sequence-realized, so the support point outside the closed set is not an abstract compactification artifact. □

Definition 9.42 (Log-annular compactification)**.** Set $\rho = \log|x|$ and write $\overline{\mathbb{R}}_{\log} = [-\infty, +\infty]$ for the two-point compactification of the log-radius axis. Fix the countable family of log-window cutoffs

$$\chi_{\rho_0,A}(\rho) := \chi\Big(\tfrac{\rho-\rho_0}{A}\Big), \qquad \rho_0 \in \mathbb{Q},\ A \in \mathbb{Q}_{>0},$$

where χ is the fixed compactly supported even cutoff of Section 9.1. The log-annular space $\overline{\mathbb{R}}_{\log}$ is endowed with the metrizable vague topology generated by these cutoffs. A sequence Λ_n of finite Borel measures on $\overline{\mathbb{R}}_{\log}$ converges to Λ vaguely if

$$\int \chi_{\rho_0,A}(\rho)\, d\Lambda_n(\rho) \longrightarrow \int \chi_{\rho_0,A}(\rho)\, d\Lambda(\rho)$$

for every cutoff in the countable family above and additionally for the test functions

$$\rho \mapsto \chi(\rho/A), \quad \rho \mapsto \chi((\rho \mp A)/1)$$

witnessing the masses at $\pm\infty$.

Definition 9.43 (Young-measure convergence at the macroscopic critical scale)**.** Let $\overline{B_M} \subset \mathbb{R}^3$ be the closed velocity ball of radius M, and set

$$M_{\text{tail}} := 2M.$$

The ungauged exact PDE fields V_n take values in $\overline{B_M}$, while the affine-normalized fields $W_n = V_n - c_n(s)$, with $|c_n(s)| \le M$, take values in $\overline{B_{M_{\text{tail}}}}$. A sequence of affine-normalized Young measures $\mathcal{Y}_n$ on the macroscopic spacetime annulus $\{a < |x| < b\} \times I$ with values in $\overline{B_{M_{\text{tail}}}}$ converges to $\mathcal{Y}$ if for every $\psi \in C_c^\infty(\{a < |x| < b\} \times I)$ and every $F \in C(\overline{B_{M_{\text{tail}}}})$,

$$\iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{n,x,s}(\xi)\, dx\, ds \longrightarrow \iint \psi(x,s) \int F(\xi)\, d\mathcal{Y}_{x,s}(\xi)\, dx\, ds.$$

This is the fundamental theorem of Young measures applied to the bounded sequence of macroscopic representatives W_n, with associated Young measures $\mathcal{Y}_{n,x,s}$ recording the unresolved bounded oscillations at (x,s). The countable critical-tail pressure and viscous test family in Definition 9.44 remains on $\overline{B_M}$, because those identities are tested against the ungauged exact PDE variables V_n. Equivalently, $\mathcal{Y}_n \to \mathcal{Y}$ in the $\sigma(L^\infty, L^1)$-topology induced by tensor test functions.

Definition 9.44 (Distributional pressure-gradient convergence at critical scale)**.** Fix once and for all a countable dense family $\mathfrak{F}_{\text{crit}}$ of test pairs (F,ψ), where $F \in C^2(\overline{B_M})$ and $\psi \in C_c^\infty(\{a < |x| < b\} \times I)$ has rational coefficients on the chosen countable annular exhaustion. Let $\mathfrak{F}_{\text{crit}}^{\text{conv}}$ denote the subfamily of pairs in this exhaustion for which $\psi \ge 0$ and $\nabla_\xi^2 F \ge 0$ on $\overline{B_M}$, enlarged if necessary so that, for every rational vector $q \in \mathbb{Q}^3$, it contains the convex quadratic observable

$$F_q(\xi) = \frac{1}{2}(q \cdot \xi)^2, \qquad \nabla_\xi^2 F_q = q \otimes q.$$

The countable family $\{q \otimes q : q \in \mathbb{Q}^3\}$ determines every positive semidefinite matrix-valued Radon defect measure by density and polarization. Thus vanishing of all positive convex coordinates implies vanishing of the full viscous defect matrix, and hence all signed viscous coordinates vanish.

Let $\widehat{Q}_n^{\text{act}}$ be actual macroscopic pressure representatives, defined modulo time-only gauges on the same annular cylinder, and let

$$Q_n^\#(x,s) := \widehat{Q}_n^{\text{act}}(x,s) - b_n(s) \cdot x$$

be their affine-quotient representatives. We say that $\nabla \widehat{Q}_n^{\text{act}}$ converges *annularly distributionally* to ∇Q, written $\nabla \widehat{Q}_n^{\text{act}} \to \nabla Q$ in $\mathcal{D}'_{\text{ann}}$, if for every compactly supported divergence-free test vector field $\Psi \in C_c^\infty(\{a < |x| < b\} \times I, \mathbb{R}^3)$

$$\langle \nabla \widehat{Q}_n^{\text{act}}, \Psi \rangle = -\langle \widehat{Q}_n^{\text{act}}, \nabla \cdot \Psi \rangle = 0$$

and for every general compactly supported $\Phi \in C_c^\infty(\{a < |x| < b\} \times I, \mathbb{R}^3)$ one has

$$\langle \nabla \widehat{Q}_n^{\text{act}}, \Phi \rangle \to \langle \nabla Q, \Phi \rangle.$$

In addition, for every $(F,\psi) \in \mathfrak{F}_{\text{crit}}$ the scalar defect coordinates

$$\widehat{\mathscr{P}}_{n,F}[\psi] := \iint \psi\, \nabla_\xi F(V_n) \cdot \nabla \widehat{Q}_n^{\text{act}}$$

and

$$\mathscr{D}_{n,F}[\psi] := R_n^{-2} \iint \psi\, \nabla V_n^T \nabla_\xi^2 F(V_n) \nabla V_n$$

are required to converge along the selected compactification subsequence. If one passes to the affine-quotient representative $Q_n^\# = \widehat{Q}_n^{\mathrm{act}} - b_n(s)\cdot x$, then

$$\widehat{\mathscr{P}}_{n,F}[\psi] = \mathscr{P}^\#_{n,F}[\psi] + \mathscr{B}_{n,F}[\psi],$$

where

$$\mathscr{P}^\#_{n,F}[\psi] := \iint \psi\, \nabla_\xi F(V_n) \cdot \nabla Q_n^\#$$

and

$$\mathscr{B}_{n,F}[\psi] := \iint \psi\, \nabla_\xi F(V_n) \cdot b_n(s).$$

A nonzero retained affine-force coordinate $\mathscr{B}_{n,F}[\psi]$ is assigned to $\mathcal{C}_{\mathrm{aff}}$. Outside $\mathcal{C}_{\mathrm{aff}}$, the critical-tail Young identity is written using the actual pressure coordinates $\widehat{\mathscr{P}}_{n,F}[\psi]$.

Lemma 9.45 (Critical-tail Young balance with pressure and viscous defects)**.** *Let $V_n \in L^\infty$ be uniformly bounded by M and satisfy the exact critical-tail blow-down equation*

$$\partial_s V_n - R_n^{-2}\Delta V_n + \frac{1}{2}x\cdot\nabla V_n + \frac{1}{2}V_n + R_n^{-1}V_n\cdot\nabla V_n + \nabla \widehat{Q}_n^{\mathrm{act}} = 0, \qquad \nabla\cdot V_n = 0 \tag{9.2}$$

on a fixed macroscopic annular cylinder $\{a<|x|<b\}\times I$, with $R_n\to\infty$, and pressure gradients satisfying the time-gauge convention of Definition 9.44. Let $(\bar{W},\mathcal{Y},\Lambda)$ be the limiting critical-tail profile in $\mathfrak{T}_M$, where $\mathcal{Y}$ is the Young measure generated by the affine-normalized fields $W_n = V_n - c_n(s)$. Let $\mathcal{Y}^V_{x,s}$ denote the ungauged Young measure generated by V_n. After passing to the subsequence defining $\mathfrak{T}_M$, the actual pressure and viscous pairings

$$\widehat{\mathscr{P}}_{n,F}[\psi] := \iint \psi(x,s)\, \nabla_\xi F(V_n(x,s)) \cdot \nabla \widehat{Q}_n^{\mathrm{act}}(x,s)\, dx\, ds$$

$$\mathscr{D}_{n,F}[\psi] := R_n^{-2} \iint \psi(x,s)\, \nabla_x V_n(x,s)^T \nabla_\xi^2 F(V_n(x,s)) \nabla_x V_n(x,s)\, dx\, ds$$

$$\mathscr{E}_{n,F}[\psi] := -R_n^{-2} \iint F(V_n)\, \Delta\psi\, dx\, ds - R_n^{-1} \iint F(V_n) V_n\cdot\nabla\psi\, dx\, ds.$$

converge for every $\psi\in C_c^\infty$ and every $F\in C^2(\overline{B_M})$ in the countable dense test class used to define the critical-tail topology. Denote the limits by $\widehat{\mathscr{P}}_F[\psi]$ and $\mathscr{D}_F[\psi]$. Define also the explicit error $\mathscr{E}_{n,F}[\psi]$ above. For each finite n, the exact identity

$$\begin{aligned} &\iint \Big(\partial_s\psi + \tfrac{1}{2}x\cdot\nabla_x\psi + \tfrac{3}{2}\psi\Big) F(V_n)\, dx\, ds \\ &\quad = \frac{1}{2}\iint \psi\, \nabla_\xi F(V_n)\cdot V_n\, dx\, ds + \widehat{\mathscr{P}}_{n,F}[\psi] + \mathscr{D}_{n,F}[\psi] + \mathscr{E}_{n,F}[\psi]. \end{aligned} \tag{9.3}$$

holds. Moreover $\mathscr{E}_{n,F}[\psi]\to 0$ as $n\to\infty$. If one rewrites the pressure with the affine-quotient representative $Q_n^\# = \widehat{Q}_n^{\mathrm{act}} - b_n(s)\cdot x$, then $\widehat{\mathscr{P}}_{n,F}[\psi] = \mathscr{P}^\#_{n,F}[\psi] + \mathscr{B}_{n,F}[\psi]$; a nonzero retained affine-force coordinate $\mathscr{B}_{n,F}[\psi]$ is routed to $\mathcal{C}_{\mathrm{aff}}$. Consequently the Young measure $\mathcal{Y}^V$ satisfies the exact weak balance

$$\begin{aligned} &\iint \Big(\partial_s\psi + \tfrac{1}{2}x\cdot\nabla_x\psi + \tfrac{3}{2}\psi\Big) \int F(\xi)\, d\mathcal{Y}^V_{x,s}(\xi)\, dx\, ds \\ &\quad = \frac{1}{2}\iint \psi \int \xi\cdot\nabla_\xi F(\xi)\, d\mathcal{Y}^V_{x,s}(\xi)\, dx\, ds + \widehat{\mathscr{P}}_F[\psi] + \mathscr{D}_F[\psi]. \end{aligned} \tag{9.4}$$

with the usual approximation from C^2 to bounded continuous observables on $\overline{B_M}$ whenever the velocity derivatives and the two defect pairings are represented by uniform limits. In particular, if F is homogeneous of degree p and $\widehat{\mathscr{P}}_F[\psi] = \mathscr{D}_F[\psi] = 0$, then

$$\iint \Big(\partial_s\psi + \tfrac{1}{2}x\cdot\nabla_x\psi + \tfrac{3-p}{2}\psi\Big) \int F(\xi)\, d\mathcal{Y}^V_{x,s}(\xi)\, dx\, ds = 0.$$

For the affine-normalized Young measure $\mathcal{Y}$, the corresponding observable is $F(\xi + c(s))$, not $F(\xi)$, unless the homogeneous gauge has already been removed in the parasitic-free representative. The log-scale component Λ, which records the covariant positions of the selected annular windows rather than a pressure-renormalized velocity observable, is invariant under the induced log-translation $\rho \mapsto \rho + \frac{1}{2}s$ on $\overline{\mathbb{R}}_{\log}$. If one also records the affine-normalized Young measure $\mathcal{Y}$ generated by W_n, then outside $\mathcal{C}_{\mathrm{aff}}$ the bridge lemma Lemma 9.90 identifies $\mathcal{Y}^V$ and $\mathcal{Y}$ by deterministic spatially homogeneous translation, so their Dirac/non-Dirac and hidden-scale alternatives agree.

Proof. Test (9.2) against $\psi(x,s)F'(V_n(x,s))$ for $F \in C^2(\overline{B_M})$ and $\psi \in C_c^\infty$. Each term is integrated by parts inside the chosen macroscopic annular cylinder where ψ has compact support. The viscous term contributes

$$-R_n^{-2}\iint \psi\, \nabla_\xi F(V_n)\cdot \Delta V_n\, dx\, ds.$$

Integrating by parts gives

$$\begin{aligned} &- R_n^{-2}\iint \psi\, \nabla_\xi F(V_n)\cdot \Delta V_n\, dx\, ds \\ &\quad = R_n^{-2}\iint \psi\, \nabla_x V_n^T \nabla_\xi^2 F(V_n)\nabla_x V_n\, dx\, ds + R_n^{-2}\iint \nabla_x F(V_n)\cdot\nabla\psi\, dx\, ds. \end{aligned}$$

We are in the retained non-affine branch of Lemma 9.69; otherwise the sequence has already been routed to the affine or pressure/noncompact alternative and no $\mathfrak{T}_M$ Young-balance extraction is being performed. Since F' and F'' are bounded on $\overline{B_M}$, V_n is bounded by M, and the local energy bound from Lemma 9.69 gives $R_n^{-2}\iint |\nabla V_n|^2\psi\, dx\, ds \le C$, the first term is exactly $\mathscr{D}_{n,F}[\psi]$ and is uniformly bounded on the fixed countable test family. The second term equals $-R_n^{-2}\iint F(V_n)\Delta\psi\, dx\, ds$, hence is $O(R_n^{-2})$ and tends to 0.

The convective term, after the standard chain-rule manipulation, equals

$$R_n^{-1}\iint F'(V_n)(V_n\cdot\nabla V_n)\psi\, dx\, ds = R_n^{-1}\iint (V_n\cdot\nabla)F(V_n)\,\psi\, dx\, ds = -R_n^{-1}\iint F(V_n)V_n\cdot\nabla\psi\, dx\, ds,$$

using $\nabla\cdot V_n = 0$. Since F and V_n are bounded, this term is $O(R_n^{-1})$ and tends to 0.

The drift term $\frac{1}{2}x\cdot\nabla V_n + \frac{1}{2}V_n$ tested against $F'(V_n)\psi$ gives, after the chain rule and integration by parts in x,

$$\iint \Big(\tfrac{1}{2}x\cdot\nabla F(V_n) + \tfrac{1}{2}F'(V_n)V_n\Big)\psi\, dx\, ds = -\iint F(V_n)\Big(\tfrac{1}{2}\nabla\cdot(x\psi)\Big)\, dx\, ds + \iint \tfrac{1}{2}F'(V_n)V_n\,\psi\, dx\, ds,$$

where the boundary term vanishes by compact support of ψ. Using $\frac{1}{2}\nabla\cdot(x\psi) = \frac{3}{2}\psi + \frac{1}{2}x\cdot\nabla\psi$ gives

$$-\iint F(V_n)\Big(\tfrac{3}{2}\psi + \tfrac{1}{2}x\cdot\nabla\psi\Big)\, dx\, ds + \iint \tfrac{1}{2}F'(V_n)V_n\,\psi\, dx\, ds.$$

Combining with the remaining ∂_s term, which gives $-\iint F(V_n)\partial_s\psi\, dx\, ds$ after integration by parts in time, and keeping the pressure term explicitly, we obtain the exact finite-n identity

$$\begin{aligned} &\iint F(V_n)\Big(\partial_s\psi + \tfrac{1}{2}x\cdot\nabla\psi + \tfrac{3}{2}\psi\Big)\, dx\, ds - \iint \tfrac{1}{2}F'(V_n)V_n\,\psi\, dx\, ds \\ &\quad = \widehat{\mathscr{P}}_{n,F}[\psi] + \mathscr{D}_{n,F}[\psi] + \mathscr{E}_{n,F}[\psi], \end{aligned}$$

which is exactly (9.3). The remainder $\mathscr{E}_{n,F}[\psi]$ tends to zero because the two terms estimated above are $O(R_n^{-2})$ and $O(R_n^{-1})$, respectively. Since the two terms on the left-hand side are uniformly bounded by $M\|\psi\|_{C^1}$ and $\mathscr{D}_{n,F}[\psi]$ is uniformly bounded as above, the finite-n identity also gives a uniform bound for the actual pressure pairings $\widehat{\mathscr{P}}_{n,F}[\psi]$ on the fixed countable family. These scalar coordinates are therefore available for diagonal extraction in the definition of $\mathfrak{T}_M$.

Choosing $F(\xi) = \xi \cdot e_k$ recovers the linear barycenter equation; choosing more general F and passing to the ungauged Young-measure limit using Definition 9.43 gives the exact weak identity

$$\iint \Big(\partial_s\psi + \tfrac{1}{2}x \cdot \nabla\psi + \tfrac{3}{2}\psi\Big) \int F(\xi)\, d\mathcal{Y}^V_{x,s}(\xi)\, dx\, ds$$
$$= \iint \tfrac{1}{2}\psi \int F'(\xi)\xi\, d\mathcal{Y}^V_{x,s}(\xi)\, dx\, ds + \widehat{\mathscr{P}}_F[\psi] + \mathscr{D}_F[\psi].$$

This is (9.4). The homogeneous-observable special case follows from Euler's identity $\xi \cdot \nabla_\xi F = pF$ whenever the pressure and viscous defect pairings vanish for that observable. If either pairing does not vanish, the nonzero defect is retained as pressure-gradient or Young/viscous defect data in $\mathfrak{T}_M$ and is assigned to the pressure/noncompact, affine quotient, or Young/defect critical-tail alternatives rather than silently discarded.

The same calculation applied to the radial pushforward $\rho \mapsto \log|x|$ of V_n, or directly to the dyadic annular Λ_n, gives the invariance of Λ under the induced log-translation $\rho \mapsto \rho + \frac{1}{2}s$. $\square$

9.1. **Log-annular virial identities for recurrent critical defects.** Throughout this subsection let $\chi \in C_c^\infty((-2,2))$ satisfy $0 \le \chi \le 1$ and $\chi \equiv 1$ on $(-1,1)$. For $\rho \in \mathbb{R}$ and $A \ge 1$, set

$$\phi_{\rho,A}(y) := \chi\left(\frac{\log|y| - \rho}{A}\right), \qquad R := e^\rho.$$

The cutoff is used only on annuli avoiding the origin. When differentiating with respect to ρ, A is kept fixed.

Lemma 9.46 (Log-annular oscillation virial identity)**.** *Let $(U,\Pi) \in \mathfrak{X}_M$ be a smooth bounded solution of (1.2). Fix $A \ge 1$, $\rho \in \mathbb{R}$, and put $\phi = \phi_{\rho,A}$. Define the weighted affine gauge*

$$c_{\rho,A}(\tau) := \frac{\int \phi(y)U(y,\tau)\, dy}{\int \phi(y)\, dy}, \qquad V_{\rho,A}(y,\tau) := U(y,\tau) - c_{\rho,A}(\tau),$$

so that

$$\int \phi V_{\rho,A}\, dy = 0.$$

Set

$$E_{\rho,A}(\tau) := \int \phi|V_{\rho,A}|^2\, dy, \qquad \mathcal{V}_{\rho,A}(\tau) := R^{-1}E_{\rho,A}(\tau),$$
$$D_{\rho,A}(\tau) := \int \phi|\nabla U|^2\, dy, \qquad \mathcal{D}_{\rho,A}(\tau) := R^{-1}D_{\rho,A}(\tau).$$

Then $\mathcal{V}_{\rho,A}$ is locally absolutely continuous in τ, and for a.e. τ, after subtracting an arbitrary pressure gauge $\pi(\tau)$, one has

$$\begin{aligned}\partial_\tau\mathcal{V}_{\rho,A} + 2\mathcal{D}_{\rho,A} + \frac{1}{2}\partial_\rho\mathcal{V}_{\rho,A} = R^{-1}&\int (\Delta\phi)|V_{\rho,A}|^2\, dy \\ &+ R^{-1}\int |V_{\rho,A}|^2 U \cdot \nabla\phi\, dy \\ &+ 2R^{-1}\int (\Pi - \pi(\tau))V_{\rho,A} \cdot \nabla\phi\, dy.\end{aligned} \tag{9.5}$$

Moreover

$$|\nabla\phi_{\rho,A}(y)| \le \frac{C_\chi}{A|y|}, \qquad |\Delta\phi_{\rho,A}(y)| \le \frac{C_\chi}{A|y|^2} + \frac{C_\chi}{A^2|y|^2}, \tag{9.6}$$

and

$$y\cdot\nabla\phi_{\rho,A}(y) = \frac{1}{A}\chi'\Big(\frac{\log|y|-\rho}{A}\Big), \qquad \partial_\rho\phi_{\rho,A} = -y\cdot\nabla\phi_{\rho,A}.$$

Proof. Since $c_{\rho,A}(\tau)$ is the weighted least-squares affine gauge,

$$\int \phi V_{\rho,A}\,dy = 0.$$

Hence the derivative of the gauge drops out:

$$\frac{d}{d\tau}\int \phi|U-c_{\rho,A}|^2 = 2\int \phi V_{\rho,A}\cdot\partial_\tau U.$$

Substitute the centered equation

$$\partial_\tau U = \Delta U - (U\cdot\nabla)U - \frac{1}{2}y\cdot\nabla U - \frac{1}{2}U - \nabla\Pi.$$

The diffusion term gives

$$2\int \phi V\cdot\Delta U = -2\int \phi|\nabla U|^2 + \int(\Delta\phi)|V|^2.$$

The convective term gives, using $U\cdot\nabla U = U\cdot\nabla V$ and $\nabla\cdot U = 0$,

$$-2\int \phi V\cdot(U\cdot\nabla U) = -\int \phi U\cdot\nabla|V|^2 = \int |V|^2 U\cdot\nabla\phi.$$

The centered drift and linear terms give

$$\begin{aligned}-\int \phi V\cdot(y\cdot\nabla U) - \int \phi V\cdot U &= -\frac{1}{2}\int \phi y\cdot\nabla|V|^2 - \int\phi|V|^2\\ &= \frac{1}{2}\int y\cdot\nabla\phi|V|^2 + \frac{1}{2}\int\phi|V|^2.\end{aligned}$$

The terms involving $c_{\rho,A}$ vanish because $\int\phi V = 0$. Finally, for any scalar gauge $\pi(\tau)$,

$$-2\int\phi V\cdot\nabla\Pi = 2\int(\Pi-\pi(\tau))V\cdot\nabla\phi,$$

because $\nabla\cdot V = 0$ and

$$\int \pi(\tau)V\cdot\nabla\phi\,dy = \pi(\tau)\int\nabla\cdot(\phi V)\,dy = 0.$$

Combining these identities yields

$$\begin{aligned}\partial_\tau E_{\rho,A} + 2D_{\rho,A} = &\int(\Delta\phi)|V|^2 + \int|V|^2U\cdot\nabla\phi + 2\int(\Pi-\pi)V\cdot\nabla\phi\\ &+\frac{1}{2}\int\phi|V|^2 + \frac{1}{2}\int y\cdot\nabla\phi|V|^2.\end{aligned}$$

Since

$$\partial_\rho\phi_{\rho,A} = -y\cdot\nabla\phi_{\rho,A}$$

and the weighted least-squares condition also removes the $\partial_\rho c_{\rho,A}$ term,

$$\partial_\rho E_{\rho,A} = \int\partial_\rho\phi_{\rho,A}|V|^2 = -\int y\cdot\nabla\phi_{\rho,A}|V|^2.$$

Therefore

$$\frac{1}{2}R^{-1}\left(\int \phi|V|^2 + \int y\cdot\nabla\phi|V|^2\right) = -\frac{1}{2}\partial_\rho(R^{-1}E_{\rho,A}) = -\frac{1}{2}\partial_\rho\mathcal{V}_{\rho,A}.$$

Thus, after multiplying the energy identity by R^{-1}, the centered drift and zeroth-order contribution is $-\frac{1}{2}\partial_\rho\mathcal{V}_{\rho,A}$. Moving it to the left gives (9.5). The cutoff bounds follow by direct differentiation in polar coordinates. □

Remark 9.47 (Virial normalization)*.* The normalization $R^{-1}\int\phi|U-c|^2$ is the correct scale for an actual spatial tail of size $|y|^{-1}$. It is not automatically a compact observable for an arbitrary bounded residual profile. In the critical-tail reduction below, one uses it either on selected scale-critical defect windows or as an observable on the compact recurrent defect hull after the required scale-critical virial tightness has been verified. Without that compactness, the virial identity remains valid for smooth annular cutoffs but cannot by itself close the critical-tail alternatives.

Definition 9.48 (Scale-critical virial-tight recurrent subhull)**.** A realized recurrent critical-tail defect lies in the scale-critical virial-tight recurrent subhull if, along a realizing recurrent sequence and for every rational log-window (ρ, A),

$$\sup_n\left(|\mathcal{V}^{(n)}_{\rho,A}| + |\mathcal{D}^{(n)}_{\rho,A}| + |\mathcal{F}^{(n)}_{\rho,A}|\right) < \infty,$$

where $\mathcal{F}^{(n)}_{\rho,A}$ denotes the sum of the three right-hand-side flux contributions in (9.5). On this subhull the virial, dissipation, and flux quantities are adjoined to the realized recurrent compactification as bounded scalar coordinates on the fixed rational family of log-windows.

Lemma 9.49 (Realized log-window virial compactness)**.** *Let $\mathcal{D}\in\mathfrak{T}_M$ be a recurrent critical-tail defect generated by a normalized terminal sequence realized from the original blow-up sequence after concentration-set removal and affine normalization, and assume that $\mathcal{D}$ lies in the scale-critical virial-tight recurrent subhull of Definition 9.48. Then, along the selected defect subsequence and after the admissible local pressure gauges of Proposition 1.21, the annular virial observables $\mathcal{V}_{\rho,A}$, the local dissipation $\mathcal{D}_{\rho,A}$, and the right-hand side flux contributions*

$$R^{-1}\int(\Delta\phi)|V|^2, \qquad R^{-1}\int|V|^2U\cdot\nabla\phi, \qquad 2R^{-1}\int(\Pi-\pi)V\cdot\nabla\phi$$

appearing in (9.5) *are* not *asserted to be continuous for the weak velocity-pressure topology alone. Instead, on the fixed countable rational family of log-windows (ρ, A), these virial, dissipation, and pressure-flux quantities are included as scalar coordinates of the realized recurrent critical-tail compactification. Along every realized sequence they are uniformly bounded, hence admit simultaneous subsequential limits by diagonal extraction. The resulting coordinate functions are continuous by definition on the corresponding compact virial hull. In the strong-compactness subcase, these coordinate limits agree with the ordinary annular integrals of the limiting coherent profile.*

Proof. The required boundedness is precisely the defining property of the scale-critical virial-tight subhull. Therefore, on the fixed countable rational family of log-windows, the virial, dissipation, and flux observables admit simultaneous subsequential limits by diagonal extraction. These quantities are not claimed to be continuous under weak L^∞-velocity and distributional-pressure convergence alone: quadratic energy is only weakly lower semicontinuous, dissipation is not weakly continuous, and the pressure flux requires more than distributional convergence against fixed test fields. Instead, on the virial-tight subhull one adjoins these bounded scalar coordinates to the recurrent compactification. By construction the resulting coordinate maps are continuous on the corresponding compact virial hull. If at any stage the homogeneous gauge sequence loses compactness, Lemma 9.79 assigns the descendant to $\mathcal{C}_{\mathrm{aff}}$. In the complementary case the observability hypothesis of Lemma 9.50 is verified by these recorded coordinates, and the reduction provided by the minimal mesoscopic-scale theorem Theorem 9.86 closes the diagram. When the annular blow-downs converge strongly, the recorded coordinate limits coincide with the usual virial integrals of the limiting coherent profile. □

Lemma 9.50 (Recurrent virial equality case and reduction)**.** *Let $\mathcal{D} \in \mathfrak{T}_M$ be a recurrent critical-tail defect generated by a bounded normalized terminal sequence after concentration-set removal and affine normalization. Assume that $\mathcal{D}$ lies in the scale-critical virial-tight recurrent subhull of Definition 9.48. Thus the log-annular virial, dissipation, and flux coordinates in Lemma 9.46 are bounded continuous observables on the chosen recurrent hull. If the defect is not in this virial-tight subhull, the failure of virial tightness is assigned, by definition of the ordered critical-tail compactification, to the log-diffuse, pressure/noncompact, or hidden-scale variance branch before the recurrent virial equality is invoked. Let $\mathbb{P}$ be an invariant probability measure on that hull for the joint time-translation and log-translation flow. Assume also that the averaged pressure gauges are chosen by a local annular pressure decomposition, so that the three right-hand side terms in (9.5) have limits as defect observables. Then the following conclusions hold.*

(i) Zero-flux equality case. *If*

$$\lim_{A\to\infty} \int \left| R^{-1} \int (\Delta\phi)|V|^2 + R^{-1} \int |V|^2 U \cdot \nabla\phi + 2R^{-1} \int (\Pi - \pi) V \cdot \nabla\phi \right| d\mathbb{P} = 0, \tag{9.7}$$

then every local tangent in the support of $\mathbb{P}$ has zero spatial gradient. Consequently

$$\mathcal{D} \in \mathcal{C}_{\mathrm{aff}}.$$

(ii) Tight Dirac flux reduction. *Suppose (9.7) fails, and suppose the nonzero limiting flux is neither concentrated on a positive-threshold unit concentrating observer nor lost to logarithmic infinity. Suppose moreover that the annular blow-downs selected by this nonzero flux generate a Dirac Young measure. Then the selected annular blow-downs converge strongly in L^3_{loc} on macroscopic annular cylinders to a nonzero coherent critical tail. Hence*

$$\mathcal{D} \in \mathcal{C}_{\mathrm{cohcrit}}.$$

The minimal mesoscopic-scale reduction theorem proved below supplies the tight-Dirac alternative for realized recurrent defects. Combined with that theorem, the preceding alternatives give the inclusion

$$\mathcal{D} \in \mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

Proof. Because we work on the virial-tight recurrent hull, the observables $\mathcal{V}_{\rho,A}$, $\mathcal{D}_{\rho,A}$, and $\mathcal{F}_{\rho,A}$ are bounded and continuous. Hence averaging against the invariant measure $\mathbb{P}$ is legitimate. We do not invoke infinitesimal generator-domain theory for $\mathcal{V}_{\rho,A}$. Instead, fix a nonnegative smooth cutoff $\eta \in C_c^\infty(\mathbb{R}^2)$ with $\eta \equiv 1$ on $[-1,1]^2$ and support in $[-2,2]^2$, and set

$$\eta_T(\tau,\zeta) := \eta(\tau/T, \zeta/T), \qquad N_T := \iint_{\mathbb{R}^2} \eta_T(\tau,\zeta)\, d\tau\, d\zeta.$$

Translate the finite-level identity (9.5) by the joint time/log action, multiply by η_T, integrate over $(\tau,\zeta) \in \mathbb{R}^2$, and then integrate over the recurrent hull against $\mathbb{P}$. Because the virial coordinates are recorded as bounded continuous observables on the virial-tight hull, this averaged identity passes to the hull level. Moving the derivative terms onto η_T gives

$$\frac{1}{N_T}\iint \big(\partial_\tau \mathcal{V}_{\rho+\zeta,A} + \tfrac{1}{2}\partial_\rho \mathcal{V}_{\rho+\zeta,A}\big)\,\eta_T(\tau,\zeta)\,d\tau\,d\zeta$$
$$= -\frac{1}{N_T}\iint \mathcal{V}_{\rho+\zeta,A}\Big(\partial_\tau \eta_T(\tau,\zeta) + \tfrac{1}{2}\partial_\zeta \eta_T(\tau,\zeta)\Big)\,d\tau\,d\zeta.$$

After integrating also over $\mathbb{P}$, invariance of $\mathbb{P}$ under the joint translation action makes the average of each translated observable equal to its $\mathbb{P}$-mean. Since $\sup|\mathcal{V}_{\rho,A}| < \infty$ on the virial-tight hull and $\|\nabla\eta_T\|_{L^1}/N_T \to 0$, the averaged derivative contribution tends to zero as $T\to\infty$. Therefore the averaged identity yields

$$2\int \mathcal{D}_{\rho,A}\,d\mathbb{P} = \int \mathcal{F}_{\rho,A}\,d\mathbb{P},$$

where $\mathcal{F}_{\rho,A}$ denotes the sum of the three right-hand side flux terms in (9.5). Under (9.7) and by nonnegativity of the dissipation,

$$\lim_{A\to\infty}\int \mathcal{D}_{\rho,A}\,d\mathbb{P} = 0.$$

Choose a countable exhausting family of ordinary unit covariant cylinders. For each cylinder, choose A large enough that the corresponding cutoff is identically one on a fixed enlargement. The preceding limit then implies that, for $\mathbb{P}$-a.e. profile, the local dissipation $\int_Q |\nabla U|^2$ vanishes on every cylinder in the exhausting family. Hence $\nabla U = 0$ distributionally on every compact cylinder. Thus every local tangent is spatially constant. By the affine normalization dichotomy, this is precisely the affine/parasitic lower stratum $\mathcal{C}_{\mathrm{aff}}$.

We now prove the tight Dirac flux reduction. Since the averaged flux does not vanish and is not concentrated on a unit concentrating observer, there are logarithmic annuli with nonzero normalized flux but with no positive-threshold unit constant-in-time oscillatory subcylinder. Since the flux is not lost to logarithmic infinity, these annuli may be selected with a tight macroscopic scale $R_n\to\infty$. Passing to the corresponding macroscopic variables gives bounded blown-down fields

$$\widetilde{U}_n := \mathcal{R}_{(y_n,\sigma_n)}U_n, \qquad V_n(x,s) = \widetilde{U}_n(R_n x, s), \qquad W_n(x,s) = \widetilde{U}_n(R_n x,s) - c_n(s)$$

on every fixed annular cylinder, with pressure and affine gauges normalized by the exact blow-down definitions of Definition 9.89. Terminal compactness gives a weak-$*$ L^∞_{loc} limit and an associated Young measure. By the Dirac hypothesis on the selected flux, the Young measure is δ_W almost everywhere. Uniform boundedness then upgrades convergence to strong L^3_{loc} convergence on annular cylinders for the affine-normalized representatives, and for the ungauged representatives up to the deterministic homogeneous translation supplied by the gauge bridge. Passing to the limit in the exact ungauged equation (9.23), the viscous and nonlinear terms vanish by the factors R_n^{-2} and R_n^{-1}, and the normalized pressure gradients converge in annular distributions. The ungauged limit therefore satisfies

$$\partial_s V + \frac{1}{2}x\cdot\nabla V + \frac{1}{2}V + \nabla Q^{\mathrm{act}} = 0, \qquad \nabla\cdot V = 0,$$

on macroscopic annular cylinders. The field $W = V - c(s)$ is the corresponding affine-quotient representative; any term $\big(c'(s) + \frac{1}{2}c(s)\big)\cdot x$ is understood as an affine pressure distribution and is not used as an ordinary pressure without additional time regularity. The selected flux is nonzero, so the quotient representative has nonzero spatial quotient oscillation. Thus $(W,Q)\in\mathcal{C}_{\mathrm{cohcrit}}$ in the quotient sense of Definition 9.39. ∎

9.2. **Logarithmic escape as a hidden-scale obstruction.** The logarithmic escape alternative is not ruled out below by a separate global tail estimate. The point of the next reduction is narrower and more useful: once tight annular descendants have no hidden-scale variance, a nonzero logarithmic escape component cannot survive. Thus the log-diffuse alternative and the Young alternative are both reduced to the same minimal mesoscopic-scale theorem.

Definition 9.51 (Log-diffuse recurrent critical defect)**.** A realized recurrent critical-tail defect is log-diffuse if, after projecting the normalized diffuse oscillation measures to the logarithmic radial variable $\rho = \log|y|$, their log marginals λ_n satisfy

$$\sup_{\rho_0 \in \mathbb{R}} \lambda_n([\rho_0 - A, \rho_0 + A]) \longrightarrow 0 \qquad \text{for every fixed } A < \infty.$$

Equivalently, the critical-tail compactification has a nonzero logarithmic escape component Λ. This alternative is denoted by $\mathcal{C}_{\log\mathrm{diff}}$.

Definition 9.52 (Log-window normalized support profile)**.** Let λ_n be the normalized logarithmic defect marginals of a log-diffuse critical defect. A log-window normalized support profile, also called a log-window support profile, is obtained by choosing numbers $A \geq 1$, centers ρ_n, and intervals

$$I_n = [\rho_n - A, \rho_n + A]$$

with $\lambda_n(I_n) > 0$, restricting the defect measures to I_n, and renormalizing:

$$\widetilde{\lambda}_n := \frac{\lambda_n \lfloor I_n}{\lambda_n(I_n)}.$$

After applying the logarithmic recentering $\rho \mapsto \rho - \rho_n$, the renormalized measures are tight in the fixed interval $[-A, A]$. Any critical-tail profile obtained as a subsequential limit of these normalized restrictions is called a log-window normalized support profile of the original log-diffuse defect. It is a retained descendant only if a separate unit velocity lower bound is proved.

Lemma 9.53 (Nonzero logarithmic escape has tight log-window support profiles)**.** *Let $\mathcal{D} \in \mathcal{C}_{\log\mathrm{diff}}$ be a nonzero realized recurrent critical-tail defect after concentration-set removal and affine normalization. Then, for some fixed $A \geq 1$, $\mathcal{D}$ has a nonzero log-window normalized support profile. This profile is a tight macroscopic annular critical-tail realization and retains normalized spatial quotient oscillation mass.*

Proof. The logarithmic marginals λ_n are probability measures. Fix any $A \geq 1$. Cover $\mathbb{R}_\rho$ by the intervals

$$I_{k,A} := [2Ak - A, 2Ak + A], \qquad k \in \mathbb{Z}.$$

For each n, at least one interval has positive λ_n-mass. Choose one such interval and denote it by $I_n = [\rho_n - A, \rho_n + A]$. Restricting to I_n, renormalizing, and translating ρ_n to the origin gives probability measures supported in the fixed compact interval $[-A, A]$. Compactness of the observer boundary profile class and the local Seregin compactness of $\mathfrak{X}_M$ give a subsequential limit. By construction, the limiting profile has total normalized defect mass one, hence is nonzero.

The logarithmic recentering fixes the selected multiplicative annulus, so the resulting critical-tail realization is tight in logarithmic scale. The original diffuse defect was measured after affine normalization by the oscillation defect measure; the normalized restriction preserves total oscillation mass equal to one. Hence the normalized support profile carries nonzero spatial quotient oscillation. No positive lower bound for the unnormalized mass $\lambda_n(I_n)$ is required, because the profile is a normalized boundary profile. $\square$

Lemma 9.54 (Support transfer for log-window defect compactification)**.** *Let $X_{\log}$ be the compact metrizable log-window defect space obtained by restricting the critical-tail compactification $\mathfrak{T}_M$ to normalized logarithmic windows, and let*

$$\lambda_n = \frac{\Lambda_n \lfloor I_n}{\Lambda_n(I_n)} \qquad (\Lambda_n(I_n) > 0)$$

be normalized realized log-window restrictions. If $\lambda_n \rightharpoonup \lambda \in \mathcal{P}(X_{\log})$ and $A \subset X_{\log}$ is closed, then support of λ in A removes all surviving log-window defect mass outside A; any positive surviving mass outside A gives a normalized log-window support profile outside A.

Proof. Apply Lemma 1.14 in the compact metrizable space $X_{\log}$. The normalization by $\Lambda_n(I_n)$ is exactly the realized support-profile normalization; the size of $\Lambda_n(I_n)$ is irrelevant after passing to probability measures. □

Lemma 9.55 (Renormalized log-window heredity)**.** *Let $\mathcal{D} \in \mathcal{C}_{\log \text{diff}}$ be a log-diffuse defect realized from the original blow-up sequence, and let $\mathcal{F} \subset \mathfrak{T}_M$ be a closed union of alternatives which is closed under retained descendants and normalized support profiles. Suppose that every nonzero log-window normalized support profile of $\mathcal{D}$ belongs to $\mathcal{F}$. Then the original log-diffuse defect itself has no surviving mass outside $\mathcal{F}$. In particular, if $\mathcal{F}$ has already been excluded by the lower-stratum closure theorems, then $\mathcal{D}$ is impossible.*

Proof. Let Λ be the compactified logarithmic defect measure associated with $\mathcal{D}$. Suppose that $\Lambda(\mathfrak{T}_M \setminus \mathcal{F}) > 0$. Since $\mathcal{F}$ is closed, choose a compact set

$$K \subset \mathfrak{T}_M \setminus \mathcal{F}$$

with $\Lambda(K) > 0$. Pick a support point $x_* \in K \cap \operatorname{supp} \Lambda$. By the definition of support in the compact log-window defect space and by Lemma 9.54, there are logarithmic windows I_n with $\Lambda_n(I_n) > 0$ whose normalized restrictions converge to a support profile represented by x_*. This limit profile lies in $\mathfrak{T}_M \setminus \mathcal{F}$, contradicting the hypothesis that every nonzero log-window normalized support profile of $\mathcal{D}$ belongs to $\mathcal{F}$. Hence Λ is supported on $\mathcal{F}$. Since $\mathcal{F}$ is closed under retained descendants and normalized support profiles, the original log-diffuse defect has no surviving mass outside $\mathcal{F}$. □

Theorem 9.56 (Logarithmic escape reduces to hidden-scale compactness)**.** *Assume Theorem 9.88. Let $\mathcal{D} \in \mathcal{C}_{\log \text{diff}}$ be a nonzero realized log-diffuse critical-tail defect generated after concentration-set removal and affine normalization. Then every nonzero log-window normalized support profile of $\mathcal{D}$ belongs to*

$$\mathcal{C}_{\text{conc}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Consequently, after the concentration, affine/parasitic, and coherent critical-tail alternatives have been removed, no log-diffuse alternative remains.

Proof. By Lemma 9.53, nonzero tight log-window normalized support profiles exist. Let $\widetilde{\mathcal{D}}$ be any such profile. If this support profile regenerates a positive-threshold unit concentrating observer, then it belongs to $\mathcal{C}_{\text{conc}}$. If its spatial quotient oscillation vanishes, then Theorem 8.21 places it in $\mathcal{C}_{\text{aff}}$. It remains to consider the case in which it is neither concentrating nor affine.

Because $\widetilde{\mathcal{D}}$ was obtained by restriction to a fixed logarithmic window, its logarithmic escape component is zero. The only remaining possible obstructions in the critical-tail compactification are a non-Dirac annular Young defect or a nonzero pressure/viscous defect coordinate. By Theorem 9.88, every tight annular blow-down generated by the descendant has no hidden-scale variance. Hence Lemma 9.59 makes the Young measure Dirac. The pressure/viscous defect-coordinate alternatives vanish in the tight Dirac branch by Lemma 9.95, after the already closed affine, pressure/noncompact, concentration, and hidden-scale branches have been removed. In the remaining full $\mathfrak{T}_{\mathrm{coh}}$ case, the coherent extraction Corollary 9.96 then gives a nonzero coherent critical tail; thus $\widetilde{\mathcal{D}} \in \mathcal{C}_{\mathrm{cohcrit}}$.

Since $\widetilde{\mathcal{D}}$ was arbitrary, every nonzero log-window normalized support profile of $\mathcal{D}$ lies in the closed excluded set

$$\mathcal{F} := \mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

The set $\mathcal{F}$ is closed in the log-window compactification by the closedness of the positive-threshold activity observables, the closed affine-normalized oscillation functional, and the coherent-tail extraction topology used in Corollary 9.96. By Lemma 9.55, if every log-window normalized support profile of the original log-diffuse defect lies in $\mathcal{F}$, then the original defect has no surviving mass outside $\mathcal{F}$. Once the concentration, affine/parasitic, and coherent critical-tail alternatives are removed, this makes the original log-diffuse defect impossible. □

Corollary 9.57 (Log-diffuse alternative excluded by minimal mesoscopic scale reduction)**.** *Inside the retained residual decomposition,*

$$\mathcal{C}_{\mathrm{log\,diff}} = \varnothing.$$

Equivalently, after affine/parasitic lower-stratum removal, logarithmic hidden escape is excluded by the same hidden-scale compactness theorem that removes the Young alternative.

Proof. By Theorem 9.86, Theorem 9.88 applies to all tight annular descendants. Hence Theorem 9.56 reduces every nonzero log-diffuse defect to a descendant in $\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}$. The concentration alternative has been removed by terminal concentration-set extraction, the affine stratum by the routing theorem Theorem 9.99, and the coherent critical-tail alternative by Corollary 9.111. Hence no log-diffuse alternative remains. □

9.3. **Hidden-scale Young variance.**

Definition 9.58 (Hidden-scale variance)**.** Let W_n be a macroscopic critical-tail blow-down on a compact annular cylinder

$$K \Subset \{0 < |x| < \infty\} \times I.$$

For $0 < \delta < 1$, set

$$\mathfrak{H}_K(W_n, \delta) := \sup_{\substack{|h|<\delta \\ |\theta|<\delta}} \|W_n(\cdot + h, \cdot + \theta) - W_n\|_{L^3(K)},$$

where the translated functions are evaluated on a fixed slightly larger annular cylinder containing all $(x + h, s + \theta)$ with $(x, s) \in K$ and $|h| + |\theta| < \delta$. The sequence has no hidden-scale variance on K if

$$\lim_{\delta \downarrow 0} \limsup_{n\to\infty} \mathfrak{H}_K(W_n, \delta) = 0.$$

Because the blow-down is spatial/logarithmic rather than parabolic, the time shift θ is measured in the centered variable itself; only the spatial displacement can later be magnified into a mesoscopic spatial radius by multiplication with R_n. It has no hidden-scale variance if this holds on every compact macroscopic annular cylinder.

Lemma 9.59 (Hidden-scale variance is the Young obstruction)**.** *Let W_n be uniformly bounded in L^∞_{loc} on macroscopic annular cylinders and suppose*

$$W_n \rightharpoonup^* \bar{W} \qquad \text{in } L^\infty_{\mathrm{loc}}.$$

On a compact annular cylinder K, the following are equivalent:

(i) *$W_n \to \bar{W}$ strongly in $L^3(K)$;*

(ii) *the generated Young measure is Dirac on K:*

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x, s) \in K;$$

(iii) *W_n has no hidden-scale variance on K.*

Consequently the non-Dirac Young-measure subcase of a realized Young/defect critical tail is precisely a failure of the hidden-scale variance condition on some compact annular cylinder; the pressure/viscous subcases are closed separately by Lemma 9.95.

Proof. The uniform $L^\infty(K)$ bound gives a uniform $L^3(K)$ bound. The Fréchet–Kolmogorov compactness criterion on $L^3(K)$, applied after extending the functions to a fixed slightly larger cylinder, says that this bounded family is relatively compact in $L^3(K)$ if and only if the translation modulus in Definition 9.58 tends to zero. Thus (iii) is equivalent to relative compactness in $L^3(K)$. Any strong subsequential limit must agree with the weak-* limit $\bar W$, so relative compactness is equivalent to strong convergence to $\bar W$. This proves (i)⇔(iii).

Strong $L^3(K)$ convergence implies convergence in measure, hence the associated Young measure is $\delta_{\bar W}$. Conversely, if the generated Young measure is Dirac and $\{W_n\}$ is uniformly bounded in $L^\infty(K)$, then $W_n \to \bar W$ in measure on K. Uniform boundedness gives uniform integrability of $|W_n - \bar W|^3$, so Vitali convergence yields strong $L^3(K)$ convergence. Hence (i)⇔(ii). □

Definition 9.60 (Multiscale affine oscillation)**.** Let (U_n, Π_n) be a terminal sequence. For a covariant observer $z = (\xi, \sigma)$ and $r \geq 1$, define the critical-tail cylinder

$$\mathcal{Q}^{\mathrm{cov}}_{r,\mathrm{ct}}(z) := \mathcal{R}_z\big(B_r(0) \times (-1,1)\big).$$

Define

$$\mathcal{A}_n(z,r) := r^{-3}\,\mathrm{osc}_{3,\mathrm{sp}}\big(\mathcal{R}_z U_n; B_r(0) \times (-1,1)\big).$$

The exponent 3 is the spatial dimension; the centered time interval is kept at fixed size because critical-tail blow-downs are spatial/logarithmic blow-downs, not parabolic rescalings. Allowing the minimizing constant to depend on time removes the harmless spatially homogeneous affine mode. Unit concentration-set removal is formulated with $\mathrm{osc}_{3,\mathrm{ct}}$; after the affine/parasitic quotient has been applied, the mesoscopic residual analysis is formulated with $\mathrm{osc}_{3,\mathrm{sp}}$, and $\sup_z \mathcal{A}_n(z,1)$ is below the retained spatial-oscillation threshold on the residual exterior window, after the fixed cover constants are absorbed into the threshold.

Definition 9.61 (Minimal mesoscopic oscillation scale)**.** Let $R_n \to \infty$ be the macroscopic scale of a tight critical-tail blow-down. A minimal mesoscopic oscillation scale is a sequence of covariant centers z_n, radii r_n, and a number $\varepsilon_0 > 0$ such that

$$1 \ll r_n \ll R_n, \qquad \mathcal{A}_n(z_n, r_n) \geq \varepsilon_0,$$

and, after replacing r_n by a dyadic comparable scale if necessary,

$$\sup_z \sup_{1 \leq r \leq r_n/2} \mathcal{A}_n(z,r) < \varepsilon_0/2$$

on the selected residual exterior window. Thus all smaller retained subscales are affine-flat, while the scale r_n is the first scale at which non-affine oscillation is visible.

Lemma 9.62 (Finite-overlap control of affine oscillation)**.** *Let $I \Subset \mathbb{R}$, let $f \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times I)$ with $\|f\|_{L^\infty} \leq M$, and let*

$$B_1(a_0), B_1(a_1), \ldots, B_1(a_N)$$

be a finite chain of unit balls such that

$$|B_1(a_j) \cap B_1(a_{j+1})| \geq \kappa > 0 \qquad (j = 0, \ldots, N-1).$$

Assume that for each j there exists $c_j \in L^\infty(I;\mathbb{R}^3)$, $|c_j| \leq M$, such that

$$\int_I \int_{B_1(a_j)} |f(y,s) - c_j(s)|^3\,dy\,ds \leq \delta.$$

Then, for $0 \le j,k \le N$,

$$\int_I |c_j(s) - c_k(s)|^3\,ds \le C(N,\kappa)\delta, \tag{9.8}$$

and

$$\int_I \int_{B_1(a_j)} |f(y,s) - c_k(s)|^3\,dy\,ds \le C(N,\kappa)\delta. \tag{9.9}$$

Consequently, if all unit cylinders in a bounded chain have affine-normalized oscillation $o(1)$*, then all their local affine gauges agree up to* $o(1)$ *in* $L^3(I)$.

Proof. For neighboring indices put $E_j = B_1(a_j) \cap B_1(a_{j+1})$. Since $|E_j| \ge \kappa$, the pointwise inequality

$$|c_j(s) - c_{j+1}(s)|^3 \le 4\kappa^{-1} \int_{E_j} \big(|f(y,s) - c_j(s)|^3 + |f(y,s) - c_{j+1}(s)|^3\big)\,dy$$

holds for a.e. s. Integrating in s gives

$$\int_I |c_j - c_{j+1}|^3\,ds \le C_\kappa \delta.$$

Summing along the chain and using

$$\left|\sum_{\ell=j}^{k-1} (c_\ell - c_{\ell+1})\right|^3 \le |j-k|^2 \sum_{\ell=j}^{k-1} |c_\ell - c_{\ell+1}|^3$$

proves (9.8). Finally,

$$|f - c_k|^3 \le 4|f - c_j|^3 + 4|c_j - c_k|^3$$

on $B_1(a_j) \times I$, and (9.9) follows from (9.8). □

Lemma 9.63 (Spacetime finite-overlap translation control)**.** *Let* $K \Subset \mathbb{R}^3 \times I$*, let* $(h_n, \theta_n) \to 0$*, and let* f_n *be uniformly bounded in* L^∞ *on a fixed neighborhood of*

$$K \cup \big(K + (h_n,\theta_n)\big).$$

Assume that every unit covariant cylinder in a finite spacetime chain covering the segment joining $(x,s) \in K$ *to* $(x + h_n, s + \theta_n)$ *has spatial quotient oscillation* $o(1)$*. Then there are spatially homogeneous functions* $a_n, b_n \in L^\infty(I;\mathbb{R}^3)$ *such that*

$$\|f_n - a_n\|_{L^3(K)} + \|f_n(\cdot + h_n, \cdot + \theta_n) - b_n\|_{L^3(K)} = o(1),$$

and

$$\|f_n(\cdot + h_n, \cdot + \theta_n) - f_n - (b_n - a_n)\|_{L^3(K)} = o(1).$$

Consequently, if the translated difference has a positive $L^3(K)$ *lower bound while no unit concentrating observer is present, then that lower bound is carried by the spatially homogeneous time-dependent mode* $b_n - a_n$*. In the ordered residual decomposition this descendant belongs to the affine/parasitic stratum* $\mathcal{C}_{\mathrm{aff}}$*.*

Proof. Cover K and $K + (h_n, \theta_n)$ by finitely many unit spacetime cylinders contained in the fixed neighborhood. Refine the cover so that any two successive cylinders in the chain have spatial overlap bounded below and time overlap bounded below by constants depending only on K. On each cylinder choose a near-minimizing spatially homogeneous gauge. Applying Lemma 9.62 on each time-overlap interval and summing along the finite chain shows that all gauges on the chain agree in L^3 on their common time intervals up to $o(1)$. Gluing the finitely many local gauges by this overlap relation gives global-in-space functions $a_n(s)$ on the part of the chain covering

K and $b_n(s)$ on the part covering $K+(h_n,\theta_n)$. The first displayed estimate follows from the local small oscillation and finite overlap. Subtracting the two approximations gives the second displayed estimate.

If the translated difference remains bounded below, then $\|b_n - a_n\|_{L^3(\mathrm{proj}_s K)}$ stays bounded below. The corresponding descendant has zero $\mathrm{osc}_{3,\mathrm{sp}}$ on every compact cylinder but a nonzero homogeneous time-dependent component. That is the affine/parasitic lower stratum in the ordered profile decomposition. ☐

Lemma 9.64 (Translation defect produces concentration, affine collapse, or a mesoscopic radius)**.** *Let $K \Subset \{0<|x|<\infty\}\times I$, let*

$$\widetilde{U}_n := \mathcal{R}_{(y_n,\sigma_n)}U_n, \qquad W_n(x,s) = \widetilde{U}_n(R_n x, s) - c_n(s), \qquad R_n \to \infty,$$

be a bounded macroscopic blow-down, and suppose that for some $(h_n,\theta_n)\to 0$ and $\eta_0>0$,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

Since $(h_n,\theta_n)\to 0$, the time displacement remains inside a uniformly finite chain of centered unit time windows. Only the spatial displacement is magnified by the spatial/logarithmic blow-down. Set

$$\ell_n := R_n|h_n|.$$

Then, after passing to a subsequence, exactly one of the following alternatives is selected by the ordered priority rule.

(i) *A positive-threshold unit concentrating observer is regenerated.*
(ii) *The defect is carried only by a spatially homogeneous time-dependent affine mode; equivalently the descendant is assigned to $\mathcal{C}_{\mathrm{aff}}$.*
(iii) *$1 \ll \ell_n \ll R_n$, and there are covariant centers z_n, radii $r_n \sim \ell_n$, and $\varepsilon_0 = \varepsilon_0(\eta_0,K)>0$ such that*

$$\mathcal{A}_n(z_n,r_n) \geq \varepsilon_0.$$

Proof. Since $(h_n,\theta_n)\to 0$, the time displacement remains inside a uniformly finite chain of centered unit time windows, while $\ell_n/R_n = |h_n| \to 0$. First assume ℓ_n is bounded. The two points compared in the original centered variables are then connected, on the compact set K, by a uniformly finite spacetime chain of covariant unit cylinders. If a unit cylinder in that chain has a positive constant-in-time oscillation lower bound, alternative (i) holds. Otherwise the spatial quotient oscillation on every cylinder in the chain tends to zero. By Lemma 9.63, the displayed translation defect can persist only through a spatially homogeneous time-dependent mode. This is alternative (ii). Thus, if neither (i) nor (ii) occurs, $\ell_n\to\infty$.

In that remaining case $1 \ll \ell_n \ll R_n$. Pulling back a finite macroscopic cover of $K \cup (K+(h_n,\theta_n))$ gives centered cylinders with spatial radius comparable to ℓ_n; the time shift θ_n contributes only a uniformly finite overlap in the centered time variable. If all such cylinders had spatial quotient oscillation below a sufficiently small $\varepsilon_0=\varepsilon_0(\eta_0,K)$, then Lemma 9.63, after rescaling the finite overlap constants, would again force the translated fields to differ by only an affine/parasitic time mode. This is alternative (ii), contrary to the present case. Therefore one such cylinder has $\mathcal{A}_n(z_n,r_n)\geq\varepsilon_0$ with $r_n\sim\ell_n$, proving (iii). ☐

Lemma 9.65 (Failure of hidden-scale compactness produces a minimal mesoscopic scale)**.** *Let $\widetilde{U}_n = \mathcal{R}_{(y_n,\sigma_n)}U_n$ and*

$$W_n(x,s) = \widetilde{U}_n(R_n x, s) - c_n(s)$$

be a tight macroscopic annular blow-down after concentration-set removal. If W_n fails the hidden-scale variance condition on some compact annular cylinder K, then either a positive-threshold *unit concentrating observer is regenerated, the affine stratum $\mathcal{C}_{\mathrm{aff}}$ is reached by a descendant, or there is a minimal mesoscopic oscillation scale in the sense of Definition 9.61.*

Proof. Failure of hidden-scale variance gives a compact annular cylinder K, a number $\eta_0 > 0$, and translations $(h_n, \theta_n) \to 0$ such that, after passing to a subsequence,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

By Lemma 9.64, the translation defect either regenerates a unit concentrating observer, reaches $\mathcal{C}_{\mathrm{aff}}$, or gives centers z_n, radii ρ_n with $1 \ll \rho_n \ll R_n$, and $\varepsilon_0 > 0$ such that

$$\mathcal{A}_n(z_n, \rho_n) \geq \varepsilon_0.$$

In particular, pure time-translation defects are already routed to the first two alternatives and do not create a fictitious mesoscopic spatial radius. In the last case, let r_n be the smallest dyadic radius in $[1, C\rho_n]$ for which

$$\sup_z \mathcal{A}_n(z, r_n) \geq \varepsilon_0$$

on the selected residual exterior window. Unit concentration-set removal excludes $r_n = O(1)$, so $r_n \to \infty$. Since $r_n \lesssim \rho_n$ and $\rho_n / R_n \to 0$, one has $r_n / R_n \to 0$. By dyadic minimality, all smaller scales up to $r_n/2$ have oscillation below $\varepsilon_0/2$, after slightly lowering ε_0. This is a minimal mesoscopic oscillation scale. □

Definition 9.66 (Minimal-scale spatial blow-down)**.** Let z_n and r_n be a minimal mesoscopic oscillation scale. Write

$$U_n^{z_n} := \mathcal{R}_{z_n} U_n, \qquad \Pi_n^{z_n} := \mathcal{R}_{z_n} \Pi_n.$$

Choose measurable velocity gauges $c_n : (-T, T) \to \mathbb{R}^3$, with $|c_n| \leq M$, which are minimizing, or within $o(1)$ of minimizing, the affine oscillation on each compact time interval. The minimal-scale spatial blow-down is

$$V_n(x, s) := U_n^{z_n}(r_n x, s), \qquad W_n(x, s) := V_n(x, s) - c_n(s).$$

Thus V_n is the ungauged blow-down and W_n is the affine-normalized velocity used to measure non-affine oscillation. The associated scaled pressure is

$$Q_n^{\mathrm{act}}(x, s) := r_n^{-1} \Pi_n^{z_n}(r_n x, s),$$

defined modulo time-only gauges. For the affine-quotient topology one writes

$$Q_n^{\#}(x, s) := Q_n^{\mathrm{act}}(x, s) - L_n(x, s), \qquad L_n(x, s) = b_n(s) \cdot x + a_n(s),$$

but this affine subtraction is used only in the quotient topology, not inside the local energy inequality. No pointwise time derivative of c_n is required. This is not the Navier–Stokes parabolic scaling. It is the spatial/logarithmic tail blow-down adapted to the centered residual equation. The actual pressure normalization is chosen so that $\nabla_x Q_n^{\mathrm{act}}$ is the rescaled pressure-gradient term appearing in the centered blow-down equation.

Remark 9.67 (First-bad pressure and gauge convention). Throughout the minimal mesoscopic-scale argument, all distributional equations, pressure Poisson equations, and local energy inequalities are written for the ungauged field V_n and its genuine scaled pressure Q_n^{act}. The field $W_n = V_n - c_n(s)$ is used only for quotient quantities: spatial oscillation, Young measures, and variance after the homogeneous gauge sequence has either compactified in time or has been assigned to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$. Thus an "affine pressure paired with c_n" is quotient notation. It is not asserted to be an $L^{3/2}_{\mathrm{loc}}$ pressure, and no pointwise derivative $\partial_s c_n$ is used unless compactness of c_n has already been obtained.

Lemma 9.68 (Minimal-scale blow-down equation)**.** *Let $V_n, W_n, Q_n^{\mathrm{act}}, c_n$ be the minimal-scale blow-downs of Definition 9.66. On every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, V_n and W_n are divergence-free, $\|V_n\|_{L^\infty(K)} \le M$, $\|W_n\|_{L^\infty(K)} \le 2M$, and the ungauged field satisfies*

$$\partial_s V_n - r_n^{-2}\Delta V_n + \frac{1}{2}x\cdot\nabla V_n + \frac{1}{2}V_n + r_n^{-1}(V_n\cdot\nabla)V_n + \nabla Q_n^{\mathrm{act}} = 0 \tag{9.10}$$

in distributions. The affine-normalized field $W_n = V_n - c_n(s)$ is used for oscillation and Young-measure compactness. Its equation is understood only in the quotient sense made precise in Lemma 9.70; no pointwise derivative of c_n and no $L^{3/2}_{\mathrm{loc}}$ affine pressure for W_n is assumed. Moreover

$$-\Delta Q_n^{\mathrm{act}} = r_n^{-1}\partial_i\partial_j\big(V_{n,i}V_{n,j}\big) \tag{9.11}$$

in distributions, after the same gauges.

Proof. The covariant recentering $\mathcal{R}_{z_n}$ is an exact symmetry of the centered equation, so $(U_n^{z_n}, \Pi_n^{z_n})$ solves (1.2). Since centered time is not rescaled,

$$\partial_s V_n = (\partial_\tau U_n^{z_n})(r_n x, s), \qquad \Delta_y U_n^{z_n} = r_n^{-2}\Delta_x V_n,$$
$$y\cdot\nabla_y U_n^{z_n} = x\cdot\nabla_x V_n, \qquad (U_n^{z_n}\cdot\nabla_y)U_n^{z_n} = r_n^{-1}(V_n\cdot\nabla_x)V_n.$$

If $Q_n^{\mathrm{act}} = r_n^{-1}\Pi_n^{z_n}(r_n x, s)$, then

$$\nabla_x Q_n^{\mathrm{act}} = (\nabla_y \Pi_n^{z_n})(r_n x, s).$$

Substituting these identities gives the scaled equation for V_n. We do not treat $W_n = V_n - c_n(s)$ as carrying an ordinary pressure unless the gauges have additional time regularity. Instead, all PDE identities below are first written for the ungauged field V_n, and the passage to W_n is made only after quotienting out spatially homogeneous modes. Taking the divergence of the ungauged scaled equation gives (9.11); the linear drift has zero divergence contribution because V_n is divergence-free. The L^∞ bounds follow from $\|U_n\|_{L^\infty} \le M$ and $|c_n| \le M$. □

Lemma 9.69 (Local energy inequality under spatial/logarithmic blow-down)**.** *Let (U_n, Π_n) be locally suitable centered profiles realized from the original blow-up sequence in the sense of Definition 1.11, let z_n be covariant centers, and let $r_n \to \infty$. Define*

$$V_n(x,s) = U_n^{z_n}(r_n x, s), \qquad Q_n^{\mathrm{act}}(x,s) = r_n^{-1}\Pi_n^{z_n}(r_n x, s),$$

Then $(V_n, Q_n^{\mathrm{act}})$ satisfy the exact local energy inequality under spatial/logarithmic blow-down on every relatively compact spacetime domain $K \Subset \mathbb{R}^3 \times \mathbb{R}$: for every nonnegative $\phi \in C_c^\infty(K)$,

$$\begin{aligned} r_n^{-2}\iint |\nabla V_n|^2\phi\,dx\,ds &\le \iint \frac{|V_n|^2}{2}\Big(\partial_s\phi + r_n^{-2}\Delta\phi + \frac{1}{2}x\cdot\nabla\phi + \frac{1}{2}\phi\Big)\,dx\,ds \\ &\quad + r_n^{-1}\iint \frac{|V_n|^2}{2}V_n\cdot\nabla\phi\,dx\,ds + \iint Q_n^{\mathrm{act}}V_n\cdot\nabla\phi\,dx\,ds. \end{aligned} \tag{9.12}$$

Moreover, exactly one of the following alternatives holds:

(i) *a retained affine-pressure descendant or pressure/noncompact defect is generated and the sequence is routed to the corresponding lower stratum;*
(ii) *after time-only pressure gauges, the non-affine branch satisfies*

$$r_n^{-2}\iint |\nabla V_n|^2\phi\,dx\,ds \le C(M, \|\phi\|_{C^2}, |K|) \tag{9.13}$$

with a finite constant uniform in n, and the pressure flux has the stated distributional limit in the local energy inequality.

All later uses of (9.13) *are in alternative* (ii)*, after the affine and pressure/noncompact alternatives have been routed away.*

Proof. The setup paper local energy inequality (recalled inside the imported results Theorem 1.3 (3)) states that for every locally suitable centered profile (U, Π) and every nonnegative $\Phi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$,

(9.14)
$$\iint |\nabla_y U|^2 \Phi \, dy \, d\tau \le \iint \frac{|U|^2}{2} \big(\partial_\tau \Phi + \Delta_y \Phi + \tfrac{1}{2} y \cdot \nabla_y \Phi + \tfrac{1}{2}\Phi\big) \, dy \, d\tau + \iint \Big(\tfrac{|U|^2}{2} + \Pi\Big) U \cdot \nabla_y \Phi \, dy \, d\tau.$$

This identity is preserved by the exact covariant recentering $\mathcal{R}_{z_n}$, so it applies to $U_n^{z_n}$ and $\Pi_n^{z_n}$ with the same constants.

Apply (9.14) to $U_n^{z_n}, \Pi_n^{z_n}$ with the test function

$$\Phi(y, \tau) := r_n^{-3} \, \phi(y/r_n, \tau)$$

where $\phi \in C_c^\infty(K)$ is fixed. Change variables $y = r_n x$, keeping $\tau = s$ (centered time is not rescaled). Then

$$\iint |\nabla_y U_n^{z_n}|^2 \Phi \, dy \, d\tau = \iint |\nabla_y U_n^{z_n}(r_n x, s)|^2 r_n^{-3} \phi(x, s) \, r_n^3 \, dx \, ds = r_n^{-2} \iint |\nabla_x V_n|^2 \phi \, dx \, ds,$$

because $|\nabla_y U_n^{z_n}|^2(r_n x, s) = r_n^{-2} |\nabla_x V_n|^2(x, s)$. Similarly,

$$\iint \frac{|U_n^{z_n}|^2}{2} \partial_\tau \Phi \, dy \, d\tau = \iint \frac{|V_n|^2}{2} \partial_s \phi \, dx \, ds,$$
$$\iint \frac{|U_n^{z_n}|^2}{2} \Delta_y \Phi \, dy \, d\tau = r_n^{-2} \iint \frac{|V_n|^2}{2} \Delta_x \phi \, dx \, ds,$$
$$\iint \frac{|U_n^{z_n}|^2}{2} \tfrac{1}{2} y \cdot \nabla_y \Phi \, dy \, d\tau = \iint \frac{|V_n|^2}{2} \tfrac{1}{2} x \cdot \nabla_x \phi \, dx \, ds,$$
$$\iint \frac{|U_n^{z_n}|^2}{2} \tfrac{1}{2} \Phi \, dy \, d\tau = \iint \frac{|V_n|^2}{2} \tfrac{1}{2} \phi \, dx \, ds.$$

For the convective and pressure-flux terms,

$$\iint \Big(\tfrac{|U_n^{z_n}|^2}{2} + \Pi_n^{z_n}\Big) U_n^{z_n} \cdot \nabla_y \Phi \, dy \, d\tau = r_n^{-1} \iint \tfrac{|V_n|^2}{2} V_n \cdot \nabla_x \phi \, dx \, ds + \iint Q_n^{\mathrm{act}} V_n \cdot \nabla_x \phi \, dx \, ds,$$

using $Q_n^{\mathrm{act}} = r_n^{-1} \Pi_n^{z_n}(r_n x, s)$, $\nabla_y \Phi = r_n^{-4} \nabla_x \phi(\cdot/r_n, \cdot)$, and $dy = r_n^3 dx$. Substituting all changes into (9.14) gives (9.12).

The exact inequality (9.12) holds before routing. To obtain the uniform bound, use the affine pressure coefficient dichotomy for spatial/logarithmic blow-downs, Lemma 9.75. If a retained affine force or pressure/noncompact component occurs, we are in alternative (i). Otherwise, in alternative (ii), $Q_n^{\mathrm{act}} = q_n^{\mathrm{na}} + a_n(s) + o(1)$ with the affine spatial coefficient vanishing and the non-affine part converging strongly to zero after time-only gauges. The time-only gauge contributes nothing to the pressure flux because $\nabla \cdot V_n = 0$. The remaining right-hand side is then bounded by the L^∞ bound $|V_n| \le M$, by $r_n^{-1} \le 1$ and $r_n^{-2} \le 1$ for all large n, and by Hölder's inequality on the compact support of ϕ, with a constant depending only on M, $\|\phi\|_{C^2}$, and $|K|$. Since $r_n \to \infty$, the viscous and convective terms carry the vanishing factors r_n^{-2} and r_n^{-1}. After passing to the time-only pressure-gauge subsequence and to the bounded-velocity subsequence, the remaining drift and pressure-flux terms pass to their distributional limits. This applies both to minimal mesoscopic blow-downs and to macroscopic critical-tail blow-downs, with $r_n = R_n$ in the latter case. □

Lemma 9.70 (Minimal-scale quotient formulation)**.** *Let $V_n, W_n, c_n, Q_n^{\mathrm{act}}, Q_n^{\#}$ be as in Lemma 9.68. The minimal-scale analysis uses the following quotient formulation.*

(i) *The distributional equation and local energy inequality are always applied to the ungauged field V_n and its actual pressure Q_n^{act}, modulo time-only gauges.*
(ii) *The affine-normalized field $W_n = V_n - c_n(s)$ is used only for oscillation functionals and Young measures, which are invariant under spatially homogeneous shifts after the gauge sequence is either compact or assigned to $\mathcal{C}_{\mathrm{aff}}$.*
(iii) *The affine-quotient pressure $Q_n^{\#} = Q_n^{\mathrm{act}} - L_n$, with $L_n(x, s) = b_n(s) \cdot x + a_n(s)$, is used only in the quotient topology after the affine component has either been routed to $\mathcal{C}_{\mathrm{aff}}$ or discarded from the non-affine quotient description.*
(iv) *If c_n is not compact in L^3_{loc} in time, the homogeneous gauge sequence itself generates an affine/parasitic descendant. If $c_n \to c$ strongly in L^3_{loc}, all Young-measure variances and strong compactness statements pass from V_n to W_n by deterministic translation of the generated Young measures.*

Thus the symbol "affine pressure paired with c_n" means a quotient functional, not an asserted $L^{3/2}_{\mathrm{loc}}$ pressure inside the local energy inequality unless the time regularity of c_n has been separately proved.

Proof. Statement (i) is the construction in Lemma 9.68. The local energy inequality is stable under the minimal-scale spatial blow-down for V_n, because V_n is obtained from an actual centered solution by an exact covariant recentering and a spatial scaling; no subtraction of c_n is involved.

For (ii) and (iii), observe that subtracting a spatially homogeneous function does not change any $\mathrm{osc}_{3,\mathrm{sp}}$-type seminorm in which the same class of homogeneous gauges is minimized over. If the sequence c_n is not compact in L^3_{loc}, Lemma 9.79 places the resulting homogeneous descendant in $\mathcal{C}_{\mathrm{aff}}$. If $c_n \to c$ strongly, then Lemma 9.80 identifies the Young measure of W_n with the deterministic translate of the Young measure of V_n, and the variance densities agree. Hence all compactness and variance conclusions used for W_n follow from identities proved for V_n, without requiring $\partial_s c_n$ to be a function. □

Lemma 9.71 (Covariance of the whole-space Riesz pressure)**.** *At the finite-energy Leray level one may choose the standard whole-space pressure representative*

$$p(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_i u_j)(\cdot, t)$$

modulo a function of time. This representative is preserved, modulo the corresponding time gauge, by physical Type I scaling, the centered self-similar change of variables, centered time translation, spatial translation, and the covariant translations used in this paper. Local pressure representatives are restrictions of this whole-space representative, followed only by the admissible addition of functions of time.

Proof. Let K_{ij} be the Calderón–Zygmund pressure kernel. It is homogeneous of degree -3:

$$K_{ij}(\lambda x) = \lambda^{-3} K_{ij}(x), \qquad \lambda > 0.$$

Therefore, if

$$u^\lambda(x, t) = \lambda u(x_0 + \lambda x, t_0 + \lambda^2 t), \qquad p^\lambda(x, t) = \lambda^2 p(x_0 + \lambda x, t_0 + \lambda^2 t),$$

then

$$p^\lambda(\cdot, t) = \mathcal{R}_i \mathcal{R}_j (u_i^\lambda u_j^\lambda)(\cdot, t)$$

modulo the scaled time gauge. Spatial translations commute with convolution by K_{ij}, and time translations only relabel the time variable. The centered change of variables is a time-dependent parabolic scaling with the pressure factor $P(y, \tau) = (-t)p(x, t)$; hence the same homogeneity gives the centered Riesz representative at each fixed centered time. The covariant translations used below are compositions of these exact spatial/time changes and the admissible pressure gauges recorded in the centered equation. Since all additional gauges are functions of time, their gradients vanish and their restrictions to compact cylinders are precisely the local pressure gauges used in the Seregin topology. □

Lemma 9.72 (Harmonic remainders in normalized pressure limits with blow-down coefficient)**.** *Let (U_n, Π_n) be a sequence of normalized profiles obtained from the whole-space representative in Lemma 9.71, before applying any homogeneous velocity quotient. Suppose that, before the affine quotient, each pressure representative has the form*

$$\Pi_n(\cdot, s) = \kappa_n \mathcal{R}_i \mathcal{R}_j (U_{n,i} U_{n,j})(\cdot, s) + \alpha_n(s),$$

where $\kappa_n \to \kappa \in \{0, 1\}$, with $\kappa = 1$ for natural-scale profiles and $\kappa = 0$ for spatial/logarithmic blow-downs. If $U_n \to U$ locally smoothly and, after time-dependent gauges, $\Pi_n \to \Pi$ locally in $L^{3/2}$ or weakly in $L^{3/2}$, then

$$\Pi(\cdot, s) - \kappa \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

is a time-dependent constant. After the affine/parasitic quotient is applied, the only additional admissible harmonic pressure direction is a linear function $\beta(s) \cdot y$; this linear part is the pressure paired with the spatially homogeneous parasitic velocity mode and is absent in the parasitic-free representative.

Proof. Before the affine quotient, set

$$P_n^*(\cdot, s) := \kappa_n \mathcal{R}_i \mathcal{R}_j (U_{n,i} U_{n,j})(\cdot, s).$$

Since the velocities are uniformly bounded in L^∞, these representatives have uniformly bounded BMO seminorms on $\mathbb{R}^3$:

$$\|P_n^*(\cdot, s)\|_{\mathrm{BMO}} \le CM^2.$$

The BMO bound is stable under the exact scalings and translations in Lemma 9.71, and $|\kappa_n| \le 1$. Hence, after passing to a subsequence at each fixed time interval and using a diagonal extraction, the representatives P_n^* converge weakly-* in local BMO-duality to a whole-space BMO pressure P^*, modulo time gauges. The local $L^{3/2}$-pressure limit Π agrees with P^* on each compact cylinder, up to the same time-dependent gauge, because both are local limits of the same normalized representatives. Passing to the limit in the pressure Poisson equation gives

$$-\Delta P^*(\cdot, s) = \kappa\, \partial_i \partial_j (U_i U_j)(\cdot, s)$$

in distributions. Thus the difference

$$h(\cdot, s) := P^*(\cdot, s) - \kappa \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

is harmonic in space and belongs to $\mathrm{BMO}(\mathbb{R}^3)$ modulo an additive constant. An entire harmonic BMO function is constant: indeed, for each ball $B_R(x)$, the interior estimate for harmonic functions and the John–Nirenberg control of mean oscillation give

$$|\nabla h(x, s)| \le CR^{-1} \|h(\cdot, s)\|_{\mathrm{BMO}(\mathbb{R}^3)}$$

with C independent of R; letting $R \to \infty$ yields $\nabla h(\cdot, s) = 0$. Since Π and P^* differ only by the local time gauge, before affine quotienting the difference between Π and $\kappa \mathcal{R}_i \mathcal{R}_j (U_i U_j)$ is spatially constant.

Equivalently, if a non-affine harmonic component were present on some compact ball and time interval, elliptic regularity would give a point where a nonzero spatial derivative of order at least one remains. Testing against compactly supported functions with zero spatial mean, and with zero first moments when isolating the affine quotient, would detect this component.

Such a component cannot arise from the uniformly BMO whole-space Riesz representatives plus time-only gauges described in Lemma 9.71.

The affine/parasitic normalization is the only subsequent operation that changes the velocity by a spatially homogeneous function. For a smooth homogeneous mode $b(s)$, substitution into the centered equation gives the affine pressure gradient

$$\nabla_y\big[(-\partial_s b - \tfrac{1}{2}b)\cdot y\big].$$

For merely bounded measurable homogeneous gauges this expression is used only as the quotient functional specified in Definition 8.16, Remark 9.67, and Lemma 9.70. Thus the normalized profile class contains no non-affine harmonic pressure remainders: only time gauges and, inside $\mathcal{C}_{\mathrm{aff}}$, the explicit affine parasitic pressure direction are allowed. □

Proposition 9.73 (Normalized pressure representation with blow-down coefficient)**.** *Every normalized pressure representative used in the residual profile class has the form*

$$\Pi(\cdot,s) = \kappa\,\mathcal{R}_i\mathcal{R}_j(U_iU_j)(\cdot,s) + \beta(s)\cdot y + \alpha(s), \tag{9.15}$$

where α is a time gauge, $\beta(s)\cdot y$ is the affine pressure paired with the spatially homogeneous parasitic velocity mode, and the scalar κ is determined by the operation which produced the profile.

For natural-scale Seregin profiles, terminal profiles, exact covariant descendants, and genuine parabolic concentrating descendants, $\kappa = 1$. For finite spatial/logarithmic blow-down representatives, the non-affine Riesz part carries coefficient r_n^{-1} or R_n^{-1}. For their limiting blow-down profiles, $\kappa = 0$. Outside the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$, the representative is taken with $\beta = 0$. In particular, for spatial/logarithmic blow-downs the canonical non-affine Riesz part carries coefficient r_n^{-1} or R_n^{-1} and therefore vanishes in the limit. The actual pressure representative used in the local energy inequality may still contain a spatially affine harmonic component. Such an affine component is not subtracted inside the local energy inequality; it is either shown to vanish by the affine-coefficient dichotomy or routed to $\mathcal{C}_{\mathrm{aff}}$.

Proof. At natural scale the coefficient is 1 by the whole-space Riesz pressure representative and its covariance, recorded in Lemma 9.71. The exclusion of non-affine harmonic pressure remainders in normalized limits, with the limiting coefficient tracked explicitly, is Lemma 9.72. Under the spatial/logarithmic blow-down $x \mapsto r_n x$, the actual pressure is normalized as

$$Q_n^{\mathrm{act}}(x,s) = r_n^{-1}\Pi_n(r_n x, s),$$

so the non-affine Riesz part is

$$r_n^{-1}\mathcal{R}_i\mathcal{R}_j(V_{n,i}V_{n,j})(x,s).$$

The same formula with R_n^{-1} holds for macroscopic critical-tail blow-downs. The BMO bound for the Riesz representative gives local $L^{3/2}$-compactness modulo constants, and the prefactor tending to zero forces the non-affine Riesz part to vanish on compact blow-down windows. The only possible surviving harmonic direction is affine, and that direction is precisely the affine/parasitic quotient. This yields (9.15) with the stated values of κ. □

Lemma 9.74 (Scaled pressure compactness modulo affine gauges)**.** *Let (U_n, Π_n) be bounded centered solutions with $\|U_n\|_{L^\infty} \le M$, let z_n be covariant centers, let $r_n \to \infty$, and set*

$$\widetilde{U}_n(y,s) := (\mathcal{R}_{z_n}U_n)(y,s), \qquad \widetilde{\Pi}_n(y,s) := (\mathcal{R}_{z_n}\Pi_n)(y,s).$$

For

$$\widetilde{Q}_n(x,s) := r_n^{-1}\widetilde{\Pi}_n(r_n x, s)$$

and every compact cylinder $K = B_R \times I \Subset \mathbb{R}^3 \times \mathbb{R}$, there are affine functions

$$L_n(x,s) = b_n(s)\cdot x + a_n(s)$$

such that the non-affine pressure part

$$q_n^{\mathrm{na}} := \widetilde{Q}_n - L_n$$

satisfies, for every finite $1 \le p < \infty$,

$$\|q_n^{\mathrm{na}}\|_{L^p(K)} \le C_{p,K,M} r_n^{-1}. \tag{9.16}$$

In particular,

$$\widetilde{Q}_n - L_n \to 0 \qquad \text{strongly in } L^p(K) \quad \text{for every finite } p. \tag{9.17}$$

The same statement holds for the actual minimal-scale pressures Q_n^{act}; after subtracting the affine gauges, it becomes the quotient statement for $Q_n^{\#}$ in the sense of Lemma 9.70. The same finite-p non-affine estimate also holds for macroscopic critical-tail actual pressures $\widehat{Q}_n^{\mathrm{act}}$, with r_n replaced by the critical-tail scale R_n, on every fixed annular cylinder. If the affine parts L_n carry a nonzero retained descendant, then that descendant lies in the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$. Otherwise the affine parts may be discarded in the affine-normalized quotient, and the non-affine pressure converges strongly to zero.

Proof. For each n, the covariant recentering is an exact symmetry, so $(\widetilde{U}_n, \widetilde{\Pi}_n)$ is again a bounded centered solution with $\|\widetilde{U}_n\|_{L^\infty} \le M$. For a.e. s, the pressure Poisson equation is

$$-\Delta \widetilde{\Pi}_n(\cdot, s) = \partial_i \partial_j (\widetilde{U}_{n,i} \widetilde{U}_{n,j})(\cdot, s)$$

in $\mathbb{R}^3$. Let

$$\Pi_n^{\mathrm{loc}}(\cdot, s) := \mathcal{R}_i \mathcal{R}_j (\widetilde{U}_{n,i} \widetilde{U}_{n,j})(\cdot, s)$$

be the whole-space Calderón–Zygmund representative. Since $\|\widetilde{U}_n\|_{L^\infty} \le M$,

$$\|\Pi_n^{\mathrm{loc}}(\cdot, s)\|_{\mathrm{BMO}(\mathbb{R}^3)} \le CM^2$$

uniformly in s. Choose the measurable constant representative $a_{n,\rho}(s) = (\Pi_n^{\mathrm{loc}})_{B_\rho}(s)$, the spatial average on B_ρ. Hence, by the John–Nirenberg inequality, for every ball B_ρ and every finite $1 \le p < \infty$, the same constants $a_{n,\rho}(s)$ satisfy

$$\|\Pi_n^{\mathrm{loc}}(\cdot, s) - a_{n,\rho}(s)\|_{L^p(B_\rho)} \le C_p M^2 |B_\rho|^{1/p}.$$

After the scaling $x \mapsto r_n x$, this gives

$$\|r_n^{-1} \Pi_n^{\mathrm{loc}}(r_n \cdot, s) - a_n(s)\|_{L^p(B_R)} \le C_{p,R,M} r_n^{-1} \to 0.$$

where $a_n(s) = r_n^{-1} a_{n, r_n R}(s)$. The bound is uniform for $s \in I$, hence the same measurable averages $a_n(s)$ obey, for every finite p,

$$\|r_n^{-1} \Pi_n^{\mathrm{loc}}(r_n \cdot, s) - a_n(s)\|_{L^p(B_R \times I)} \le C_{p,R,M,I} r_n^{-1} \to 0.$$

By Proposition 9.73, after the spatial blow-down the non-affine Riesz part carries the explicit coefficient r_n^{-1}, while the only possible surviving harmonic component is affine. Thus for each n there exist measurable functions $\beta_n : I \to \mathbb{R}^3$ and $\alpha_n : I \to \mathbb{R}$ such that

$$\widetilde{\Pi}_n(\cdot, s) = \Pi_n^{\mathrm{loc}}(\cdot, s) + \beta_n(s) \cdot y + \alpha_n(s)$$

on $\mathbb{R}^3$. After scaling,

$$r_n^{-1}\big(\beta_n(s) \cdot r_n x + \alpha_n(s)\big) = \beta_n(s) \cdot x + r_n^{-1} \alpha_n(s),$$

which is an affine function of x. Taking

$$L_n(x, s) := \beta_n(s) \cdot x + a_n(s) + r_n^{-1} \alpha_n(s),$$

the remaining pressure is the scaled non-affine Riesz representative already estimated above. This proves (9.16) and (9.17). The minimal-scale pressure is used only up to the affine quotient described in Lemma 9.70; in that quotient the same estimate applies. The macroscopic critical-tail statement is identical, with the spatial blow-down parameter denoted R_n, because the normalized pressure representation of Proposition 9.73 gives the same explicit non-affine coefficient R_n^{-1} on fixed annular windows. □

Lemma 9.75 (Affine pressure coefficient dichotomy for spatial/logarithmic blow-downs)**.** *Assume that on every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ the actual pressure at a spatial/logarithmic blow-down scale $r_n \to \infty$ has the affine expansion*

$$Q_n^{\mathrm{act}}(x,s) = b_n(s)\cdot x + a_n(s) + o_{L^{3/2}(K)}(1)$$

as supplied by scaled pressure compactness. Then exactly one of the following alternatives occurs:

(i) *for some compact time interval $I \Subset \mathbb{R}$,*

$$\limsup_{n\to\infty} \|b_n\|_{L^{3/2}(I)} > 0,$$

and the associated affine pressure/velocity quotient descendant is assigned to $\mathcal{C}_{\mathrm{aff}}$;

(ii) *for every compact time interval $I \Subset \mathbb{R}$,*

$$b_n \to 0 \qquad \text{in } L^{3/2}(I;\mathbb{R}^3),$$

and hence, after time-only gauges,

$$Q_n^{\mathrm{act}} - a_n(s) \to 0 \qquad \text{strongly in } L^{3/2}_{\mathrm{loc}}.$$

Proof. Scaled pressure compactness gives that, on each compact cylinder, the actual pressure differs from an affine function $b_n(s)\cdot x + a_n(s)$ by a term converging strongly to zero in $L^{3/2}$. At minimal mesoscopic scales this is Lemma 9.74; at macroscopic critical-tail scales it is the affine part of the actual/quotient pressure split in Definition 9.44. If b_n does not converge to zero on some compact time interval I, then a nontrivial affine pressure force remains on that interval. After the same affine normalization used in the spatial/logarithmic quotient, this produces a nonzero spatially homogeneous pressure-force descendant; by the affine/parasitic routing convention this descendant belongs to $\mathcal{C}_{\mathrm{aff}}$. This is alternative (i).

If alternative (i) does not occur, then for every compact interval I one must have $b_n \to 0$ in $L^{3/2}(I)$. On a compact cylinder $K = B_R \times I$,

$$\|b_n(s)\cdot x\|_{L^{3/2}(K)} \le C_R \|b_n\|_{L^{3/2}(I)} \to 0,$$

so the affine spatial part vanishes in $L^{3/2}(K)$. The remaining term is the time gauge $a_n(s)$, proving alternative (ii). □

Lemma 9.76 (Pressure–velocity product convergence)**.** *Let $K \Subset \mathbb{R}^3 \times \mathbb{R}$. Suppose*

$$Q_n \to Q \quad \text{strongly in } L^{3/2}(K), \qquad W_n \rightharpoonup^* W \quad \text{in } L^\infty(K),$$

with $\sup_n \|W_n\|_{L^\infty(K)} < \infty$. Then

$$Q_n W_n \to QW \qquad \text{in } \mathcal{D}'(K).$$

Proof. Let $\varphi \in C_c^\infty(K;\mathbb{R}^3)$. Then

$$\iint_K (Q_n W_n - QW)\cdot\varphi = \iint_K (Q_n - Q) W_n \cdot \varphi + \iint_K Q(W_n - W)\cdot\varphi.$$

The first term tends to zero by the strong $L^{3/2}$ convergence of Q_n and the uniform L^∞ bound on W_n. Since $Q\varphi \in L^1(K)$, the second term tends to zero by weak-* convergence in $L^\infty(K)$. □

Lemma 9.77 (Pressure compactness at a minimal mesoscopic scale)**.** *Let $Q_n^{\#}$ be the affine-quotient pressures associated to the actual scaled pressures Q_n^{act} from Lemma 9.68. Assume in addition the* normalized whole-space Riesz-pressure hypothesis*: each Q_n^{act} is the scaled whole-space Riesz-pressure representative, with blow-down coefficient r_n^{-1}, of the minimal-scale rescaled solution, normalized modulo time-only functions in the sense of the local pressure atlas imported in Theorem 1.3 (3); equivalently, each Q_n^{act} is the canonical scaled representative produced by Lemma 9.6 before affine subtraction, and $Q_n^{\#}$ is obtained from Q_n^{act} only after passing to the affine quotient topology. For every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, exactly one of the following holds:*

(i) *the affine pressure/velocity gauges produce a retained descendant in $\mathcal{C}_{\mathrm{aff}}$;*

(ii) *after discarding those affine gauges in the affine-normalized quotient, there exists a pressure limit Q such that*

$$Q_n^{\#} \to Q \qquad \textit{strongly in } L^{3/2}(K).$$

In fact, in the non-affine quotient one may take $Q = 0$. Consequently, if $W_n \rightharpoonup^* \bar{W}$ in L^∞_{loc}, then

$$Q_n^{\#} W_n \rightharpoonup Q\bar{W} \qquad \textit{in distributions on } K.$$

Proof. This is Lemma 9.74 applied first to the actual minimal-scale pressures Q_n^{act}, and only afterwards to the affine-quotient representatives $Q_n^{\#}$. If the affine gauges retain nonzero descendant concentration, we are in alternative (i). Otherwise the affine gauges are precisely the quotient directions removed by affine normalization, and the remaining pressure converges strongly to 0 in $L^{3/2}(K)$. Strong convergence of the pressure and weak-* convergence of the uniformly bounded velocity give the product convergence in distributions by Lemma 9.76. □

Lemma 9.78 (Actual pressure in the minimal-scale energy inequality)**.** *In the minimal-scale blow-down argument, the local energy inequality is always applied to the ungauged field V_n and the actual scaled pressure Q_n^{act}, modulo time-only gauges. Exactly one of the following holds:*

(i) *the affine pressure/velocity component produces a retained affine-parasitic descendant, hence the profile is assigned to $\mathcal{C}_{\mathrm{aff}}$;*

(ii) *after time-only gauge normalization, the actual non-affine pressure satisfies*

$$Q_n^{\mathrm{act}} - a_n(s) \to 0 \qquad \textit{strongly in } L^{3/2}_{\mathrm{loc}},$$

and the pressure flux in the local energy inequality satisfies

$$\iint Q_n^{\mathrm{act}} V_n \cdot \nabla\phi \, dx\, ds \longrightarrow 0 \qquad \textit{for every } \phi \in C_c^\infty.$$

Equivalently,

$$\mathrm{div}(Q_n^{\mathrm{act}} V_n) \to 0 \qquad \textit{in distributions}$$

in the non-affine branch.

No spatially affine pressure subtraction is used inside the local energy inequality unless its contribution has first been routed to $\mathcal{C}_{\mathrm{aff}}$.

Proof. The actual scaled pressure $Q_n^{\mathrm{act}} = r_n^{-1}\Pi_n^{z_n}(r_n\cdot, s)$ is the pressure entering (9.10) and the local energy inequality. Its non-affine part is the scaled whole-space Riesz pressure with coefficient r_n^{-1}, so the BMO/John–Nirenberg estimate in Lemma 9.74 and the affine-coefficient dichotomy of Lemma 9.75 imply exactly the required two-way alternative. If an affine pressure/velocity mode survives after the quotient normalization, then by definition that mode is routed to $\mathcal{C}_{\mathrm{aff}}$, which is alternative (i).

In the complementary case there are measurable time gauges $a_n(s)$ such that

$$Q_n^{\mathrm{act}} - a_n(s) \to 0 \qquad \text{strongly in } L^{3/2}_{\mathrm{loc}}.$$

Because V_n is uniformly bounded in L^∞_{loc}, $(Q_n^{\mathrm{act}} - a_n(s))V_n \to 0$ in distributions by Lemma 9.76. This product convergence is used only after pairing with $\nabla\phi$ in the local energy pressure flux. The time-only gauge itself does not contribute to that flux, since for every compactly supported scalar ϕ,

$$\iint a_n(s) V_n \cdot \nabla\phi = -\iint a_n(s)\phi\, \nabla \cdot V_n = 0.$$

Thus the pressure flux is determined by the actual non-affine pressure alone, and $\operatorname{div}(Q_n^{\mathrm{act}} V_n) \to 0$ distributionally in the non-affine quotient. This is alternative (ii). □

Lemma 9.79 (Homogeneous gauges: compactness modulo the affine stratum)**.** *Let $c_n : I \to \mathbb{R}^3$ be uniformly bounded in $L^\infty(I)$ on every compact interval $I \Subset \mathbb{R}$. Then, after passing to a subsequence, exactly one of the following alternatives is used in the ordered minimal-scale argument:*

(i) *$c_n \to c$ strongly in $L^3_{\mathrm{loc}}(\mathbb{R}; \mathbb{R}^3)$;*
(ii) *the spatially homogeneous sequence*

$$C_n(y, s) := c_n(s)$$

has a noncompact homogeneous quotient defect. This defect has zero spatial quotient oscillation on every compact cylinder and is assigned to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$.

Proof. If c_n has a strongly convergent subsequence in L^3_{loc}, we pass to that subsequence and are in alternative (i). If no such subsequence exists, then the family is not relatively compact in $L^3(I)$ for some compact interval $I \Subset \mathbb{R}$. Since the $L^\infty(I)$ bound gives uniform boundedness and tightness on this fixed interval, the only possible failure in the Fréchet–Kolmogorov criterion is time-translation equicontinuity. Hence there are translations $\theta_n \to 0$ and $\eta > 0$ such that

$$\|c_n(\cdot + \theta_n) - c_n\|_{L^3(I)} \geq \eta.$$

The associated spatially homogeneous profiles $C_n(y, s) = c_n(s)$ have zero spatial quotient oscillation on every compact cylinder. The displayed lower bound records noncompactness only in the homogeneous time-dependent quotient; it does not assert the existence of a strongly convergent nonzero homogeneous descendant. By the affine/parasitic routing convention and Lemma 8.18, this quotient defect belongs to $\mathcal{C}_{\mathrm{aff}}$. This is alternative (ii). □

Lemma 9.80 (Variance is invariant under homogeneous velocity gauges)**.** *Let V_n be uniformly bounded in L^∞_{loc}, let $c_n \to c$ strongly in $L^3_{\mathrm{loc}}(\mathbb{R})$, and set*

$$W_n(x, s) := V_n(x, s) - c_n(s).$$

If V_n generates a Young measure $\nu_{x,s}$ with barycenter $\bar{V}$, then W_n generates the translated Young measure

$$\widetilde{\nu}_{x,s}(A) = \nu_{x,s}(A + c(s))$$

with barycenter $\bar{W} = \bar{V} - c(s)$. The variance densities agree:

$$\langle |\xi - \bar{V}(x, s)|^2 \rangle_{\nu_{x,s}} = \langle |\xi - \bar{W}(x, s)|^2 \rangle_{\widetilde{\nu}_{x,s}}.$$

Consequently any distributional variance inequality proved for V_n passes unchanged to the affine-normalized fields W_n.

Proof. After passing to a subsequence, $c_n(s) \to c(s)$ a.e. and in measure on every compact time interval. For every continuous compactly supported function $F(\xi)$ and every compactly supported test function $\varphi(x,s)$,

$$\iint \varphi F(W_n)\,dx\,ds = \iint \varphi F(V_n - c_n(s))\,dx\,ds \to \iint \varphi \int F(\xi - c(s))\,d\nu_{x,s}(\xi)\,dx\,ds.$$

This is precisely the translated Young measure. Translation by a deterministic vector preserves centered second moments, giving the variance identity. The distributional inequality involves only this variance density and therefore is unchanged. □

Lemma 9.81 (Support transfer for the Young variance defect)**.** *Let $K \Subset \mathbb{R}^3 \times \mathbb{R}$ be a compact macroscopic cylinder and let $M_{\mathrm{tail}} = 2M$. Set $X_Y(K) = K \times \mathcal{P}(\overline{B_{M_{\mathrm{tail}}}})$, where $\overline{B_{M_{\mathrm{tail}}}}$ is the compact velocity ball for the affine-normalized fields $W_n = V_n - c_n(s)$. Let*

$$\lambda_n = \frac{\mu_n^Y \lfloor E_n}{\mu_n^Y(E_n)} \qquad (\mu_n^Y(E_n) > 0)$$

be normalized realized Young-measure variance defects on $X_Y(K)$. If $\lambda_n \rightharpoonup \lambda$ and $A \subset X_Y(K)$ is closed, then support of λ in A removes all surviving Young variance defect outside A; positive surviving variance outside A produces a normalized Young support profile outside A.

Proof. The space $X_Y(K)$ is compact metrizable because K is compact and $\mathcal{P}(\overline{B_{M_{\mathrm{tail}}}})$ is compact metrizable in the weak topology. The conclusion is Lemma 1.14 applied to this space. This is the only measure-theoretic transfer used when the hidden-scale Young defect is removed after the variance inequality. □

Lemma 9.82 (Renormalization for the coherent barycenter equation)**.** *Let $Z \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$, $Q \in L^{3/2}_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$, and $\nabla \cdot Z = 0$. Suppose*

$$\partial_s Z + \frac{1}{2} x \cdot \nabla Z + \frac{1}{2} Z + \nabla Q = 0$$

in distributions. Then

$$\partial_s \frac{|Z|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|Z|^2}{2} + \frac{1}{2}|Z|^2 + \mathrm{div}(QZ) = 0$$

in distributions.

Proof. Let ϱ_ε be a standard space–time mollifier and write $Z_\varepsilon = \varrho_\varepsilon * Z$, $Q_\varepsilon = \varrho_\varepsilon * Q$ on a compact cylinder K, after extending across a slightly larger compact cylinder. Since $x/2$ is smooth and Lipschitz on K,

$$\varrho_\varepsilon * ((x/2) \cdot \nabla Z) - (x/2) \cdot \nabla Z_\varepsilon \to 0 \qquad \text{in } L^p(K)$$

for every finite p. The mollified equation is therefore

$$\partial_s Z_\varepsilon + \frac{1}{2} x \cdot \nabla Z_\varepsilon + \frac{1}{2} Z_\varepsilon + \nabla Q_\varepsilon = R_\varepsilon, \qquad R_\varepsilon \to 0 \quad \text{in } L^p_{\mathrm{loc}}.$$

Multiplying by Z_ε gives

$$\partial_s \frac{|Z_\varepsilon|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|Z_\varepsilon|^2}{2} + \frac{1}{2}|Z_\varepsilon|^2 + Z_\varepsilon \cdot \nabla Q_\varepsilon = R_\varepsilon \cdot Z_\varepsilon.$$

Because $\nabla \cdot Z = 0$, also $\nabla \cdot Z_\varepsilon = 0$, and hence

$$Z_\varepsilon \cdot \nabla Q_\varepsilon = \mathrm{div}(Q_\varepsilon Z_\varepsilon).$$

Now $Z_\varepsilon \to Z$ in L^p_{loc} for every finite p, while $Z \in L^\infty_{\mathrm{loc}}$, and $Q_\varepsilon \to Q$ in $L^{3/2}_{\mathrm{loc}}$. Thus $Q_\varepsilon Z_\varepsilon \to QZ$ in distributions, and $R_\varepsilon \cdot Z_\varepsilon \to 0$ in distributions. Letting $\varepsilon \downarrow 0$ gives the claimed renormalized identity. □

Lemma 9.83 (Barycenter equation and variance inequality)**.** *Let $V_n, W_n, Q_n^{\mathrm{act}}, Q_n^{\#}, c_n$ be a minimal-scale blow-down, where Q_n^{act} is the actual scaled pressure entering (9.10) and $Q_n^{\#}$ is its affine-quotient representative. Then either the sequence has a retained affine/parasitic descendant in $\mathcal{C}_{\mathrm{aff}}$, or, after passing to a subsequence, $c_n \to c$ strongly in L^3_{loc},*

$$V_n \rightharpoonup^* \bar{V}, \qquad W_n \rightharpoonup^* \bar{W} \qquad \text{in } L^\infty_{\mathrm{loc}},$$

and V_n, W_n generate Young measures $\nu_{x,s}$ and $\widetilde{\nu}_{x,s}$, respectively. Set

$$E(x,s) := \langle |\xi|^2 \rangle_{\widetilde{\nu}_{x,s}}, \qquad \mathcal{V}(x,s) := E(x,s) - |\bar{W}(x,s)|^2.$$

Then $\bar{V}$ satisfies the coherent transport equation with pressure Q. The affine-normalized barycenter $\bar{W} = \bar{V} - c(s)$ satisfies the same equation only as a quotient modulo spatially homogeneous modes; after passing to the parasitic-free representative outside $\mathcal{C}_{\mathrm{aff}}$, this quotient equation is represented as

$$\partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \qquad \nabla \cdot \bar{W} = 0, \tag{9.18}$$

and the nonnegative variance density $\mathcal{V} \in L^\infty_{\mathrm{loc}}$ satisfies

$$\partial_s \mathcal{V} + \frac{1}{2} x \cdot \nabla \mathcal{V} + \mathcal{V} \le 0 \tag{9.19}$$

in distributions.

Proof. By Lemma 9.79, either the homogeneous gauge defect belongs to $\mathcal{C}_{\mathrm{aff}}$, or $c_n \to c$ strongly in L^3_{loc} after passing to a subsequence. We work in the second case. The weak-* compactness of V_n, W_n and generation of Young measures are standard consequences of the uniform L^∞_{loc} bound. Testing (9.10) against a compactly supported divergence-free vector field, the viscous term is bounded by

$$C r_n^{-2} \|V_n\|_{L^\infty},$$

and the nonlinear term by

$$C r_n^{-1} \|V_n\|^2_{L^\infty}.$$

Both tend to zero. The quotient-pressure compactness in Lemma 9.78 and the coefficient form of Proposition 9.73 give either a retained affine descendant in $\mathcal{C}_{\mathrm{aff}}$, or time gauges $a_n(s)$ such that $Q_n^{\mathrm{act}} - a_n(s) \to 0$ strongly in $L^{3/2}_{\mathrm{loc}}$. Since time gauges have zero spatial gradient, passing to the limit in (9.10) first yields the barycenter equation for $\bar{V}$ with the actual pressure representative $Q = 0$ in the non-affine quotient. Only after this step do we pass to the affine-quotient description recorded by $Q_n^{\#}$. Thus $\bar{V}$ satisfies the coherent transport equation. Since $\bar{W} = \bar{V} - c(s)$, passing from $\bar{V}$ to $\bar{W}$ changes only the spatially homogeneous quotient component. We do not differentiate c as a function. Instead, Lemma 9.70 says that this is an affine/parasitic quotient direction. If it has retained contribution, the profile is in $\mathcal{C}_{\mathrm{aff}}$; otherwise the canonical parasitic-free representative removes it, and the representative satisfies (9.18).

We next pass to the local energy inequality for the ungauged fields V_n. The appropriate form is the scaled suitable inequality of Lemma 9.69, applied to nonnegative compactly supported test functions. This avoids using a formal pointwise multiplication of the scaled equation by V_n. There is no affine pressure work term in this inequality; the homogeneous gauge has not been subtracted. The local energy inequality is applied to V_n with the actual pressure Q_n^{act}, not with the affine-quotient representative. By Lemma 9.78, either the affine pressure component has already routed the sequence to $\mathcal{C}_{\mathrm{aff}}$, or $Q_n^{\mathrm{act}} - a_n(s) \to 0$ strongly in $L^{3/2}_{\mathrm{loc}}$, and the time-only gauge contributes zero to $\operatorname{div}(Q_n^{\mathrm{act}} V_n)$. In the latter case the pressure flux converges distributionally to $\operatorname{div}(Q\bar{V})$, with $Q = 0$ in the non-affine quotient. In (9.12), the viscous diffusion and nonlinear-flux terms vanish in the limit because of the factors r_n^{-2} and r_n^{-1}, while the dissipation term is nonnegative and may be dropped. The strong actual-pressure convergence modulo time-only gauges gives

$$\operatorname{div}(Q_n^{\mathrm{act}}V_n) \to \operatorname{div}(Q\bar{V}).$$

Therefore

$$\partial_s \frac{E_V}{2} + \frac{1}{2}x \cdot \nabla \frac{E_V}{2} + \frac{1}{2}E_V + \operatorname{div}(Q\bar{V}) \le 0$$

in distributions, where

$$E_V(x,s) := \langle |\xi|^2 \rangle_{\nu_{x,s}}$$

is the second moment of the Young measure generated by V_n. On the other hand, applying Lemma 9.82 to the barycenter equation for $\bar{V}$ gives

$$\partial_s \frac{|\bar{V}|^2}{2} + \frac{1}{2}x \cdot \nabla \frac{|\bar{V}|^2}{2} + \frac{1}{2}|\bar{V}|^2 + \operatorname{div}(Q\bar{V}) = 0.$$

Subtracting gives the variance inequality for the variance of V_n. By Lemma 9.80, the same variance density is $\mathcal{V}$, the variance of W_n. This proves (9.19). □

Lemma 9.84 (Bounded ancient variance Liouville theorem)**.** *Let $F \in L^\infty_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R})$ satisfy*

$$0 \le F \le C < \infty$$

and

$$\partial_s F + \frac{1}{2}x \cdot \nabla F + F \le 0$$

in distributions on $\mathbb{R}^3 \times \mathbb{R}$. Then $F \equiv 0$.

Proof. Step 1: renormalization. Set $G(x,s) := e^s F(x,s)$, so that $0 \le G \le Ce^s$. Since $G = e^s F$, the product rule gives, in distributions,

$$\partial_s G + \frac{1}{2}x \cdot \nabla G = e^s \left(\partial_s F + \frac{1}{2}x \cdot \nabla F + F \right) \le 0.$$

Equivalently, for every smooth nonnegative compactly supported test function ψ, the distributional hypothesis applied to $e^s\psi \ge 0$ yields

$$\partial_s G + \frac{1}{2}x \cdot \nabla G \le 0 \qquad \text{in distributions on } \mathbb{R}^3 \times \mathbb{R}. \tag{9.20}$$

Step 2: backward adjoint semigroup. The forward characteristic flow of $\widetilde{L} := \partial_s + \frac{1}{2}x \cdot \nabla$ starting from (x_0, s_0) is $x(\sigma) = e^{(\sigma - s_0)/2}x_0$. Define, for $T \ge 0$, the backward adjoint operator on test functions

$$(\mathcal{S}_T\psi)(x,s) := e^{-3T/2}\psi\big(e^{-T/2}x,\ s - T\big).$$

The prefactor $e^{-3T/2}$ is the Jacobian of $x \mapsto e^{-T/2}x$ on $\mathbb{R}^3$, so $\int \mathcal{S}_T\psi \, dx\, ds = \int \psi \, dx \, ds$ and the family $\{\mathcal{S}_T\}_{T \ge 0}$ is the adjoint of the forward characteristic flow on $L^1(\mathbb{R}^3 \times \mathbb{R})$. In particular, by the substitution $x = e^{T/2}x'$, $s = s' + T$ (Jacobian $e^{3T/2}$),

$$\langle G, \mathcal{S}_T\psi \rangle = \iint G(e^{T/2}x', s' + T)\, \psi(x', s')\, dx'\, ds'. \tag{9.21}$$

Step 3: monotonicity in T. Fix $\psi \ge 0$ smooth and compactly supported. A direct computation using the chain rule shows that

$$\frac{d}{dT}\mathcal{S}_T\psi = \widetilde{L}^*(\mathcal{S}_T\psi)$$

where $\widetilde{L}^* = -\partial_s - \frac{1}{2}x \cdot \nabla - \frac{3}{2}$. Therefore

$$\frac{d}{dT}\langle G, \mathcal{S}_T\psi \rangle = \langle G, \widetilde{L}^*(\mathcal{S}_T\psi) \rangle = \langle \widetilde{L}G, \mathcal{S}_T\psi \rangle.$$

Since $\mathcal{S}_T\psi \geq 0$ and $\widetilde{L}G \leq 0$ distributionally by (9.20),

$$\frac{d}{dT}\langle G, \mathcal{S}_T\psi\rangle \leq 0. \tag{9.22}$$

Step 4: test-function vanishing and conclusion. Let $\varphi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ be nonnegative and, for $T \geq 0$, define

$$\psi_T(x,s) := e^{3T/2}\varphi(e^{T/2}x, s+T).$$

Then $\psi_T \geq 0$, $\psi_T \in C_c^\infty$, and a direct computation gives $\mathcal{S}_T\psi_T = \varphi$. Applying (9.22) with $\psi = \psi_T$ and evaluating at time T yields

$$\langle G, \varphi\rangle = \langle G, \mathcal{S}_T\psi_T\rangle \leq \langle G, \psi_T\rangle.$$

Since $0 \leq G(x,s) \leq Ce^s$,

$$\langle G, \psi_T\rangle \leq C\iint e^s e^{3T/2}\varphi(e^{T/2}x, s+T)\, dx\, ds.$$

Make the change of variables $x' = e^{T/2}x$, $s' = s+T$. Then $dx\, ds = e^{-3T/2}dx'\, ds'$, so

$$\langle G, \psi_T\rangle \leq Ce^{-T}\iint e^{s'}\varphi(x', s')\, dx'\, ds' \xrightarrow[T\to\infty]{} 0.$$

Hence $\langle G, \varphi\rangle = 0$ for every nonnegative test function φ. Therefore $G = 0$ distributionally, so also almost everywhere. Since $G = e^sF$, it follows that $F \equiv 0$. □

Lemma 9.85 (Minimal-scale blow-downs are strongly compact)**.** *Every minimal-scale blow-down W_n either has a retained affine/parasitic descendant in $\mathcal{C}_{\mathrm{aff}}$, or has, after passing to a subsequence, a limit $\bar{W}$ such that*

$$W_n \to \bar{W} \qquad \textit{strongly in } L^3_{\mathrm{loc}}(\mathbb{R}^3 \times \mathbb{R}).$$

The limit satisfies (9.18).

Proof. By Lemma 9.83, either the affine/parasitic alternative has occurred, or the Young-measure variance $\mathcal{V}$ satisfies the hypotheses of Lemma 9.84. Hence $\mathcal{V} \equiv 0$. Thus

$$\widetilde{\nu}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x,s).$$

Uniform L^∞_{loc} boundedness then upgrades convergence in measure to strong convergence in L^3_{loc}. The limit equation has already been proved in Lemma 9.83. □

Theorem 9.86 (Minimal mesoscopic oscillation scale reduction)**.** *Every minimal mesoscopic oscillation scale generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has a descendant in*

$$\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

Proof. Let z_n, r_n be a minimal mesoscopic oscillation scale. If, before passing to the minimal-scale blow-down, some unit covariant observer in the selected residual window carries a positive constant-in-time oscillation threshold, then the concentration alternative is regenerated and the descendant lies in $\mathcal{C}_{\mathrm{conc}}$. It remains to consider the case in which unit concentration has been removed at all fixed positive thresholds.

By construction, this theorem is invoked only after Lemmas 9.64 and 9.65 have already produced a genuine spatial mesoscopic scale $1 \ll r_n \ll R_n$; pure time-translation defects were routed earlier to regenerated concentration or to the affine/parasitic stratum.

Form the minimal-scale blow-down W_n of Definition 9.66. By Lemma 9.85, either the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$ is reached, or, after passing to a subsequence,

$$W_n \to W \qquad \text{strongly in } L^3_{\mathrm{loc}},$$

and (W, Q) satisfies the coherent critical-tail transport equation

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \qquad \nabla \cdot W = 0.$$

The minimal-scale lower bound and strong convergence give

$$\operatorname{osc}_{3,\mathrm{sp}}\big(W; B_1(0) \times (-1,1)\big) \geq \varepsilon_0.$$

Thus the limit is not spatially affine/parasitic on the unit critical-tail window. If this non-affine lower bound were lost in a further descendant, the descendant would be assigned to $\mathcal{C}_{\mathrm{aff}}$ by Theorem 8.21. Otherwise the strongly realized nonzero limit is a coherent critical-tail profile on annular cylinders. Consider its compact joint log/time translation hull inside the critical-tail compactification. If some hull descendant regenerates local concentration or collapses to the affine/parasitic quotient, then the branch is assigned to $\mathcal{C}_{\mathrm{conc}}$ or $\mathcal{C}_{\mathrm{aff}}$, respectively. Otherwise, by compactness of the hull and descendant heredity Lemma 10.11 and Theorem 10.12, the hull contains a sequence-realized recurrent support profile with the same nonzero spatial quotient oscillation lower bound. This recurrent support profile is a coherent critical-tail descendant, hence belongs to $\mathcal{C}_{\mathrm{cohcrit}}$. Therefore every minimal mesoscopic oscillation scale is assigned to $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, or $\mathcal{C}_{\mathrm{cohcrit}}$, as claimed. ☐

Remark 9.87 (Critical-tail reduction theorem)*.* Theorem 9.86 is the critical-tail reduction theorem proved above. It is a local multiscale rigidity statement: oscillation first appearing at an intermediate scale $1 \ll r_n \ll R_n$ must either regenerate a concentrating observer, collapse into the affine/parasitic lower stratum, or produce a coherent critical tail. The log-diffuse alternative has been reduced above to the same hidden-scale compactness statement through tight log-window normalized support profiles.

Theorem 9.88 (No hidden-scale variance from minimal mesoscopic scale reduction)**.** *Every tight macroscopic annular blow-down generated by a realized recurrent critical-tail extraction after concentration-set removal and affine normalization has no hidden-scale variance on compact macroscopic annular cylinders.*

Proof. Suppose that hidden-scale variance fails. By Lemma 9.65, either a unit concentrating observer is regenerated, the affine stratum is reached, or a minimal mesoscopic oscillation scale exists. The first alternative contradicts concentration-set removal. The affine alternative is routed by Theorem 9.99 to $\mathcal{C}_{\mathrm{aff}}$, hence it does not produce a retained normalized residual representative. In the third alternative, Theorem 9.86 gives a descendant in $\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}$. The concentration case again contradicts concentration-set removal; the affine case is routed by Theorem 9.99 to $\mathcal{C}_{\mathrm{aff}}$, hence it is not a retained normalized representative; and the coherent critical-tail case is closed by Corollary 9.111. Thus all alternatives are impossible, and hidden-scale variance cannot fail. ☐

Definition 9.89 (Macroscopic critical-tail blow-down map)**.** Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$, let $z_n = (y_n, \sigma_n)$ be covariant observer centers, and let $R_n \to \infty$ be a selected macroscopic scale. Put

$$\widetilde{U}_n := \mathcal{R}_{z_n} U_n, \qquad \widetilde{\Pi}_n := \mathcal{R}_{z_n} \Pi_n.$$

The ungauged spatial/logarithmic blow-down is

$$V_n(x,s) := \widetilde{U}_n(R_n x, s), \qquad \widehat{Q}_n^{\mathrm{act}}(x,s) := R_n^{-1}\big(\widetilde{\Pi}_n(R_n x, s) - a_n(s)\big),$$

where $a_n(s)$ is the time-dependent pressure gauge. If an affine pressure component

$$L_n(y,s) = b_n(s) \cdot y + a_n(s)$$

is removed in the affine quotient, the corresponding quotient pressure is

$$Q_n^{\#}(x,s) := R_n^{-1}\big(\widetilde{\Pi}_n(R_n x,s) - L_n(R_n x,s)\big).$$

Velocity oscillations are measured after a spatially homogeneous velocity gauge

$$W_n(x,s) := V_n(x,s) - c_n(s), \qquad |c_n(s)| \le M.$$

The map is not the Navier–Stokes parabolic scaling: centered time is not rescaled. The scaling identities are

term	blow-down size
$\partial_s V_n$	1
ΔV_n	R_n^{-2}
$(V_n\cdot\nabla)V_n$	R_n^{-1}
$\frac{1}{2}y\cdot\nabla_y U_n$	$\frac{1}{2}x\cdot\nabla_x V_n$
$\nabla_y\widetilde{\Pi}_n$	$\nabla_x\widehat{Q}_n^{\mathrm{act}}.$

Thus V_n satisfies the exact distributional equation

$$\partial_s V_n - R_n^{-2}\Delta V_n + \frac{1}{2}x\cdot\nabla V_n + \frac{1}{2}V_n + R_n^{-1}(V_n\cdot\nabla)V_n + \nabla\widehat{Q}_n^{\mathrm{act}} = 0. \tag{9.23}$$

If one passes to the affine-quotient representative $Q_n^{\#} = \widehat{Q}_n^{\mathrm{act}} - b_n(s)\cdot x$, then

$$\nabla\widehat{Q}_n^{\mathrm{act}} = \nabla Q_n^{\#} + b_n(s).$$

Thus the exact equation is not written with $Q_n^{\#}$ alone unless the affine force $b_n(s)$ has already been routed to $\mathcal{C}_{\mathrm{aff}}$ or removed by the canonical affine normalization. The field W_n is used for oscillation and Young-measure compactness, not as an ungauged Navier–Stokes solution unless the time regularity of c_n has separately been obtained.

Lemma 9.90 (Critical-tail homogeneous gauge bridge)**.** *Let V_n be ungauged macroscopic critical-tail blow-downs on fixed annular cylinders, and let*

$$W_n(x,s) := V_n(x,s) - c_n(s)$$

be the affine-normalized representatives with $|c_n(s)| \le M$ on compact time intervals. After passing to a subsequence, exactly one of the following alternatives occurs:

(i) *the homogeneous gauge sequence has a noncompact quotient defect, and by Lemma 9.79 the corresponding descendant is assigned to $\mathcal{C}_{\mathrm{aff}}$;*

(ii) *$c_n \to c$ strongly in $L^3_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^3)$. If $\mathcal{Y}^V_{x,s}$ and $\mathcal{Y}^W_{x,s}$ are the Young measures generated by V_n and W_n, respectively, then*

$$\mathcal{Y}^V_{x,s} = (\xi\mapsto\xi + c(s))_{\#}\mathcal{Y}^W_{x,s}, \qquad \mathcal{Y}^W_{x,s} = (\xi\mapsto\xi - c(s))_{\#}\mathcal{Y}^V_{x,s}.$$

In particular, Dirac/non-Dirac status, hidden-scale variance, and spatial quotient oscillation pass between the ungauged and affine-normalized representatives modulo the affine quotient.

Proof. Apply Lemma 9.79 to the spatially homogeneous gauge sequence c_n. This gives the dichotomy between compactness in L^3_{loc} and routing to $\mathcal{C}_{\mathrm{aff}}$. In the compact case, Lemma 9.80 gives exactly the deterministic translation relation between the Young measures generated by V_n and W_n, together with preservation of variance inequalities. Since the spatial quotient oscillation functionals are defined modulo spatially homogeneous velocity gauges, they are unchanged by adding or subtracting $c(s)$. □

Theorem 9.91 (Critical-tail compactification extraction)**.** *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a bounded normalized terminal residual sequence, $\|U_n\|_{L^\infty} \le M$. Assume that, after removal of all affine-normalized concentrating covariant observers, there remains a nonzero diffuse exterior oscillation defect. Then, after passing to a subsequence, exactly one of the following alternatives occurs:*

(i) *a retained affine-pressure or homogeneous-gauge descendant is generated and the branch is routed to $\mathcal{C}_{\mathrm{aff}}$;*

(ii) *a pressure/noncompact descendant is generated and routed to the corresponding pressure/noncompact branch;*

(iii) *after alternatives* (i)–(ii) *have been routed away, the retained critical-tail data generate a nonzero element*

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda; \widehat{\mathscr{P}}, \mathscr{D}, \mathfrak{q}_{\mathrm{aff}}, \mathfrak{v}_{\mathrm{vir}}, \mathfrak{s}) \in \mathfrak{T}_M.$$

This retained alternative is read in priority order: if $\Lambda \neq 0$, it is the log-diffuse full state, whose annular coordinates are inactive until log-window selection; if $\Lambda = 0$, it is an annular retained state whose pressure and viscous defect coordinates are uniformly bounded on the fixed countable test family.

In the annular retained subcase of alternative (iii)*, the ungauged barycentric velocity satisfies the linear critical-tail transport law*

$$\partial_s \bar{V} + \frac{1}{2} x \cdot \nabla \bar{V} + \frac{1}{2} \bar{V} + \nabla Q^{\mathrm{act}} = 0, \qquad \nabla \cdot \bar{V} = 0$$

in distributions on macroscopic annular cylinders. The affine-normalized barycenter $\bar{W} = \bar{V} - c(s)$ is the corresponding quotient representative; its equation is understood modulo spatially homogeneous velocity modes and affine pressure forces, where Q is obtained by first passing to the limit in the actual pressure gradients modulo time gauges and only afterwards projecting to the non-affine affine-quotient class. The compactification stores the affine-normalized Young/log state $(\mathcal{Y}, \Lambda)$ generated by W_n, together with the pressure, viscous, affine-quotient, virial, and support coordinates in the displayed tuple. The ungauged Young measure $\mathcal{Y}^V$ generated by the exact PDE variables V_n satisfies the balance identity of Lemma 9.45; outside $\mathcal{C}_{\mathrm{aff}}$, Lemma 9.90 identifies $\mathcal{Y}^V$ and $\mathcal{Y}$ by deterministic translation, so their Dirac/non-Dirac and hidden-scale alternatives agree. In particular the log-diffuse subcase is exactly the case $\Lambda \neq 0$, and the annular Young subcase satisfies the induced log-dilation balance while nonlinear velocity observables retain the pressure and viscous defect pairings explicitly recorded in (9.4)*.*

Proof. Let ν_n^{sp} be the spatial quotient oscillation defect measure after concentration-set removal, as in Lemma 9.22. By the mixed-diffuse hypothesis there are escaping observer windows E_n and $m_0 > 0$ such that

$$\nu_n^{\mathrm{sp}}(E_n) \ge m_0.$$

Since all positive-threshold constant-in-time oscillatory observers have been removed and

$$\mathrm{osc}_{3,\mathrm{sp}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z)\big) \le \mathrm{osc}_{3,\mathrm{ct}}\big(U; \mathcal{Q}_1^{\mathrm{cov}}(z)\big),$$

the atom size of ν_n^{sp} on E_n tends to zero. Applying Lemma 9.31 to $\gamma_n = \nu_n^{\mathrm{sp}}$, after passing to a subsequence we may replace E_n by an escaping finite net-cell subset, still denoted E_n, such that

$$m_0 \le \nu_n^{\mathrm{sp}}(E_n) \le 2m_0.$$

Normalize

$$\lambda_n := \frac{\nu_n^{\mathrm{sp}} \lfloor E_n}{\nu_n^{\mathrm{sp}}(E_n)}.$$

By Lemma 9.22 and the proof of Theorem 9.29, after passing to a subsequence these probability measures converge to a boundary diffuse-defect profile. Because all positive-threshold concentrating observers have been removed in the stronger $\mathrm{osc}_{3,\mathrm{ct}}$-sense, the limit gives no mass to ordinary unit concentrating observers in Z.

We now compactify the critical boundary and split according to annular tightness. Apply Prokhorov compactness first in the observer compactification and then in the logarithmic radial compactification.

First suppose the normalized spatial quotient defect is not tight in any fixed multiplicative annulus. Then its logarithmic pushforward has nonzero mass at the logarithmic boundary. We define the critical-tail state with $\Lambda \neq 0$; this is the log-diffuse critical-tail alternative inside $\mathfrak{T}_M$. In this coarse log-boundary representative the annular Young, pressure, and viscous slots are not invoked to construct the branch; the annular coordinates enter only after passing to normalized log-window support profiles. The subsequent log-window selection is handled by Lemmas 9.53 and 9.55. This gives the log-diffuse subcase of alternative (iii).

It remains to consider the annular-tight case. Then select macroscopic scales $R_n \to \infty$, observers $z_n = (y_n, \sigma_n)$, affine velocity gauges $c_n(s)$, and pressure gauges. On fixed annular cylinders use the exact blow-down map of Definition 9.89; in this case no mass is assigned to the logarithmic boundary, so $\Lambda = 0$:

$$V_n(x,s) = \widetilde{U}_n(R_n x, s), \qquad W_n(x,s) = V_n(x,s) - c_n(s),$$

with $\widetilde{U}_n = \mathcal{R}_{z_n} U_n$. Before comparing the affine-normalized compactness variable W_n with the exact PDE variable V_n, apply Lemma 9.90 to the homogeneous gauges c_n. If the gauge sequence has a noncompact homogeneous time defect, then the corresponding descendant is assigned to $\mathcal{C}_{\mathrm{aff}}$, so this branch is routed away before entering the retained critical-tail compactification; this is alternative (i) of the present theorem, and there is nothing further to prove in that branch. Hence we work in the complementary case, after passing to a subsequence, in which

$$c_n \to c \qquad \text{strongly in } L^3_{\mathrm{loc}}(\mathbb{R};\mathbb{R}^3).$$

By Lemma 9.102, the affine-normalized gauges may be chosen with $|c_n(s)| \leq M$ on compact time intervals. Hence

$$|W_n| \leq 2M$$

on every selected macroscopic cylinder; since also $|V_n| \leq M$, Banach–Alaoglu gives weak-* compactness of both V_n and W_n in L^∞_{loc}. The fundamental theorem of Young measures gives, after passing to a further subsequence, ungauged barycenter/Young data $(\bar{V}, \mathcal{Y}^V)$ for V_n and affine-normalized barycenter/Young data $(\bar{W}, \mathcal{Y})$ for W_n. By Lemma 9.90, $\mathcal{Y}^V$ and $\mathcal{Y}$ differ only by deterministic translation. If the affine-normalized sequence is strongly compact in L^3_{loc}, then $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$; otherwise $\mathcal{Y}$ records the unresolved oscillation defect. Apply Theorem 9.40 to the selected critical-tail family. If its affine-pressure or pressure/noncompact alternative occurs, the branch is precisely alternative (i) or (ii) of the present theorem, and we stop. Otherwise we are in the retained bounded-coordinate branch. In this branch the pressure-defect coordinates $\widehat{\mathscr{P}}_{n,F}[\psi]$, the viscous-defect coordinates $\mathscr{D}_{n,F}[\psi]$, and, on the realized recurrent scale-critical virial-tight subhull of Definition 9.48, the virial coordinates are uniformly bounded on the fixed countable dense families entering the compactification. Diagonal extraction therefore gives the full tuple in $\mathfrak{T}_M$. Since the normalized defect mass is one, the retained limit satisfies $\mathcal{T} \neq 0$, giving alternative (iii).

It remains, in the annular retained subcase of alternative (iii), to identify the equation satisfied by the barycenter. The ungauged fields V_n satisfy the exact scaled equation (9.23). Passing from V_n to $W_n = V_n - c_n(s)$ is used only to measure oscillation. We do not differentiate the homogeneous gauge $c_n(s)$. Instead, the exact equation is passed to the limit for V_n and only afterwards projected to the affine/parasitic quotient. The noncompact homogeneous-gauge alternative has already been routed to $\mathcal{C}_{\mathrm{aff}}$ by Lemma 9.90; hence in the present branch $\bar{W} = \bar{V} - c(s)$ differs from $\bar{V}$ only by a deterministic spatially homogeneous mode. For every compactly supported test function, the viscous and nonlinear contributions are bounded respectively by

$$CR_n^{-2}\|V_n\|_{L^\infty}, \qquad CR_n^{-1}\|V_n\|_{L^\infty}^2,$$

and hence both tend to zero. The pressure gauges give distributional compactness of the remaining gradients on annuli. Passing to the weak-* barycentric limit of V_n yields the coherent transport equation for $\bar{V}$ with the actual pressure representative Q^{act}; projecting only after this step gives the affine-quotient transport equation for the non-affine barycenter $\bar{W}$, modulo spatially homogeneous velocity modes and affine pressure forces. No derivative of the limiting gauge $c(s)$ is used unless it is absorbed into that affine pressure quotient. Applying Lemma 9.45 gives the exact Young/log balance for the ungauged Young measure $\mathcal{Y}^V$. By Lemma 9.90, the affine-normalized Young measure $\mathcal{Y}$ stored in $\mathfrak{T}_M$ differs from $\mathcal{Y}^V$ only by deterministic translation, so the branch decomposition sees the same Dirac/non-Dirac and hidden-scale alternatives. The log-scale component satisfies the induced log-dilation balance, while nonlinear velocity observables retain the pressure and viscous defect pairings displayed in (9.4). This construction is local on macroscopic annuli and uses no global critical norm, CKN budget, or energy estimate. ☐

Definition 9.92 (Young/defect critical-tail branch)**.** A tight critical-tail profile belongs to $\mathcal{C}_{\mathrm{Young}}$ if $\Lambda = 0$ and at least one of the following holds:

(i) the affine-normalized Young measure $\mathcal{Y}$ is non-Dirac on a set of positive macroscopic measure;

(ii) for some $(F, \psi) \in \mathfrak{F}_{\mathrm{crit}}$,
$$\widehat{\mathscr{P}}_F[\psi] \neq 0;$$

(iii) for some positive convex test pair $(F, \psi) \in \mathfrak{F}_{\mathrm{crit}}^{\mathrm{conv}}$,
$$\mathscr{D}_F[\psi] > 0.$$

Thus $\mathcal{C}_{\mathrm{Young}}$ records every tight critical-tail obstruction to coherence not already routed to regenerated concentration, affine/parasitic collapse, or a pressure/noncompact branch. The viscous subcase is stated using positive convex tests because only those coordinates have a sign. Once all positive convex coordinates vanish, the associated positive matrix-valued viscous defect measure vanishes, and the remaining signed coordinates vanish by polarization.

Lemma 9.93 (No pure viscous defect in the tight Dirac critical-tail branch)**.** *Let $\mathcal{T} \in \mathfrak{T}_M$ be tight, with*

$$\Lambda = 0, \qquad \mathcal{Y} = \delta_{\bar{W}},$$

and suppose the realized profile is outside $\mathcal{C}_{\mathrm{conc}}$, outside $\mathcal{C}_{\mathrm{aff}}$, outside the pressure/noncompact branch, and has no hidden-scale variance on compact annular cylinders. Then for every $(F, \psi) \in \mathfrak{F}_{\mathrm{crit}}^{\mathrm{conv}}$,

$$\mathscr{D}_F[\psi] = 0.$$

Proof. Work on a compact annular cylinder and in the non-affine, non-pressure/noncompact branch. The Dirac Young measure and Lemmas 9.59 and 9.90 give strong L^3_{loc} convergence of the affine-normalized fields, and of the ungauged fields up to the deterministic homogeneous translation. By the general affine-pressure coefficient dichotomy Lemma 9.75, the actual pressure is normalized so that

$$\widehat{Q}_n^{\mathrm{act}} - a_n(s) \to 0 \qquad \text{strongly in } L^{3/2}_{\mathrm{loc}},$$

and the time-only gauge contributes zero to the local-energy pressure flux. In the present branch, the affine-pressure coefficient has either been routed to $\mathcal{C}_{\mathrm{aff}}$ or tends to zero, and the non-affine pressure part tends to zero strongly after time-only gauges. Hence the ungauged barycenter equation is represented with an $L^{3/2}_{\mathrm{loc}}$ pressure representative; in fact, in the non-affine quotient one may take $Q = 0$.

Apply the blow-down local energy inequality Lemma 9.69 with nonnegative cutoffs supported in the annular cylinder. Passing to the limit uses the strong velocity convergence, the pressure-flux convergence just described, and the vanishing factors in the viscous diffusion and convective flux. The limiting ungauged coherent transport profile satisfies the renormalized inviscid energy identity from Lemma 9.82. Subtracting the limiting energy identity from the limiting local energy inequality leaves only the nonnegative trace of the scaled viscous defect measure. Therefore that trace is zero on every compact annular cylinder.

The positive convex coordinates $\mathscr{D}_F[\psi]$ are contractions of this positive matrix-valued defect measure with the nonnegative matrix $\nabla_\xi^2 F$ and the nonnegative cutoff ψ. Since the trace measure is zero, every such contraction is zero. This proves the claim without using a Poincaré lower bound from gradient energy to velocity oscillation. □

Lemma 9.94 (No pressure defect after viscous-defect removal)**.** *Under the hypotheses of Lemma 9.93, after affine-pressure routing and pressure/noncompact routing one has*

$$\widehat{\mathscr{P}}_F[\psi] = 0 \qquad \textit{for every } (F, \psi) \in \mathfrak{F}_{\mathrm{crit}}.$$

Proof. Use the actual/time-gauge representative supplied by Lemma 9.75 and the non-affine estimate of Lemma 9.74. On the fixed annular cylinder write

$$\widehat{Q}_n^{\mathrm{act}} = q_n^{\mathrm{na}} + b_n(s) \cdot x + a_n(s), \qquad \|q_n^{\mathrm{na}}\|_{L^p(K)} \le C_{p,K,M} R_n^{-1}$$

for every finite p. The time-only gauge has zero spatial gradient and therefore drops out of the pressure-defect pairing. The affine spatial piece is kept in gradient form:

$$\iint \psi \, \nabla_\xi F(V_n) \cdot \nabla(b_n(s) \cdot x) = \iint \psi \, \nabla_\xi F(V_n) \cdot b_n(s).$$

If b_n has a retained nonzero component, the branch is routed to $\mathcal{C}_{\mathrm{aff}}$. In the present non-affine branch, $b_n \to 0$ locally in $L^{3/2}_s$, and the last display tends to zero because $\nabla_\xi F(V_n)$ is bounded and ψ has compact support.

It remains to treat the non-affine pressure. Integrating only q_n^{na} by parts gives

$$\begin{aligned} \iint \psi \, \nabla_\xi F(V_n) \cdot \nabla q_n^{\mathrm{na}} = &- \iint q_n^{\mathrm{na}} \, \nabla\psi \cdot \nabla_\xi F(V_n) \\ &- \iint q_n^{\mathrm{na}} \, \psi \, \nabla_\xi^2 F(V_n) : \nabla V_n. \end{aligned}$$

The first term tends to zero by $\|q_n^{\mathrm{na}}\|_{L^{3/2}(K)} = O(R_n^{-1})$ and the uniform velocity bound. For the second term, Cauchy–Schwarz and the L^2 case of (9.16) give

$$\left| \iint q_n^{\mathrm{na}} \psi \, \nabla_\xi^2 F(V_n) : \nabla V_n \right| \le C \Big(R_n^{-2} \iint_{\operatorname{supp} \psi} |\nabla V_n|^2 \Big)^{1/2}.$$

The factor on the right is the trace viscous defect on the support of ψ, which vanishes by Lemma 9.93. Therefore all pressure-defect coordinates vanish. □

Lemma 9.95 (Critical-tail defect-coordinate closure)**.** *Let $\mathcal{T} \in \mathfrak{T}_M$ be tight and Dirac in the affine-normalized Young measure. After regenerated concentration, affine/parasitic, pressure/noncompact, and hidden-scale branches have been removed, all pressure and viscous defect coordinates vanish:*

$$\widehat{\mathscr{P}} = 0, \qquad \mathscr{D} = 0.$$

Proof. The non-Dirac Young-measure case is exactly the hidden-scale variance obstruction by Lemma 9.59. A genuinely retained positive L^3-oscillation defect is routed to the minimal mesoscopic branch by Theorem 9.86. In the remaining tight Dirac branch, pure viscous defects vanish by Lemma 9.93. Once the positive convex viscous coordinates vanish, all signed viscous coordinates vanish by the polarization statement in Definition 9.44. The pressure coordinates then vanish by Lemma 9.94. Thus no pressure or viscous critical-tail defect coordinate survives in the retained residual class. □

Corollary 9.96 (Coherent extraction under annular tightness and strong compactness)**.** *If the critical-tail extraction is in alternative* (iii) *of Theorem 9.91, producing a retained defect $\mathcal{T} \in \mathfrak{T}_M$, and if this defect satisfies $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{W(x,s)}$ for a.e. (x, s), with $\widehat{\mathscr{P}} = 0$, $\mathscr{D} = 0$, and no retained affine or pressure/noncompact defect, then there exist macroscopic scales, covariant observers, affine gauges, and pressure gauges for which the ungauged blown-down fields converge, up to the deterministic homogeneous gauge bridge, to an ungauged representative (V, Q^{act}) satisfying the literal coherent transport equation on punctured annular cylinders. The affine-normalized fields converge strongly in L^3_{loc} and weak-* in L^∞_{loc} to the corresponding nonzero quotient coherent critical tail $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$, with $W = V - c(s)$.*

Proof. This is the coherent subcase of the retained bounded-coordinate alternative in Theorem 9.91. The condition $\Lambda = 0$ rules out loss of mass across logarithmic scale, and the Dirac-valued Young measure gives strong L^3_{loc} convergence by uniform L^∞ boundedness. The vanishing of $\widehat{\mathscr{P}}$, $\mathscr{D}$, and retained affine/pressure defects is precisely the extra coordinate condition in $\mathfrak{T}_{\mathrm{coh}}$, so no residual pressure/viscous coordinate remains outside the coherent transport equation. The literal equation is carried by the ungauged representative; the displayed affine-normalized tail is the quotient representative described in Definition 9.39. Nontriviality follows from the normalized affine oscillation mass. □

Theorem 9.97 (Coherent critical-tail classification)**.** *Let $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$ be a coherent critical tail, with W understood as the affine-normalized quotient representative of an ungauged coherent tail. The classification below uses only this quotient representative, its non-affine spatial oscillation, and its log/time translation hull. In logarithmic radial variables*

$$x = e^\rho \omega, \qquad Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s),$$

its compact log-translation/time-translation hull belongs to exactly one of

$$\mathcal{C}_{\mathrm{affine}}, \qquad \mathcal{C}_{\mathrm{homcrit}}, \qquad \mathcal{C}_{\mathrm{logper}}, \qquad \mathcal{C}_{\mathrm{ap}}.$$

Here $\mathcal{C}_{\mathrm{homcrit}}$ is the homogeneous (-1)-tail class, $\mathcal{C}_{\mathrm{logper}}$ is the nonzero log-periodic tail class, and $\mathcal{C}_{\mathrm{ap}}$ is the remaining aperiodic minimal log-translation hull class.

Proof. Let

$$\mathcal{H}(Z) := \overline{\{Z(\rho + a, \omega, s + b) : a, b \in \mathbb{R}\}}$$

be the compact logarithmic translation hull. If the affine-normalized oscillation vanishes on every compact log-cylinder, then the tail is affine/parasitic. Otherwise, if every element of $\mathcal{H}(Z)$ is independent of ρ, then W is (-1)-homogeneous and belongs to $\mathcal{C}_{\mathrm{homcrit}}$. If the log-translation orbit has a nonzero period, then W is discretely self-similar at infinity and belongs to $\mathcal{C}_{\mathrm{logper}}$. If neither alternative holds, compact-flow decomposition gives a minimal compact invariant subsystem with no nonzero log period; this is the aperiodic log-translation class $\mathcal{C}_{\mathrm{ap}}$. These alternatives exhaust compact log-translation hulls. □

Theorem 9.98 (Realized critical-tail rigidity by stratification)**.** *Let $\mathcal{T} \in \mathfrak{T}_M$ be a nonzero realized critical-tail profile generated by a bounded normalized residual Navier–Stokes terminal* *sequence after terminal concentration-set removal. Then the ordered stratification selects one of the following priority profile alternatives:*

(a) $\mathcal{T}$ *regenerates affine-normalized local concentration;*
(b) $\mathcal{T}$ *is affine/parasitic;*
(c) $\mathcal{T}$ *is log-diffuse;*
(d) $\mathcal{T}$ *is a Young/defect critical tail;*
(e) $\mathcal{T}$ *is a coherent homogeneous critical tail;*
(f) $\mathcal{T}$ *is a coherent log-periodic critical tail;*
(g) $\mathcal{T}$ *is a coherent aperiodic critical tail.*

In the ordered refined decomposition, the log-diffuse, Young, homogeneous, log-periodic, and aperiodic critical-tail alternatives are lower structured strata. Thus every nonzero realized critical-tail profile which remains after these lower strata have been removed either regenerates local concentration or collapses to the affine/parasitic lower stratum.

Proof. If the covariant observer orbit closure of $\mathcal{T}$ intersects a constant-in-time concentration set $\{U : \operatorname{osc}_{3,\mathrm{ct}}(U; Q_1^{\mathrm{cov}}) \geq \eta\}$ for some $\eta > 0$, then alternative *(a)* holds. This case is removed first.

If every spatial quotient oscillation observable vanishes on the profile, then Theorem 8.21 and Definition 8.16 place the profile in the affine/parasitic lower stratum. This is alternative *(b)*. It remains to consider the case in which the profile has nonzero non-affine diffuse content and does not regenerate retained unit-scale velocity concentration.

By Theorem 9.91, the residual normalized diffuse mass either first produces an affine-pressure or pressure/noncompact alternative, or else is represented by a full $\mathfrak{T}_M$-state. The pre-$\mathfrak{T}_M$ alternatives are already assigned to their corresponding lower strata in the priority order; since $\mathcal{T} \in \mathfrak{T}_M$ is the retained object under consideration, we work in alternative (iii) of that extraction theorem. Thus the boundary mass has only two mechanisms: mass escaping every bounded logarithmic annulus, recorded by Λ, and annular blow-down mass, recorded by $(\bar{W}, \mathcal{Y})$, together with the scalar pressure/viscous defect coordinates recorded in $\mathfrak{T}_M$. If $\Lambda \neq 0$, the profile is log-diffuse, giving *(c)*. If $\Lambda = 0$ but $\mathcal{Y}_{x,s}$ is not a Dirac mass on a set of positive macroscopic measure, the profile is a Young/defect critical tail, giving *(d)*. If $\Lambda = 0$ and $\mathcal{Y}$ is Dirac but some coordinate $\widehat{\mathscr{P}}_F[\psi]$ or $\mathscr{D}_F[\psi]$ is nonzero, then Definition 9.92 and Lemma 9.95 place the profile in the Young/defect critical-tail alternative before closure; after the already ordered concentration, affine, pressure/noncompact, hidden-scale, and minimal mesoscopic branches have been removed, those coordinates vanish and this alternative is empty.

The remaining case is coherent in the full compactification topology: $\Lambda = 0$, $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$ a.e., $\widehat{\mathscr{P}} = 0$, $\mathscr{D} = 0$, and no retained affine or pressure/noncompact defect. Then Corollary 9.96 gives a coherent critical tail $(\bar{W}, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$. Applying Theorem 9.97 gives exactly one of the homogeneous, log-periodic, or aperiodic coherent alternatives, namely *(e)*, *(f)*, or *(g)*, unless the affine/parasitic case already occurred. The alternatives are mutually exclusive after the priority order above: regenerated concentration is removed first, then affine collapse, then log-diffuse mass, then Young/defect obstruction, and finally the coherent log-translation classification. The final assertion follows from this ordered classification. □

9.4. **Complete closure of lower alternatives.** The critical-tail classification above is a classification theorem rather than an emptiness theorem. The complete residual decomposition is closed by local lower-stratum exclusions together with the minimal mesoscopic-scale theorem Theorem 9.86. First, Theorem 9.99 routes affine/parasitic profiles to the quotient stratum $\mathcal{C}_{\mathrm{aff}}$. Next, Corollary 9.111 closes the coherent critical-tail row by bounded-origin realization and log-hull minimality, independently of the log-diffuse and hidden-scale arguments. The minimal mesoscopic-scale theorem then gives hidden-scale compactness through Theorem 9.88. The non-Dirac subcase of the Young alternative is reduced to that compactness statement by the Fréchet–Kolmogorov/Young-measure equivalence in Lemma 9.59; its pressure/viscous subcases are closed by Lemma 9.95. The log-diffuse alternative is reduced by log-window support-profile selection together with the same hidden-scale theorem and the already closed coherent row; see Theorem 9.56 and Corollary 9.57. Each statement is applied only to the selected realized residual datum and its named local windows. The proof introduces no $L^\infty_t L^3_x$ entry assumption, finite-energy tail budget, or CKN budget.

Critical-tail theorem-production directory. The following directory is the critical-tail portion of Table 4. It records logical theorem production, not the order in which the statements appear in the file. Each row is conditional on the current profile or defect datum having entered the state named in its first column.

TABLE 13. Local theorem-production order for the selected critical-tail datum. The last column contains consumers of the produced local output and supplies no premise to the producing row.

Selected local owner or input	**Existing producer and exact prerequisite**	**Local output carried forward**	**Later consumer only**
Current pre-quotient retained profile.	Theorem 9.99 applies after the earlier local classes have been removed for this profile.	The current profile is routed to $\mathcal{C}_{\mathrm{aff}}$, or its canonical representative retains positive non-affine oscillation.	The affine route is consumed in Theorem 9.88 and Corollary 9.112.
Current realized coherent critical tail.	The selected tail carries the retained non-affine annular activity and domain interface required by Theorem 9.104; Corollary 9.111 closes that selected coherent row.	The homogeneous, log-periodic, or aperiodic alternative selected for this tail is excluded by its existing bounded-origin argument.	The established coherent closure is consumed by Theorem 9.88 and Corollaries 9.57 and 9.112.
Minimal scale (z_n, r_n) selected from the current realized recurrent critical-tail extraction.	The first-bad construction uses the actual pressure in its local energy chart as in Lemma 9.78; only after that step does Lemma 9.77 pass to the affine-quotient pressure. These interfaces produce the input of Theorem 9.86.	The theorem produces a realized descendant route in $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, or $\mathcal{C}_{\mathrm{cohcrit}}$. Diagonal lifting and heredity are the named realization interfaces used in its proof; no additional retained velocity bound is assigned by this directory.	Theorem 9.88 consumes this reduction.
Selected tight macroscopic annular blow-down.	Theorem 9.88 uses the first-bad reduction, affine routing, and the already established coherent closure.	Hidden-scale variance vanishes on the compact macroscopic annular cylinders of this selected blow-down.	Corollary 9.100 and Theorem 9.56 consume this local compactness output independently.
Current realized Young or defect critical-tail alternative.	For its non-Dirac annular subcase, Corollary 9.100 uses hidden-scale exclusion and Lemma 9.59. Its pressure and viscous coordinates use Lemma 9.95.	The Young or defect alternative selected for the current profile is excluded.	Corollary 9.112 consumes this closure; the log-diffuse proof does not use the Young closure as a separate premise.
Current realized log-diffuse defect and its selected normalized log windows.	Lemmas 9.54 and 9.55 provide the contrapositive support interface. Then Theorem 9.56 and Corollary 9.57 use hidden-scale exclusion and the already established coherent closure.	If positive selected-window mass survived outside the closed active, affine, and coherent routes, the support theorem would produce a support point there. The cited reduction and closures leave no such surviving mass.	Corollary 9.112 consumes the log-diffuse closure.

Selected local owner or input	**Existing producer and exact prerequisite**	**Local output carried forward**	**Later consumer only**
Current realized critical-tail profile after terminal concentration removal and affine routing.	Corollary 9.112 uses regenerated-concentration routing, affine routing, log-diffuse closure, Young closure, and coherent closure in the displayed order.	No critical-tail profile selected from this terminal extraction remains.	Theorems 10.66 and 11.3 consume this conclusion; they are not premises of any row above.

The directory introduces no profile class, compactification, hypothesis, or closure argument. Realized ancestry, retained velocity, normalized defect mass, and produced support points are recorded only in the rows whose cited theorems supply those interfaces. Pressure remains attached to the local blow-down chart in which its compactness theorem is applied. Mildness, sequence-L^3, finite-shift mildness, the setup residual theorem, and final assembly are downstream of this directory.

Theorem 9.99 (Oscillatory entry normalization and affine routing)**.** *Let (U, Π) be a retained residual profile after the earlier lower classes recorded in the companion setup paper have been removed. Then every pre-quotient retained residual candidate satisfies exactly one of the following:*

(i) *after canonical affine/parasitic normalization and an admissible affine symmetry S_c, the profile retains positive constant-in-time non-affine oscillation:*

$$\Phi(S_c U) > 0;$$

(ii) *all spatial oscillations $\operatorname{osc}_{3,\mathrm{sp}}(U; Q)$ vanish on compact cylinders, and the profile is assigned to the affine/parasitic quotient stratum $\mathcal{C}_{\mathrm{aff}}$.*

Since $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ is defined after this affine/parasitic routing, the retained normalized residual decomposition has

$$\mathcal{C}_{\mathrm{aff}} \cap \mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing.$$

Proof. The dichotomy is Theorem 8.21. If all spatial oscillations $\operatorname{osc}_{3,\mathrm{sp}}(U; Q)$ vanish on compact cylinders, then the profile is a spatially homogeneous quotient mode and is routed to $\mathcal{C}_{\mathrm{aff}}$. If this alternative does not occur, the parasitic-free representative has $\Phi(S_c U) > 0$ for an admissible affine symmetry S_c. Thus every pre-quotient candidate is either retained with positive non-affine oscillation or assigned to $\mathcal{C}_{\mathrm{aff}}$, and the normalized residual class $\mathcal{R}^{\#}(\mathcal{S}; I, J)$ contains no representative lying in $\mathcal{C}_{\mathrm{aff}}$. □

Corollary 9.100 (Young alternative exclusion from minimal mesoscopic scale reduction)**.**

$$\mathcal{C}_{\mathrm{Young}} = \varnothing.$$

Proof. Let a retained residual profile generate a nonzero Young/defect critical-tail alternative. By definition of $\mathcal{C}_{\mathrm{Young}}$, its logarithmic component is tight: $\Lambda = 0$, and one of the subcases in Definition 9.92 occurs. In the non-Dirac Young measure subcase, let W_n be the corresponding annular blow-down and $\mathcal{Y}_{x,s}$ its Young measure. By Theorems 9.86 and 9.88, W_n has no hidden-scale variance on compact annular cylinders. Hence Lemma 9.59 gives

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \qquad \text{for a.e. } (x, s),$$

contradicting that subcase. In the pressure/viscous defect-coordinate subcases, Lemmas 9.93 to 9.95 show that, after affine, pressure/noncompact, regenerated-concentration, and hidden-scale branches are removed, no nonzero defect coordinate remains. Therefore $\mathcal{C}_{\mathrm{Young}} = \varnothing$. □

Theorem 9.101 (Critical-tail rigidity from minimal mesoscopic scale reduction)**.** *Every nonzero recurrent critical-tail defect generated by a realized residual terminal sequence belongs to*

$$\mathcal{C}_{\mathrm{conc}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\mathrm{cohcrit}}.$$

Equivalently, after regenerated concentration, affine/parasitic collapse, and coherent critical tails have been removed, no recurrent critical-tail defect remains.

Proof. Let $\mathcal{D} \in \mathfrak{T}_M$ be a nonzero recurrent critical-tail defect. By Theorem 9.98, the ordered priority alternative is one of: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young/defect critical tail, or coherent critical tail. The first two alternatives are precisely $\mathcal{C}_{\mathrm{conc}}$ and $\mathcal{C}_{\mathrm{aff}}$. The log-diffuse alternative is empty by Corollary 9.57. The Young alternative is empty by Corollary 9.100. The only remaining nonempty critical-tail alternative is therefore the coherent alternative $\mathcal{C}_{\mathrm{cohcrit}}$. This proves the inclusion. Once $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, and $\mathcal{C}_{\mathrm{cohcrit}}$ are removed, the inclusion leaves no recurrent critical-tail defect. $\square$

Lemma 9.102 (Bounded velocity gauges for realized blow-downs)**.** *Let $f : E \to \mathbb{R}^3$ be measurable on a finite positive-measure set E, and assume $|f| \le M$ a.e. Then the minimization problem*

$$\inf_{c\in\mathbb{R}^3} \int_E |f - c|^3$$

has a minimizer in $\overline{B_M(0)}$. Consequently every affine-normalized blow-down of a sequence satisfying $\|U_n\|_{L^\infty} \le M$ may be represented, without increasing any local oscillation functional, by velocity gauges c_n with $|c_n| \le M$. If the normalization is performed time-slice by time-slice, the same conclusion holds for a measurable choice $c_n(s)$, with $|c_n(s)| \le M$ for a.e. s.

Proof. The functional is continuous and coercive in c, so a minimizer exists. Let $\pi : \mathbb{R}^3 \to \overline{B_M(0)}$ be the nearest-point projection. Since $\overline{B_M(0)}$ is closed and convex, for every $v \in \overline{B_M(0)}$ and every $c \in \mathbb{R}^3$,

$$|v - \pi(c)| \le |v - c|.$$

Taking $v = f(y)$ and integrating gives

$$\int_E |f - \pi(c)|^3 \le \int_E |f - c|^3.$$

Thus any minimizing sequence may be projected into $\overline{B_M(0)}$, and a minimizer can be chosen there. Applying this to the local velocity fields used to define the affine-normalized observer oscillations gives the asserted bounded representatives. The time-slice version is the same argument on each slice, with measurability obtained by the usual measurable-selection theorem for a continuous coercive finite-dimensional minimization problem. The projection step is pointwise in s, so the bound $|c_n(s)| \le M$ is preserved. $\square$

Lemma 9.103 (Domain realization dichotomy for critical-tail windows)**.** *Every critical-tail window used in the bounded-origin realization theorem is of exactly one of the following two admissible types.*

(i) *It is a window of a bounded ancient descendant $(U, \Pi) \in \mathfrak{X}_M$. Then every fixed macroscopic blow-down cylinder is contained in the full domain $\mathbb{R}^3 \times \mathbb{R}$.*
(ii) *It is a first-generation local window satisfying the local-window condition of Definition 9.7 on every fixed blow-down cylinder used in the extraction.*

If neither condition holds, the window is not used in bounded-origin realization. It is assigned by Theorem 9.17 to the exterior/diffuse, pressure/noncompact, or critical-tail boundary branch before bounded-origin realization is invoked.

Proof. Such windows are obtained after an ancient bounded profile has already been extracted, so their domains are global in centered variables. For a first-generation window, the local setup estimates are available precisely on the physical image of the local Type I cylinder. If every fixed blow-down cylinder remains in that image, the local-window condition gives the required compactness and pressure atlas. If some fixed blow-down cylinder leaves the local window, the sequence has escaped the first-generation setup domain; by Lemma 9.10, it must be assigned through the expanding-region protocol rather than treated as a bounded-origin compactness input. □

Theorem 9.104 (Bounded-origin realization of coherent critical tails)**.** *Let a coherent critical tail be generated by a bounded normalized residual Navier–Stokes terminal sequence with:*

(i) *locally bounded velocity representatives;*
(ii) *pressure gauges bounded modulo time and affine functions on each fixed time window;*
(iii) *convergence on every fixed annulus/log-window;*
(iv) *the normalized support-profile property realized from the original blow-up sequence;*
(v) *the domain alternative in Lemma 9.103.*

Then it is realized, after subsequence extraction and admissible gauges, as a weak-$*$ L^∞_{loc} limit on full macroscopic compact cylinders, including cylinders meeting the macroscopic origin. In particular, every coherent critical tail has a representative which is locally bounded near $x = 0$ on compact macroscopic time intervals, and whose restrictions to all tail windows reproduce the coherent tail.

Remark 9.105 (Use of the bounded-origin representative)*.* The bounded-origin representative is used only as a locally bounded extension of the annular coherent tail. The coherent transport equation and pressure normalization are required only on punctured annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$. None of the homogeneous, log-periodic, or aperiodic exclusions below uses a distributional equation through $x = 0$.

Proof. Let $(W, Q) \in \mathcal{C}_{\mathrm{cohcrit}}$ be a coherent critical tail generated by a bounded terminal sequence $(U_n, \Pi_n) \subset \mathfrak{X}_M$, with $\|U_n\|_{L^\infty} \le M$. By coherent extraction, after passing to a subsequence there are macroscopic scales $R_n \to \infty$, covariant observer centers (y_n, σ_n), affine velocity gauges, and pressure gauges such that on every compact annular cylinder

$$A_{r_0,r_1,I} := \{r_0 < |x| < r_1\} \times I, \qquad 0 < r_0 < r_1 < \infty,$$

the fields

$$\widetilde{U}_n := \mathcal{R}_{(y_n,\sigma_n)} U_n, \qquad W_n(x, s) := \widetilde{U}_n(R_n x, s) - c_n(s)$$

converge to W strongly in $L^3(A_{r_0,r_1,I})$ and weak-$*$ in $L^\infty(A_{r_0,r_1,I})$. By Lemma 9.103, the fixed macroscopic cylinders used below are contained in the domain either because the sequence is an already extracted ancient profile, or because the first-generation local-window condition holds. If this domain condition failed, the window would already belong to an exterior/diffuse or noncompact branch and would not be an input to bounded-origin realization.

By Lemma 9.102, the affine-normalized blow-down may be represented by gauges satisfying

$$|c_n(s)| \le M$$

for a.e. s on every compact time interval, without increasing the oscillation functional. Hence on every full macroscopic cylinder

$$K_{R,I} := B_R(0) \times I,$$

one has

$$|W_n(x, s)| \le |\widetilde{U}_n(R_n x, s)| + |c_n(s)| \le 2M.$$

Thus Banach–Alaoglu gives a weak-$*$ L^∞_{loc} limit on full cylinders, including cylinders meeting $x = 0$.

The domain realization dichotomy Lemma 9.103 is the hypothesis which ensures that W_n is defined on $K_{R,I}$ for all sufficiently large n.

Choose an exhaustion of $\mathbb{R}^3 \times \mathbb{R}$ by full compact macroscopic cylinders $K_j = B_j(0) \times [-j, j]$. Banach–Alaoglu gives, by a diagonal extraction that does not change notation, a field

$$W^{\text{full}} \in L^\infty_{\text{loc}}(\mathbb{R}^3 \times \mathbb{R}),$$

such that

$$W_n \rightharpoonup^* W^{\text{full}} \qquad \text{in } L^\infty(K_j)$$

for every j. The subsequence still has the annular strong L^3_{loc} convergence to the coherent tail W. Therefore, for every annular cylinder $A_{r_0,r_1,I}$,

$$W^{\text{full}} = W \qquad \text{a.e. on } A_{r_0,r_1,I}.$$

Since the annuli exhaust $(\mathbb{R}^3 \setminus \{0\}) \times \mathbb{R}$, the annular coherent tail is the restriction of the full weak-$*$ limit W^{full} away from $x = 0$.

We take W^{full} as the full-cylinder representative of the coherent tail. It is locally bounded on every compact macroscopic cylinder, including cylinders meeting $x = 0$, and it realizes the same annular tail because it agrees with W on all punctured compact cylinders. Pressure gauges are not needed for the boundedness conclusion; if one records the pressure as well, the local pressure-gauge compactness in the terminal profile class gives distributional pressure-gradient limits on the same full cylinders after the usual time-dependent gauges. The construction uses only the uniform local boundedness inherited from the Type I terminal profile class and finite-dimensional affine normalization; it uses no global critical norm or global tail budget. $\square$

Lemma 9.106 (Failure of amplitude-normalized bounded-origin heredity is routed)**.** *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be generated by a bounded terminal sequence and realized by Theorem 9.104. Write*

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Consider amplitude-normalized translated windows

$$Z(\rho + a_j, \omega, s + b_j).$$

If, after passing to a subsequence, these translated windows do not admit a sequence-realized bounded-origin representative stable under the amplitude-normalized translation, then the ordered critical-tail protocol assigns a sequence-realized support profile to one of the already stratified lower alternatives

$$\mathcal{C}_{\text{conc}}, \qquad \mathcal{C}_{\text{aff}}, \qquad \mathcal{C}_{\log\text{diff}}, \qquad \mathcal{C}_{\text{Young}}.$$

Proof. The translated windows are still generated from the original blow-up sequence, so every realized subsequential profile extracted from them is a realized critical-tail profile. Apply the ordered critical-tail extraction theorem Theorem 9.98 to such a realized translated profile. Its alternatives are regenerated concentration, affine/parasitic collapse, log-diffuse escape, Young/defect critical-tail defect, or coherent critical-tail realization. The last alternative is exactly the successful bounded-origin heredity case in which the translated amplitude-normalized windows admit a bounded-origin representative stable under the normalized translation. Therefore failure of successful bounded-origin heredity leaves only the first four alternatives, namely $\mathcal{C}_{\text{conc}}$, $\mathcal{C}_{\text{aff}}$, $\mathcal{C}_{\log\text{diff}}$, and $\mathcal{C}_{\text{Young}}$. The corresponding profiles are sequence-realized by the support-transfer statements already built into the critical-tail compactification, so the translated family is formally routed to one of these lower strata. $\square$

Lemma 9.107 (Realized amplitude-normalized log-hull heredity)**.** *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be generated by a bounded terminal sequence and be realized by Theorem 9.104. Write*

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Let Z_* be a nonzero element of the amplitude-normalized log/time hull, obtained as a local limit

$$Z_*(\rho,\omega,s)=\lim_{j\to\infty} Z(\rho+a_j,\omega,s+b_j).$$

Then one of the following alternatives occurs:

(i) *the translated windows are realized by admissible critical-tail extractions from the original blow-up sequence, with bounded-origin representatives stable under the amplitude-normalized translation, and*

$$Z_*(\rho,\cdot,\cdot)\to 0 \qquad \text{as } \rho\to-\infty$$

in the local logarithmic topology;

(ii) *the failure of such realized bounded-origin heredity is routed by Lemma 9.106 to one of the already stratified lower alternatives*

$$\mathcal{C}_{\mathrm{conc}}, \qquad \mathcal{C}_{\mathrm{aff}}, \qquad \mathcal{C}_{\log\mathrm{diff}}, \qquad \mathcal{C}_{\mathrm{Young}}.$$

Proof. A translated log profile $Z(\rho+a_j,\omega,s+b_j)$ represents

$$e^{a_j}W(e^{a_j}x,s+b_j),$$

not merely $W(e^{a_j}x,s+b_j)$. Thus it is not admitted into the coherent critical-tail closure solely by changing the macroscopic scale in the no-amplitude blow-down. It is admitted only when the amplitude-normalized profile is itself realized by the normalized critical-tail extraction from the original terminal sequence.

Assume first that this realized amplitude-normalized extraction exists along the translated windows. The bounded-origin realization theorem applied to these realized representatives gives L^∞_{loc} bounds on every full compact macroscopic cylinder, including cylinders meeting $x=0$, with constants stable under the local limit. Hence Z_* corresponds to a velocity

$$W_*(r\omega,s)=r^{-1}Z_*(\log r,\omega,s)$$

which is locally bounded near $r=0$.

For $r=e^\rho\downarrow 0$, local boundedness gives

$$|Z_*(\rho,\omega,s)|=r\,|W_*(r\omega,s)|\le C_K r$$

on every compact angular-time set K. Thus $Z_*(\rho,\cdot,\cdot)\to 0$ locally as $\rho\to-\infty$, proving alternative (i).

It remains to identify the complementary case. If the amplitude-normalized translated windows cannot be realized with the same bounded-origin control, then Lemma 9.106 applies and gives alternative (ii). □

Lemma 9.108 (Homogeneous tails are incompatible with bounded-origin retention)**.** *Let $(W,Q)\in\mathcal{C}_{\mathrm{homcrit}}$ be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries the retained non-affine annular activity required in the definition of the coherent tail stratum. Then no such W exists.*

Proof. In the homogeneous critical-tail class,

$$W(r\omega,s)=r^{-1}A(\omega,s).$$

The bounded-origin realization gives a locally bounded representative on $\{0<r<r_0\}$ for each compact angular-time set. Hence

$$|A(\omega,s)|=r|W(r\omega,s)|\le C_K r$$

on every compact angular-time set K. Letting $r\downarrow 0$ gives $A=0$, and therefore $W\equiv 0$ on punctured compact cylinders. This is incompatible with the retained non-affine annular activity that defines a nonzero coherent critical tail. □

Lemma 9.109 (Log-periodic tails are incompatible with bounded-origin retention)**.** *Let* $(W, Q) \in \mathcal{C}_{\mathrm{logper}}$ *be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries retained non-affine annular activity. Then no such* W *exists.*

Proof. Write

$$Z(\rho, \omega, s) = e^{\rho} W(e^{\rho}\omega, s).$$

Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \to 0$ in the local logarithmic topology as $\rho \to -\infty$. If the log-period is $L \neq 0$, then

$$Z(\rho, \omega, s) = Z(\rho - kL, \omega, s) \qquad (k \in \mathbb{Z}).$$

For $L > 0$ take $k \to +\infty$, and for $L < 0$ take $k \to -\infty$. In both cases $\rho - kL \to -\infty$, so the right-hand side converges to zero. Thus $Z \equiv 0$, hence $W \equiv 0$ on punctured annular cylinders, contrary to retained non-affine annular activity. $\square$

Lemma 9.110 (Aperiodic coherent tails are incompatible with bounded-origin retention)**.** *Let* $(W, Q) \in \mathcal{C}_{\mathrm{ap}}$ *be a coherent critical tail obtained from the bounded-origin realization theorem. Assume it carries retained non-affine annular activity. Then no such* W *exists.*

Proof. Let $Z(\rho, \omega, s) = e^{\rho} W(e^{\rho}\omega, s)$, and let $\mathcal{M}$ be the nonzero minimal aperiodic logarithmic subsystem in the definition of $\mathcal{C}_{\mathrm{ap}}$, taken inside the realized amplitude-normalized hull. Choose $Z_* \in \mathcal{M}$. By Lemma 9.107, either a lower stratum has already occurred, or Z_* has a bounded-origin representative and hence

$$Z_*(\rho, \cdot, \cdot) \to 0 \qquad \text{as } \rho \to -\infty.$$

By the ordered definition of the coherent aperiodic stratum, the object under consideration is already inside the realized bounded-origin coherent hull. Hence the routing alternative of Lemma 9.106 is not admissible for an object genuinely lying in $\mathcal{C}_{\mathrm{ap}}$: if it occurred, the construction would have been assigned earlier to $\mathcal{C}_{\mathrm{conc}}$, $\mathcal{C}_{\mathrm{aff}}$, $\mathcal{C}_{\mathrm{log\,diff}}$, or $\mathcal{C}_{\mathrm{Young}}$, and the profile would not have entered the coherent aperiodic row. Therefore, for a profile genuinely lying in $\mathcal{C}_{\mathrm{ap}}$, only alternative (i) of Lemma 9.107 remains. In that case $0 \in \mathcal{H}(Z_*)$. Minimality gives $\mathcal{H}(Z_*) = \mathcal{M}$. Since the zero profile is a closed invariant subsystem, $\mathcal{M} = \{0\}$. This contradicts the retained non-affine annular activity that makes the aperiodic coherent tail a nonzero residual tail. $\square$

Corollary 9.111 (Coherent critical-tail alternative closure)**.** *One has*

$$\mathcal{C}_{\mathrm{homcrit}} = \varnothing, \qquad \mathcal{C}_{\mathrm{logper}} = \varnothing, \qquad \mathcal{C}_{\mathrm{ap}} = \varnothing.$$

Proof. Lemma 9.108 excludes $\mathcal{C}_{\mathrm{homcrit}}$, Lemma 9.109 excludes $\mathcal{C}_{\mathrm{logper}}$, and Lemma 9.110 excludes $\mathcal{C}_{\mathrm{ap}}$. Each contradiction uses the same retained non-affine annular activity that makes the coherent tail a nonzero realized residual object, so no coherent critical-tail lower stratum remains. $\square$

Corollary 9.112 (Critical-tail exclusion, complete form)**.** *No nonzero realized critical-tail profile remains after terminal concentration-set removal and affine/parasitic lower-stratum removal.*

Proof. Let a nonzero realized critical-tail profile remain after terminal concentration-set removal and affine/parasitic lower-stratum removal. The extraction first either routes to an affine-pressure or pressure/noncompact branch, or else produces a full $\mathcal{T} \in \mathfrak{T}_M$ in alternative (iii) of Theorem 9.91. The pre-$\mathfrak{T}_M$ branches are assigned by affine routing and the pressure/noncompact terminal protocol to lower strata, so they cannot remain in the retained residual class. In the remaining case, by Theorem 9.98, the ordered priority alternative is one of: regenerated concentration, affine/parasitic collapse, log-diffuse defect, Young/defect critical tail, coherent homogeneous tail, coherent log-periodic tail, or coherent aperiodic tail.

Regenerated concentration contradicts terminal concentration-set removal. The affine/parasitic alternative is routed by Theorem 9.99 to $\mathcal{C}_{\mathrm{aff}}$, hence it is not a retained normalized residual representative. The log-diffuse alternative is impossible by Corollary 9.57, and the Young alternative is impossible by Corollary 9.100. The three coherent alternatives are impossible by Corollary 9.111. All alternatives are therefore impossible, so no such $\mathcal{T}$ remains. ☐

Theorem 9.113 (No mixed compact–diffuse residual profile)**.** *Let (U, Π) be a retained terminal profile in the refined residual class, after the affine normalization of Definition 8.16. Then U cannot have a diffuse exterior oscillation remainder coexisting with its compact retained core. Equivalently, the paired terminal configurations*

$$\textit{one compact concentration core} + \mathcal{R}_{\mathrm{diff}},$$

and their finite, infinite, or continuum concentration-set analogues do not occur in the refined residual class.

Proof. Assume that such a diffuse exterior remainder exists. By the paired concentration-set decomposition Theorem 9.24, remove neighborhoods of the concentration set at every positive constant-in-time oscillation threshold. The mixed assumption says that a nonzero diffuse oscillation residual remains. Apply Theorem 9.38.

If regenerated concentration occurs, then a positive oscillatory observer survives after all positive-threshold concentration sets were removed, contradicting the definition of the diffuse residual. If affine/parasitic collapse occurs, then Definition 8.16 and Theorem 10.12 places the relevant descendant hull in an already closed lower stratum. This cannot occur inside $\mathcal{C}_{\mathrm{gen}}$.

The only remaining alternative is a critical diffuse tail. By Theorem 9.91, it either routes first to an affine-pressure or pressure/noncompact branch, or else generates a nonzero retained $\mathcal{T} \in \mathfrak{T}_M$. The routed branches have already been assigned to lower terminal strata and cannot remain inside the refined residual class; the retained $\mathfrak{T}_M$ case contradicts Corollary 9.112. Hence no mixed compact–diffuse residual profile exists. If the mixed configuration were witnessed by an escaping retained recentering, Lemma 9.114 would first route that recentering to the appropriate terminal branch; the contradiction occurs only after the corresponding branch closure theorem is applied. ☐

Lemma 9.114 (Escaping retained recenterings are routed by the terminal protocol)**.** *An escaping retained recentering coexisting with a compact retained core is not an immediate contradiction. By Corollary 9.11, such a recentering is assigned to one of the following: a locally compact retained descendant, a diffuse exterior defect, a pressure/noncompact defect, or a critical-tail defect. The locally compact case is recorded as a separated retained successor or a concentration-chain successor. The diffuse, pressure/noncompact, and critical-tail branches are then handled by the corresponding terminal closure arguments.*

Proof. This is the routing content of Corollary 9.11. Alternatives (i) and (ii) there produce locally compact descendants, which are precisely the separated-family or chain successors used in the terminal protocol. Alternative (iii) routes the escaping recentering to the diffuse, pressure/noncompact, or critical-tail terminal branch. No contradiction is claimed at the classification stage; the contradiction appears only after the corresponding branch closure theorem is applied. ☐

Definition 9.115 (Finite separated family of retained velocity concentration profiles)**.** A finite separated family of retained velocity concentration profiles is a finite family of recentering sequences. For the original terminal extraction these are the parabolic recenterings of Definition 9.1; for descendants of a centered profile they are the covariant tail recenterings of Definition 8.4. We write them uniformly as $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

(i) the recentering sequences are mutually separated;

(ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile (U_j, Π_j);
(iii) each limiting profile has retained local velocity concentration:

$$\mathcal{V}_1(U_j; 0) \geq \eta_* \qquad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

10. Separated concentration profiles, indecomposability, and sequence-L^3

We now close the terminal alternatives carrying retained local velocity concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only local profile outcomes: an indecomposable retained velocity concentration profile, a mixed compact–diffuse or compact–noncompact successor, or a chain of retained concentrating descendants. The mixed compact–diffuse successors are excluded by Theorem 9.113, while the compact–noncompact successors are routed by Lemma 9.114; infinite chains of retained concentrating descendants and finite loops are excluded by the compact recurrence theorem.

Definition 10.1 (Tail limits and terminal indecomposability)**.** Let (U, Π) be a terminal profile with retained local velocity concentration. A retained concentrating tail limit is a limit of covariant tail recenterings

$$\mathscr{T}_{(x_n,\tau_n)}(U, \Pi), \qquad |x_n| + |\tau_n|^{1/2} \to \infty,$$

in the local smooth velocity topology and the local $L^{3/2}$ pressure topology modulo gauges, with positive retained local velocity concentration at the origin, equivalently with retained local velocity concentration at some rational level $\eta \in \mathbb{Q}_{>0}$. An ordinary fixed spatial shift is not used in this definition; by Remark 8.3 and Definition 8.4, it would introduce an unbounded spurious drift for escaping centers. Diffuse exterior concentration means that there are $\tau_n \to -\infty$, $R_n \to \infty$, and $\eta > 0$, such that

$$\int_{|y|>R_n} |U(y, \tau_n)|^3 \, dy \geq \eta,$$

while every unit-scale exterior covariant recentering sequence based in the parabolic exterior windows

$$\{|y| > R_n\} \times (\tau_n - 1, \tau_n + 1)$$

has velocity concentration below the velocity threshold, unless that recentering sequence has loss of local compactness. If a pressure-inclusive CKN lower bound remains without velocity concentration, it is assigned by Lemma 9.5 to pressure loss, the affine/parasitic quotient, or neighboring velocity concentration. Thus the word "diffuse" is always parabolic: localization is tested by unit parabolic cylinders, not only by spatial balls on the distinguished time slices. An escaping non-compact recentering sequence is a covariant tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained velocity concentration profile is terminally indecomposable if it has no covariant tail recentering limit carrying positive retained local velocity concentration at any rational level $\eta \in \mathbb{Q}_{>0}$, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

Lemma 10.2 (Compact state-space time-slice concentration)**.** *Let $\mathcal{K} \subset \mathfrak{X}_M^{\#}$ be compact in the normalized local profile topology, with pressure representatives taken from the fixed setup pressure* *atlas and with affine/parasitic modes already removed by the canonical representative or assigned to $\mathcal{C}_{\mathrm{aff}}$. For every $\varepsilon_0 > 0$ there are*

$$\delta_{\mathrm{sl}} = \delta_{\mathrm{sl}}(\mathcal{K}, \varepsilon_0) \in (0,1), \qquad c_{\mathrm{sl}} = c_{\mathrm{sl}}(\mathcal{K}, \varepsilon_0) > 0,$$

such that every $(U, \Pi) \in \mathcal{K}$ satisfying

$$\int_{B_1(0)} |U(y,0)|^3 \, dy \geq \varepsilon_0 > 0,$$

also satisfies, after the standard local pressure gauge,

$$\iint_{B_1(0) \times (-\delta_{\mathrm{sl}}^2, 0)} |U|^3 \, dy \, d\tau \geq c_{\mathrm{sl}}, \qquad \text{and hence} \qquad \mathcal{V}_1(U; 0) \geq c_{\mathrm{sl}}.$$

The same constants apply to any covariant tail recentering family whose normalized pressure-atlas closure is contained in the same compact set $\mathcal{K}$.

Proof. For $(U, \Pi) \in \mathcal{K}$ define

$$F(U, \tau) := \int_{B_1} |U(y, \tau)|^3 \, dy.$$

The topology of $\mathfrak{X}_M^{\#}$ gives local C^0 convergence of the velocity. Hence $(U, \tau) \mapsto F(U, \tau)$ is continuous on $\mathcal{K} \times [-1, 1]$. Since this set is compact, the continuity is uniform. Choose $\delta_{\mathrm{sl}} \in (0,1)$ so small that, for every $(U, \Pi) \in \mathcal{K}$,

$$|F(U,\tau) - F(U,0)| \leq \frac{1}{2}\varepsilon_0 \qquad \text{whenever } -\delta_{\mathrm{sl}}^2 < \tau < 0.$$

If $F(U, 0) \geq \varepsilon_0$, then $F(U, \tau) \geq \varepsilon_0/2$ throughout that time interval. Therefore

$$\iint_{B_1 \times (-\delta_{\mathrm{sl}}^2, 0)} |U|^3 \, dy \, d\tau \geq \frac{1}{2} \varepsilon_0 \delta_{\mathrm{sl}}^2.$$

Set $c_{\mathrm{sl}} := \varepsilon_0 \delta_{\mathrm{sl}}^2 / 2$. The pressure contribution is not used to certify nontriviality; the pressure atlas is used only to define the compact normalized state space in which the uniform time modulus is valid.

The dependence on $\mathcal{K}$, rather than only on M, is essential. A spatially homogeneous parasitic family $U_N(y, \tau) = b_N(\tau)$, paired with the affine pressure $-(\partial_\tau b_N + \frac{1}{2} b_N) \cdot y$, can have a uniform velocity L^∞ bound and fixed time-slice mass while losing every uniform backward time modulus. Such a family is not compact in the normalized pressure-atlas topology unless the homogeneous modes have a convergent modulus, and nonzero homogeneous modes are assigned to the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$. □

Lemma 10.3 (Quantitative exterior slice-to-defect alternative)**.** *Let $(U, \Pi) \in \mathfrak{X}_M^{\#}$ be represented after canonical affine/parasitic normalization. Let $\tau_n \to -\infty$, $R_n \to \infty$, and*

$$E_n := \{|y| > R_n\}$$

satisfy

$$\int_{E_n} |U(y, \tau_n)|^3 \, dy \geq m_0 > 0.$$

If the exterior covariant recenterings in the windows

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

do not have compact closure in the normalized local topology with the setup pressure atlas fixed, then alternative (i) *below occurs. Otherwise, after passing to a subsequence, let $\mathcal{K}_{\mathrm{ext}} \subset \mathfrak{X}_M^{\#}$ be a compact set containing all these recentered profiles. Fix $\alpha > 0$, and let $c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}}, \alpha) > 0$ be* *the parabolic lower bound given by Lemma 10.2 for the compact state class $\mathcal{K}_{\mathrm{ext}}$ and unit ball time-slice mass α. Define*

$$m_n^{\mathrm{vel}} := \sup_{\substack{|x|>R_n-2 \\ |s-\tau_n|\le 1}} \mathcal{V}_1\big(\mathscr{T}_{(x,s)}U;0\big).$$

If $m_n^{\mathrm{vel}} < c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha)$, then every unit ball with center in the exterior collar satisfies

$$\int_{B_1(x)} |U(y,\tau_n)|^3\,dy < \alpha \qquad (|x| > R_n - 1).$$

Consequently there are measurable exterior subsets $F_n \subset E_n$, escaping every compact set, such that

$$m_0 \le \int_{F_n} |U(y,\tau_n)|^3\,dy \le 2m_0,$$

and, after restricting F_n to a bounded-overlap unit cover, the normalized exterior time-slice measures have no unit-observer atom larger than $C_{\mathrm{ov}}\alpha/m_0$, where C_{ov} is the overlap constant.

Moreover, at least one of the following alternatives occurs after passing to a subsequence:

(i) *an exterior covariant recentering has loss of local compactness or pressure-gauge compactness;*
(ii) *a pressure-inclusive exterior unit cylinder has a lower bound not carried by velocity, and is assigned by Lemma 9.5;*
(iii) *the exterior mass is critical only after the logarithmic or normalized-tail compactification and enters the corresponding critical-tail alternative;*
(iv) *the remaining exterior mass is a diffuse exterior concentration in the sense of Definition 10.1.*

Proof. If the exterior recentering family has no compact closure in the normalized pressure-atlas topology, this is precisely the local compactness or pressure gauge failure alternative (i). We therefore work under the compact closure assumption and use the compact set $\mathcal{K}_{\mathrm{ext}}$ from the statement.

Since $|U(\cdot,\tau_n)|^3dy$ is a nonatomic measure absolutely continuous with respect to Lebesgue measure, the lower bound on E_n allows us to choose a measurable subset $F_n \subset E_n$ with

$$m_0 \le M_n := \int_{F_n} |U(y,\tau_n)|^3\,dy \le 2m_0.$$

For instance, take an exterior truncation with mass larger than m_0 and, if necessary, cut it by a level set of a smooth radial coordinate to reduce the mass below $2m_0$. Because $F_n \subset E_n$ and $R_n \to \infty$, these sets escape every compact set.

Let $\{B_1(x_{n,j})\}_j$ be a measurable unit-ball cover of F_n, chosen with centers $|x_{n,j}| > R_n - 1$ and with overlap at most C_{ov} after enlarging the balls by a fixed harmless factor. Suppose that for some j,

$$\int_{B_1(x_{n,j})} |U(y,\tau_n)|^3\,dy \ge \alpha.$$

Apply Lemma 10.2 to the covariantly recentered solution $\mathscr{T}_{(x_{n,j},\tau_n)}(U,\Pi)$, which belongs to $\mathcal{K}_{\mathrm{ext}}$. At local time 0 the ball $B_1(0)$ is the original ball $B_1(x_{n,j})$. Hence

$$\mathcal{V}_1\Big(\mathscr{T}_{(x_{n,j},\tau_n)}U;0\Big) \ge c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha),$$

so the parabolic velocity part of the raw atlas defect measure on that unit cylinder has mass at least $c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha)$. This is the only "thickening" used here: it applies to a unit ball with a quantitative time-slice lower bound, not to an arbitrarily diffuse exterior slice as a whole. The displayed lower bound contradicts $m_n^{\mathrm{vel}} < c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}}, \alpha)$. This proves the stated unit-ball time-slice bound.

Define discrete exterior slice measures on the observer centers by

$$\lambda_n^{\mathrm{sl}} := \frac{1}{M_n} \sum_j \left(\int_{B_1(x_{n,j}) \cap F_n} |U(y, \tau_n)|^3 \, dy \right) \delta_{(x_{n,j}, \tau_n)}.$$

Their total masses are bounded above and below by constants depending only on the overlap convention, and each unit observer ball has mass at most $C_{\mathrm{ov}} \alpha / m_0$. Since $R_n \to \infty$ and $\tau_n \to -\infty$, these normalized measures escape every compact subset of the covariant observer space. Passing to the observer compactification gives a nonzero boundary measure unless one of the local extraction inputs fails along some exterior observer sequence.

If local compactness or pressure-gauge compactness fails along such a sequence, we are in alternative (i). If compactness holds but a lower bound in a unit exterior cylinder is pressure-only, then Lemma 9.5 gives alternative (ii). If the mass is not seen at unit scale but becomes critical only after the logarithmic or normalized-tail compactification, it is the already listed critical-tail alternative (iii). In the complementary case the escaping exterior slice measure persists, all unit exterior velocity observers are below the velocity threshold fixed by $c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}}, \alpha)$, pressure-only observers have been assigned to the pressure alternatives, and no noncompact recentering remains. The result is the parabolic diffuse exterior concentration alternative in Definition 10.1. The argument is local on unit covariant cylinders and uses no global L^3 or CKN budget. □

Lemma 10.4 (One retained tail limit gives two separated concentration profiles)**.** *Let U be a retained velocity concentration terminal profile. If U has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained velocity concentration profiles.*

Proof. The origin recentering sequence carries retained velocity concentration:

$$\mathcal{V}_1(U; 0) \geq \eta_*.$$

Let W be a retained concentrating tail limit. By definition there are covariant relative tail recentering parameters $\zeta_n = (x_n, \tau_n)$, with $|x_n| + |\tau_n|^{1/2} \to \infty$, such that $\mathscr{T}_{\zeta_n}(U, \Pi) \to (W, \Pi_W)$ in the terminal topology and

$$\mathcal{V}_1(W; 0) \geq \eta_*.$$

By strong local convergence of the velocity,

$$\mathcal{V}_1(\mathscr{T}_{\zeta_n} U; 0) \geq \frac{1}{2} \eta_*$$

for all sufficiently large n. Thus, for large n, the origin recentering sequence and the covariant tail sequence ζ_n are separated and have retained local velocity concentration. They form a finite separated two-profile concentration family. □

Proposition 10.5 (No finite separated family and local exclusions imply indecomposability)**.** *After finite separated-family, mixed compact–diffuse, and noncompact escaping alternatives have been excluded, every single retained velocity concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

Proof deferred. The proof is given immediately after Theorem 10.65. This proposition is not used in the proof of that theorem. □

Lemma 10.6 (Indecomposable sequence-L^3 completeness)**.** *Let $(U, \Pi) \in \mathfrak{X}_M^{\#}$ be represented after canonical affine/parasitic normalization. Suppose U has retained local velocity concentration* *and is terminally indecomposable in the sense of Definition 10.1. Then there exists a sequence $\tau_k \to -\infty$ such that*

$$\sup_k \|U(\cdot,\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y,\tau)|^3\,dy \in [0,\infty].$$

If the conclusion fails, then

$$\liminf_{\tau\to-\infty} A(\tau) = \infty.$$

Choose $\tau_n \to -\infty$ such that $A(\tau_n) \to \infty$, allowing the value $+\infty$. Since U is bounded by some M, for each finite R,

$$\int_{|y|\le R} |U(y,\tau_n)|^3\,dy \le M^3|B_R|.$$

Choose $R_n \to \infty$ so slowly that, after discarding finitely many n,

$$\int_{|y|>R_n} |U(y,\tau_n)|^3\,dy \ge 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose R_n with $M^3|B_{R_n}| \le A(\tau_n)/2$.

Define the parabolic exterior velocity concentration level

$$m_n^{\mathrm{vel}} := \sup_{\substack{|x|>R_n-2\\ |s-\tau_n|\le 1}} \mathcal{V}_1\big(\mathscr{T}_{(x,s)}U;0\big).$$

This is the quantity which can be passed to a limiting profile by strong local L^3 convergence. Pressure-inclusive CKN mass in the same exterior windows is handled separately by Lemma 9.5.

If some unit exterior covariant recentering sequence in the windows

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

has loss of local compactness or pressure-gauge compactness, terminal indecomposability is contradicted by the noncompact defect alternative. Hence we may assume that the normalized pressure-atlas closure of all such exterior recentering profiles is compact; denote one compact closure by $\mathcal{K}_{\mathrm{ext}} \subset \mathfrak{X}_M^{\#}$.

If $\limsup_{n\to\infty} m_n^{\mathrm{vel}} = \gamma > 0$, then, passing to a subsequence, choose (x_n,s_n), with

$$|x_n| > R_n - 2, \qquad |s_n - \tau_n| \le 1,$$

such that

$$\mathcal{V}_1\big(\mathscr{T}_{(x_n,s_n)}U;0\big) \ge \frac{\gamma}{2}.$$

Since $R_n \to \infty$ and $\tau_n \to -\infty$, these are escaping covariant tail recenterings. If they have loss of local compactness, terminal indecomposability is contradicted. Otherwise local compactness gives a subsequential terminal limit (W,Π_W), and strong local L^3 convergence gives

$$\mathcal{V}_1(W;0) \ge \frac{\gamma}{4} > 0.$$

Choose a rational $0 < \eta < \gamma/4$. Thus W is a covariant tail limit with retained local velocity concentration at level η, contradicting the all-positive-threshold definition of terminal indecomposability. Therefore

$$m_n^{\mathrm{vel}} \to 0.$$

Fix a small $\alpha_0 > 0$, and let

$$c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}},\alpha_0) > 0$$

be the compact-state constant from Lemma 10.2, as used in Lemma 10.3. Since $m_n^{\mathrm{vel}} \to 0$, for all large n,

$$m_n^{\mathrm{vel}} < c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}}, \alpha_0).$$

If the velocity concentration is small but the pressure-inclusive CKN mass is not, Lemma 9.5 assigns the recentering either to pressure compactness failure, to the affine/parasitic pressure quotient, or to a neighboring unit cylinder with positive velocity concentration. Each case is already excluded by terminal indecomposability or by lower-stratum removal. If the exterior mass is critical only after log-window normalization, the resulting normalized support profile is a critical-tail defect and is one of the already excluded terminal alternatives. If no such recentering loses compactness and no critical-tail normalization is selected, then the exterior time-slice lower bound

$$\int_{|y|>R_n} |U(y,\tau_n)|^3\,dy \geq 1$$

together with the parabolic velocity smallness $m_n^{\mathrm{vel}} < c_{\mathrm{sl}}(\mathcal{K}_{\mathrm{ext}}, \alpha_0)$ is now handled quantitatively by Lemma 10.3. Apply that lemma with $m_0 = 1$, $\alpha = \alpha_0$, and the displayed inequality. If the lemma produces local compactness loss, pressure-gauge loss, pressure-only mass, or a critical-tail normalization, the branch is one of the already covered noncompact, pressure, affine/parasitic, or critical-tail alternatives. In the remaining case it produces diffuse exterior concentration in the sense of Definition 10.1. This again contradicts terminal indecomposability. Both parabolic alternatives are impossible, so the desired sequence exists. □

Proposition 10.7 (Stability of the bounded mild formulation)**.** *Let (U_m, Π_m) be normalized parasitic-free centered profiles whose physical pullbacks are bounded mild Navier–Stokes solutions on every finite terminal interval. Assume $U_m \to U$ locally smoothly in centered variables and that the U_m are uniformly bounded on compact centered time intervals. Then the physical pullback of U is also bounded mild on every finite terminal interval. Let*

$$u(x,t) := (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t<0,$$

be the physical pullback. For each fixed $T<0$, define the finite terminal shift

$$u^T(x,s) := u(x,T+s) = (-(T+s))^{-1/2} U\left(\frac{x}{\sqrt{-(T+s)}}, -\log(-(T+s))\right), \qquad s<0,$$

Then, after translating the interval so that its terminal time is 0, one has, for all $-\infty < s < t < 0$,

$$u^T(t) = e^{(t-s)\Delta} u^T(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P}\nabla\cdot(u^T \otimes u^T)(\sigma)\,d\sigma \tag{10.1}$$

in the bounded mild duality sense used in the setup paper. The same conclusion is preserved by centered time translations, covariant translations, retained tail limits, concentrating descendants, recurrent-core limits, and affine normalization outside $\mathcal{C}_{\mathrm{aff}}$.

Proof. For each m, the normalized setup representative satisfies the Duhamel formula on every finite physical interval. Fix $T<0$ and $-\infty < s < t < 0$. Pull the centered convergence back to the compact physical time interval $[s,t]$ after the finite terminal shift. The velocities converge locally smoothly and are uniformly bounded; hence

$$u_m^T \otimes u_m^T \to u^T \otimes u^T$$

locally in every finite L^q, and in particular in distributions. Testing the Duhamel identity for u_m^T against a compactly supported smooth vector field and using the heat-kernel/Leray-projection duality gives convergence of the linear heat term and of the bilinear Duhamel term. The bounded mild duality formulation used in the setup paper then extends the identity from compactly supported tests to the standard bounded mild class, yielding (10.1) for u^T.

Centered time translations and covariant translations are exact symmetries of (1.2); in physical variables they are compositions of translations, scalings, and finite time shifts, which preserve the Duhamel identity. Retained tail limits, concentrating descendants, and recurrent-core limits are locally smooth limits of such normalized representatives, so the preceding limit argument applies. The affine normalization either removes a spatially homogeneous parasitic component or assigns the profile to $\mathcal{C}_{\mathrm{aff}}$; outside $\mathcal{C}_{\mathrm{aff}}$ no homogeneous parasitic forcing remains, and the same Duhamel identity holds for the canonical representative. □

Theorem 10.8 (Parasitic-free mildness inheritance for normalized profiles)**.** *Let* (U, Π) *be a retained terminal profile, retained covariant tail descendant, concentrating descendant limit, or recurrent-core limit represented in the normalized residual profile class generated by the Seregin extraction in the setup paper. Then exactly one of the following alternatives applies.*

(i) (U, Π) *belongs to the affine/parasitic lower stratum of Definition 8.16;*

(ii) *after the canonical affine normalization already built into the residual representative, the physical pullback*

$$u(x,t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \qquad t < 0,$$

is parasitic-free and has the following property: for every $T < 0$*, the finite-terminal shift*

$$u^T(x,s) := u(x, T+s), \qquad s < 0,$$

is a bounded mild ancient Navier–Stokes solution in the Albritton–Barker sense on $\mathbb{R}^3 \times (-\infty, 0)$*.*

Proof. Affine-constant centered profiles are precisely the homogeneous parasitic profiles removed by Definition 8.16; they are assigned to alternative (i). We therefore consider a representative outside that lower stratum.

The setup paper constructs normalized Seregin limits as centered bounded mild ancient representatives, after removal of the spatially homogeneous parasitic mode. The stability statement Proposition 10.7 shows that the limiting operations used in this paper preserve the bounded mild formulation: exact centered symmetries preserve the Duhamel identity, local limits preserve it by bounded mild duality, and affine normalization either removes the homogeneous parasitic mode or places the profile in $\mathcal{C}_{\mathrm{aff}}$. Thus retained terminal profiles, covariant tail descendants, concentrating descendant limits, and recurrent-core limits remain parasitic-free mild unless they have fallen into the affine/parasitic lower stratum.

Finally fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the physical pullback

$$u(x,t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right)$$

is smooth and bounded by $(-T)^{-1/2}\|U\|_{L^\infty}$, and it solves the physical Navier–Stokes equations by the direct self-similar change of variables. By the inherited parasitic-free mild formulation, for $-\infty < s < t \le T$,

$$u(t) = e^{(t-s)\Delta} u(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P}\nabla \cdot (u \otimes u)(\sigma)\, d\sigma.$$

After translating the finite terminal time T to 0, this is the mild ancient formulation for u^T on every finite forward interval $(-\infty, 0)$. The bound $\|u^T\|_{L^\infty(\mathbb{R}^3 \times (-\infty,0))} \le (-T)^{-1/2}\|U\|_{L^\infty}$ is finite. Thus u^T is a genuine bounded mild ancient solution in the sense required by the Albritton–Barker endpoint theorem. □

Theorem 10.9 (Albritton–Barker endpoint Liouville theorem)**.** *Let $u \in L^\infty(\mathbb{R}^3 \times (-\infty, 0))$ be a bounded mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \to -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem in the sequence-L^3 form proved by Albritton and Barker [1]. □

Lemma 10.10 (No retained indecomposable concentration profile)**.** *No retained terminally indecomposable concentration profile exists.*

Proof. Let U have retained local velocity concentration and be terminally indecomposable. By Lemma 10.6, choose $\tau_k \to -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \to -\infty$, the sequence satisfies $t_k \to -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By Theorem 10.8, either U is in the affine/parasitic lower stratum, which is not part of the normalized residual class, or else for each $T < 0$

$$u^T(x, s) := u(x, T + s), \qquad s < 0,$$

is a bounded mild ancient solution. Choose $T < 0$ and discard finitely many indices so that $t_k < T$. Then $s_k := t_k - T \to -\infty$, and

$$\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} \le \sup_j \|U(\cdot, \tau_j)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Theorem 10.9 gives $u^T \equiv 0$. Since $T < 0$ was arbitrary, the physical pullback vanishes on $\mathbb{R}^3 \times (-\infty, 0)$, and therefore $U \equiv 0$. This contradicts retained local velocity concentration. □

Lemma 10.11 (Diagonal lifting of retained concentrating tail limits)**.** *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \ge 2$, be a finite retained velocity concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let (U_j, Π_j) be the terminal profile obtained in the j-th recentering. If (U_j, Π_j) has a retained concentrating tail limit (W, Π_W), then there is a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_n)$, in the sense of Definition 9.2, for the original terminal sequence such that:*

- *(i) $\mathfrak{d}$ is asymptotically separated from g_j;*
- *(ii) the recentered original sequence at $\mathfrak{d}$ converges locally to (W, Π_W), after admissible pressure gauges;*
- *(iii) $\mathfrak{d}$ has retained local velocity concentration.*

Proof. By definition of retained concentrating tail limit, there are relative covariant tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \to \infty,$$

such that

$$\mathscr{T}_{\zeta_m}(U_j, \Pi_j) \to (W, \Pi_W)$$

locally in the terminal topology, and

$$\mathcal{V}_1(W; 0) \ge \eta_*.$$

Fix m. Since $V_n^{g_j} \to U_j$ locally smoothly and $(P_n^{g_j}) \to \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed covariant tube

$$\{(y + e^{s/2}\xi_m, s + \sigma_m) : (y, s) \in Q_m(0, 0)\}.$$

Therefore one can choose $n(m)$ so large that the covariantly recentered original fields

$$V_{n(m)}(y + y_{j,n(m)} + e^{s/2}\xi_m,\, s + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within m^{-1} of the $\mathscr{T}_{\zeta_m}$-recentered fields of (U_j, Π_j) on $Q_m(0, 0)$.

Set

$$d_m := (g_{j,n(m)}; \xi_m, \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a combined parabolic-covariant descendant recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since the covariant parabolic distance from d_m to $g_{j,n(m)}$ is $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the j-th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to (W, Π_W), after the admissible pressure gauges used above. The diagonal convergence gives strong local L^3-convergence of the velocity, so

$$\liminf_{m\to\infty} \mathcal{V}_1\big(V_{n(m)}^{\mathfrak{d}}; 0\big) \geq \mathcal{V}_1(W; 0) \geq \eta_*.$$

Thus $\mathfrak{d}$ has retained local velocity concentration. The pressure gauges are inherited through the same diagonal construction, but pressure-inclusive $\mathcal{E}_1$ is not used to certify retained nonzero concentration. □

Theorem 10.12 (Heredity of retained descendants). *Let (U, Π) be a normalized retained residual profile in $\mathfrak{X}_M$, and let (W, Π_W) be a retained covariant tail descendant of (U, Π). Then $(W, \Pi_W) \in \mathfrak{X}_M$, it carries retained local velocity concentration at the descendant threshold fixed in Definition 3.4, and it has the same parasitic-free normalization as a terminal profile generated directly by the Seregin extraction in the setup paper. Consequently exactly one of the following ordered outcomes occurs:*

(i) (W, Π_W) *belongs to one of the previously closed lower strata, in which case that descendant is already excluded;*

(ii) (W, Π_W) *belongs to the normalized residual terminal profile class, so the concentrating descendant and terminal-indivisibility arguments may be iterated from W.*

Proof. The descendant is obtained as a limit of covariant tail recenterings $\mathscr{T}_{\zeta_n}(U, \Pi)$. By Lemma 8.9, the recentered sequence is locally precompact in $\mathfrak{X}_M$, and the limit remains a bounded smooth locally suitable centered solution with the same L^∞ bound and compatible pressure gauges. Retained local velocity concentration is part of the definition of retained descendant; when the descendant is produced through a limit passage, the velocity semicontinuity in Lemma 9.4 fixes the threshold loss once and for all through Definition 3.4. The affine/parasitic normalization is canonical for the residual representative and is preserved by T_a, Θ_s, and their composition $\mathscr{T}_{\zeta_n}$; if a limit is affine-parasitic, it is assigned to the affine-constant lower stratum of Definition 8.16. Otherwise it is represented in the same normalized residual profile class. The ordered residual decomposition then places the descendant either in a previously closed lower class or in the generic residual profile class. This is a local profile reduction and does not use a global critical norm or global tail budget. □

Lemma 10.13 (Escaping recenterings in a separated family). *Let $A = \{\mathfrak{s}_1, \ldots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained velocity concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, the following ordered alternatives exhaust the possible successor profiles:*

(i) the profile (U_j, Π_j) *is terminally indecomposable;*
(ii) (U_j, Π_j) *has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;*
(iii) (U_j, Π_j) *has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local velocity concentration.*

Proof. This is the definition of terminal indecomposability plus the diagonal lifting lemma, with no disjointness assertion. If (U_j, Π_j) is terminally indecomposable, (i) holds. If it is not terminally indecomposable, at least one of the three excluded alternatives in Definition 10.1 occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, Lemma 10.11 lifts the limit to a recentering sequence of the original terminal sequence with retained local velocity concentration, and Theorem 10.12 places the descendant either in a closed lower stratum or back in the normalized residual profile class, giving (iii) in the iterative residual case. If several successor profiles coexist, they are all recorded; the next lemma removes the nonconcentrating ones. □

Lemma 10.14 (No diffuse or non-compact escaping recentering from a retained family)**.** *In the setting of Lemma 10.13, alternative* (ii) *cannot occur.*

Proof. Suppose that (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in Lemma 10.11, this tail profile is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by the imported setup results. The fact that the diffuse or noncompact profile is not the sole source of the original compact concentration does not exclude coexistence with the compact core (U_j, Π_j). In the diffuse case, Theorem 9.113 rules out exactly this coexistence in the refined residual class. In the escaping noncompact case, Lemma 9.114 shows that the profile is not an immediate contradiction but is routed to a locally compact retained descendant, a diffuse defect, a pressure/noncompact defect, or a critical-tail defect. The locally compact routed case is recorded as a successor, so it does not belong to alternative *(ii)*. The diffuse, pressure/noncompact, and critical-tail routed cases are closed by the corresponding terminal branch arguments. Therefore alternative *(ii)* cannot occur for a refined residual family with retained local velocity concentration. □

Definition 10.15 (Retained velocity concentration successor relation)**.** For a fixed threshold $\eta > 0$, let

$$\mathfrak{A}_\eta := \mathfrak{A}_\eta^\# = \{U \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

be the compact non-affine oscillation set of Corollary 8.20. We write

$$U \,\mathcal{R}_\eta\, V$$

if $U, V \in \mathfrak{A}_\eta$ and V is a retained concentrating covariant tail descendant of U.

Remark 10.16 (Why the profile-only relation is not declared closed)*.* Local convergence $U_m \to U$ controls U_m only on fixed compact cylinders. A witness for $U_m \mathcal{R}_\eta V_m$ may be centered at a point escaping with m, outside every cylinder on which $U_m \to U$ is known. Hence the diagonal argument for closedness requires the observer witnesses, their pressure charts, and their realization from the original sequence as part of the state. The argument below uses the given realized chain directly and therefore requires no profile-only closedness assertion. Loss of compactness of the witnesses is already routed by Corollary 9.11; loss of covariant balance after the witnesses remain compact is recorded in Definition 10.24.

Theorem 10.17 (Compact event laws for a realized active chain)**.** *Let*

$$U_0\mathcal{R}_\eta U_1\mathcal{R}_\eta U_2\mathcal{R}_\eta\cdots \tag{10.2}$$

be an actual realized chain. The event-count measures

$$\nu_N := \frac{1}{N}\sum_{j=0}^{N-1}\delta_{U_j} \qquad (N\geq 1) \tag{10.3}$$

have weak- limit points in $\mathcal{P}(\mathfrak{X}^{\#}_M)$, and every such limit ν satisfies*

$$\nu(\mathfrak{A}_\eta)=1. \tag{10.4}$$

The same conclusion holds for the shift empirical measures of the single given path in the closure of its shift orbit. No closedness of $\mathcal{R}_\eta$ is used.

Proof. The normalized state space $\mathfrak{X}^{\#}_M$ is compact and metrizable by Lemma 8.11. Hence its probability measures are weak-* compact. Every ν_N is supported on the closed set $\mathfrak{A}_\eta$; the Portmanteau theorem therefore gives $\nu(\mathfrak{A}_\eta)=1$ for every weak-* limit. The closure of the shift orbit of the one-sided sequence $(U_j)_{j\geq 0}$ is compact in $(\mathfrak{X}^{\#}_M)^{\mathbb{N}}$. Averaging the Dirac masses of its shifts and using the endpoint identity

$$\frac{1}{N}\sum_{j=0}^{N-1}\big(F(S^{j+1}p)-F(S^jp)\big)=\frac{F(S^Np)-F(p)}{N}$$

gives a shift-invariant limit. Its zeroth marginal is a limit of (10.3) and hence satisfies (10.4). A limit path need not retain the successor edges, which is why no assertion about the closed path space is made. □

Lemma 10.18 (Concatenation of retained concentrating descendants)**.** *Let*

$$U_0\,\mathcal{R}_\eta\,U_1\,\mathcal{R}_\eta\cdots\mathcal{R}_\eta\,U_N$$

be a finite chain of retained concentrating descendants. Then U_N is obtained from U_0 by a single diagonal covariant recentering sequence, after admissible pressure gauges, and it remains in $\mathfrak{A}_\eta$. The pressure gauges in the intermediate limits can be chosen compatibly with the final diagonal limit.

Proof. For $N=1$ this is the definition of $\mathcal{R}_\eta$. Assume it holds up to $N-1$. The relation $U_{N-1}\mathcal{R}_\eta U_N$ gives escaping covariant observers for U_{N-1} which converge locally to U_N. Compose these observers with the diagonal observers which realize U_{N-1} from U_0. The semidirect covariance identity in Lemma 8.5 shows that the composition is again a covariant observer sequence for U_0. A diagonal choice on $Q_m(0,0)$, together with the pressure-gauge compatibility in Lemma 9.4, gives local convergence to U_N. Since each link lies in $\mathfrak{A}_\eta$, the endpoint does too. □

Definition 10.19 (Covariant activity density)**.** Let $G=\mathbb{R}^3\rtimes\mathbb{R}$ be the covariant observer group, with left Haar measure dg, acting on $\mathfrak{X}^{\#}_M$ by $g\mapsto\mathscr{G}_g$. For $U\in\mathfrak{X}^{\#}_M$, define the upper covariant activity density

$$d^*_\eta(U):=\sup_{(F_\alpha)}\limsup_{\alpha}\frac{\big|\{g\in F_\alpha:\mathscr{G}_gU\in\mathfrak{A}_\eta\}\big|}{|F_\alpha|}, \tag{10.5}$$

where the supremum is over all left Følner nets of precompact measurable sets of positive Haar measure. Thus $d^*_\eta(U)=0$ means that every ordinary covariant occupation average loses the rooted activity target.

Lemma 10.20 (Invariant probability laws cannot retain non-affine activity)**.** *Let μ be a G-invariant Borel probability measure on a compact G-invariant subset of $\mathfrak{X}_M^{\#}$ carrying the fixed normalized pressure atlas and affine/parasitic normalization. Then*

$$\Phi(U) = 0 \qquad \text{for } \mu\text{-almost every } U. \tag{10.6}$$

In particular, $\mu(\mathfrak{A}_\eta) = 0$ for every $\eta > 0$.

Proof. Apply the proof of Theorem 8.29 on $\operatorname{supp}\mu$. Minimality and full support were used there only to promote an almost-everywhere conclusion to every point of a minimal core. The averaged identities themselves require only invariance. They give

$$\int |\nabla U(0,0)|^2 \, d\mu(U) = 0, \qquad \int |u - u^0|^2 \, d\mu = 0.$$

Invariance moves the first identity to every rational spacetime point; smoothness and a countable intersection give $\nabla U \equiv 0$ for μ-almost every profile. For such a profile write $U(y,\tau) = B(\tau)$. By Theorem 10.8, either it already belongs to the affine/parasitic lower stratum or its normalized physical pullback is bounded mild on every finite terminal interval. In the latter case the nonlinear Duhamel term vanishes and the centered Stokes semigroup gives

$$B(\tau) = e^{-(\tau-\sigma)/2} B(\sigma) \qquad (\sigma < \tau).$$

Boundedness as $\sigma \to -\infty$ gives $B(\tau) = 0$. In the former case the fixed affine/parasitic normalization also gives the zero representative. Thus $\Phi(U) = 0$ for μ-almost every profile by Theorem 8.19 and Remark 8.17. This proves (10.6). □

Proposition 10.21 (Positive-density active orbits are excluded)**.** *If $U \in \mathfrak{A}_\eta$ and $d^*_\eta(U) > 0$, then U cannot occur as a normalized retained residual profile.*

Proof. Choose a left Følner net (F_α) and $\delta > 0$ for which the limsup in (10.5) is at least δ. Set

$$\mu_\alpha := \frac{1}{|F_\alpha|} \int_{F_\alpha} \delta_{\mathscr{G}_g U} \, dg.$$

Compactness of $\mathfrak{X}_M^{\#}$ gives a weak-* convergent subnet. The Følner calculation in Proposition 9.36 shows that its limit μ is G-invariant. Since $\mathfrak{A}_\eta$ is closed, the Portmanteau theorem gives

$$\mu(\mathfrak{A}_\eta) \geq \limsup_\alpha \mu_\alpha(\mathfrak{A}_\eta) \geq \delta.$$

This contradicts Lemma 10.20. □

Definition 10.22 (Event covariance defect and moving-root current)**.** For the event measures ν_N in (10.3), $g \in G$, and $F \in C(\mathfrak{X}_M^{\#})$, set

$$\mathfrak{C}_N(g, F) := \int \big(F(\mathscr{G}_g U) - F(U) \big) \, d\nu_N(U). \tag{10.7}$$

Choose a countable dense subgroup $G_0 \subset G$ and a countable uniformly dense subalgebra $\mathscr{F}_0 \subset C(\mathfrak{X}_M^{\#})$. The chain is *eventwise covariantly balanced* along a subsequence N_ℓ if

$$\mathfrak{C}_{N_\ell}(g, F) \longrightarrow 0 \qquad (g \in G_0, \ F \in \mathscr{F}_0). \tag{10.8}$$

For a differentiable local observable and a spatial generator D_k, the infinitesimal version is

$$\mathfrak{J}_{N,k}(F) := \int D_k F \, d\nu_N. \tag{10.9}$$

When $q_k = D_k h$ is the nonzero spatial pressure mode, choosing $F = h u_k$ gives

$$\int u_k q_k \, d\nu_N = \mathfrak{J}_{N,k}(h u_k), \tag{10.10}$$

because $D_k u_k = 0$ after summation in k by incompressibility. Thus (10.9) is the pressure correction created by following the selected roots.

Lemma 10.23 (Vanishing event covariance defect closes the chain)**.** *No active chain* (10.2) *has a subsequence along which* (10.8) *holds.*

Proof. Pass to a further subsequence such that $\nu_{N_\ell} \rightharpoonup \nu$. By Theorem 10.17, $\nu(\mathfrak{A}_\eta) = 1$. Passing to the limit in (10.7) shows

$$\int F \circ \mathscr{G}_g \, d\nu = \int F \, d\nu \qquad (g \in G_0,\ F \in \mathscr{F}_0).$$

Density of G_0, uniform density of $\mathscr{F}_0$, and continuity of the action extend this identity to every $g \in G$ and every continuous F. Hence ν is G-invariant, contradicting Lemma 10.20 and $\nu(\mathfrak{A}_\eta) = 1$. □

Definition 10.24 (Sparse moving-root-current branch)**.** The class $\mathcal{C}_{\text{sparse-root}}$ consists of normalized residual candidates whose realized terminal extraction produces a threshold $\eta > 0$ and an active chain (10.2) with the following two properties:

- (S1) $d^*_\eta(U_j) = 0$ for every j; thus every ordinary covariant occupation average loses the active roots;
- (S2) for every subsequence along which ν_N converges, eventwise covariant balance fails. Equivalently, after a further subsequence there are $g \in G_0$, $F \in \mathscr{F}_0$, and $\delta > 0$ such that

$$|\mathfrak{C}_N(g, F)| \geq \delta. \tag{10.11}$$

  For differentiable pressure-energy observables this defect contains the moving-root current (10.10).

All diffuse, pressure-atlas-loss, critical-tail, affine/parasitic, and positive-density outputs are removed before a candidate is placed in $\mathcal{C}_{\text{sparse-root}}$. Thus $\mathcal{C}_{\text{sparse-root}}$ records one strict failure: a zero-density active chain sustained by a nonvanishing covariant root current.

Theorem 10.25 (Infinite-chain reduction to the sparse moving-root-current branch)**.** *Every infinite chain of retained concentrating descendants in the refined residual class is excluded or places its original residual candidate in $\mathcal{C}_{\text{sparse-root}}$. More precisely, positive covariant activity density and vanishing event covariance defect are both impossible; hence every surviving chain satisfies* (S1)–(S2) *of Definition 10.24.*

Proof. Suppose an infinite chain of retained concentrating descendants exists. If $\limsup_j \Phi(U_j) = 0$, local compactness gives a subsequential limit $U_{j_k} \to U_\infty$ with $\Phi(U_\infty) = 0$. The diagonal lifting used in Lemma 10.11 shows that this limit is still a retained tail descendant of the original residual profile. By Theorem 8.21, U_∞ belongs to the affine/parasitic lower stratum, contradicting descendant heredity in the refined residual class. Hence $\limsup_j \Phi(U_j) > 0$. Choose $\eta > 0$ and an infinite ordered subchain contained in $\mathfrak{A}_\eta$. By transitivity of retained tail descendants through Lemma 10.11, this ordered subchain is an $\mathcal{R}_\eta$-chain.

For each coordinate U_j, either $d^*_\eta(U_j) > 0$, in which case Proposition 10.21 gives a contradiction, or $d^*_\eta(U_j) = 0$. If the event-count measures have a covariantly balanced subsequence, Lemma 10.23 gives a contradiction. The only remaining case satisfies (S1) and (S2) in Definition 10.24; by definition the original candidate lies in $\mathcal{C}_{\text{sparse-root}}$. □

10.1. **Critical shell accounting for sparse active chains.** The preceding event law identifies the loss of covariant balance but does not assign a Navier–Stokes sign to that loss. We therefore retain the physical local energy identity on every selected scale and normalize its production by that scale. The resulting summation-by-parts formula rules out a positive mean production with sublinear interscale current. It does not, by itself, identify the expense paid along a moving root. The moving-chart identity below does so without assigning a false sign to pressure, translation, or dilation. It closes sublinear signed supply. The structural reductions below route bounded-depth transverse escape and noncompact moving marks into earlier data or into the single persistent current-saturation class.

Definition 10.26 (Critical parabolic excess)**.** Let (u,p) be a locally suitable physical representative, let $z=(x_0,t_0)$, and put

$$Q_r(z) := B_r(x_0)\times(t_0-r^2,t_0).$$

The velocity and pressure excesses are

$$\mathscr{C}(u;z,r) := r^{-2}\inf_{c\in L^3(t_0-r^2,t_0;\mathbb{R}^3)}\iint_{Q_r(z)}|u-c(t)|^3\,dx\,dt, \tag{10.12}$$

$$\mathscr{D}(p;z,r) := r^{-2}\iint_{Q_r(z)}|p-(p)_{B_r}(t)|^{3/2}\,dx\,dt. \tag{10.13}$$

Both quantities are invariant under the Navier–Stokes scaling. Fix nested cutoffs $\psi_{z,r}\in C_c^\infty(Q_r(z))$, equal to one on $Q_{r/2}(z)$, obtained from one fixed cutoff by parabolic scaling.

A finite family

$$Q_{r_N}(z_N)\Subset\cdots\Subset Q_{r_1}(z_1)\Subset Q_{r_0}(z_0),\qquad r_{j+1}\le r_j/4, \tag{10.14}$$

is a *coherent critical active-shell family at level* η if

$$\mathscr{C}(u;z_j,r_j)\ge\eta\qquad(0\le j\le N) \tag{10.15}$$

and the cutoffs are chosen so that $0\le\psi_{z_{j+1},r_{j+1}}\le\psi_{z_j,r_j}$. The latter condition is automatic for the fixed cutoff convention after requiring $Q_{r_{j+1}}(z_{j+1})\Subset Q_{r_j/2}(z_j)$. We write $(z',r')\prec(z,r)$ when $r'\le r/4$ and $Q_{r'}(z')\Subset Q_{r/2}(z)$. Thus a coherent family is a chain for the strict partial order $\prec$.

Lemma 10.27 (Exact localized energy balance)**.** *For every locally suitable representative* (u,p)*, there is a nonnegative Radon measure* m_u *such that, for every nonnegative* $\psi\in C_c^\infty$*,*

$$\mathscr{P}_u[\psi]=\mathscr{J}_u[\psi], \tag{10.16}$$

where

$$\mathscr{P}_u[\psi] := 2\iint|\nabla u|^2\psi\,dx\,dt+\int\psi\,dm_u \tag{10.17}$$

and

$$\mathscr{J}_u[\psi] := \iint|u|^2(\partial_t\psi+\Delta\psi)\,dx\,dt+\iint(|u|^2+2p)u\cdot\nabla\psi\,dx\,dt. \tag{10.18}$$

Both functionals are linear in ψ*, and* $\mathscr{P}_u[\psi]\ge0$ *when* $\psi\ge0$*. A function of time added to the pressure makes no contribution to* $\mathscr{J}_u$*.*

Proof. For compactly supported nonnegative ψ, the physical local energy inequality is

$$2\iint|\nabla u|^2\psi\,dx\,dt\le\mathscr{J}_u[\psi].$$

Move the two sides into one distribution. Its nonnegativity on nonnegative tests makes it a Radon measure m_u. Writing the inequality as an equality with this measure gives (10.16). The distributional equation and the kinetic-energy pairing are linear in the test function, hence so are $\mathscr{P}_u$ and $\mathscr{J}_u$. If p is replaced by $p + a(t)$, the additional term is

$$\iint a(t) u \cdot \nabla\psi \, dx \, dt = 0$$

by incompressibility. This proves the gauge assertion. □

Lemma 10.28 (Zero-cost critical shells)**.** *Let (u_n, p_n) be normalized locally suitable representatives on Q_2, and let $\chi_n = \psi_n^+ - \psi_n^- \geq 0$ be differences of nested cutoffs. Suppose that the local compactness and pressure-atlas hypotheses of the retained branch hold and*

$$\mathscr{P}_{u_n}[\chi_n] \longrightarrow 0. \tag{10.19}$$

After passage to a subsequence, on every compact set K for which there is $c_K > 0$ such that $\chi_n \geq c_K$ on K for all sufficiently large n,

$$\nabla u_n \longrightarrow 0 \quad \text{in } L^2(K), \qquad \nabla U = 0 \quad \text{a.e. on } K, \tag{10.20}$$

where U is the local limit. Thus a zero-cost shell is spatially homogeneous on its open shell region. This conclusion does not force an active profile enclosed by the inner cutoff to vanish.

Proof. By Lemma 10.27,

$$2 \iint |\nabla u_n|^2 \chi_n \, dx \, dt \leq \mathscr{P}_{u_n}[\chi_n] \longrightarrow 0.$$

The lower bound $\chi_n \geq c_K > 0$ on K gives the strong convergence in (10.20). Local compactness identifies its limit with ∇U. The final assertion records the precise scope of the argument: the weight χ_n vanishes behind the inner cutoff, so no estimate on that enclosed region follows from (10.19). □

Lemma 10.29 (Weighted critical shell identity)**.** *Let (10.14) be a coherent critical active-shell family. Write $\psi_j = \psi_{z_j, r_j}$, $w_j = r_j^{-1}$, $J_j = \mathscr{J}_u[\psi_j]$, and*

$$P_j := w_j \mathscr{P}_u[\psi_j - \psi_{j+1}] \qquad (0 \leq j < N).$$

Then

$$\sum_{j=0}^{N-1} P_j = w_0 J_0 - w_{N-1} J_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) J_j. \tag{10.21}$$

Each P_j is nonnegative, and every term in (10.21) is dimensionless under the Navier–Stokes scaling.

Proof. By Lemma 10.27 and linearity,

$$\mathscr{P}_u[\psi_j - \psi_{j+1}] = J_j - J_{j+1}.$$

Therefore

$$\sum_{j=0}^{N-1} P_j = \sum_{j=0}^{N-1} w_j (J_j - J_{j+1}).$$

Discrete summation by parts gives (10.21). Since $\psi_j - \psi_{j+1} \geq 0$, nonnegativity follows from (10.17). The scale assertion follows by changing variables under the Navier–Stokes scaling: both $\mathscr{P}_u[\psi_j - \psi_{j+1}]$ and $\mathscr{J}_u[\psi_j - \psi_{j+1}]$ scale like r_j. □

Corollary 10.30 (No positive-production shell family with sublinear current)**.** *There is no sequence of coherent critical active-shell families for which*

$$\liminf_{N\to\infty} \frac{1}{N}\sum_{j=0}^{N-1} P_j > 0 \tag{10.22}$$

and, simultaneously,

$$w_0 J_0 - w_{N-1}J_N + \sum_{j=1}^{N-1}(w_j - w_{j-1})J_j = o(N). \tag{10.23}$$

Proof. The left side of (10.21) has a positive linear lower bound by (10.22), whereas its right side is $o(N)$ by (10.23). □

Lemma 10.31 (Vanishing mean production selects a homogeneous shell)**.** *Suppose coherent critical active-shell families of lengths $N \to \infty$ satisfy*

$$\frac{1}{N}\sum_{j=0}^{N-1} P_j \longrightarrow 0.$$

There are indices $j_N < N$ such that, after rescaling the parent cylinder $Q_{r_{j_N}}(z_{j_N})$ to unit size, the corresponding normalized shell production tends to zero. If the local compactness and pressure-atlas conditions remain valid, every resulting limit is spatially homogeneous on the open region between the two selected cutoffs. Otherwise the sequence enters the corresponding previously classified compactness or pressure alternative.

Proof. Choose j_N so that

$$P_{j_N} \le \frac{1}{N}\sum_{j=0}^{N-1} P_j.$$

The right side tends to zero. Under the Navier–Stokes rescaling based at (z_{j_N}, r_{j_N}), the production between the two cutoffs is exactly P_{j_N}, because $P_{j_N} = r_{j_N}^{-1}\mathscr{P}_u[\psi_{j_N} - \psi_{j_N+1}]$. The retained compactness branch therefore satisfies the hypotheses of Lemma 10.28. Its conclusion gives spatial homogeneity on the open shell. Failure of either compactness or the pressure atlas is routed by Corollary 9.11 and Lemma 9.5. □

Definition 10.32 (Moving parabolic realization)**.** Let $I \subset \mathbb{R}$ be an interval. A moving parabolic realization is an absolutely continuous scale $r : I \to (0,\infty)$, center $x : I \to \mathbb{R}^3$, and physical time $t : I \to \mathbb{R}$ satisfying

$$\dot t = r^2 \quad \text{a.e. on } I.$$

Set

$$V(y,\tau) = r(\tau)u(x(\tau) + r(\tau)y, t(\tau)), \qquad P(y,\tau) = r(\tau)^2 p(x(\tau) + r(\tau)y, t(\tau)), \tag{10.24}$$

and write

$$a = \frac{\dot r}{r}, \qquad b = \frac{\dot x}{r}, \qquad \Lambda V = V + y\cdot\nabla V.$$

It is a *record-scale realization* when r is nonincreasing, hence $a \le 0$ almost everywhere. The functions a and b are determined by the selected centers and scales; they are not additional forces.

Lemma 10.33 (Exact equation and signed carrier in a moving realization)**.** *On every ordinary interval of a moving parabolic realization,*

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \Delta V + a\Lambda V + b\cdot\nabla V, \qquad \nabla\cdot V = 0. \tag{10.25}$$

Let $\phi\in C_c^\infty(\mathbb{R}^3)$, $\phi\ge 0$, be independent of τ, and put

$$e=\frac{|V|^2}{2}, \qquad K_\phi=\int e\phi, \qquad D_\phi=\int|\nabla V|^2\phi.$$

The pushed-forward suitable-energy defect is a nonnegative measure dM_ϕ, and

$$dK_\phi + D_\phi\,d\tau + dM_\phi = \big(F_{\mathrm{int},\phi}+F_{\mathrm{sc},\phi}+F_{\mathrm{tr},\phi}\big)d\tau, \tag{10.26}$$

where

$$F_{\mathrm{int},\phi}=\int (e+P)V\cdot\nabla\phi+\int e\Delta\phi, \tag{10.27}$$

$$F_{\mathrm{sc},\phi}=-a\int e(\phi+y\cdot\nabla\phi), \tag{10.28}$$

$$F_{\mathrm{tr},\phi}=-\int e\,b\cdot\nabla\phi. \tag{10.29}$$

All three terms are signed. In particular, the condition $a\le 0$ does not give a sign to $F_{\mathrm{sc},\phi}$ for a compactly supported cutoff.

Proof. Differentiate (10.24). The time, amplitude, dilation, and center derivatives give respectively $r^3\partial_t u$, aV, $ay\cdot\nabla V$, and $b\cdot\nabla V$. Since $\dot t = r^2$, substitution of the physical Navier–Stokes equation gives (10.25).

Test that equation against $V\phi$. Incompressibility converts convection and pressure into (10.27); integration by parts gives the viscous terms. In dimension three,

$$\int V\cdot\Lambda V\,\phi = -\int e(\phi+y\cdot\nabla\phi), \qquad \int V\cdot(b\cdot\nabla V)\phi = -\int e\,b\cdot\nabla\phi.$$

Writing the suitable local energy inequality as an equality with its nonnegative defect measure proves (10.26). The factor $\phi+y\cdot\nabla\phi$ changes sign on a general cutoff annulus, which proves the final assertion. □

Lemma 10.34 (Moving roots do not create a Type II output)**.** *Every moving cell obtained from a retained descendant inside the admissible local window is an exact recentering and parabolic rescaling of the fixed Type I extraction. It retains the scale-invariant local energy bounds and the pointwise Type I envelope on each compact subwindow. If a proposed cell leaves the admissible local window, its first exit is an exterior or diffuse recentering datum. Thus neither unbounded chart velocity (a,b) nor persistent signed supply is, by itself, a Type II singularity. A failure of the original pointwise Type I envelope occurs only at the entry test, before the residual class is formed.*

Proof. The moving chart (10.24) is a composition of spatial translation, time translation, and Navier–Stokes parabolic scaling. These operations preserve the scale-invariant local energy quantities and transform the pointwise envelope covariantly. The compact-window conclusion follows from Proposition 1.21. The expanding-window protocol and retained-recentering alternative Theorem 9.17 and Corollary 9.11 give the exhaustive alternative: a retained recentering inside the local Type I window has the same compactness, while one which leaves it is assigned to the exterior or diffuse terminal remainder. The quantities $a=\dot r/r$ and $b=\dot x/r$ describe the selected chart and do not alter the physical envelope. They therefore cannot change a Type I candidate into a Type II candidate. □

Definition 10.35 (Marked moving cell and its signed supply)**.** A marked moving cell consists of a moving parabolic realization on a compact interval $I_C = [\tau_C^-, \tau_C^+]$, a cutoff ϕ_C equal to one on the fixed cylinder on which the cell's activity floor is measured, and the retained data

$$(V, P, M_C, r, x, a, b, \phi_C, K_C^-, K_C^+).$$

The pressure is recorded modulo functions of time. Define its dissipation, defect, and signed supply by

$$D_C := \int_{I_C} D_{\phi_C}\, d\tau, \qquad\qquad M_C := M_{\phi_C}(I_C), \tag{10.30}$$

$$H_C := \int_{I_C} (F_{\mathrm{int},\phi_C} + F_{\mathrm{sc},\phi_C} + F_{\mathrm{tr},\phi_C})\, d\tau. \tag{10.31}$$

Thus $D_C, M_C \geq 0$, while H_C is signed. A marked family is *concatenated* when $K_{C_j}^+ = K_{C_{j+1}}^-$. This condition concerns only the kinetic carrier at the common event. No cancellation between the three summands in (10.31) is included in the definition.

Lemma 10.36 (Exact eventwise expense identity)**.** *Every marked moving cell satisfies*

$$D_C + M_C = K_C^- - K_C^+ + H_C. \tag{10.32}$$

Consequently, a concatenated family $C_0, \ldots, C_{N-1}$ satisfies

$$\sum_{j=0}^{N-1} (D_{C_j} + M_{C_j}) = K_{C_0}^- - K_{C_{N-1}}^+ + \sum_{j=0}^{N-1} H_{C_j}. \tag{10.33}$$

In particular, if the carriers are uniformly bounded and

$$\sum_{j=0}^{N-1} H_{C_j} = o(N), \tag{10.34}$$

then

$$\frac{1}{N} \sum_{j=0}^{N-1} (D_{C_j} + M_{C_j}) \longrightarrow 0. \tag{10.35}$$

Proof. Integrate (10.26) over I_C to obtain (10.32). Summation telescopes the carrier terms and gives (10.33). Divide by N; boundedness of the two endpoint carriers and (10.34) give (10.35). ☐

Lemma 10.37 (Observer currents are the material derivative of the cutoff)**.** *Let ϕ be the fixed cutoff in a moving realization and define the corresponding physical cutoff*

$$\psi(x, t(\tau)) := \phi\left(\frac{x - x(\tau)}{r(\tau)}\right).$$

Then

$$r(\tau)^2 \partial_t \psi(x, t(\tau)) = -b(\tau) \cdot \nabla_y \phi - a(\tau) y \cdot \nabla_y \phi, \qquad y = \frac{x - x(\tau)}{r(\tau)}. \tag{10.36}$$

Consequently $F_{\mathrm{tr},\phi}$ and the $-a \int e\, y \cdot \nabla\phi$ portion of $F_{\mathrm{sc},\phi}$ are exactly the kinetic-energy pairing with the material derivative of the selected physical cutoff. The remaining term $-a \int e\phi$ is the amplitude Jacobian of the Navier–Stokes scaling. Neither term is an external energy source, and neither has a sign.

Proof. Differentiate ψ at fixed physical x. Since $\dot{t} = r^2$, $\dot{x} = rb$, and $\dot{r} = ar$, the chain rule gives

$$r^2 \partial_t \psi = \partial_\tau \psi = -\frac{\dot{x}}{r} \cdot \nabla_y \phi - \frac{\dot{r}}{r} y \cdot \nabla_y \phi = -b \cdot \nabla_y \phi - ay \cdot \nabla_y \phi.$$

Substitution into (10.28) and (10.29) proves the assertion. □

Lemma 10.38 (Algebraic content of current saturation)**.** *For every marked cell,*

$$H_C - D_C - M_C = K_C^+ - K_C^-. \tag{10.37}$$

Hence (10.42) is precisely the stationary average of a carrier coboundary. In particular it contains no displayed nonnegative remainder controlling the distance between consecutive marked profiles modulo the covariant symmetry group. Any estimate of the form

$$\mathcal{R}_C \geq c\, d_{\mathfrak{X}/G}(U_C, U_{C^+})^2, \qquad \mathcal{R}_C \geq 0, \tag{10.38}$$

must be supplied by a further PDE identity or monotonicity formula; it is not a consequence of current saturation.

Proof. Equation (10.37) is (10.32) rearranged. Integrating it against a shift-invariant marked path law makes the mean of $K_C^+ - K_C^-$ vanish and gives exactly (10.42). No other term occurs in this calculation, which proves the assertion about its algebraic content. □

Definition 10.39 (Oriented logarithmic shell-current measure)**.** Let C_j, $0 \leq j < N$, be a coherent concatenated record-scale family, with event scales $r_{j+1} \leq r_j/4$, and set

$$\rho_j = -\log r_j, \qquad I_j = [\rho_j, \rho_{j+1}], \qquad h_j := H_{C_j}.$$

The complete oriented current of the cell is h_j; its pressure, transport, cutoff, translation, and dilation terms are combined before its positive part is taken. Define

$$\Lambda_N^{\mathrm{cur}} := \frac{1}{N} \sum_{j=0}^{N-1} h_j^+ \frac{\mathbf{1}_{I_j}(\rho)}{|I_j|}\, d\rho. \tag{10.39}$$

Since $|I_j| \geq \log 4$, the intervals have disjoint interiors and Λ_N^{cur} does not count one log location as two different shell currents.

Lemma 10.40 (Positive event price produces nonzero logarithmic current)**.** *Assume every cell of a coherent family retains the fixed non-affine activity floor and hence, by Lemma 10.46,*

$$D_{C_j} + M_{C_j} \geq d_* > 0. \tag{10.40}$$

If the endpoint carriers are uniformly bounded, then

$$\liminf_{N\to\infty} \Lambda_N^{\mathrm{cur}}(\overline{\mathbb{R}}_{\log}) \geq d_*. \tag{10.41}$$

The same conclusion holds for every sequence of consecutive blocks whose length tends to infinity.

Proof. The exact block identity (10.33) gives

$$\sum_{j<N} h_j = \sum_{j<N} (D_{C_j} + M_{C_j}) + K_{C_{N-1}}^+ - K_{C_0}^-.$$

If $0 \leq K_C^\pm \leq K_*$, then

$$\sum_{j<N} h_j^+ \geq \sum_{j<N} h_j \geq N d_* - 2K_*.$$

By (10.39), its total mass is $N^{-1} \sum_{j<N} h_j^+$. Letting $N \to \infty$ proves (10.41). The identical calculation applies after replacing the first and last indices by the endpoints of any consecutive block. □

Theorem 10.41 (Persistent coherent shell current produces a realized tail or a regenerated center)**.** *Let a compact coherent record-scale chain remain after the affine/parasitic, pressure/noncompact, diffuse, regenerated-concentration, and previously selected critical-tail alternatives have been removed. Suppose it retains a fixed non-affine activity floor in every cell. Then its logarithmic shell currents produce exactly one of the following sequence-realized outputs:*

(i) *a positive-threshold active observer in a selected bounded logarithmic window, hence a regenerated retained center;*
(ii) *a nonzero state in the critical-tail compactification $\mathfrak{T}_M$ after all such active observers have been removed from the selected windows.*

No new current coordinate is adjoined to $\mathfrak{T}_M$.

Proof. Form Λ_N^{cur} from Definition 10.39. Compactness of the marked cell space bounds the carriers and all current coordinates on the fixed countable cutoff family. By Lemma 10.40, the measures have uniformly positive total mass. Normalize their restrictions to the support of the positive current and pass to the logarithmic annular compactification of Definition 9.42.

We verify realization in the critical-tail topology. Fix a consecutive block $C_m, \dots, C_{n-1}$ and view it from its deepest child at scale r_n. The shell I_j becomes a macroscopic annulus whose inner and outer radii are r_{j+1}/r_n and r_j/r_n. Distinct shells have disjoint logarithmic interiors. The moving-chart change of variables and Lemma 10.37 identify h_j with the sum of the oriented pressure, transport, cutoff, and moving-boundary flux coordinates on that annulus. Write this finite decomposition as

$$h_j = h_j^{\mathrm{p}} + h_j^{\mathrm{nl}} + h_j^{\mathrm{cut}} + h_j^{\mathrm{mov}}.$$

Since

$$h_j^+ \le |h_j^{\mathrm{p}}| + |h_j^{\mathrm{nl}}| + |h_j^{\mathrm{cut}}| + |h_j^{\mathrm{mov}}|,$$

the positive mass in (10.41), followed by a four-way pigeonhole extraction, gives positive limiting mass to at least one of these existing annular observables. The pressure term is an actual pressure-flux pairing, the nonlinear term is the transport pairing in the annular local-energy identity, and the cutoff and moving-boundary terms are the material-cutoff/virial pairings. On the virial-tight subhull these are the bounded rational log-window coordinates of Definition 9.48 and Lemma 9.49. Failure of their boundedness is routed by the same ordered compactness protocol to a pressure/noncompact or non-virial-tight critical-tail output. Thus no scalar current coordinate is inserted into the compactification. For every finite collection of shells, Lemma 10.47 realizes all of them in one member of the original terminal extraction with compatible pressure gauges. A diagonal choice over the countable log-window and flux tests therefore makes every support limit sequence-realized.

Apply the compactness and support-transfer results Theorem 9.40 and Lemma 9.41. If a coordinate loses boundedness, their ordered alternatives give an affine pressure or pressure/noncompact descendant. Those alternatives were removed in the hypotheses. Otherwise remove from each selected log window every positive-threshold active observer. If this removal captures positive limiting shell-current mass, support transfer produces a sequence-realized active observer and hence alternative (i). If it does not, the remaining normalized current mass is still positive. The critical-tail compactness theorem gives a convergent subsequence in $\mathfrak{T}_M$, and the positive mass of one of the four displayed existing observables prevents that state from vanishing. This is alternative (ii).

The possible locations of this nonzero support are exhaustive. Escape through the ends of the logarithmic compactification gives $\Lambda \neq 0$, the log-diffuse critical-tail state. Tight support in a bounded log window gives an annular pressure/viscous/flux state. If that window contains a positive-threshold unit active observer, alternative (i) holds; if its spatial quotient oscillation vanishes, the affine/parasitic alternative is selected. In the remaining tight case the Young/defect coordinates are nonzero or the support is coherent under log translation, giving respectively a Young/defect or coherent critical tail by Theorems 9.91 and 9.98. Thus the alternatives in the statement are exhaustive. In particular, a regenerated center is not silently relabelled as a critical-tail profile after concentration removal. □

Definition 10.42 (Scale-saturation law)**.** A shift-invariant probability law ϖ on the marked path space of record-scale moving cells is *current saturated* when its one-cell marginal retains the activity floor and

$$\int \big(H_C - D_C - M_C\big)\, d\varpi(C) = 0. \tag{10.42}$$

The equality uses the distinguished activity cutoff carried by each cell. It is the stationary form of (10.33). It retains all pressure, transport, cutoff, translation, and dilation contributions in the single signed mark H_C.

Proposition 10.43 (Longitudinal current reduces to current saturation)**.** *Let coherent critical active-shell families of length $N \to \infty$ remain after the pressure/noncompact, diffuse, critical-tail, affine/parasitic, and positive-density alternatives have been removed. Then one of the following occurs:*

- (L1) *no concatenated moving-cell realization with uniformly bounded carriers and locally precompact marks exists;*
- (L2) *such realizations exist and their signed supply is sublinear, in which case* (10.35) *holds;*
- (L3) *such realizations exist and have persistent signed supply:*

$$\limsup_{N\to\infty} \frac{1}{N} \sum_{j=0}^{N-1} H_{C_j} > 0; \tag{10.43}$$

- (L4) *a stationary marked limit law exists and satisfies* (10.42) *while retaining the activity floor.*

Alternative (L2) *produces cells with arbitrarily small mean dissipation and defect. If such cells retain activity in their cutoff core, their local limits are spatially homogeneous there and are passed to the affine/parasitic alternative. If activity leaves the core, its exit location is the next retained recentering datum. Thus the unresolved longitudinal output consists of* (L1)*, or of the persistent-current configuration* (L3)–(L4)*.*

Proof. Attempt the moving-cell realization on each coherent family. Failure of uniform carrier bounds, local compactness of the marked variables, or exact concatenation is (L1). Otherwise apply Lemma 10.36. If the supply is sublinear, that lemma gives (L2). Select a cell whose cost does not exceed the mean cost. After rescaling it to the fixed chart, local compactness and lower semicontinuity give zero gradient and zero defect on the cutoff core. Retained activity in that core can then arise only from a time-dependent spatially homogeneous mode. The affine normalization dichotomy Theorem 8.21 and Remark 8.17 assigns that mode to $\mathcal{C}_{\mathrm{aff}}$. Escape from the core is recorded by the retained-recentering alternatives of Corollary 9.11. This proves the asserted closure of (L2).

If sublinear supply fails, pass to a subsequence on which (10.43) holds, giving (L3). When the marked records are precompact, take empirical measures on the orbit of the one-sided marked path under the left shift. Their weak limits are shift invariant by the endpoint calculation. Dividing (10.33) by N and passing to the limit gives (10.42). Closedness of the retained activity set preserves the activity floor, giving (L4). □

Lemma 10.44 (Bounded nesting produces transverse active antichains)**.** *Suppose realizations of the first m events have $\prec$-chain length at most L, independently of m. Then, for every J, some* *finite realization contains J pairwise $\prec$-incomparable active cylinders. For any such sequence of antichains, exactly one of the following occurs after diagonal extraction:*

(A1) *two cylinder sequences determine inequivalent retained recentering profiles and hence a separated two-profile family;*

(A2) *every fixed finite subfamily coalesces into one non-separated local profile class after recentering.*

Thus bounded nesting is structurally either branching concentration or a coalescent moving cascade; it is not a compactness conclusion.

Proof. Partially order the active cylinders in a realization by $\prec$. A finite poset of height at most L is the union of at most L antichains: assign to an element the maximum length of a chain ending there. Hence a realization with m events contains an antichain of cardinality at least m/L, which proves the first assertion.

Choose successively larger antichains and apply the local profile compactness and pressure-gauge extraction on each fixed finite collection. If two realizing sequences are inequivalent in the non-separated local profile relation of Lemma 9.14, the proof of that lemma shows that their covariant distance escapes and that they form a separated two-profile family. If no such pair exists, every fixed finite collection belongs to one equivalence class. A diagonal extraction gives (A2). These alternatives are exhaustive. □

Lemma 10.45 (Noncompact moving marks are geometric escape data)**.** *For coherent nested active families inside the fixed Type I window, the kinetic carriers are uniformly bounded and the velocity, pressure, and defect coordinates are locally precompact. Consequently, failure of precompactness of the full moving-cell marks can only occur in the center–scale coordinates. After passage to a subsequence, such a failure gives one of:*

(G1) *an unbounded normalized center displacement, routed to an exterior, diffuse, pressure/noncompact, or retained separated recentering;*

(G2) *an unbounded logarithmic scale displacement, routed to the realized critical-tail compactification;*

(G3) *tight center and logarithmic-scale increments; choosing the canonical shortest bounded-variation path in the finite-dimensional covariant observer group makes the integrated translation and dilation currents tight signed marks and produces a compact marked path law.*

In particular, marked-realization failure is not an additional terminal mechanism.

Proof. The fixed Type I envelope bounds V on the support of every fixed cutoff, so $K_C^{\pm} \leq C(M, \phi_C)$. Local velocity and pressure compactness, and weak-* compactness of the nonnegative suitable-energy defect measures, follow from Proposition 1.21 and Lemma 8.9. Thus only the observer parameters can fail tightness.

Join consecutive observer parameters by the canonical shortest bounded-variation path in the spatial-translation/logarithmic-scale coordinates. This removes artificial oscillation of (a, b) between fixed endpoints. If the normalized centers escape, apply Corollary 9.11 and Lemma 9.114; these give precisely (G1). If the center increments remain tight but the logarithmic scale increments escape every bounded interval, the logarithmic compactification in Theorems 9.40 and 9.91 gives (G2). In the complementary case both parameter increments are tight. For the canonical paths, their distributional derivatives are finite signed measures whose total variations are bounded by the endpoint displacements. Their pairings with the fixed smooth cutoff gradients in (10.28)–(10.29) are then tight scalar marks. Diagonal compactness on the countable local test family gives the marked path law in (G3). □

Lemma 10.46 (Critical activity has a fixed normalized viscous price)**.** *Let $|V| \le M$ on $B_2 \times I$, and suppose the affine/parasitic alternative has been removed. There is $c = c(M, B_1, B_2) > 0$ such that*

$$\operatorname{osc}_{3,\mathrm{sp}}(V; B_1 \times I) \ge \eta \quad \Longrightarrow \quad \int_I \int_{B_2} |\nabla V|^2 \, dy \, d\tau \ge c\eta. \tag{10.44}$$

Consequently every retained non-affine moving cell pays a positive scale-invariant viscous price depending only on the fixed activity threshold and normalized Type I bound.

Proof. For almost every τ, take $c(\tau) = (V)_{B_1}(\tau)$. Then $|c(\tau)| \le M$, hence $|V - c(\tau)| \le 2M$. Poincaré's inequality and enlargement from B_1 to B_2 give

$$\int_{B_1} |V - c(\tau)|^3 \le 2M \int_{B_1} |V - c(\tau)|^2 \le CM \int_{B_2} |\nabla V|^2.$$

Integrate in time and take the infimum over time-dependent homogeneous gauges. This proves (10.44). When $\operatorname{osc}_{3,\mathrm{sp}} = 0$, any remaining constant-in-time activity is carried by a time-dependent spatially homogeneous mode and is assigned to $\mathcal{C}_{\mathrm{aff}}$ by Theorem 8.21. □

Lemma 10.47 (Finite simultaneous realization of selected active events)**.** *Let $U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta \cdots$ be a realized retained chain, and select finitely many events together with finitely many compact normalized cylinders on which their activity is measured. There is one member of the original terminal extraction and one compatible family of composed observer maps such that all selected recentered fields approximate their corresponding events on the prescribed cylinders, with one common accuracy and compatible pressure gauges. The same conclusion holds for any finite antichain selected from one finite realization of the chain.*

Proof. For one event this is the definition of a realized descendant. Suppose the claim has been arranged for the first q selected events. Realize the next descendant from its parent on its prescribed compact cylinder, and compose that observer with the parent's observer by Lemma 8.5. Only finitely many compact observer images have been prescribed. Local convergence in each preceding descendant limit therefore permits one common terminal index large enough that all $q + 1$ approximations hold simultaneously. The pressure gauges are normalized on the overlaps and extended to their finite union by Lemma 9.4. Induction proves the assertion. A finite antichain in a finite realization is a finite selection of these composed observers, so the same argument applies. □

Theorem 10.48 (Transverse antichains reduce to geometric escape or a coalescent conveyor)**.** *Let $\mathcal{A}_N$ be simultaneously realized transverse active antichains, with $|\mathcal{A}_N| \to \infty$, obtained from the first m_N events of one retained descendant chain. After passage to a subsequence, exactly one of the following ordered outputs occurs:*

- (T1) *normalized centers escape every fixed parent window, producing an exterior, diffuse, pressure/noncompact, or retained separated recentering;*
- (T2) *logarithmic scales escape every bounded interval, producing a realized critical-tail datum;*
- (T3) *for every $\delta > 0$, all but a uniformly bounded number of the events lie in δ-clusters in center–scale space. Each cluster has one non-separated local profile class, and the empirical marked cluster laws form a compact coalescent moving cascade.*

In particular, the bounded-depth transverse alternative introduces no terminal branch: (T1)–(T2) *enter previously defined routes, while* (T3) *enters the current-saturated moving-cascade alternative after Proposition 10.43.*

Proof. Every $\mathcal{A}_N$ is selected inside one finite realization. Apply Lemma 10.47 to its members and their activity cylinders. They therefore coexist in one normalized representative. This is the simultaneous-realization point needed for additive local-energy accounting.

If the normalized center parameters are not tight, the first escaping center and its retained activity are routed by Corollary 9.11 and Lemma 9.114, giving (T1). Under center tightness, failure of logarithmic-scale tightness gives (T2) by Theorems 9.40 and 9.91. It remains to treat a fixed compact center–scale window. Thus, after one normalization, all cylinders under consideration lie in a fixed parent cylinder, and their radii lie in $[\lambda, 1]$ for some $\lambda > 0$.

Fix $\delta > 0$. Select a maximal subfamily whose core cylinders are pairwise separated by at least δ in the normalized parabolic metric. Because their radii lie in $[\lambda, 1]$, this family has overlap bounded by a constant depending only on δ and λ. After extraction, either the spatial activity has a common positive floor or it tends to zero. In the latter case local compactness produces a spatially homogeneous limit, which is routed to $\mathcal{C}_{\text{aff}}$ by the affine normalization dichotomy. In the former case every selected member carries the spatial activity floor. By Lemma 10.46, each selected core therefore costs at least $c(M, \lambda, \delta)\eta$ in normalized dissipation. Since the cores coexist with bounded overlap, their costs add up to the overlap constant, while the local Type I energy estimate Theorem 1.28 bounds their sum by the dissipation on one fixed enlargement of the parent cylinder. Consequently the cardinality of the separated subfamily is bounded by

$$N_\delta \le \frac{E(M, \lambda, \delta)}{c(M, \lambda, \delta)\eta},$$

independently of N.

Maximality places every remaining event in a δ-cluster around one of these finitely many representatives. Let $\delta_k \downarrow 0$ and diagonalize. Local state-space compactness and gauge compatibility show that each nested sequence of occupied clusters converges to one non-separated local profile class; two inequivalent limits would remain separated at some fixed δ_k and would have been counted as distinct representatives. The canonical bounded-variation observer interpolation of Lemma 10.45 makes the associated marked cluster paths compact. Their empirical shift-orbit laws therefore give the coalescent moving cascade in (T3).

Apply Proposition 10.43 to each occupied cluster class. Sublinear signed supply routes to the affine/parasitic or retained recentering alternatives; noncompact modulation gives (T1) or (T2) by Lemma 10.45; the remaining output is a current-saturated marked law. Hence no independent transverse branch remains. □

Definition 10.49 (Critical interscale-flux branch)**.** The class $\mathcal{C}_{\text{interscale-flux}}$ consists of the candidates in $\mathcal{C}_{\text{sparse-root}}$ for which, after the previously classified affine/parasitic, pressure/noncompact, diffuse, critical-tail, and positive-density alternatives have been removed, arbitrarily long concatenated moving-cell realizations exist and every surviving subsequence has persistent signed supply (10.43). Whenever the marked records are precompact, their shift-orbit laws converge to a current-saturated law in the sense of Definition 10.42. Denote this last subclass by $\mathcal{C}_{\text{scale-saturation}}$. Thus $\mathcal{C}_{\text{interscale-flux}} = \mathcal{C}_{\text{scale-saturation}}$: the transverse antichain alternative has been reduced to this class or to earlier terminal data by Theorem 10.48. The pressure in every supply mark is the actual pressure modulo functions of time. No affine spatial pressure is subtracted inside the local energy identity. By Lemma 10.34, this class is not a concealed Type II output; an actual loss of the Type I envelope belongs to the entry alternative rather than to $\mathcal{C}_{\text{interscale-flux}}$.

Corollary 10.50 (Current-saturated coherent cascades are excluded)**.** *Every candidate in $\mathcal{C}_{\text{scale-saturation}}$ whose logarithmic current has a nonzero component after positive-threshold active observers are removed is excluded by the complete realized critical-tail closure.*

Proof. Let a candidate satisfy the hypothesis. Its affine/parasitic and compactness-loss outputs have already been routed. If the spatial quotient activity had no fixed positive floor along a subsequence, local compactness would give a spatially homogeneous limit, already assigned to $\mathcal{C}_{\text{aff}}$. Thus the retained non-affine branch has a fixed floor after extraction and Lemma 10.46

supplies (10.40). Alternative (ii) of Theorem 10.41 gives a nonzero realized critical-tail state, contradicting Corollary 9.112. □

Definition 10.51 (Regenerated-current cascade)**.** The class $\mathcal{C}_{\text{regenerated-current}} \subset \mathcal{C}_{\text{scale-saturation}}$ consists of current-saturated coherent cascades for which, on every sequence of consecutive blocks with lengths tending to infinity, removal of positive-threshold active observers captures all limiting positive logarithmic shell-current mass. Equivalently, every nonzero support profile produced by Lemma 10.40 regenerates a retained active center before it enters the concentration-free critical-tail compactification. The centers remain separated by $r_{j+1} \le r_j/4$, and the complete oriented current, rather than pressure alone, is attached to each parent–child logarithmic interval.

Lemma 10.52 (Physical weight of normalized regeneration expense)**.** *For a normalized moving cell C_j based at physical radius r_j,*

$$D^{\text{phys}}_{C_j} = r_j D_{C_j}, \qquad M^{\text{phys}}_{C_j} = r_j M_{C_j}. \tag{10.45}$$

Consequently the fixed normalized price $D_{C_j} + M_{C_j} \ge d_*$ gives only

$$D^{\text{phys}}_{C_j} + M^{\text{phys}}_{C_j} \ge r_j d_*. \tag{10.46}$$

If $r_{j+1} \le r_j/4$, then

$$\sum_{j=0}^{\infty} r_j d_* \le \frac{4}{3} r_0 d_* < \infty. \tag{10.47}$$

Thus finite physical energy does not by itself contradict infinitely many geometrically contracting events with fixed normalized expense.

Proof. Under

$$y = \frac{x - x_j}{r_j}, \qquad \tau = \frac{t - t_j}{r_j^2}, \qquad V_j(y, \tau) = r_j u(x, t),$$

one has $\nabla_y V_j = r_j^2 \nabla_x u$ and $dx\, dt = r_j^5 dy\, d\tau$. Therefore

$$\iint_{C_j^{\text{phys}}} |\nabla_x u|^2 \psi_j \, dx\, dt = r_j \iint_{C_j} |\nabla_y V_j|^2 \phi_j \, dy\, d\tau = r_j D_{C_j}.$$

The suitable-energy defect has the same homogeneity, since it is the nonnegative remainder in an identity all of whose terms scale like length. This proves (10.45). The price bound gives (10.46); $r_j \le 4^{-j} r_0$ and the geometric series give (10.47). □

Proposition 10.53 (Exact parent–child ledger reduction)**.** *For a candidate in $\mathcal{C}_{\text{regenerated-current}}$, summing the physical local-energy identities between consecutive stopping surfaces cancels every common parent–child carrier once. It leaves only the endpoint carriers, the nonnegative expenses $r_j(D_{C_j} + M_{C_j})$, and complete oriented current escaping the selected regenerated-center family. Persistent escaping current is routed to the exterior, diffuse, pressure/noncompact, or critical-tail alternatives. If no such current remains, the branch reduces to a geometrically contracting active-center chain whose normalized expense stays bounded below but whose physical price has the summable weights r_j. The ledger contains no additional nonnegative term controlling the change of normalized profile.*

Proof. Apply Lemma 10.27 to the physical regions between successive stopping surfaces, with opposite orientations on shared interfaces. Linearity cancels each shared carrier exactly once. The resulting finite-block identity is the physical version of (10.33), with the weights from (10.45). Any uncancelled boundary current has sequence-realized support by Lemma 10.47; the existing center–scale and critical-tail support alternatives give the stated routing.

In the complementary case only endpoint carriers and nonnegative weighted expenses remain. Their scale lower bounds are summable by (10.47). By Lemma 10.38, the identity has no nonnegative modulation remainder. In particular, $r_j(D_{C_j} + M_{C_j}) \to 0$ cannot be divided by $r_j \to 0$ to infer vanishing normalized expense. This proves the reduction. ☐

Theorem 10.54 (Critical-currency computation on a regenerated cascade)**.** *Let $C_0, C_1, \dots$ be a coherent regenerated-current cascade, and write*

$$w_j = r_j^{-1}, \qquad J_j = \mathscr{J}_u[\psi_j], \qquad P_j = w_j \mathscr{P}_u[\psi_j - \psi_{j+1}].$$

Then $P_j \geq 0$ is invariant under Navier–Stokes scaling and, for every N,

$$\sum_{j=0}^{N-1} P_j = w_0 J_0 - w_{N-1} J_N + \sum_{j=1}^{N-1} (w_j - w_{j-1}) J_j. \tag{10.48}$$

After passage to the retained non-affine subchain, either:

(i) *the mean critical price tends to zero, and a normalized zero-cost shell is spatially homogeneous on its open shell;*

(ii) *the mean critical price has a positive lower bound, in which case the nonexact scale current on the right of (10.48) has the same positive linear growth;*

(iii) *compactness, pressure, realization, or support transfer fails and the corresponding previously defined branch is produced.*

Thus the scale-invariant price is the correct event currency. On $\mathcal{C}_{\text{regenerated-current}}$, however, its accumulated value is financed exactly by the nonexact scale current; the identity does not bound that current by physical energy.

Proof. Nonnegativity and scale invariance of P_j, as well as (10.48), are precisely the local energy equality and discrete summation by parts in Lemma 10.29. If $N^{-1} \sum_{j<N} P_j \to 0$ along a subsequence, choose a term no larger than the mean and apply Lemmas 10.28 and 10.31; this gives alternative (i), unless one of their compactness or pressure interfaces fails, which is alternative (iii).

If the mean does not tend to zero, pass to a subsequence on which it is bounded below by a positive constant. Equation (10.48) then gives identical linear growth of its right-hand side. Sublinear scale current is already impossible by Corollary 10.30. In the present class, every nonzero support window of the remaining current regenerates the next center by Definition 10.51; otherwise the support protocol yields alternative (iii). This is alternative (ii) and proves the exhaustive statement. ☐

Lemma 10.55 (Compact closed marked regeneration paths)**.** *After alternatives* (G1)–(G2) *of Lemma 10.45 have been removed, a regenerated-current cascade has a compact marked path space whose j-th coordinate records*

$$(U_j, \Pi_j, C_j, \lambda_j, \xi_j, \theta_j, \mathfrak{s}_j), \qquad 0 < \lambda_* \leq \lambda_j \leq \frac{1}{4}.$$

Here $\lambda_j = r_{j+1}/r_j$, ξ_j and θ_j are the normalized spatial and temporal increments, and $\mathfrak{s}_j$ is the sequence-realization support datum. The concatenated regenerated paths form a closed shift-invariant subset. Every shift-invariant limit law furnished by Proposition 10.43 is supported there and retains the fixed non-affine activity floor in its zeroth marginal.

Proof. Local state-space compactness gives compactness of the velocity, pressure atlas, suitable-defect, carrier, and countable cutoff coordinates. Alternative (G1) removes unbounded normalized center increments, and alternative (G2) removes logarithmic increments escaping every bounded interval. Hence $|\xi_j| + |\theta_j| \le B$ and $-\log \lambda_j$ lies in a compact subset of $[\log 4, \infty)$, which gives the displayed lower bound. The support coordinate lies in the closed sequence-realized support class.

Concatenation is equality of the outgoing and incoming carrier, observer, and pressure-gauge overlap coordinates. Regeneration is the closed positive-threshold activity condition on the child. Thus both are closed in the marked product topology. Retaining the witnesses is essential here: no witness can escape while the profile coordinates converge. Compactness of the countable marked product gives the path space. The shift preserves its local defining conditions, and closedness of the activity set preserves the activity floor under the empirical-law limit. ☐

Theorem 10.56 (Marked recurrence produces a realized critical tail)**.** *Every candidate in* $\mathcal{C}_{\text{regenerated-current}}$ *produces a nonzero sequence-realized profile in* $\mathfrak{T}_M$*. The profile is log-diffuse, Young/defect, or coherent, and hence belongs to one of the realized critical-tail alternatives already classified by Theorem 9.98.*

Proof. Let X_{mark} be the compact closed path space of Lemma 10.55, let S be its left shift, and let ϖ be a shift-invariant marked limit law. Poincaré recurrence gives a recurrent path $p \in X_{\text{mark}}$, retaining the activity floor, and integers $n_k \to \infty$ such that

$$S^{n_k} p \longrightarrow p \qquad \text{in } X_{\text{mark}}. \tag{10.49}$$

This returns every fixed finite block of velocities, pressure gauges, defects, observer increments, and realization data, not merely the zeroth profile.

Compose the first n observer maps. In physical parent coordinates their scale, center, and time parameters are

$$\Lambda_n = \prod_{j=0}^{n-1} \lambda_j, \tag{10.50}$$

$$X_n - X_0 = r_0 \sum_{j=0}^{n-1} \Big(\prod_{i=0}^{j-1} \lambda_i \Big) \xi_j, \tag{10.51}$$

$$T_n - T_0 = r_0^2 \sum_{j=0}^{n-1} \Big(\prod_{i=0}^{j-1} \lambda_i^2 \Big) \theta_j. \tag{10.52}$$

These are the physical-coordinate form of (8.1). Since $\lambda_j \le 1/4$ and the increments are bounded, the last two series converge, whereas

$$0 < \Lambda_{n_k} \le 4^{-n_k} \longrightarrow 0. \tag{10.53}$$

After one limiting spacetime recentering, (10.49) is therefore a sequence-realized return of the same non-affine marked profile through observers with logarithmic displacement tending to $+\infty$.

Retain the complete marked blocks between scales 1 and Λ_{n_k}, viewed from their deepest children. Their component shells have disjoint logarithmic interiors. By Lemma 10.47, compatible pressure gauges, and the support coordinates of X_{mark}, every finite family of annular tests is realized in one member of the original terminal extraction. A diagonal extraction produces a sequence-realized logarithmic annular state. It is nonzero because the recurrent cross-section retains the fixed spatial quotient activity floor and returns at arbitrarily large logarithmic displacements.

Apply the existing critical-tail compactification alternatives. Failure of logarithmic tightness gives $\Lambda \ne 0$, hence a log-diffuse state. Under logarithmic tightness, failure of strong annular compactness or survival of a pressure or viscous coordinate gives a Young/defect state. In the remaining case the block convergence is strong on compact annuli, the Young measure is Dirac, and the pressure and viscous defect coordinates vanish; Corollary 9.96 then gives a nonzero realized coherent critical tail. Affine, pressure/noncompact, and realization failures are already classified outputs, having been removed before entry into $\mathcal{C}_{\text{regenerated-current}}$. Thus every retained output is a nonzero realized element of $\mathfrak{T}_M$, as claimed. $\square$

Corollary 10.57 (Regenerated-current cascade exclusion)**.**

$$\mathcal{C}_{\text{regenerated-current}} = \varnothing.$$

Proof. A candidate in $\mathcal{C}_{\text{regenerated-current}}$ would produce, by Theorem 10.56, a nonzero sequence-realized critical-tail profile. This contradicts Corollary 9.112. $\square$

Remark 10.58 (Critical scale current and recurrence closure)*.* The critical ledger identifies the scale current financing regeneration, but no independent global bound for that current is needed. Compactness of the marked witnesses and recurrence of their actual successor cocycle convert persistent financing into the realized critical tail of Theorem 10.56. The rooted/full-hull problem does not recur: the proof never enlarges the path to all covariant translations. It uses only returns of the closed marked successor shift and then the established logarithmic critical-tail compactification.

Lemma 10.59 (Center–scale alternative for a realized active chain)**.** *Let* (10.2) *be a realized active chain. After a diagonal extraction, one of the following occurs:*

(i) *a previously classified pressure/noncompact, diffuse, affine/parasitic, or critical-tail output occurs;*
(ii) *coherent nesting has uniformly bounded depth: the $\prec$-chain length of every finite realization is bounded by one common integer;*
(iii) *for every N, it contains a coherent nested family of N critical active shells.*

Proof. Use Lemma 10.18 to realize every finite initial segment of the chain from one member of the original terminal sequence and attach to each realized event a cylinder on which the fixed activity floor is attained. If local compactness, pressure-gauge compatibility, or the expanding-region condition fails in this construction, then Corollary 9.11 and Theorem 9.17 gives item (i).

In the remaining case, let L_m be the maximum cardinality of a $\prec$-chain among the active cylinders in a realization of the first m events. If $\sup_m L_m < \infty$, item (ii) holds. If $\sup_m L_m = \infty$, then for every N a finite realization contains a $\prec$-chain of length N, giving item (iii). The definition of $\prec$ supplies both the scale separation and the nested cutoffs in (10.14). This alternative is exhaustive and does not identify transverse cylinders with one coherent nested family. $\square$

Theorem 10.60 (Critical-shell reduction of the sparse moving-root branch)**.** *One has*

$$\mathcal{C}_{\text{sparse-root}} \subset \mathcal{C}_{\text{interscale-flux}}. \tag{10.54}$$

More precisely, every sparse active chain either reaches an already classified terminal alternative or belongs to the current-saturated class of Definition 10.49.

Proof. Apply Lemma 10.59. Its first alternative is handled by the corresponding terminal argument. In its second alternative, Lemma 10.44 and Theorem 10.48 route the transverse antichains to prior terminal data or to a coalescent moving cascade. Apply Proposition 10.43 to the latter. In the third alternative first apply Lemma 10.29. Positive mean shell production with sublinear weighted current is excluded by Corollary 10.30; vanishing mean production selects a homogeneous shell or a previously routed compactness/pressure datum by Lemma 10.31.

Apply Proposition 10.43 to the remaining coherent families. By Lemma 10.45, failure of compact moving marks routes to an exterior/diffuse, pressure/noncompact, separated-profile, or critical-tail datum and is not a terminal subcase. Sublinear signed supply is alternative (L2) and is closed in that proposition by zero mean dissipation, unless activity exits the cutoff core, in which case Corollary 9.11 supplies the next previously classified datum. The only longitudinal output left is persistent signed supply; compact marked limits lie in $\mathcal{C}_{\text{scale-saturation}}$. This proves (10.54). ☐

Theorem 10.61 (Exact reduction of the critical interscale-flux branch)**.** *After all previously classified alternatives and the realized critical-tail output have been removed,*

$$\mathcal{C}_{\text{interscale-flux}} = \mathcal{C}_{\text{scale-saturation}} = \mathcal{C}_{\text{regenerated-current}} = \varnothing. \tag{10.55}$$

Thus the sparse moving-root configuration is excluded.

Proof. By Definition 10.49, $\mathcal{C}_{\text{interscale-flux}} = \mathcal{C}_{\text{scale-saturation}}$. Apply Theorem 10.41. Its critical-tail alternative is excluded by Corollary 10.50. Its other alternative says precisely that all surviving positive current is supported on regenerated active centers on every long consecutive block, which is the definition of $\mathcal{C}_{\text{regenerated-current}}$. The reverse inclusion is immediate from that definition. Finally $\mathcal{C}_{\text{regenerated-current}} = \varnothing$ by Corollary 10.57. This proves (10.55). ☐

Theorem 10.62 (No infinite retained velocity concentration chain)**.** *No infinite chain of retained concentrating descendants can occur in the refined residual class.*

Proof. By Theorem 10.25, every such chain is either excluded by the positive-density or covariantly balanced alternatives or places the original residual candidate in $\mathcal{C}_{\text{sparse-root}}$. The latter alternative is reduced by Theorem 10.60 to $\mathcal{C}_{\text{interscale-flux}}$. By Theorem 10.61, this class is empty. ☐

Lemma 10.63 (Finite separated families produce indecomposable profiles or concentration chains)**.** *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained velocity concentration separated family for a refined residual candidate. Then either one of the terminal profiles (U_j, Π_j) is terminally indecomposable, or the family generates an infinite chain of retained concentrating descendants. A finite recurrent loop is counted as an infinite chain by periodic repetition.*

Proof. Assume no (U_j, Π_j) is terminally indecomposable. By Lemmas 10.13 and 10.14, each (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local velocity concentration.

Starting from A, enlarge the separated family whenever a lifted retained tail descendant is separated from every profile already selected. If this enlargement can be carried out indefinitely, the successive lifted descendants already form an infinite chain of retained concentrating descendants. Otherwise the procedure stops after finitely many enlargements at a finite separated family A' which is maximal for this descendant-lifting procedure.

Form a finite directed graph whose vertices are the retained profiles in A'. Draw an edge $i \to j$ if a retained tail descendant of U_i, lifted by Lemma 10.11, is equivalent to U_j. There is no escape edge by the construction of A': an escape edge would be exactly another separated retained unit concentrating cylinder in the original sequence, so the enlargement procedure would not have stopped. Every decomposable vertex has an outgoing edge by the preceding paragraph.

A finite directed graph in which every vertex has an outgoing edge contains a directed cycle. Therefore, if no terminally indecomposable profile appears, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \to \mathfrak{s}_{j_1} \to \cdots \to \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. The nonconcentrating successor alternatives at each vertex are excluded by Lemma 10.14, while coexistence of a compact core with a diffuse remainder is handled by Theorem 9.113 and an escaping noncompact recentering is routed by Lemma 9.114. This proof is graph/recurrence based and does not use a global counting or mass budget. ☐

Lemma 10.64 (Finite recurrent loops of retained tail limits are impossible)**.** *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

Proof. Let
$$\mathfrak{s}_{j_0} \to \mathfrak{s}_{j_1} \to \cdots \to \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$
be such a loop, and write (U_ℓ, Π_ℓ) for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of U_ℓ. Thus, for each ℓ, there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Repeating the loop periodically gives an infinite chain of retained concentrating descendants. This contradicts Theorem 10.62. Therefore no finite recurrent loop of retained concentrating tail limits exists. □

Theorem 10.65 (Finite separated retained velocity concentration profile families are impossible)**.** *No refined residual candidate admits a finite separated family of retained velocity concentration profiles.*

Proof. Suppose, for contradiction, that a refined residual candidate admits a finite retained velocity concentration separated family $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$. By Lemma 10.63, either some component of the family is terminally indecomposable or the family generates an infinite chain of retained concentrating descendants. The first alternative contradicts Lemma 10.10. The second contradicts Theorem 10.62. Hence the finite retained velocity concentration separated family cannot exist. □

Proof of Proposition 10.5. Let U be a single retained velocity concentration terminal profile after the finite separated-family, mixed compact–diffuse, and noncompact escaping alternatives have been excluded. If U had a retained concentrating tail limit, Lemma 10.4 would produce a finite two-profile separated family with retained velocity concentration, contradicting Theorem 10.65. If U had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would be a mixed profile configuration consisting of the retained compact core U plus a diffuse or noncompact tail remainder. This is excluded in the diffuse case by Theorem 9.113, while the escaping noncompact case is routed by Lemma 9.114. Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and U is terminally indecomposable. □

Selected endpoint input production. The endpoint argument and the production of a terminally indecomposable state form two distinct dependency chains. The first chain conditionally excludes the current realized profile once terminal indecomposability has been obtained. The second chain shows how the terminal routing reaches that state. Every row below is instantiated only when the current profile, sequence, shift, family, or descendant chain has entered the state in its first column.

TABLE 14. Input production for the conditional exclusion of the selected terminally indecomposable profile. The last column contains consumers only.

Selected owner	**Existing producer and exact prerequisite**	**Produced fact for that owner**	**Later consumer only**
Current realized terminal profile U carrying retained local velocity concentration.	Theorem 1.13 supplies the root realization; Theorem 10.12 is applied at each realized descendant step.	This profile U carries positive retained local velocity concentration at its named threshold.	Lemma 10.10 uses this obstruction at the final contradiction.

Selected owner	Existing producer and exact prerequisite	Produced fact for that owner	Later consumer only
The same U, conditionally assumed or already proved to be terminally indecomposable.	Lemma 10.6 applies to this affine-normalized realized terminal profile with its retained concentration and terminal indecomposability.	There is a selected sequence $\tau_k \to -\infty$ such that $\sup_k \|U(\cdot,\tau_k)\|_{L^3(\mathbb{R}^3)} < \infty$.	Lemma 10.10 consumes this selected sequence.
The same U after its affine/parasitic alternative has been routed.	Theorem 10.8, through Proposition 10.7, transports the mild interface along this realized non-affine branch.	For each fixed $T < 0$, the physical pullback u^T of this profile is a bounded mild ancient solution.	Theorem 10.9 consumes this fact only after the selected sequence-L^3 input has been transported.
The selected pair (U, τ_k) and a fixed finite shift $T < 0$.	The critical-scaling calculation in Lemma 10.10 sets $t_k = -e^{-\tau_k}$ and, after discarding finitely many indices, $s_k = t_k - T$.	One has $s_k \to -\infty$ and $\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot,\tau_k)\|_{L^3(\mathbb{R}^3)}$.	Theorem 10.9 consumes this transported sequence.
The selected bounded mild pullback u^T carrying that sequence.	Theorem 10.9.	$u^T \equiv 0$.	The pullback step in Lemma 10.10.
The current centered profile U.	Lemma 10.10 uses the preceding conclusion for arbitrary finite $T < 0$.	$U \equiv 0$, contradicting the retained local velocity concentration of this same profile.	The finite separated-family exclusion and terminal routing consume the conditional no-atomic conclusion.

The sequence-L^3 bound in Table 14 is produced by Lemma 10.6; it is not an entry condition for an R2 or R3 profile. The mild row records only the non-affine alternative of Theorem 10.8. Its affine/parasitic alternative is routed to $\mathcal{C}_{\mathrm{aff}}$ for the current profile. The endpoint theorem acts on the selected finite-shift pullback u^T.

No pressure representative is transported between the rows of this endpoint directory. A pressure-loss case occurring in Lemma 10.6 uses the representative supplied on its own local chart and is routed by Lemma 9.5. Pressure is not an additional endpoint hypothesis.

TABLE 15. Production of the terminally indecomposable state for the selected terminal route. The conditional atomic exclusion is an already established consumer in the third row.

Selected input	Existing producer	Produced local conclusion	Later consumer only
An infinite retained descendant chain realized through the named lifting and heredity interfaces.	Theorems 10.17 and 10.62.	That selected chain is excluded.	Theorem 10.65 and the terminal routing.
A selected finite separated family.	Lemmas 10.13, 10.14 and 10.63 give the finite successor alternative; the already established Lemma 10.10 and Theorem 10.62 exclude its two outputs.	Theorem 10.65 excludes that selected family.	Proposition 10.5 and the terminal routing.

Selected input	**Existing producer**	**Produced local conclusion**	**Later consumer only**
The remaining selected single retained profile, after the finite separated, mixed compact–diffuse, and routed noncompact alternatives have been excluded for this profile.	Proposition 10.5; a retained concentrating tail limit would produce a finite two-profile family by Lemma 10.4.	This selected profile is terminally indecomposable.	The already proved conditional exclusion Lemma 10.10.
The selected terminal extraction after the preceding routes have been resolved for its current profile.	Theorem 10.66 consumes the preceding branch closures.	No retained terminal profile from this selected extraction remains.	The R2, R3, and generic residual closures and final assembly.

The finite separated-family theorem retains its finite directed successor graph and the independent no-infinite-chain and no-atomic inputs. Neither Proposition 10.5 nor Theorem 10.66 is a premise of that theorem. The two directories assert no atomic or residual-profile existence: their rows are conditional routes for the profile, sequence, family, or chain selected by the fixed terminal extraction. R2, R3, the setup residual theorem, and final assembly are downstream consumers of the produced conclusions.

The terminal endpoint argument uses the following acyclic order:

$$\begin{gathered}\text{concentration-set extraction}\\ \Downarrow\\ \text{diffuse, pressure, noncompact, and critical-tail closures}\\ \Downarrow\\ \text{sequence-}L^3\text{ for terminally indecomposable atomic profiles}\\ \Downarrow\\ \text{no atomic active profile by bounded-mild endpoint Liouville}\\ \Downarrow\\ \text{no finite separated family and no infinite chain}\\ \Downarrow\\ \text{terminal stratification assembly.}\end{gathered}$$

In particular, Lemma 10.6 does not use Theorem 10.66; it uses only terminal indecomposability and the already covered exterior, diffuse, pressure/noncompact, and critical-tail alternatives.

TABLE 16. Acyclic dependency table for the terminal concentration argument.

Result	**May use**	**Must not use**
Concentration-set extraction	Local Seregin compactness, pressure atlas, and the defect measures μ_n.	Terminal assembly theorem.
Diffuse-defect compactness	Observer compactification, concentration-neighborhood removal, and amenable averaging.	Atomic Liouville contradiction.
Critical-tail closure	Mesoscopic reduction, affine quotient, bounded-origin realization, and coherent tail classification.	Terminal assembly theorem.
Mixed compact–diffuse exclusion	Concentration-set extraction, diffuse-defect compactness, and critical-tail closure.	Finite separated-family theorem.

Result	May use	Must not use
Sequence-L^3 lemma	Terminal indecomposability and absence of retained velocity, diffuse, noncompact, pressure-only, and critical-tail descendants.	Albritton–Barker endpoint theorem.
Atomic contradiction	Sequence-L^3, finite-shift bounded mildness, and the Albritton–Barker endpoint theorem.	Terminal assembly theorem.
Finite separated-family exclusion	No atomic profile and no infinite retained velocity concentration chain.	Terminal assembly theorem.
Terminal stratification assembly theorem	All previous rows.	Itself.

Theorem 10.66 (Terminal stratification assembly theorem)**.** *Let (V_n, P_n) be a terminal residual sequence realized from the original blow-up sequence and satisfying the imported setup admissibility hypotheses, the pressure atlas hypotheses, and a retained unit velocity concentration lower bound*

$$\liminf_{n\to\infty} \mathcal{V}_1(V_n; z_n) \geq \eta_* > 0.$$

Assume all lower classes already excluded by the setup paper have been removed. The compact subthreshold regular residual alternative (b) *of Theorem 9.17 is excluded before the terminal stratification by Lemma 9.18 (CKN ε-regularity); hence it does not appear as a terminal residual alternative below. Then the paired terminal extraction of Theorem 9.24 gives the following ordered alternatives:*

(1) *a compact concentrating descendant occurs;*
(2) *a finite separated family of retained velocity concentration profiles occurs;*
(3) *an infinite chain of retained concentrating descendants occurs;*
(4) *a diffuse exterior defect occurs after concentration-neighborhood removal;*
(5) *a noncompact or pressure-gauge defect occurs;*
(6) *a critical-tail defect occurs;*
(7) *affine/parasitic routing occurs into $\mathcal{C}_{\mathrm{aff}}$;*
(8) *an indecomposable atomic concentration profile occurs and satisfies a sequence-L^3 bound.*

Alternative (1) *is a reduction alternative: by descendant heredity it is iterated inside the same normalized profile class until one reaches* (2)*,* (3)*, a mixed diffuse/noncompact/critical-tail branch, affine routing, or the atomic case* (8)*. Alternatives* (2)*–*(7) *are excluded by the preceding branch closures and local terminal arguments. Alternative* (8) *is excluded by the endpoint bounded-mild ancient Liouville theorem after finite terminal shift. Hence no terminal residual profile satisfying the retained unit velocity concentration lower bound remains.*

Proof. This is the assembly of the local results and explicit profile-class arguments in Sections 9 and 10. The local concentration decomposition is the paired concentration-set/remainder decomposition Theorem 9.24. Closed or continuum active loci are reduced to the single-profile, finite-separated, or infinite-chain cases by Lemma 9.14, so no additional continuum active-locus branch is hidden in the compact concentration set. This theorem does not introduce an additional compactness principle; it only combines the preceding extraction, reduction, and closure results in the order recorded in Table 16. The pure radiative alternatives enter only as sole-source exclusions, and are used only in that form through Theorem 9.27. Coexistence of a compact concentration core with a diffuse tail remainder is ruled out by Theorem 9.113, while escaping noncompact recenterings are routed by Lemma 9.114. Pressure-only or pressure/noncompact defects are assigned by Lemma 9.5. Critical tails are closed by Theorem 9.98 and Corollary 9.112.

Infinite concentration chains and finite recurrent loops are ruled out by Theorem 10.62, while finite separated concentration families are ruled out by Theorem 10.65. Therefore any retained concentration terminal profile that survives the ordered lower-class removals is single and has no retained concentrating descendant or diffuse/noncompact tail remainder. By Proposition 10.5 it is terminally indecomposable, and then Lemmas 10.6 and 10.10 and Theorem 10.8 give the sequence-L^3, parasitic-free finite-shift mildness, and endpoint contradiction. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence-L^3 conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. □

Remark 10.67 (No global budget in the terminal assembly theorem)*.* The terminal stratification assembly theorem assumes no global L^3, global CKN budget, finite energy bound on expanding regions, or global counting measure for concentration profiles. Compactness and recurrence are local, normalized, or realized from the original blow-up sequence.

Corollary 10.68 (Rotational residual closure)**.** *The ordered rotational relative-equilibrium residual class is empty:*

$$\mathcal{C}_{\mathrm{rot}}(\mathcal{S}; I, J) = \varnothing.$$

Proof. By Proposition 5.8, every R2 residual either belongs to a previously closed lower class or produces a terminal concentration profile realized from the original blow-up sequence with retained unit velocity concentration. The lower classes are empty by the imported setup results, and the terminal concentration case is empty by Theorem 10.66. Hence no R2 residual remains. □

Corollary 10.69 (Stationary-hull residual closure)**.** *For every admissible stationary phase space* $\mathfrak{B}$*,*

$$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}(\mathcal{S}; I, J; \mathfrak{B}) = \varnothing.$$

Proof. By Propositions 6.12 and 7.6, every R3 stationary-hull residual either lies in a lower class already excluded by the setup paper or produces a stationary terminal concentration profile realized from the original blow-up sequence. The lower classes are empty by the imported setup results. The stationary terminal profile satisfies the retained unit velocity concentration hypotheses of Theorem 10.66, so it is reduced to the atomic alternative only after all separated, diffuse, noncompact, critical-tail, and affine alternatives have been removed. In that atomic case Lemmas 10.6 and 10.10 and Theorem 10.8 give the sequence-L^3 bound, finite-shift bounded mildness, and the endpoint Albritton–Barker contradiction. Hence no stationary terminal concentration profile remains. This proves the deferred stationary concentration and degenerate stationary-family occurrence statements Remarks 7.4 and 7.5 and Proposition 6.10, and therefore the R3 class is empty. □

11. Closure of the generic terminal residual

Definition 11.1 (Generic residual profile)**.** A generic residual profile is a smooth bounded normalized Seregin ancient solution of (1.2) which carries persistent compact velocity L^3 concentration and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$, and none of the structured or tight classes removed earlier, including the ordered critical-tail lower strata $\mathcal{C}_{\log\mathrm{diff}}, \mathcal{C}_{\mathrm{Young}}, \mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$.

Theorem 11.2 (Generic terminal exhaustion)**.** *Let* $(U, \Pi) \in \mathcal{C}_{\mathrm{gen}}$ *be represented after canonical affine/parasitic normalization and after removal of the lower alternatives already excluded by the setup paper. Then a terminal extraction of* (U, Π)*, with the priority order used throughout the terminal section, selects one of the following priority alternatives:*

(1) *a retained compact active descendant;*

(2) *a finite separated family of retained velocity profiles;*
(3) *an infinite chain of retained active descendants;*
(4) *a diffuse exterior defect after active-neighborhood removal;*
(5) *a noncompact or pressure-gauge defect;*
(6) *a critical-tail defect;*
(7) *a single terminally indecomposable retained velocity profile.*

This is a concentration-compactness exhaustion of the realized terminal sequence, not a convention in the definition of $\mathcal{C}_{\text{gen}}$.

Proof. The generic representative is first put in the affine-normalized quotient. If all non-affine oscillation disappears, then Theorem 8.21 places the profile in the affine/parasitic lower stratum, contradicting the assumption that we are in $\mathcal{C}_{\text{gen}}$. Thus a positive non-affine oscillation threshold survives, while the setup results supply retained compact velocity concentration.

Apply the paired terminal concentration-set/remainder decomposition Theorem 9.24 together with the compact-window to expanding-region protocol Theorem 9.17. The retained velocity mass forces at least one selected active or residual alternative by Corollary 9.25. If active neighborhoods persist, the priority order first records a compact retained descendant; iterating the same realized-descendant construction either produces a finite separated retained family, an infinite active chain, or terminates at a single retained profile with no further retained active descendant. If active neighborhoods do not exhaust the residual measure on the selected terminal regions, the protocol selects a diffuse exterior defect, a loss of local compactness or pressure-gauge compactness, or a scale-critical tail window. These alternatives are (4)–(6). If none of the separated, chained, diffuse, noncompact, or critical-tail remainders occurs, then the remaining single profile is terminally indecomposable by Proposition 10.5. The priority order makes the selected alternative unique. □

Theorem 11.3 (Closure of the generic residual class by local concentration analysis)**.** *One has*

$$\mathcal{C}_{\text{gen}} = \varnothing.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{gen}}$. Apply Theorem 11.2. A retained compact active descendant is not a terminal stopping case: by descendant heredity it is reinserted into the same ordered terminal extraction until it gives a finite separated family, an infinite active chain, a diffuse/noncompact/critical-tail remainder, or a single indecomposable retained profile. Finite separated families are excluded by Theorem 10.65, and infinite active chains by Theorem 10.62. Diffuse remainders are excluded by Theorems 9.27 and 9.113. Escaping noncompact remainders are routed by Corollary 9.11 and Lemma 9.114 to the diffuse, pressure/noncompact, or critical-tail branches; those routed branches are closed by the corresponding theorems cited here. Critical tails are excluded by Theorem 9.98 and Corollary 9.112. The pressure-only case is assigned by Lemma 9.5 into the pressure/noncompact or affine/parasitic lower stratum, both already removed. In the remaining single-profile case, Proposition 10.5 gives terminal indecomposability and Lemma 10.10 gives the endpoint contradiction. Every selected alternative in the generic terminal exhaustion is impossible, so $\mathcal{C}_{\text{gen}} = \varnothing$. □

12. Final assembly

The following display is the closure view of the refined residual states. It records the eventual closing mechanism for each state, rather than the order in which the required theorems are proved. The acyclic theorem-production order is given in Tables 4 and 16, and the state-to-state interfaces are given in Table 2. The rotational relative-equilibrium arrow is the localized Coriolis-flux reduction followed by terminal assembly; the terminal concentration-set compactness, finite separated-family exclusion, and indecomposability steps are proved in this paper from local estimates:

$$\mathcal{C}_{\mathrm{ax}} \xrightarrow{\text{r-Liouville}} \varnothing,$$
$$\mathcal{C}_{\mathrm{rot}} \xrightarrow{\text{local flux reduction + terminal assembly}} \varnothing,$$
$$\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}} \xrightarrow{\text{stationary-hull reduction + terminal assembly}} \varnothing,$$
$$\mathcal{C}_{\mathrm{aff}} \xrightarrow{\text{affine/parasitic routing}} \text{no retained normalized representative},$$
$$\mathcal{C}_{\log\mathrm{diff}} \xrightarrow{\text{log-window support profiles + hidden-scale closure}} \varnothing,$$
$$\mathcal{C}_{\mathrm{Young}} \xrightarrow{\text{hidden-scale variance exclusion}} \varnothing,$$
$$\mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} \xrightarrow{\text{bounded-origin realization + log-hull minimality}} \varnothing,$$
$$\mathcal{C}_{\mathrm{gen}} \xrightarrow{\text{critical-tail closure + terminal concentration analysis}} \varnothing.$$

The normalized profile selected from the fixed extraction carries retained local velocity concentration. A descendant row uses this obstruction only after the cited realization and heredity interfaces transport it to that descendant.

Final selected-owner directory. The following table records the closure view for the fixed incoming profile and its setup ancestry ledger. Each row is conditional on the current selected datum entering the state in the first column. The dependency-status and consumer columns record no premise for an earlier row.

TABLE 17. Selected-owner directory for final residual assembly. Obstructions are recorded only for the datum whose cited producer supplies them.

Current owner or state	**Entry or producer**	**Owned obstruction**	**Closure or reduction**	**Status**	**Later consumer only**
Arbitrary incoming candidate V carrying the fixed setup ledger, under the antecedent $V \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S})$.	Proposition 1.5.	The root retained local velocity concentration supplied by the fixed extraction.	Affine routing, or entry of the same candidate's canonical representative into the refined decomposition.	Conditional handoff.	Applied contrapositively after the affine and refined rows are closed.
Current canonical representative selected into $\mathcal{C}_{\mathrm{ax}}$.	Definition 1.10, instantiated to this representative.	The retained obstruction already recorded for this profile.	Theorem 4.4.	Independent closure.	Theorem 1.16.
Current profile selected into $\mathcal{C}_{\mathrm{rot}}$.	Proposition 5.8.	Retained velocity only on the selected realized terminal sequence, through its cited ancestry and heredity interfaces.	Theorem 10.66, followed by Corollary 10.68.	Terminal output; no sequence-L^3 entry input.	Theorem 1.16.

Current owner or state	**Entry or producer**	**Owned obstruction**	**Closure or reduction**	**Status**	**Later consumer only**
Current profile selected into $\mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}$.	Propositions 6.12 and 7.6.	Retained velocity only on the selected realized terminal sequence.	Theorem 10.66, followed by Corollary 10.69.	Terminal output; the stationary sequence-L^3 subcase is produced there.	Theorem 1.16.
Current affine/parasitic representative.	Theorem 8.21.	Its selected non-affine oscillation, when retained.	Theorem 9.99.	Quotient routing.	The refined closure; no retained normalized representative remains in this row.
Current realized coherent critical tail.	Corollary 9.96 and Theorem 9.104.	Its selected annular non-affine activity and realized bounded-origin interface.	Corollary 9.111.	Produced before hidden-scale and log-diffuse closure.	The hidden-scale, log-diffuse, and complete critical-tail rows.
Current realized Young or pressure/viscous defect alternative.	Theorem 9.88 for its selected tight annular blow-down and Lemma 9.95 for its chartwise coordinates.	The Young variance or named defect coordinate produced for this selected datum.	Corollary 9.100.	Selected-datum closure.	Corollary 9.112.
Current realized log-diffuse defect and its selected normalized log windows.	Lemmas 9.54 and 9.55.	Positive normalized window mass and a support point only when the cited theorem produces them.	Corollary 9.57.	Uses hidden-scale exclusion and the already closed coherent row.	Corollary 9.112.
Current realized critical-tail profile.	Theorem 9.98, instantiated to that profile.	Only the selected tail obstruction recorded by its producer.	Corollary 9.112.	Completed tail closure.	The terminal and generic routes.
Current generic profile and its selected terminal extraction.	Theorem 11.2.	Retained velocity transported along the named realization chain; sequence-L^3 and mildness only at the selected indecomposable endpoint.	The named branch closures in the proof of Theorem 11.3.	Sibling of the R2/R3 terminal consumer.	Theorem 1.16.
Refined decomposition of the current canonical representative.	The ordered routing in Definition 1.10 and the preceding row closures.	No new obstruction is assigned at assembly.	Theorem 1.16.	All producer rows already proved.	The exact raw-residual handoff.
The same arbitrary incoming candidate V, under the antecedent $V \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S})$.	Proposition 1.5, used contrapositively after affine and refined closure.	The original fixed-ledger retained obstruction.	Theorem 12.1 proves $V \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S}) \Rightarrow \perp$.	Literal raw-residual closure.	Corollary 12.2.

Current owner or state	**Entry or producer**	**Owned obstruction**	**Closure or reduction**	**Status**	**Later consumer only**
The proved setup residual statement.	Corollary 12.2.	No new analytic input.	The literal setup raw-residual statement.	Downstream handoff complete.	Corollary 12.3 and the setup final assembly.

The obstruction column is not a uniform inheritance column. Root velocity concentration belongs to the fixed extraction. A descendant receives retained velocity only through its cited realization and heredity interfaces. A defect window receives normalized mass and a support point only through its named normalization and support-transfer theorems. A terminal profile receives sequence-L^3 and finite-shift mildness only through Table 14. Pressure remains attached to the local chart used in the corresponding row.

Equivalently, the final exclusion is the composition of eight ordered steps:

(1) *Entry step:* The setup paper supplies a positive local velocity concentration sequence; the endpoint weak Serrin local Type I bound Theorem 1.26 gives the terminal local energy/no-escape estimate in Theorem 1.28. Thus the sequence is admissible in the sense of Definition 1.19, and Theorem 1.7 turns it into a normalized Seregin profile with retained compact velocity concentration.

(2) *Stratification step:* the ordered residual decomposition places every remaining retained profile in the complete list of alternatives

$$\mathcal{C}_{\mathrm{ax}} \cup \mathcal{C}_{\mathrm{rot}} \cup \mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}} \cup \mathcal{C}_{\mathrm{aff}} \cup \mathcal{C}_{\log\mathrm{diff}} \cup \mathcal{C}_{\mathrm{Young}} \cup \mathcal{C}_{\mathrm{homcrit}} \cup \mathcal{C}_{\mathrm{logper}} \cup \mathcal{C}_{\mathrm{ap}} \cup \mathcal{C}_{\mathrm{gen}}.$$

(3) *Closed-strata step:* the small, stationary L^3, tight, closed structured/decay, axisymmetric, rotational, and stationary-hull strata are excluded by the results cited above.

(4) *Affine-routing step:* oscillatory entry normalization routes every affine/parasitic representative to $\mathcal{C}_{\mathrm{aff}}$, so no retained normalized representative lies in that quotient row.

(5) *Coherent critical-tail step:* the coherent rows $\mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}$ are excluded by bounded-origin realization and log-hull minimality, independently of the later hidden-scale/log-diffuse arguments.

(6) *Hidden-scale step:* minimal mesoscopic reduction yields Theorem 9.88. This excludes the Young branch $\mathcal{C}_{\mathrm{Young}}$ by Corollary 9.100 and, together with log-window support-profile selection and the already closed coherent rows, excludes $\mathcal{C}_{\log\mathrm{diff}}$ by Corollary 9.57.

(7) *Residual terminal step:* the generic residual is reduced by the paired concentration-set decomposition, diffuse-defect compactness, diffuse-defect trichotomy, critical-tail compactification, realized critical-tail rigidity, descendant heredity, active-chain exclusion, finite separated-family exclusion, and terminal indecomposability.

(8) *Endpoint Liouville step:* a terminally indecomposable retained profile is either routed to the affine/parasitic lower stratum or has a parasitic-free bounded mild physical pullback with a backward L^3-bounded sequence. In the latter case it is zero by Albritton–Barker.

Proof of Theorem 1.16, revisited. Fix an arbitrary centered profile V carrying the fixed setup ancestry ledger. To prove $\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$, it is enough to establish

$$V \in \mathcal{R}^{\#}(\mathcal{S}; I, J) \quad \Longrightarrow \quad \perp.$$

Assume the antecedent. By the definition of the retained canonical class and the setup-paper export interface, this same V carries its retained compact velocity obstruction, the normalization data, the setup ancestry, and the pressure representative appropriate to each local chart. The refined decomposition is a routing datum for this same profile: it does not change the setup export, and any later descendant or quotient profile is linked to V by the cited realization and inheritance theorem. The refined ordered decomposition selects for V one of

$$\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}, \mathcal{C}_{\mathrm{aff}}, \mathcal{C}_{\mathrm{log\,diff}}, \mathcal{C}_{\mathrm{Young}}, \mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}, \mathcal{C}_{\mathrm{gen}}.$$

If V is selected into $\mathcal{C}_{\mathrm{ax}}$, Theorem 4.4 excludes this profile. If it is selected into $\mathcal{C}_{\mathrm{rot}}$, Proposition 5.8 produces the terminal route for this profile and Corollary 10.68 excludes it. If it is selected into $\mathcal{C}_{\mathrm{stat}}^{\mathrm{ret}}$, Propositions 6.12 and 7.6 produce its terminal route and Corollary 10.69 excludes it. If its canonical representative is selected into the affine/parasitic quotient row $\mathcal{C}_{\mathrm{aff}}$, Theorem 9.99 leaves no retained normalized representative in that row. If V is selected into the log-diffuse or Young row, the corresponding closure theorem excludes the selected datum by Corollaries 9.57 and 9.100. If it is selected into a coherent critical-tail row, Corollary 9.111 excludes the realized tail. Finally, if it is selected into $\mathcal{C}_{\mathrm{gen}}$, Theorem 11.3 excludes its selected generic terminal route.

Every possible selection contradicts the antecedent. Since V was arbitrary, $\mathcal{R}^{\#}(\mathcal{S}; I, J) = \varnothing$. Together with the affine-row closure, Proposition 1.5 now applies contrapositively to any incoming profile W carrying the fixed setup ledger:

$$W \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S}) \quad \Longrightarrow \quad \perp.$$

Consequently $\mathcal{R}_{\mathrm{setup}}(\mathcal{S}) = \varnothing$. The proof therefore uses residual membership only as the antecedent of an implication and closes the row selected for that same profile; it neither assumes a residual object nor replaces the raw setup residual by a derived handoff set. $\square$

Theorem 12.1 (Type I residual closure)**.** *Fix a candidate Type I singular point, its incoming normalized Seregin profile, and the single setup ancestry ledger determining the hull*

$$\mathcal{S} = \mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*).$$

Then the residual set for this fixed local ledger satisfies

$$\mathcal{R}_{\mathrm{setup}}(\mathcal{S}) = \mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big) = \varnothing.$$

For every incoming centered profile V carrying this fixed ancestry ledger, the proof establishes

$$V \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S}) \quad \Longrightarrow \quad \perp.$$

Thus no residual-existence premise is used. Membership is only the antecedent of the implication: a false antecedent is already harmless, while a true antecedent is excluded by the local routing and closure theorems.

Proof. Fix an arbitrary incoming centered profile V carrying the fixed setup ledger. It suffices to prove

$$V \in \mathcal{R}_{\mathrm{setup}}(\mathcal{S}) \quad \Longrightarrow \quad \perp.$$

Assume the antecedent. By Proposition 1.5, exactly one of the two local routing alternatives occurs for this same V. The affine/parasitic alternative is closed by the quotient-row conclusion of that proposition, with Definition 8.16 and Remark 8.17; it cannot be a retained normalized residual profile. In the other alternative, its canonical representative belongs to $\mathcal{R}^{\#}(\mathcal{S}; I_{\mathrm{setup}}, J_{\mathrm{res}})$, which is empty by Theorem 1.16. Both alternatives are impossible. Thus the antecedent implies a contradiction. Since V was arbitrary, $\mathcal{R}_{\mathrm{setup}}(\mathcal{S}) = \varnothing$. $\square$

Corollary 12.2 (Verification of the setup residual-closure theorem)**.** *The companion residual-closure statement Theorem 16.1 of the setup paper holds with its literal raw-residual meaning.*

Proof. Fix the incoming profile, raw generated hull, and normalization data as in Definition 1.4. The exact conclusion of Theorem 12.1 is

$$\mathcal{R}\big(\mathcal{S}_{\text{raw-gen}}(u, p, z_*; R_0, \eta_*)\big) = \varnothing.$$

The hull and normalization data are exactly those in Theorem 16.1 of the setup paper; therefore the companion statement is proved without an additional premise. $\square$

Corollary 12.3 (Local pointwise Type-I exclusion from the two-paper chain)**.** *By the setup paper's final assembly Theorem 33.1 of the setup paper and Corollary 12.2, no local pointwise Type I singularity exists under the stated admissibility hypotheses.*

Proof. This is the downstream setup assembly listed in Table 1, with the residual-closure theorem supplied by Corollary 12.2. It is invoked only after that producer has been proved. ☐

Remark 12.4 (Inherited inputs and new arguments). The paper inherits the base setup, including local concentration, the local Type I Type II dichotomy, the Type I Seregin extraction theorem, and the initial structural decomposition, from the setup paper summarized in Theorem 1.3. All remaining case-closure steps are proved in this paper using local concentration and terminal dynamics:

- terminal concentration-set extraction;
- compactness of descendant/tail families and pressure controls;
- recurrence for chains of retained velocity concentration profiles;
- finite separated-family exclusion;
- oscillatory-entry normalization;
- hidden-scale compactness reduction for critical tails;
- bounded-origin realization and critical-tail alternative exclusion;
- and the compactness and alternative exclusions culminating in Theorem 11.3.

Remark 12.5 (No hidden global Liouville or critical-norm input). The residual closure does not invoke any global critical-norm bound on the original residual class, any global Coriolis–Leray Liouville theorem for bounded centered profiles, or any global Liouville theorem for bounded ancient solutions of the centered equation. The only external Liouville input used after residual closure is the Albritton–Barker endpoint Liouville theorem Theorem 10.9, applied in exactly two places. First, the local weak-Serrin Type I estimate Theorems 1.26 and 1.28 upgrades the local pointwise Type I envelope to terminal velocity no-escape; this is an entry implication, optional inside Paper II in the sense of Section 1.9. Second, in Lemma 10.10, the endpoint ancient Liouville theorem is applied to a parasitic-free bounded mild physical pullback that carries a backward L^3-bounded sequence supplied by indecomposable sequence-L^3 completeness Lemma 10.6. No other global Liouville theorem and no global critical-norm estimate is used in any of the closure steps for $\mathcal{C}_{\mathrm{ax}}, \mathcal{C}_{\mathrm{rot}}, \mathcal{C}^{\mathrm{ret}}_{\mathrm{stat}}, \mathcal{C}_{\mathrm{aff}}, \mathcal{C}_{\mathrm{log\,diff}}, \mathcal{C}_{\mathrm{Young}}, \mathcal{C}_{\mathrm{homcrit}}, \mathcal{C}_{\mathrm{logper}}, \mathcal{C}_{\mathrm{ap}}, \mathcal{C}_{\mathrm{gen}}$. In particular, the rotational closure Corollary 10.68 proceeds through the localized Coriolis-flux identity and terminal concentration-set decomposition, and the variance/critical-tail closures proceed through the minimal mesoscopic-scale reduction theorem Theorem 9.86 together with bounded-origin realization Theorem 9.104.

Appendix A. Setup Paper Label Dictionary

The imported setup results in Theorem 1.3, together with the downstream setup consumers in Table 1, are cited by their companion-paper theorem numbers. This table records that dictionary in one place so that Paper II can be read using the public theorem numbering and the imported statements displayed here.

Theorem 5.7 of the setup paper	Positive local concentration at a Type I singular point.
Corollary 5.8 of the setup paper	Positive velocity concentration at a singular terminal point.
Theorem 8.4 of the setup paper	Local weak-Serrin/Type I implication.
Corollary 8.5 of the setup paper	Admissibility of local Type I sequences.
Lemma 8.7 of the setup paper	Local Seregin extraction.
Theorem 9.1 of the setup paper	Normalized Seregin extraction theorem.
Proposition 11.12 of the setup paper	Ancient suitable limit.

Proposition 12.2 of the setup paper	Nonzero retained velocity limit.
Lemma 11.2 of the setup paper	Pressure compactness.
Lemma 11.3 of the setup paper	Pressure gauge compatibility.
Lemma 11.4 of the setup paper	Pressure atlas.
Definition 13.14 of the setup paper	Setup mild-stratum definition.
Lemma 13.17 of the setup paper	Stability of the setup mild stratum under time translations and uniformly bounded locally smooth limits.
Lemma 15.3 of the setup paper	Projected mild identity on the setup mild stratum.
Lemma 20.4 of the setup paper	Stability of mildness for uniformly bounded locally smooth limits of mild profiles.
Lemma 25.4 of the setup paper	Finite-shift physical mildness for a bounded mild centered profile.
Theorem 15.7 of the setup paper	Small and stationary lower classes.
Theorem 15.8 of the setup paper	Uniformly tight raw-state class closure.
Theorem 15.9 of the setup paper	Known structured/decay classes.
Theorem 26.1 of the setup paper	No uniformly tight normalized collection.
Theorem 25.8 of the setup paper	Tight Liouville theorem.
Theorem 25.5 of the setup paper	Bounded mild ancient endpoint theorem.
Theorem 16.1 of the setup paper	Raw residual-closure theorem proved here.
Theorem 33.1 of the setup paper	Local final assembly in the setup paper.

Paper II uses these results only through the imported theorem statements and the companion-paper numbering displayed in this appendix.

Appendix B. Generator calculus on compact recurrent cores

This appendix collects the operator-theoretic facts used implicitly in the amenable-averaging and pressure-observable arguments of the body of the paper. The setting is a compact metrizable invariant minimal core (M, μ, Φ) of the centered translation hull, where μ is a G-invariant Borel probability measure (existence follows from compactness together with the amenable-averaging protocol of Proposition 9.36; uniqueness is not used). The acting group is $G := \mathbb{R}^3 \rtimes \mathbb{R}$ (covariant translations and parabolic time shift).

B.1. **Hilbert space and unitary representation.** Let $H := L^2(M, \mu; \mathbb{C})$. The Koopman representation $U_g f := f \circ \Phi_g^{-1}$ ($g \in G$) is a strongly continuous unitary representation of G on H, by G-invariance of μ and the dominated convergence theorem. In the pressure argument Π_0 denotes the orthogonal projection onto the subspace invariant under the three spatial covariant translations X_1, X_2, X_3. This is the zero spatial-frequency projection used in Lemma 8.28. It is not the projection onto the full spacetime-invariant subspace; time translations are used only to average total time derivatives. The arguments below require only this spatial projection, not ergodicity or uniqueness of μ.

For each one-parameter subgroup $\{e^{tX_k}\}_{t\in\mathbb{R}}$ of G ($k = 0, 1, 2, 3$: time translation X_0 and three covariant spatial translations X_1, X_2, X_3), Stone's theorem gives a self-adjoint generator A_k on a dense domain $\mathcal{D}(A_k) \subset H$ with $U_{e^{tX_k}} = e^{itA_k}$. The pressure inverse uses only the spatial generators. Define the non-negative self-adjoint operator $\mathcal{L}_{\mathrm{sp}}$ by the closed quadratic form

$$\mathfrak{q}_{\mathrm{sp}}(f) := \sum_{k=1}^{3} \|A_k f\|^2_{L^2(M,\mu)}, \qquad f \in \bigcap_{k=1}^{3} \mathcal{D}(A_k).$$

Thus $\mathcal{L}_{\mathrm{sp}}$ is the Friedrichs generator associated with the formal operator

$$\mathcal{L}_{\mathrm{sp}} = \sum_{k=1}^{3} A_k^2,$$

and $\mathcal{L}_{\mathrm{sp}}\Pi_0 = \Pi_0\mathcal{L}_{\mathrm{sp}} = 0$. All pressure inverses below are understood as resolvent limits of $(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1}(I - \Pi_0)$. Equivalently, with the skew-adjoint infinitesimal generators

$$D_k := iA_k,$$

the non-negative spatial generator is

$$\mathcal{L}_{\mathrm{sp}} = -\sum_{k=1}^{3} D_k^2$$

on the common cylinder core. This is the operator denoted by $-\sum_{k=1}^{3} D_k^2$ in the recurrent-core pressure argument.

B.2. **Pressure observables and bounded continuous core.** The pressure observables used in Definition 9.33 (iv)–(v) (compatible-gauge averaged pressure differences $\Pi(\cdot) - a(\cdot)$ integrated against compactly supported test functions, plus the affine quotient observables $[\Pi] \in D_1$) define bounded measurable functions $b_{ij}^{(\varphi,a)} : M \to \mathbb{R}$ for each test function $\varphi \in C_c^\infty$ and each gauge selection a compatible with Definition 8.14. These observables generate a unital C^*-subalgebra $\mathcal{B} \subset C(M)$ which is dense in $L^2(M,\mu)$. The restriction of the unitary representation U_g to $\mathcal{B}$ is the Koopman action by composition. Affine pressure modes have already been separated into the affine/parasitic stratum $\mathcal{C}_{\mathrm{aff}}$ before this calculus is invoked; the observables below are therefore taken modulo the affine quotient fixed in Definition 8.14.

B.3. **Regularized inverse and distributional pressure identity.** For each $\varepsilon > 0$ and each spatial generator A_k, $k = 1, 2, 3$, define the regularized distributional pressure functional by

$$\left\langle q_{k,\varepsilon}^{(\varphi,a)}, F \right\rangle := \left\langle (I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}, (\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1} D_k^* F \right\rangle$$

for cylinder test observables $F \in \mathcal{B}$. Here $D_k = iA_k$ is the skew-adjoint generator, $D_k^* = -D_k$, and $D_i D_j b_{ij}$ is interpreted in the weak generator sense on the cylinder core. The operator $(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1}$ is bounded on H with norm $\le \varepsilon^{-1}$ and commutes with Π_0; each D_k is closed on $\mathcal{D}(A_k)$, and $(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1} D_k^* F$ lies in the form domain. Therefore the displayed pairing is well-defined for every $\varepsilon > 0$, φ, a, and $F \in \mathcal{B}$. If the weak source $(I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$ is represented by an L^2 function, then this functional is represented by the corresponding H-element; the arguments in the paper need only the weak pairing. In the formal notation used in the body this is

$$q_{k,\varepsilon}^{(\varphi,a)} = D_k(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1}(I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$$

understood through the preceding weak pairing.

Lemma B.1 (Distributional pressure identity in the generator calculus)**.** *For every test function $\varphi \in C_c^\infty$ and every compatible gauge a, the family $\{q_{k,\varepsilon}^{(\varphi,a)}\}_{\varepsilon>0}$ has a distributional limit along the cylinder core as $\varepsilon \downarrow 0$. The limit is independent of the chosen gauge a in the affine quotient of Definition 8.14 and represents the centered distributional pressure identity $\Pi(\cdot) - a(\cdot) = R_i R_j(\chi u_i u_j) + h$ of Lemma 9.6 averaged against μ.*

Proof. The spectral theorem applied to $\mathcal{L}_{\mathrm{sp}}$ gives

$$D_k(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1}(I - \Pi_0) \xrightarrow{\varepsilon \downarrow 0} D_k \mathcal{L}_{\mathrm{sp}}^{-1}(I - \Pi_0)$$

in the weak resolvent sense on the form domain, after projecting away the invariant kernel. For $F \in \mathcal{B}$, the test $(\varepsilon + \mathcal{L}_{\mathrm{sp}})^{-1} D_k^* F$ converges in the form topology to $\mathcal{L}_{\mathrm{sp}}^{-1} D_k^* F$ modulo $\Pi_0 H$. Hence the displayed weak pairing with $(I - \Pi_0) D_i D_j b_{ij}^{(\varphi,a)}$ converges and defines the distributional limit $q_k^{(\varphi,a)}$. Gauge independence in the affine quotient is the statement that two compatible gauges a, a'

differ by an affine pressure mode whose distributional contribution to $D_iD_jb_{ij}$ vanishes (because affine functions are killed by two derivatives). The identification with the centered distributional pressure identity is the statement that, in a fixed local covariant chart, the operators D_k on M restrict to the distributional spatial derivatives along the chart and $\mathcal{L}_{\mathrm{sp}}^{-1}(I - \Pi_0)$ restricts to the inverse Laplacian on the centered ancient solution modulo the spatially invariant pressure kernel; this restriction is the content of the local pressure atlas (Theorem 1.3 (3)). □

Remark B.2 (Gauge independence in the affine quotient)*.* The limit $q_k^{(\varphi,a)}$ depends on the gauge a only through its affine quotient $[a] \in D_1$ of Definition 8.14. If two compatible gauges a, a' give the same affine quotient, then $a - a'$ is the time-only contribution allowed by the gauge atlas, and the pressure identity in Lemma B.1 differs by a quantity in the kernel of D_iD_j, which is killed in the regularized formula and therefore does not contribute to the limit. This is the precise statement underlying the use of pressure observables in the amenable-averaging arguments of Theorem 9.34 and Theorem 9.29: averaging is performed on the gauge-quotient pressure observables, and the regularized inverse $q_{k,\varepsilon}$ provides a regularized weak representative for each fixed ε before the limit $\varepsilon \downarrow 0$ gives the gauge-independent pressure identity.

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